



Thesis Project by Group CLS

THESIS 1-3

Thesis Project

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Section 1. Introduction

1.1 Problem Overview

Large-scale global optimization (LSGO) presents a critical challenge in computational intelligence, where the "curse of dimensionality" leads to an explosive increase in local optima and a high risk of premature convergence. While foundational algorithms like Differential Evolution (DE) are robust, their dependence on crossover and random mutation may restrict guided exploration — reducing particle diversity over time and leading to premature convergence.

This notebook implements and evaluates the proposed **Diversity-Preserving Level-Based Learning Differential Evolution (DP-LLDE)** algorithm, tested against the **CEC 2017 Large-Scale Global Optimization Benchmark** across 29 benchmark functions (F1-F30, excluding deprecated F2) at **10 dimensions**.

1.2 Research Objectives

1. Design and implement DP-LLDE, integrating sequential relative particle mutation and dynamic level-based learning into the Differential Evolution framework.
2. Evaluate DP-LLDE on the CEC 2017 benchmark suite (F1-F30) at 10 dimensions.
3. Examine convergence behavior, solution quality, and the effect of diversity-preserving strategies across unimodal, multimodal, hybrid,

and composition functions.

1.3 Algorithm Overview - DP-LLDE

DP-LLDE extends LBLDE (Qiao et al., 2022) with three core contributions:

1. **Sequential Level Assignment** — each individual is assigned a learning source level drawn randomly from $\{2, \dots, NL\}$ on the first generation, decaying toward the elite level (level 1) in subsequent generations.
2. **Diversity-Aware Exemplar Selection** — if two exemplars e_1, e_2 fall below a diversity threshold, one is replaced with the most distant candidate in that level, preventing exemplar collapse.
3. **Normalized Mutation Pressure (Eq. 10):**

$$V_i = e_1 + F_i \times (e_2 - X_i) \times \frac{NL - k}{NL}$$

The factor $(NL-k)/NL$ decays to zero as $k \rightarrow NL$, naturally reducing mutation pressure as learning quality improves.

Parameter	Value
Population Size (NP)	100
Number of Levels (NL)	4
Initial μ_{CR}	0.35
Max FES	$10,000 \times D$
Dimensions (D)	10
Seed	42

Section 2. Benchmark Function Description

2.1 Overview

The **CEC 2017 Large-Scale Global Optimization Benchmark** (Awad et al., 2017) is a standardized suite of 30 test functions designed to evaluate optimization algorithms on complex, high-dimensional search spaces. The benchmark tests an

algorithm's ability to avoid false local optima, convergence speed, and solution accuracy.

All functions are defined over the search space $[-100, 100]^D$ with their respective global optima offset by a function-specific bias value.

Library used: `cec2017` by Duncan Tilley — provides the official shift vectors, rotation matrices, and shuffle permutations from the CEC 2017 specification.

2.2 Unimodal Function Overview

Unimodal functions contain a **single global optimum and no local optima**. They are used to test the convergence speed and accuracy of an algorithm in a straightforward landscape.

No.	Function Name	Global Optimum
F1	Shifted and Rotated Bent Cigar	100
F2	Shifted and Rotated Sum of Different Power (<i>Deprecated</i>)	200
F3	Shifted and Rotated Zakharov	300

Note: F2 is officially deprecated from the CEC 2017 benchmark but is included here for completeness as it remains implemented in the `cec2017` library.

2.3 Multimodal Function Overview

Multimodal functions contain **multiple local optima**, making them significantly harder to solve. They are used to evaluate an algorithm's capacity to avoid premature convergence and escape false local optima.

No.	Function Name	Global Optimum
F4	Shifted and Rotated Rosenbrock's	400
F5	Shifted and Rotated Rastrigin's	500
F6	Shifted and Rotated Expanded Schaffer's F6	600
F7	Shifted and Rotated Lunacek Bi-Rastrigin	700
F8	Shifted and Rotated Non-Continuous Rastrigin's	800

No.	Function Name	Global Optimum
F9	Shifted and Rotated Levy	900
F10	Shifted and Rotated Schwefel's	1000

2.4 Hybrid Function Overview

Hybrid functions divide the search space into several regions, each governed by a **different basic function**. They assess an algorithm's adaptability to heterogeneous landscapes and mixed search behaviors.

No.	Component Functions	Global Optimum
F11	Zakharov; Rosenbrock; Rastrigin	1100
F12	Elliptic; Schwefel; Bent Cigar	1200
F13	Bent Cigar; Rosenbrock; Bi-Rastrigin	1300
F14	Elliptic; Ackley; Schaffer F7; Rastrigin	1400
F15	Bent Cigar; HGBat; Rastrigin; Rosenbrock	1500
F16	Schaffer F6; HGBat; Rosenbrock; Schwefel	1600
F17	Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin	1700
F18	Elliptic; Ackley; Rastrigin; HGBat; Discus	1800
F19	Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstrass; Schaffer F6	1900
F20	HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer F7	2000

2.5 Composite Function Overview

Composition functions **combine multiple shifted and rotated basic functions** into a single search space, creating highly complex and deceptive optimization landscapes. They are the most challenging category in the CEC 2017 suite and significantly test an algorithm's overall robustness and global search capability.

No.	Component Functions	Global Optimum
F21	Rosenbrock; Elliptic; Rastrigin	2100
F22	Rastrigin; Griewank; Schwefel	2200
F23	Rosenbrock; Ackley; Schwefel; Rastrigin	2300

No.	Component Functions	Global Optimum
F24	Ackley; Elliptic; Griewank; Rastrigin	2400
F25	Rastrigin; HappyCat; Ackley; Discus; Rosenbrock	2500
F26	Schaffer F6; Schwefel; Griewank; Rosenbrock; Rastrigin	2600
F27	HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic; Schaffer F6	2700
F28	Ackley; Griewank; Discus; Rosenbrock; HappyCat; Schaffer F6	2800
F29	F15; F16; F17	2900
F30	F15; F18; F19	3000

Section 3. List of Required Libraries

3.1 List of Libraries and Versions

The following libraries and modules were used for optimization, benchmarking, and visualization in this project:

Library / Module	Purpose	Version
<code>python</code>	Core programming language	3.10+
<code>numpy</code>	Numerical computing and array operations	2.0.2
<code>matplotlib</code>	Plotting and saving convergence graphs	3.10.0
<code>cec2017</code>	CEC 2017 benchmark functions (simple, hybrid, composition)	Latest (pip)
<code>cec2017.simple</code>	Simple benchmark functions F1-F10	—
<code>cec2017.hybrid</code>	Hybrid benchmark functions F11-F20	—
<code>cec2017.composition</code>	Composition benchmark functions F21-F30	—
<code>dplldc</code>	DP-LLDE optimization algorithm (uploaded locally)	Custom
<code>google.colab.files</code>	File upload utility for loading <code>dplldc.py</code> into the Colab environment	Built-in
<code>sys</code>	Clears cached module before re-importing <code>dplldc.py</code>	Built-in

Section 4. Environment Setup and Validation

4.1 Installation

Install `cec2017`

```
In [7]: # !pip install cec2017 --quiet
```

4.2 File Validation

Import `simple.py`, `hybrid.py`, and `composition.py` python files for benchmark functions

```
In [8]: from cec2017 import simple, hybrid, composition
print('cec2017 loaded!')
```

cec2017 loaded!

Import `sys` control over warning messages for verification that `lblde.py` exists

```
In [9]: from google.colab import files
import sys

uploaded = files.upload()

if 'lblde.py' in uploaded:
    if 'lblde' in sys.modules:
        del sys.modules['lblde']
    import lblde
    from lblde import LBLDE
    print('lblde.py uploaded and imported successfully!')
else:
    print('ERROR: Please upload a file named lblde.py')
```

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving `lblde.py` to `lblde.py`

lblde.py uploaded and imported successfully!

```
In [10]: import lblde
from lblde import LBLDE
```

Section 5. Configuration and Helper Functions

5.1 Library Imports

Import libraries `numpy`, `matplotlib`, and `cec2017` for numerical computation, visualization, and CEC 2017 benchmark functions.

```
In [11]: import numpy as np
import matplotlib.pyplot as plt
from cec2017 import simple, hybrid, composition
```

5.2 Global Parameters

Define global parameters

```
In [12]: D = 10 # Change this when testing different dimensions
```

```
In [13]: bounds = np.array([[-100.0, 100.0]] * D)
MAX_FES = 10000 * D
NP, NL, MU_CR = 100, 4, 0.35
SEED = 42
```

Store all results for summary table under `all_results` dict

```
In [14]: all_results = {}
```

5.3 Function Definitions

Define `make_wrapper` to adapt CEC 2017 functions to accept a 1D input array from DP-LLDE, returning a large penalty value if the output is invalid (nan/inf).

```
In [15]: def make_wrapper(fn):
    """Wrap a cec2017 function (expects (1,D) input) to accept (D,) input."""
    def wrapper(x):
        val = fn(x.reshape(1, -1))
        if hasattr(val, '__len__'):
            val = float(val[0])
        else:
            val = float(val)
        return val if np.isfinite(val) else 1e18
    return wrapper
```

Define `run_and_plot` to execute a single DP-LLDE optimization run on a given benchmark function, print the results, store them for the summary table, and save the convergence graph.

```
In [16]: def run_and_plot(fn_name, fn_eval, f_global, fn_label):
    """Run LBLDE_s on a function and plot convergence + diversity graphs."""
    print('=' * 80)
    print(f'LBLDE_s on CEC 2017 {fn_name} – Single Run')
    print('=' * 80)
    print(f'Function      : {fn_label} ({D}D)')
    print(f'Bounds        : [-100, 100]')
    print(f'Global optimum : {f_global}')

    opt = LBLDE(
        objective_func=fn_eval,
        bounds=bounds,
        NP=NP, NL=NL, mu_CR_ini=MU_CR,
        max_fes=MAX_FES,
        seed=SEED
    )
    # ← Unpack 4 values now
    best_sol, best_fit, history, diversity_history = opt.optimize(verbose=True)

    error = abs(best_fit - f_global)
    print(f'\n{"="*80}')
    print('Results:')
    print(f'  Best Fitness      : {best_fit:.6e}')
    print(f'  Global Optimum    : {f_global:.6e}')
    print(f'  Error (|best-optimum|) : {error:.6e}')
    print(f'  Best Solution (5 dim) : {best_sol[:5]}')
    print(f'  D^(0) (init. diversity): {opt.D_0:.4e}')
    print(f'{"="*80}')

    # Store for summary
    all_results[fn_name] = {
        'label': fn_label,
        'best_fit': best_fit,
        'f_global': f_global,
        'error': error
    }

    fes_axis = np.linspace(0, MAX_FES, len(history))
    gen_axis = np.arange(len(diversity_history)) # Generations (0, 1, 2, ...)

    # — Two-panel figure —
    fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 10))

    # Convergence curve (top) – FEs on x-axis
    ax1.plot(fes_axis, history, color='steelblue', linewidth=1.5)
    ax1.set_ylabel('Best Fitness', fontsize=12)
    ax1.set_xlabel('Function Evaluations (FEs)', fontsize=12)
    ax1.set_title(
        f'LBLDE – CEC 2017 {fn_name}: {fn_label} ({D}D)',
```



```

        fontsize=12
    )
    ax1.grid(True, which='both', linestyle='--', alpha=0.6)

    # Diversity curve (bottom) – Generations on x-axis
    ax2.plot(gen_axis, diversity_history, color='darkorange', linewidth=1.5)
    ax2.set_xlabel('Generation', fontsize=12)
    ax2.set_ylabel('Population Diversity\n(Mean Std Dev)', fontsize=12)
    ax2.set_title(f'Diversity Across Generations – {fn_name}', fontsize=12)
    ax2.grid(True, which='both', linestyle='--', alpha=0.6)

    plt.tight_layout()
    fname = f'convergence_{fn_name}.png'
    plt.savefig(fname, dpi=150)
    plt.show()
    print(f'Graph saved as {fname}\n')

```

Section 6. CEC2017 Benchmark Functions

6.1 Unimodal Functions (F1-F3)

```

In [17]: # =====
# F1 – Shifted and Rotated Bent Cigar
# =====
run_and_plot('F1', make_wrapper(simple.f1), 100.0, 'Shifted and Rotated Bent C

```

```

=====
=
LBLDE_s on CEC 2017 F1 – Single Run
=====
=
Function      : Shifted and Rotated Bent Cigar (10D)
Bounds       : [-100, 100]
Global optimum : 100.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 1.871674e+11 | Diversity = 5.7928e+01
Generation 10: Best = 9.104012e+09 FES = 1100 Diversity = 2.3166e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 9.778238e+08 FES = 2100 Diversity = 1.0699e+01 k_t
= [4, 3, 4, 2]
Generation 30: Best = 1.598092e+08 FES = 3100 Diversity = 7.2494e+00 k_t
= [1, 1, 1, 1]
Generation 40: Best = 4.573241e+07 FES = 4100 Diversity = 5.5896e+00 k_t
= [3, 3, 4, 4]
Generation 50: Best = 7.507583e+06 FES = 5100 Diversity = 4.5954e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.296961e+06 FES = 6100 Diversity = 4.2108e+00 k_t
= [4, 2, 2, 4]
Generation 70: Best = 8.622001e+05 FES = 7100 Diversity = 5.1309e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.938880e+05 FES = 8100 Diversity = 5.7239e+00 k_t
= [4, 3, 2, 4]
Generation 90: Best = 9.255775e+04 FES = 9100 Diversity = 5.8105e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.571741e+04 FES = 10100 Diversity = 6.4196e+00 k_t
= [2, 2, 2, 4]
Generation 110: Best = 7.873788e+03 FES = 11100 Diversity = 5.9010e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 4.262973e+03 FES = 12100 Diversity = 4.0185e+00 k_t
= [4, 2, 2, 2]
Generation 130: Best = 2.433009e+03 FES = 13100 Diversity = 3.1634e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.236598e+03 FES = 14100 Diversity = 2.8365e+00 k_t
= [4, 4, 4, 3]
Generation 150: Best = 4.424090e+02 FES = 15100 Diversity = 1.9977e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.088023e+02 FES = 16100 Diversity = 1.3564e+00 k_t
= [3, 4, 3, 4]
Generation 170: Best = 1.536327e+02 FES = 17100 Diversity = 7.4260e-01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.200596e+02 FES = 18100 Diversity = 4.8520e-01 k_t
= [3, 4, 4, 3]
Generation 190: Best = 1.070544e+02 FES = 19100 Diversity = 3.5876e-01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.040391e+02 FES = 20100 Diversity = 2.3050e-01 k_t
= [4, 4, 2, 2]
Generation 210: Best = 1.017522e+02 FES = 21100 Diversity = 1.4843e-01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.004058e+02 FES = 22100 Diversity = 8.5570e-02 k_t
= [3, 4, 4, 4]

```

Generation 230:	Best = 1.001723e+02	FES = 23100	Diversity = 6.0886e-02	k_t = [1, 1, 1, 1]
Generation 240:	Best = 1.000621e+02	FES = 24100	Diversity = 3.6923e-02	k_t = [2, 3, 4, 3]
Generation 250:	Best = 1.000299e+02	FES = 25100	Diversity = 2.1790e-02	k_t = [1, 1, 1, 1]
Generation 260:	Best = 1.000196e+02	FES = 26100	Diversity = 1.2489e-02	k_t = [2, 4, 2, 4]
Generation 270:	Best = 1.000094e+02	FES = 27100	Diversity = 9.3211e-03	k_t = [1, 1, 1, 1]
Generation 280:	Best = 1.000024e+02	FES = 28100	Diversity = 6.3566e-03	k_t = [3, 2, 3, 3]
Generation 290:	Best = 1.000015e+02	FES = 29100	Diversity = 3.6006e-03	k_t = [1, 1, 1, 1]
Generation 300:	Best = 1.000008e+02	FES = 30100	Diversity = 2.6619e-03	k_t = [4, 4, 3, 3]
Generation 310:	Best = 1.000001e+02	FES = 31100	Diversity = 2.0897e-03	k_t = [1, 1, 1, 1]
Generation 320:	Best = 1.000001e+02	FES = 32100	Diversity = 1.2343e-03	k_t = [4, 2, 2, 4]
Generation 330:	Best = 1.000000e+02	FES = 33100	Diversity = 7.9304e-04	k_t = [1, 1, 1, 1]
Generation 340:	Best = 1.000000e+02	FES = 34100	Diversity = 5.7589e-04	k_t = [3, 4, 3, 4]
Generation 350:	Best = 1.000000e+02	FES = 35100	Diversity = 3.4008e-04	k_t = [1, 1, 1, 1]
Generation 360:	Best = 1.000000e+02	FES = 36100	Diversity = 2.2015e-04	k_t = [3, 2, 3, 3]
Generation 370:	Best = 1.000000e+02	FES = 37100	Diversity = 1.2171e-04	k_t = [1, 1, 1, 1]
Generation 380:	Best = 1.000000e+02	FES = 38100	Diversity = 6.2900e-05	k_t = [2, 4, 3, 3]
Generation 390:	Best = 1.000000e+02	FES = 39100	Diversity = 4.6409e-05	k_t = [1, 1, 1, 1]
Generation 400:	Best = 1.000000e+02	FES = 40100	Diversity = 2.9473e-05	k_t = [2, 3, 4, 3]
Generation 410:	Best = 1.000000e+02	FES = 41100	Diversity = 1.7243e-05	k_t = [1, 1, 1, 1]
Generation 420:	Best = 1.000000e+02	FES = 42100	Diversity = 1.1854e-05	k_t = [2, 3, 3, 3]
Generation 430:	Best = 1.000000e+02	FES = 43100	Diversity = 7.0701e-06	k_t = [1, 1, 1, 1]
Generation 440:	Best = 1.000000e+02	FES = 44100	Diversity = 4.6577e-06	k_t = [3, 3, 3, 4]
Generation 450:	Best = 1.000000e+02	FES = 45100	Diversity = 2.5930e-06	k_t = [1, 1, 1, 1]
Generation 460:	Best = 1.000000e+02	FES = 46100	Diversity = 1.5700e-06	k_t = [4, 3, 3, 3]
Generation 470:	Best = 1.000000e+02	FES = 47100	Diversity = 1.0700e-06	k_t = [1, 1, 1, 1]
Generation 480:	Best = 1.000000e+02	FES = 48100	Diversity = 6.7195e-07	k_t = [4, 4, 2, 4]
Generation 490:	Best = 1.000000e+02	FES = 49100	Diversity = 4.8000e-07	k_t = [1, 1, 1, 1]

Generation 500:	Best = 1.000000e+02	FES = 50100	Diversity = 2.6210e-07	k_t = [2, 4, 3, 3]
Generation 510:	Best = 1.000000e+02	FES = 51100	Diversity = 1.7858e-07	k_t = [1, 1, 1, 1]
Generation 520:	Best = 1.000000e+02	FES = 52100	Diversity = 1.0570e-07	k_t = [3, 3, 3, 2]
Generation 530:	Best = 1.000000e+02	FES = 53100	Diversity = 7.5024e-08	k_t = [1, 1, 1, 1]
Generation 540:	Best = 1.000000e+02	FES = 54100	Diversity = 5.0977e-08	k_t = [2, 3, 2, 4]
Generation 550:	Best = 1.000000e+02	FES = 55100	Diversity = 3.1881e-08	k_t = [1, 1, 1, 1]
Generation 560:	Best = 1.000000e+02	FES = 56100	Diversity = 2.2911e-08	k_t = [2, 3, 2, 4]
Generation 570:	Best = 1.000000e+02	FES = 57100	Diversity = 1.6357e-08	k_t = [1, 1, 1, 1]
Generation 580:	Best = 1.000000e+02	FES = 58100	Diversity = 1.1664e-08	k_t = [3, 3, 2, 4]
Generation 590:	Best = 1.000000e+02	FES = 59100	Diversity = 9.1147e-09	k_t = [1, 1, 1, 1]
Generation 600:	Best = 1.000000e+02	FES = 60100	Diversity = 7.8038e-09	k_t = [3, 4, 2, 4]
Generation 610:	Best = 1.000000e+02	FES = 61100	Diversity = 6.6298e-09	k_t = [1, 1, 1, 1]
Generation 620:	Best = 1.000000e+02	FES = 62100	Diversity = 3.9556e-09	k_t = [2, 3, 4, 2]
Generation 630:	Best = 1.000000e+02	FES = 63100	Diversity = 4.2298e-09	k_t = [1, 1, 1, 1]
Generation 640:	Best = 1.000000e+02	FES = 64100	Diversity = 4.0029e-09	k_t = [4, 4, 4, 4]
Generation 650:	Best = 1.000000e+02	FES = 65100	Diversity = 3.9815e-09	k_t = [1, 1, 1, 1]
Generation 660:	Best = 1.000000e+02	FES = 66100	Diversity = 3.9723e-09	k_t = [2, 2, 3, 4]
Generation 670:	Best = 1.000000e+02	FES = 67100	Diversity = 3.8981e-09	k_t = [1, 1, 1, 1]
Generation 680:	Best = 1.000000e+02	FES = 68100	Diversity = 3.8930e-09	k_t = [4, 2, 4, 2]
Generation 690:	Best = 1.000000e+02	FES = 69100	Diversity = 4.2006e-09	k_t = [1, 1, 1, 1]
Generation 700:	Best = 1.000000e+02	FES = 70100	Diversity = 4.0705e-09	k_t = [4, 2, 4, 4]
Generation 710:	Best = 1.000000e+02	FES = 71100	Diversity = 4.2331e-09	k_t = [1, 1, 1, 1]
Generation 720:	Best = 1.000000e+02	FES = 72100	Diversity = 3.9860e-09	k_t = [3, 3, 3, 4]
Generation 730:	Best = 1.000000e+02	FES = 73100	Diversity = 4.2632e-09	k_t = [1, 1, 1, 1]
Generation 740:	Best = 1.000000e+02	FES = 74100	Diversity = 4.4664e-09	k_t = [3, 2, 4, 4]
Generation 750:	Best = 1.000000e+02	FES = 75100	Diversity = 4.3687e-09	k_t = [1, 1, 1, 1]
Generation 760:	Best = 1.000000e+02	FES = 76100	Diversity = 4.3209e-09	k_t = [4, 4, 4, 3]

Generation 770:	Best = 1.000000e+02	FES = 77100	Diversity = 4.0628e-09	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.000000e+02	FES = 78100	Diversity = 4.1985e-09	k_t
= [3, 2, 4, 3]				
Generation 790:	Best = 1.000000e+02	FES = 79100	Diversity = 4.1964e-09	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.000000e+02	FES = 80100	Diversity = 4.2221e-09	k_t
= [4, 3, 3, 2]				
Generation 810:	Best = 1.000000e+02	FES = 81100	Diversity = 4.3216e-09	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.000000e+02	FES = 82100	Diversity = 4.2262e-09	k_t
= [2, 2, 4, 2]				
Generation 830:	Best = 1.000000e+02	FES = 83100	Diversity = 4.2659e-09	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.000000e+02	FES = 84100	Diversity = 4.5157e-09	k_t
= [4, 3, 2, 4]				
Generation 850:	Best = 1.000000e+02	FES = 85100	Diversity = 4.6355e-09	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.000000e+02	FES = 86100	Diversity = 4.4758e-09	k_t
= [2, 4, 4, 3]				
Generation 870:	Best = 1.000000e+02	FES = 87100	Diversity = 4.2632e-09	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.000000e+02	FES = 88100	Diversity = 4.5049e-09	k_t
= [3, 2, 4, 3]				
Generation 890:	Best = 1.000000e+02	FES = 89100	Diversity = 4.2922e-09	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.000000e+02	FES = 90100	Diversity = 4.3141e-09	k_t
= [2, 2, 2, 3]				
Generation 910:	Best = 1.000000e+02	FES = 91100	Diversity = 4.2268e-09	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.000000e+02	FES = 92100	Diversity = 4.0179e-09	k_t
= [2, 3, 4, 4]				
Generation 930:	Best = 1.000000e+02	FES = 93100	Diversity = 4.1703e-09	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.000000e+02	FES = 94100	Diversity = 4.3713e-09	k_t
= [4, 2, 4, 2]				
Generation 950:	Best = 1.000000e+02	FES = 95100	Diversity = 4.4512e-09	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.000000e+02	FES = 96100	Diversity = 4.2494e-09	k_t
= [3, 3, 4, 4]				
Generation 970:	Best = 1.000000e+02	FES = 97100	Diversity = 4.2887e-09	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.000000e+02	FES = 98100	Diversity = 4.1389e-09	k_t
= [2, 4, 2, 4]				
Generation 990:	Best = 1.000000e+02	FES = 99100	Diversity = 4.1096e-09	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 1.000000e+02

Total Function Evaluations = 100000

=====

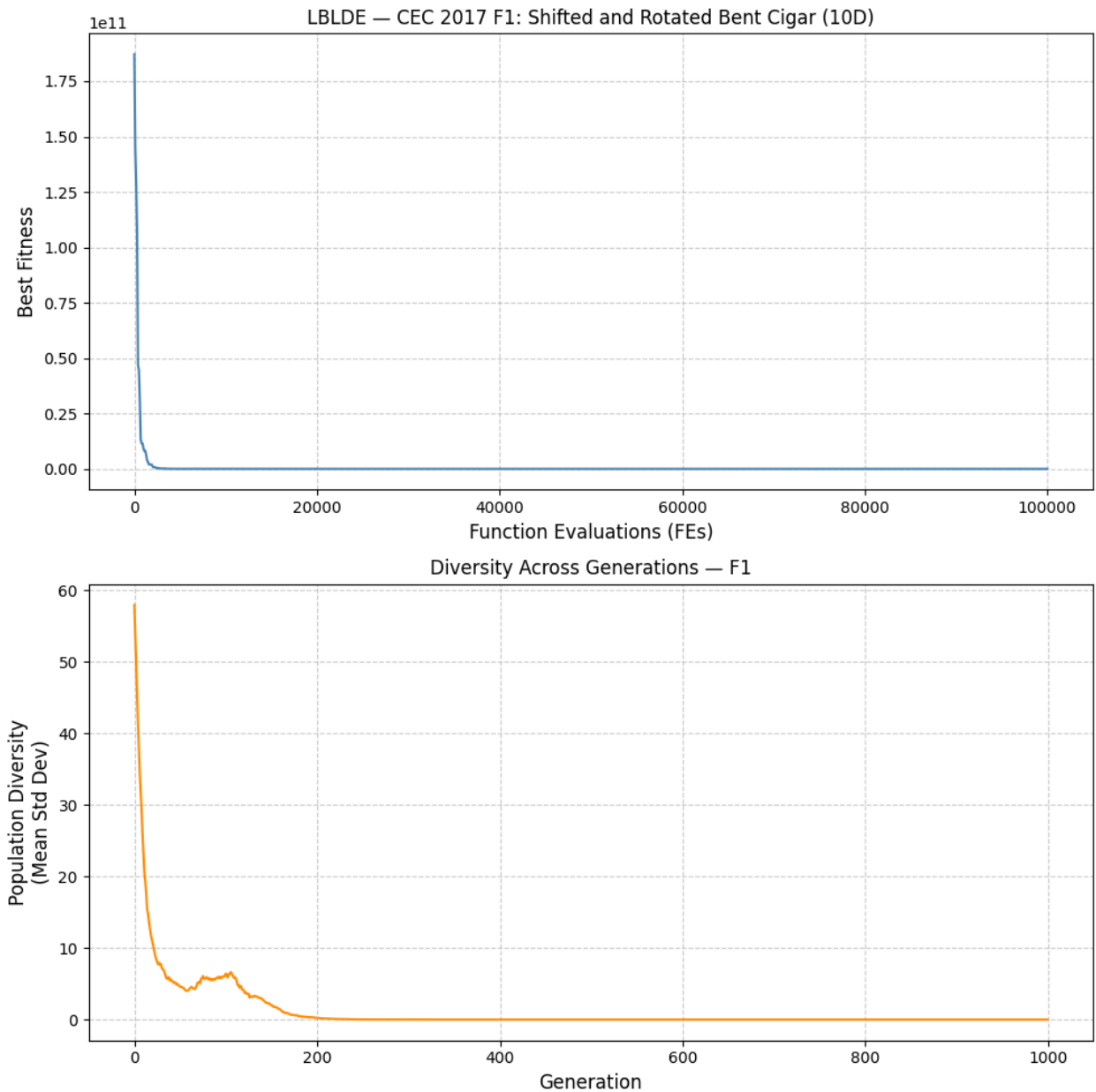
=

Results:

```

Best Fitness      : 1.000000e+02
Global Optimum    : 1.000000e+02
Error (|best-optimum|) : 0.000000e+00
Best Solution (5 dim) : [-55.27639853 -70.42955972 -29.61018187 -58.32676328
22.08960188]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F1.png

```

In [18]: # =====
# F2 — Shifted and Rotated Sum of Different Power
# =====
run_and_plot('F2', make_wrapper(simple.f2), 200.0, 'Shifted and Rotated Sum of

```

```

=====
=
LBLDE_s on CEC 2017 F2 – Single Run
=====
=
Function      : Shifted and Rotated Sum of Different Power (Deprecated) (10D)
Bounds       : [-100, 100]
Global optimum : 200.0
WARNING: f2 has been deprecated from the CEC 2017 benchmark suite
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 2.603565e+14 | Diversity = 5.7928e+01
Generation 10: Best = 2.850883e+08 FES = 1100 Diversity = 3.7934e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 4.791893e+06 FES = 2100 Diversity = 2.8998e+01 k_t
= [4, 3, 4, 2]
Generation 30: Best = 4.459102e+05 FES = 3100 Diversity = 2.5691e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 6.286562e+04 FES = 4100 Diversity = 2.0623e+01 k_t
= [3, 3, 4, 4]
Generation 50: Best = 7.226093e+03 FES = 5100 Diversity = 1.7152e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.160037e+03 FES = 6100 Diversity = 1.3260e+01 k_t
= [2, 3, 2, 3]
Generation 70: Best = 1.124237e+03 FES = 7100 Diversity = 8.2758e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 5.800741e+02 FES = 8100 Diversity = 7.1021e+00 k_t
= [3, 3, 2, 3]
Generation 90: Best = 2.446579e+02 FES = 9100 Diversity = 5.3656e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.225641e+02 FES = 10100 Diversity = 3.7914e+00 k_t
= [2, 2, 2, 4]
Generation 110: Best = 2.019374e+02 FES = 11100 Diversity = 1.4408e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.001189e+02 FES = 12100 Diversity = 5.0403e-01 k_t
= [2, 4, 2, 2]
Generation 130: Best = 2.000059e+02 FES = 13100 Diversity = 2.1466e-01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.000023e+02 FES = 14100 Diversity = 1.5114e-01 k_t
= [4, 4, 3, 2]
Generation 150: Best = 2.000005e+02 FES = 15100 Diversity = 1.1889e-01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.000002e+02 FES = 16100 Diversity = 1.0212e-01 k_t
= [4, 2, 4, 4]
Generation 170: Best = 2.000000e+02 FES = 17100 Diversity = 9.1775e-02 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.000000e+02 FES = 18100 Diversity = 7.1348e-02 k_t
= [3, 2, 4, 2]
Generation 190: Best = 2.000000e+02 FES = 19100 Diversity = 6.4437e-02 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.000000e+02 FES = 20100 Diversity = 5.6159e-02 k_t
= [4, 3, 4, 4]
Generation 210: Best = 2.000000e+02 FES = 21100 Diversity = 5.1066e-02 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.000000e+02 FES = 22100 Diversity = 4.4509e-02 k_t

```

```

= [4, 2, 3, 3]
Generation 230: Best = 2.000000e+02  FES = 23100  Diversity = 3.9284e-02  k_t
= [1, 1, 1, 1]
Generation 240: Best = 2.000000e+02  FES = 24100  Diversity = 3.4086e-02  k_t
= [4, 4, 2, 2]
Generation 250: Best = 2.000000e+02  FES = 25100  Diversity = 3.1701e-02  k_t
= [1, 1, 1, 1]
Generation 260: Best = 2.000000e+02  FES = 26100  Diversity = 2.9504e-02  k_t
= [2, 3, 2, 3]
Generation 270: Best = 2.000000e+02  FES = 27100  Diversity = 2.5166e-02  k_t
= [1, 1, 1, 1]
Generation 280: Best = 2.000000e+02  FES = 28100  Diversity = 2.1029e-02  k_t
= [3, 2, 3, 2]
Generation 290: Best = 2.000000e+02  FES = 29100  Diversity = 1.9859e-02  k_t
= [1, 1, 1, 1]
Generation 300: Best = 2.000000e+02  FES = 30100  Diversity = 1.7442e-02  k_t
= [4, 4, 3, 4]
Generation 310: Best = 2.000000e+02  FES = 31100  Diversity = 1.5623e-02  k_t
= [1, 1, 1, 1]
Generation 320: Best = 2.000000e+02  FES = 32100  Diversity = 1.4113e-02  k_t
= [3, 2, 3, 2]
Generation 330: Best = 2.000000e+02  FES = 33100  Diversity = 1.2783e-02  k_t
= [1, 1, 1, 1]
Generation 340: Best = 2.000000e+02  FES = 34100  Diversity = 1.1846e-02  k_t
= [3, 4, 3, 4]
Generation 350: Best = 2.000000e+02  FES = 35100  Diversity = 1.0206e-02  k_t
= [1, 1, 1, 1]
Generation 360: Best = 2.000000e+02  FES = 36100  Diversity = 8.8792e-03  k_t
= [2, 4, 2, 2]
Generation 370: Best = 2.000000e+02  FES = 37100  Diversity = 7.4330e-03  k_t
= [1, 1, 1, 1]
Generation 380: Best = 2.000000e+02  FES = 38100  Diversity = 6.6011e-03  k_t
= [4, 4, 2, 3]
Generation 390: Best = 2.000000e+02  FES = 39100  Diversity = 5.6457e-03  k_t
= [1, 1, 1, 1]
Generation 400: Best = 2.000000e+02  FES = 40100  Diversity = 5.2148e-03  k_t
= [2, 4, 3, 4]
Generation 410: Best = 2.000000e+02  FES = 41100  Diversity = 4.5064e-03  k_t
= [1, 1, 1, 1]
Generation 420: Best = 2.000000e+02  FES = 42100  Diversity = 4.3150e-03  k_t
= [4, 2, 4, 3]
Generation 430: Best = 2.000000e+02  FES = 43100  Diversity = 3.9662e-03  k_t
= [1, 1, 1, 1]
Generation 440: Best = 2.000000e+02  FES = 44100  Diversity = 4.0435e-03  k_t
= [3, 2, 3, 2]
Generation 450: Best = 2.000000e+02  FES = 45100  Diversity = 4.0220e-03  k_t
= [1, 1, 1, 1]
Generation 460: Best = 2.000000e+02  FES = 46100  Diversity = 3.9689e-03  k_t
= [2, 4, 3, 2]
Generation 470: Best = 2.000000e+02  FES = 47100  Diversity = 4.0183e-03  k_t
= [1, 1, 1, 1]
Generation 480: Best = 2.000000e+02  FES = 48100  Diversity = 3.8808e-03  k_t
= [4, 2, 4, 4]
Generation 490: Best = 2.000000e+02  FES = 49100  Diversity = 4.0460e-03  k_t

```



```

= [1, 1, 1, 1]
Generation 500: Best = 2.000000e+02 FES = 50100 Diversity = 4.0021e-03 k_t
= [4, 3, 2, 3]
Generation 510: Best = 2.000000e+02 FES = 51100 Diversity = 3.9828e-03 k_t
= [1, 1, 1, 1]
Generation 520: Best = 2.000000e+02 FES = 52100 Diversity = 4.0447e-03 k_t
= [4, 3, 4, 3]
Generation 530: Best = 2.000000e+02 FES = 53100 Diversity = 3.8407e-03 k_t
= [1, 1, 1, 1]
Generation 540: Best = 2.000000e+02 FES = 54100 Diversity = 3.8908e-03 k_t
= [4, 2, 3, 2]
Generation 550: Best = 2.000000e+02 FES = 55100 Diversity = 3.9656e-03 k_t
= [1, 1, 1, 1]
Generation 560: Best = 2.000000e+02 FES = 56100 Diversity = 4.0734e-03 k_t
= [3, 3, 4, 3]
Generation 570: Best = 2.000000e+02 FES = 57100 Diversity = 4.0049e-03 k_t
= [1, 1, 1, 1]
Generation 580: Best = 2.000000e+02 FES = 58100 Diversity = 3.8770e-03 k_t
= [2, 2, 4, 4]
Generation 590: Best = 2.000000e+02 FES = 59100 Diversity = 4.0192e-03 k_t
= [1, 1, 1, 1]
Generation 600: Best = 2.000000e+02 FES = 60100 Diversity = 4.0115e-03 k_t
= [4, 2, 2, 3]
Generation 610: Best = 2.000000e+02 FES = 61100 Diversity = 3.9713e-03 k_t
= [1, 1, 1, 1]
Generation 620: Best = 2.000000e+02 FES = 62100 Diversity = 3.9806e-03 k_t
= [4, 4, 2, 4]
Generation 630: Best = 2.000000e+02 FES = 63100 Diversity = 4.0884e-03 k_t
= [1, 1, 1, 1]
Generation 640: Best = 2.000000e+02 FES = 64100 Diversity = 4.1227e-03 k_t
= [4, 2, 4, 4]
Generation 650: Best = 2.000000e+02 FES = 65100 Diversity = 4.0585e-03 k_t
= [1, 1, 1, 1]
Generation 660: Best = 2.000000e+02 FES = 66100 Diversity = 4.0299e-03 k_t
= [2, 4, 2, 4]
Generation 670: Best = 2.000000e+02 FES = 67100 Diversity = 4.0756e-03 k_t
= [1, 1, 1, 1]
Generation 680: Best = 2.000000e+02 FES = 68100 Diversity = 3.9608e-03 k_t
= [2, 2, 3, 2]
Generation 690: Best = 2.000000e+02 FES = 69100 Diversity = 4.0150e-03 k_t
= [1, 1, 1, 1]
Generation 700: Best = 2.000000e+02 FES = 70100 Diversity = 4.0946e-03 k_t
= [3, 3, 2, 2]
Generation 710: Best = 2.000000e+02 FES = 71100 Diversity = 4.0006e-03 k_t
= [1, 1, 1, 1]
Generation 720: Best = 2.000000e+02 FES = 72100 Diversity = 4.0396e-03 k_t
= [2, 3, 2, 2]
Generation 730: Best = 2.000000e+02 FES = 73100 Diversity = 3.9786e-03 k_t
= [1, 1, 1, 1]
Generation 740: Best = 2.000000e+02 FES = 74100 Diversity = 3.9689e-03 k_t
= [3, 2, 4, 2]
Generation 750: Best = 2.000000e+02 FES = 75100 Diversity = 4.1118e-03 k_t
= [1, 1, 1, 1]
Generation 760: Best = 2.000000e+02 FES = 76100 Diversity = 4.3378e-03 k_t

```

```

= [4, 2, 3, 3]
Generation 770: Best = 2.000000e+02  FES = 77100  Diversity = 4.0798e-03  k_t
= [1, 1, 1, 1]
Generation 780: Best = 2.000000e+02  FES = 78100  Diversity = 4.1436e-03  k_t
= [3, 2, 2, 3]
Generation 790: Best = 2.000000e+02  FES = 79100  Diversity = 4.0852e-03  k_t
= [1, 1, 1, 1]
Generation 800: Best = 2.000000e+02  FES = 80100  Diversity = 4.2286e-03  k_t
= [2, 4, 4, 4]
Generation 810: Best = 2.000000e+02  FES = 81100  Diversity = 4.0768e-03  k_t
= [1, 1, 1, 1]
Generation 820: Best = 2.000000e+02  FES = 82100  Diversity = 4.1475e-03  k_t
= [3, 3, 3, 3]
Generation 830: Best = 2.000000e+02  FES = 83100  Diversity = 4.0693e-03  k_t
= [1, 1, 1, 1]
Generation 840: Best = 2.000000e+02  FES = 84100  Diversity = 3.9575e-03  k_t
= [4, 3, 4, 4]
Generation 850: Best = 2.000000e+02  FES = 85100  Diversity = 4.0222e-03  k_t
= [1, 1, 1, 1]
Generation 860: Best = 2.000000e+02  FES = 86100  Diversity = 4.3589e-03  k_t
= [3, 2, 3, 3]
Generation 870: Best = 2.000000e+02  FES = 87100  Diversity = 4.2934e-03  k_t
= [1, 1, 1, 1]
Generation 880: Best = 2.000000e+02  FES = 88100  Diversity = 4.2282e-03  k_t
= [4, 2, 3, 2]
Generation 890: Best = 2.000000e+02  FES = 89100  Diversity = 4.2309e-03  k_t
= [1, 1, 1, 1]
Generation 900: Best = 2.000000e+02  FES = 90100  Diversity = 4.1842e-03  k_t
= [3, 2, 3, 2]
Generation 910: Best = 2.000000e+02  FES = 91100  Diversity = 4.2300e-03  k_t
= [1, 1, 1, 1]
Generation 920: Best = 2.000000e+02  FES = 92100  Diversity = 4.2021e-03  k_t
= [3, 2, 3, 4]
Generation 930: Best = 2.000000e+02  FES = 93100  Diversity = 4.1366e-03  k_t
= [1, 1, 1, 1]
Generation 940: Best = 2.000000e+02  FES = 94100  Diversity = 4.2219e-03  k_t
= [4, 3, 2, 3]
Generation 950: Best = 2.000000e+02  FES = 95100  Diversity = 4.2999e-03  k_t
= [1, 1, 1, 1]
Generation 960: Best = 2.000000e+02  FES = 96100  Diversity = 4.1054e-03  k_t
= [2, 4, 3, 2]
Generation 970: Best = 2.000000e+02  FES = 97100  Diversity = 4.2159e-03  k_t
= [1, 1, 1, 1]
Generation 980: Best = 2.000000e+02  FES = 98100  Diversity = 4.1397e-03  k_t
= [3, 2, 3, 3]
Generation 990: Best = 2.000000e+02  FES = 99100  Diversity = 4.2151e-03  k_t
= [1, 1, 1, 1]

```

Optimization completed!

Final Best Fitness = 2.000000e+02

Total Function Evaluations = 100000

=====

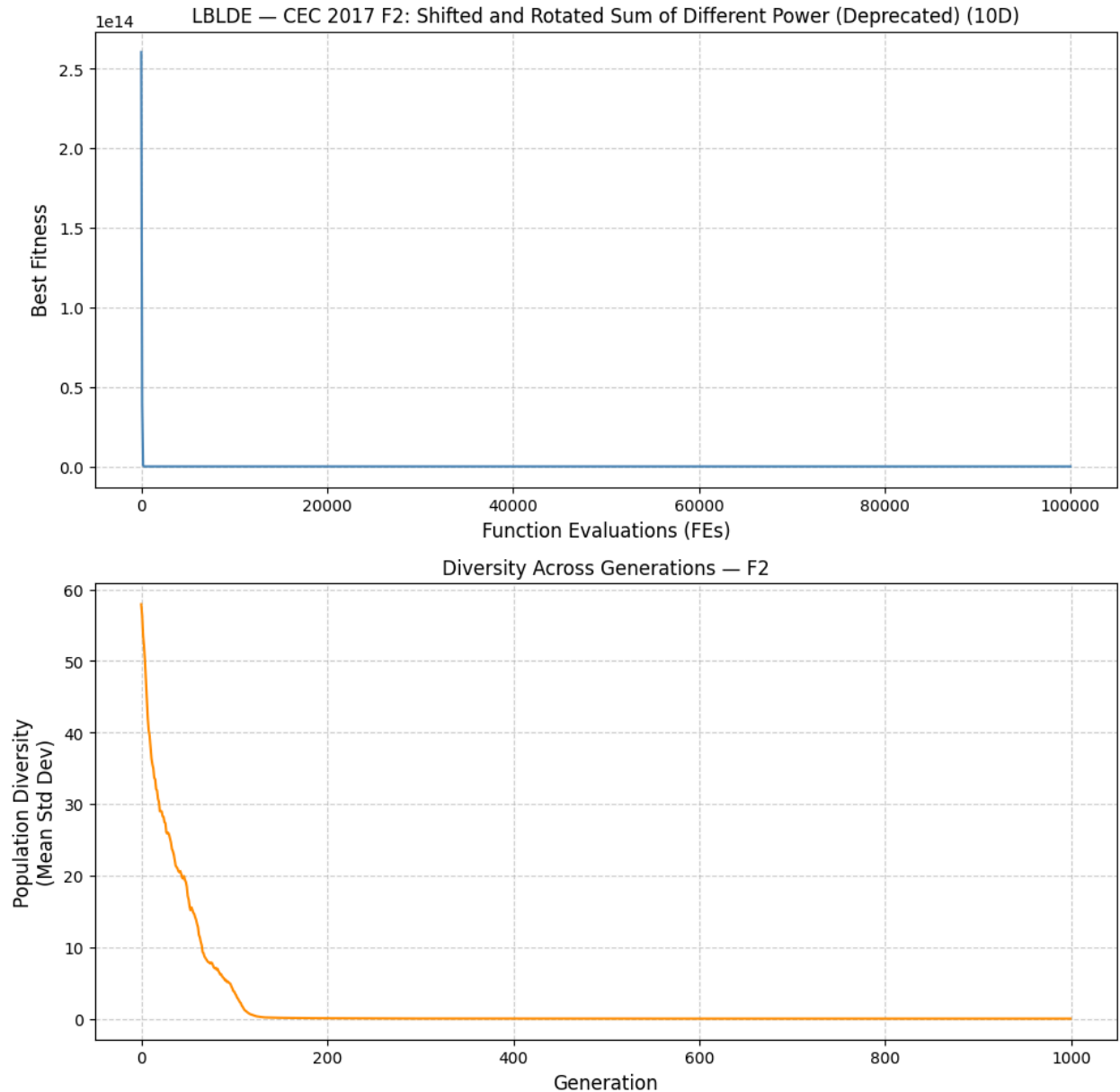
=

Results:

```
Best Fitness          : 2.000000e+02
Global Optimum        : 2.000000e+02
Error (|best-optimum|) : 0.000000e+00
Best Solution (5 dim)  : [-29.34044478 -17.0627028 -75.48318685 -0.67728745
4.24653951]
D^(0) (init. diversity): 5.7928e+01
```

=====

==



Graph saved as convergence_F2.png

```
In [19]: # =====
# F3 — Shifted and Rotated Zakharov
# =====
run_and_plot('F3', make_wrapper(simple.f3), 300.0, 'Shifted and Rotated Zakhar
```

```

=====
=
LBLDE_s on CEC 2017 F3 – Single Run
=====
=
Function      : Shifted and Rotated Zakharov (10D)
Bounds       : [-100, 100]
Global optimum : 300.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 2.625937e+04 | Diversity = 5.7928e+01
Generation 10: Best = 1.038262e+04 FES = 1100 Diversity = 5.4192e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 7.291033e+03 FES = 2100 Diversity = 4.2526e+01 k_t
= [3, 2, 2, 3]
Generation 30: Best = 3.486060e+03 FES = 3100 Diversity = 3.0151e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.264319e+03 FES = 4100 Diversity = 1.9355e+01 k_t
= [3, 3, 4, 4]
Generation 50: Best = 1.126768e+03 FES = 5100 Diversity = 1.1927e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 5.313162e+02 FES = 6100 Diversity = 8.1674e+00 k_t
= [4, 2, 2, 4]
Generation 70: Best = 4.226307e+02 FES = 7100 Diversity = 5.2536e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 3.230379e+02 FES = 8100 Diversity = 3.2029e+00 k_t
= [3, 4, 3, 4]
Generation 90: Best = 3.139294e+02 FES = 9100 Diversity = 1.8296e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 3.024776e+02 FES = 10100 Diversity = 9.6585e-01 k_t
= [4, 3, 4, 4]
Generation 110: Best = 3.008112e+02 FES = 11100 Diversity = 4.7203e-01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 3.002788e+02 FES = 12100 Diversity = 2.1817e-01 k_t
= [4, 2, 3, 3]
Generation 130: Best = 3.000435e+02 FES = 13100 Diversity = 1.2198e-01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 3.000158e+02 FES = 14100 Diversity = 5.8678e-02 k_t
= [2, 3, 2, 4]
Generation 150: Best = 3.000032e+02 FES = 15100 Diversity = 3.0294e-02 k_t
= [1, 1, 1, 1]
Generation 160: Best = 3.000006e+02 FES = 16100 Diversity = 1.7282e-02 k_t
= [3, 2, 2, 2]
Generation 170: Best = 3.000003e+02 FES = 17100 Diversity = 9.0776e-03 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.000001e+02 FES = 18100 Diversity = 4.7523e-03 k_t
= [2, 3, 2, 2]
Generation 190: Best = 3.000000e+02 FES = 19100 Diversity = 2.5094e-03 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.000000e+02 FES = 20100 Diversity = 1.3179e-03 k_t
= [4, 2, 2, 2]
Generation 210: Best = 3.000000e+02 FES = 21100 Diversity = 6.2929e-04 k_t
= [1, 1, 1, 1]
Generation 220: Best = 3.000000e+02 FES = 22100 Diversity = 3.5996e-04 k_t
= [2, 4, 2, 2]

```

Generation 230:	Best = 3.000000e+02	FES = 23100	Diversity = 2.0504e-04	k_t = [1, 1, 1, 1]
Generation 240:	Best = 3.000000e+02	FES = 24100	Diversity = 1.1462e-04	k_t = [4, 2, 2, 3]
Generation 250:	Best = 3.000000e+02	FES = 25100	Diversity = 6.5780e-05	k_t = [1, 1, 1, 1]
Generation 260:	Best = 3.000000e+02	FES = 26100	Diversity = 3.1478e-05	k_t = [3, 3, 3, 4]
Generation 270:	Best = 3.000000e+02	FES = 27100	Diversity = 1.7901e-05	k_t = [1, 1, 1, 1]
Generation 280:	Best = 3.000000e+02	FES = 28100	Diversity = 8.6130e-06	k_t = [3, 4, 2, 2]
Generation 290:	Best = 3.000000e+02	FES = 29100	Diversity = 4.0964e-06	k_t = [1, 1, 1, 1]
Generation 300:	Best = 3.000000e+02	FES = 30100	Diversity = 2.0440e-06	k_t = [2, 4, 2, 2]
Generation 310:	Best = 3.000000e+02	FES = 31100	Diversity = 1.0136e-06	k_t = [1, 1, 1, 1]
Generation 320:	Best = 3.000000e+02	FES = 32100	Diversity = 5.3312e-07	k_t = [3, 4, 4, 3]
Generation 330:	Best = 3.000000e+02	FES = 33100	Diversity = 2.7091e-07	k_t = [1, 1, 1, 1]
Generation 340:	Best = 3.000000e+02	FES = 34100	Diversity = 1.2987e-07	k_t = [3, 3, 3, 2]
Generation 350:	Best = 3.000000e+02	FES = 35100	Diversity = 7.5980e-08	k_t = [1, 1, 1, 1]
Generation 360:	Best = 3.000000e+02	FES = 36100	Diversity = 5.0947e-08	k_t = [2, 3, 2, 2]
Generation 370:	Best = 3.000000e+02	FES = 37100	Diversity = 3.4642e-08	k_t = [1, 1, 1, 1]
Generation 380:	Best = 3.000000e+02	FES = 38100	Diversity = 3.3578e-08	k_t = [4, 3, 3, 2]
Generation 390:	Best = 3.000000e+02	FES = 39100	Diversity = 3.3652e-08	k_t = [1, 1, 1, 1]
Generation 400:	Best = 3.000000e+02	FES = 40100	Diversity = 3.1829e-08	k_t = [2, 4, 4, 3]
Generation 410:	Best = 3.000000e+02	FES = 41100	Diversity = 3.1242e-08	k_t = [1, 1, 1, 1]
Generation 420:	Best = 3.000000e+02	FES = 42100	Diversity = 2.9136e-08	k_t = [4, 4, 2, 4]
Generation 430:	Best = 3.000000e+02	FES = 43100	Diversity = 2.9108e-08	k_t = [1, 1, 1, 1]
Generation 440:	Best = 3.000000e+02	FES = 44100	Diversity = 2.9592e-08	k_t = [2, 3, 3, 2]
Generation 450:	Best = 3.000000e+02	FES = 45100	Diversity = 3.0141e-08	k_t = [1, 1, 1, 1]
Generation 460:	Best = 3.000000e+02	FES = 46100	Diversity = 2.8932e-08	k_t = [3, 4, 2, 2]
Generation 470:	Best = 3.000000e+02	FES = 47100	Diversity = 2.8255e-08	k_t = [1, 1, 1, 1]
Generation 480:	Best = 3.000000e+02	FES = 48100	Diversity = 2.9234e-08	k_t = [2, 3, 2, 2]
Generation 490:	Best = 3.000000e+02	FES = 49100	Diversity = 3.0835e-08	k_t = [1, 1, 1, 1]

Generation 500:	Best = 3.000000e+02	FES = 50100	Diversity = 3.2081e-08	k_t = [4, 3, 3, 4]
Generation 510:	Best = 3.000000e+02	FES = 51100	Diversity = 3.1214e-08	k_t = [1, 1, 1, 1]
Generation 520:	Best = 3.000000e+02	FES = 52100	Diversity = 3.0615e-08	k_t = [4, 2, 2, 4]
Generation 530:	Best = 3.000000e+02	FES = 53100	Diversity = 3.0029e-08	k_t = [1, 1, 1, 1]
Generation 540:	Best = 3.000000e+02	FES = 54100	Diversity = 3.1158e-08	k_t = [3, 3, 2, 3]
Generation 550:	Best = 3.000000e+02	FES = 55100	Diversity = 3.0728e-08	k_t = [1, 1, 1, 1]
Generation 560:	Best = 3.000000e+02	FES = 56100	Diversity = 3.2244e-08	k_t = [3, 3, 4, 2]
Generation 570:	Best = 3.000000e+02	FES = 57100	Diversity = 3.3315e-08	k_t = [1, 1, 1, 1]
Generation 580:	Best = 3.000000e+02	FES = 58100	Diversity = 3.3577e-08	k_t = [4, 2, 4, 2]
Generation 590:	Best = 3.000000e+02	FES = 59100	Diversity = 3.3447e-08	k_t = [1, 1, 1, 1]
Generation 600:	Best = 3.000000e+02	FES = 60100	Diversity = 3.2103e-08	k_t = [2, 4, 2, 2]
Generation 610:	Best = 3.000000e+02	FES = 61100	Diversity = 3.2086e-08	k_t = [1, 1, 1, 1]
Generation 620:	Best = 3.000000e+02	FES = 62100	Diversity = 3.1588e-08	k_t = [2, 4, 2, 2]
Generation 630:	Best = 3.000000e+02	FES = 63100	Diversity = 3.0930e-08	k_t = [1, 1, 1, 1]
Generation 640:	Best = 3.000000e+02	FES = 64100	Diversity = 3.1126e-08	k_t = [2, 3, 3, 2]
Generation 650:	Best = 3.000000e+02	FES = 65100	Diversity = 3.2527e-08	k_t = [1, 1, 1, 1]
Generation 660:	Best = 3.000000e+02	FES = 66100	Diversity = 3.0774e-08	k_t = [3, 2, 2, 4]
Generation 670:	Best = 3.000000e+02	FES = 67100	Diversity = 2.9986e-08	k_t = [1, 1, 1, 1]
Generation 680:	Best = 3.000000e+02	FES = 68100	Diversity = 2.9539e-08	k_t = [3, 4, 4, 3]
Generation 690:	Best = 3.000000e+02	FES = 69100	Diversity = 3.0983e-08	k_t = [1, 1, 1, 1]
Generation 700:	Best = 3.000000e+02	FES = 70100	Diversity = 2.9942e-08	k_t = [2, 4, 4, 4]
Generation 710:	Best = 3.000000e+02	FES = 71100	Diversity = 2.8799e-08	k_t = [1, 1, 1, 1]
Generation 720:	Best = 3.000000e+02	FES = 72100	Diversity = 3.0561e-08	k_t = [4, 2, 2, 2]
Generation 730:	Best = 3.000000e+02	FES = 73100	Diversity = 3.1472e-08	k_t = [1, 1, 1, 1]
Generation 740:	Best = 3.000000e+02	FES = 74100	Diversity = 3.1527e-08	k_t = [4, 4, 3, 2]
Generation 750:	Best = 3.000000e+02	FES = 75100	Diversity = 3.0322e-08	k_t = [1, 1, 1, 1]
Generation 760:	Best = 3.000000e+02	FES = 76100	Diversity = 2.9878e-08	k_t = [3, 4, 3, 2]

Generation 770:	Best = 3.000000e+02	FES = 77100	Diversity = 2.9343e-08	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 3.000000e+02	FES = 78100	Diversity = 2.9945e-08	k_t
= [4, 3, 2, 4]				
Generation 790:	Best = 3.000000e+02	FES = 79100	Diversity = 2.9708e-08	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 3.000000e+02	FES = 80100	Diversity = 2.9589e-08	k_t
= [2, 3, 3, 3]				
Generation 810:	Best = 3.000000e+02	FES = 81100	Diversity = 2.9450e-08	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 3.000000e+02	FES = 82100	Diversity = 3.0396e-08	k_t
= [4, 2, 3, 2]				
Generation 830:	Best = 3.000000e+02	FES = 83100	Diversity = 3.0446e-08	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 3.000000e+02	FES = 84100	Diversity = 3.1094e-08	k_t
= [2, 2, 3, 3]				
Generation 850:	Best = 3.000000e+02	FES = 85100	Diversity = 3.0105e-08	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 3.000000e+02	FES = 86100	Diversity = 3.0267e-08	k_t
= [3, 3, 3, 3]				
Generation 870:	Best = 3.000000e+02	FES = 87100	Diversity = 3.0586e-08	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 3.000000e+02	FES = 88100	Diversity = 3.2057e-08	k_t
= [3, 2, 3, 2]				
Generation 890:	Best = 3.000000e+02	FES = 89100	Diversity = 3.1434e-08	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 3.000000e+02	FES = 90100	Diversity = 3.0499e-08	k_t
= [2, 2, 3, 4]				
Generation 910:	Best = 3.000000e+02	FES = 91100	Diversity = 3.1321e-08	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 3.000000e+02	FES = 92100	Diversity = 3.1581e-08	k_t
= [2, 3, 4, 2]				
Generation 930:	Best = 3.000000e+02	FES = 93100	Diversity = 3.3095e-08	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 3.000000e+02	FES = 94100	Diversity = 3.1493e-08	k_t
= [3, 4, 3, 4]				
Generation 950:	Best = 3.000000e+02	FES = 95100	Diversity = 3.0899e-08	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 3.000000e+02	FES = 96100	Diversity = 3.2863e-08	k_t
= [2, 2, 4, 3]				
Generation 970:	Best = 3.000000e+02	FES = 97100	Diversity = 3.2999e-08	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 3.000000e+02	FES = 98100	Diversity = 3.4685e-08	k_t
= [2, 4, 3, 2]				
Generation 990:	Best = 3.000000e+02	FES = 99100	Diversity = 3.3925e-08	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 3.000000e+02

Total Function Evaluations = 100000

=====

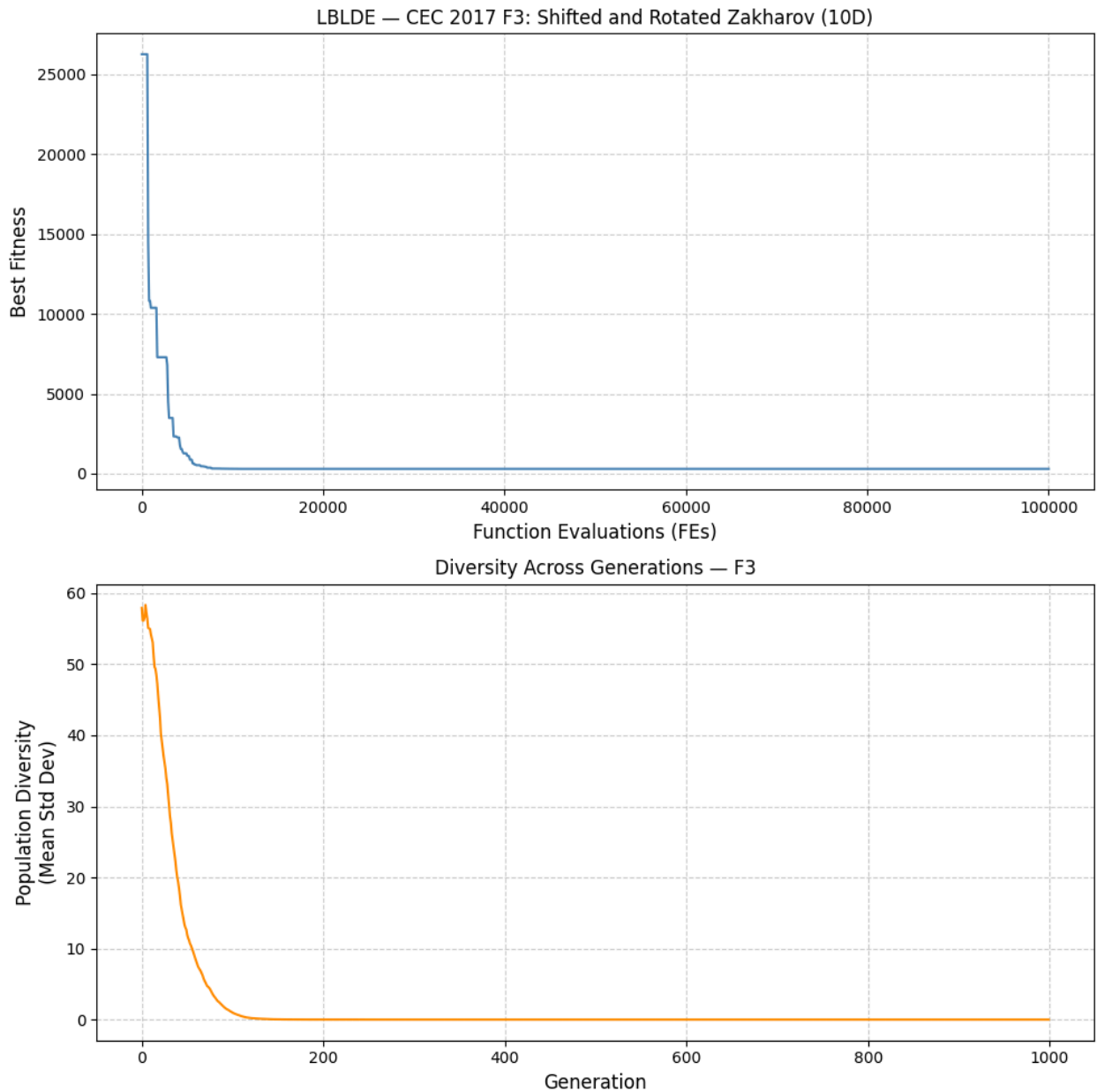
=

Results:

```

Best Fitness      : 3.000000e+02
Global Optimum    : 3.000000e+02
Error (|best-optimum|) : 0.000000e+00
Best Solution (5 dim) : [-55.93832677  4.54306536  35.30007015  8.27944087
-47.43694286]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F3.png

6.2 Multimodal Functions (F4-F10)

```

In [20]: # =====
# F4 — Shifted and Rotated Rosenbrock

```



```
# =====  
run_and_plot('F4', make_wrapper(simple.f4), 400.0, "Shifted and Rotated Rosenb
```

```

=====
=
LBLDE_s on CEC 2017 F4 – Single Run
=====
=
Function      : Shifted and Rotated Rosenbrock's (10D)
Bounds       : [-100, 100]
Global optimum : 400.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 3.004346e+03 | Diversity = 5.7928e+01
Generation 10: Best = 5.277046e+02 FES = 1100 Diversity = 2.6210e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 4.539030e+02 FES = 2100 Diversity = 1.6921e+01 k_t
= [2, 4, 2, 3]
Generation 30: Best = 4.152427e+02 FES = 3100 Diversity = 8.8723e+00 k_t
= [1, 1, 1, 1]
Generation 40: Best = 4.079778e+02 FES = 4100 Diversity = 3.7527e+00 k_t
= [2, 3, 4, 4]
Generation 50: Best = 4.073752e+02 FES = 5100 Diversity = 1.6135e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 4.068725e+02 FES = 6100 Diversity = 1.0920e+00 k_t
= [4, 3, 4, 2]
Generation 70: Best = 4.066470e+02 FES = 7100 Diversity = 7.4683e-01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 4.064796e+02 FES = 8100 Diversity = 5.9458e-01 k_t
= [4, 3, 4, 2]
Generation 90: Best = 4.062987e+02 FES = 9100 Diversity = 3.7980e-01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 4.062174e+02 FES = 10100 Diversity = 2.5484e-01 k_t
= [4, 4, 2, 4]
Generation 110: Best = 4.061308e+02 FES = 11100 Diversity = 1.9333e-01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 4.060793e+02 FES = 12100 Diversity = 1.6094e-01 k_t
= [4, 3, 4, 4]
Generation 130: Best = 4.060386e+02 FES = 13100 Diversity = 1.4480e-01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 4.059655e+02 FES = 14100 Diversity = 1.6880e-01 k_t
= [2, 4, 3, 4]
Generation 150: Best = 4.059330e+02 FES = 15100 Diversity = 1.6664e-01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 4.058654e+02 FES = 16100 Diversity = 1.7984e-01 k_t
= [3, 2, 4, 3]
Generation 170: Best = 4.057657e+02 FES = 17100 Diversity = 2.5985e-01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 4.056696e+02 FES = 18100 Diversity = 2.4607e-01 k_t
= [3, 4, 2, 4]
Generation 190: Best = 4.055989e+02 FES = 19100 Diversity = 1.9101e-01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 4.055191e+02 FES = 20100 Diversity = 2.3624e-01 k_t
= [3, 2, 2, 4]
Generation 210: Best = 4.054108e+02 FES = 21100 Diversity = 2.7544e-01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 4.053116e+02 FES = 22100 Diversity = 2.6088e-01 k_t
= [2, 2, 4, 4]

```

Generation 230:	Best = 4.051808e+02	FES = 23100	Diversity = 3.4517e-01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 4.050549e+02	FES = 24100	Diversity = 3.3467e-01	k_t = [4, 4, 2, 4]
Generation 250:	Best = 4.049497e+02	FES = 25100	Diversity = 2.3244e-01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 4.048578e+02	FES = 26100	Diversity = 2.5917e-01	k_t = [3, 4, 3, 4]
Generation 270:	Best = 4.047679e+02	FES = 27100	Diversity = 2.5164e-01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 4.046941e+02	FES = 28100	Diversity = 2.1039e-01	k_t = [2, 2, 3, 3]
Generation 290:	Best = 4.046028e+02	FES = 29100	Diversity = 2.1945e-01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 4.045451e+02	FES = 30100	Diversity = 2.0900e-01	k_t = [2, 3, 3, 3]
Generation 310:	Best = 4.044770e+02	FES = 31100	Diversity = 1.9361e-01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 4.044061e+02	FES = 32100	Diversity = 2.2457e-01	k_t = [2, 4, 3, 2]
Generation 330:	Best = 4.043190e+02	FES = 33100	Diversity = 2.4573e-01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 4.042403e+02	FES = 34100	Diversity = 2.0817e-01	k_t = [4, 2, 3, 2]
Generation 350:	Best = 4.041519e+02	FES = 35100	Diversity = 2.2199e-01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 4.040435e+02	FES = 36100	Diversity = 2.5365e-01	k_t = [3, 2, 3, 4]
Generation 370:	Best = 4.039391e+02	FES = 37100	Diversity = 2.8478e-01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 4.038318e+02	FES = 38100	Diversity = 3.2582e-01	k_t = [3, 2, 3, 2]
Generation 390:	Best = 4.037214e+02	FES = 39100	Diversity = 3.4158e-01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 4.036063e+02	FES = 40100	Diversity = 2.9442e-01	k_t = [3, 3, 4, 2]
Generation 410:	Best = 4.035305e+02	FES = 41100	Diversity = 2.3500e-01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 4.034427e+02	FES = 42100	Diversity = 2.0381e-01	k_t = [4, 4, 2, 4]
Generation 430:	Best = 4.033742e+02	FES = 43100	Diversity = 1.7773e-01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 4.032936e+02	FES = 44100	Diversity = 1.8279e-01	k_t = [2, 3, 3, 2]
Generation 450:	Best = 4.032042e+02	FES = 45100	Diversity = 2.0388e-01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 4.031356e+02	FES = 46100	Diversity = 1.9863e-01	k_t = [3, 4, 2, 2]
Generation 470:	Best = 4.030647e+02	FES = 47100	Diversity = 1.7807e-01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 4.030040e+02	FES = 48100	Diversity = 1.6556e-01	k_t = [3, 3, 4, 3]
Generation 490:	Best = 4.029282e+02	FES = 49100	Diversity = 1.5840e-01	k_t = [1, 1, 1, 1]

Generation 500:	Best = 4.028130e+02	FES = 50100	Diversity = 2.1984e-01	k_t = [4, 2, 3, 2]
Generation 510:	Best = 4.027269e+02	FES = 51100	Diversity = 1.9463e-01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 4.026326e+02	FES = 52100	Diversity = 2.0089e-01	k_t = [2, 3, 2, 3]
Generation 530:	Best = 4.025787e+02	FES = 53100	Diversity = 1.4968e-01	k_t = [1, 1, 1, 1]
Generation 540:	Best = 4.025232e+02	FES = 54100	Diversity = 1.5874e-01	k_t = [2, 4, 3, 4]
Generation 550:	Best = 4.024615e+02	FES = 55100	Diversity = 1.4279e-01	k_t = [1, 1, 1, 1]
Generation 560:	Best = 4.024031e+02	FES = 56100	Diversity = 1.3465e-01	k_t = [4, 2, 3, 4]
Generation 570:	Best = 4.023510e+02	FES = 57100	Diversity = 1.2546e-01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 4.022944e+02	FES = 58100	Diversity = 1.4856e-01	k_t = [4, 4, 4, 4]
Generation 590:	Best = 4.022281e+02	FES = 59100	Diversity = 1.3503e-01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 4.021581e+02	FES = 60100	Diversity = 1.3707e-01	k_t = [4, 3, 2, 3]
Generation 610:	Best = 4.021103e+02	FES = 61100	Diversity = 1.2400e-01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 4.020664e+02	FES = 62100	Diversity = 1.1279e-01	k_t = [4, 4, 4, 3]
Generation 630:	Best = 4.020160e+02	FES = 63100	Diversity = 1.1826e-01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 4.019553e+02	FES = 64100	Diversity = 1.4854e-01	k_t = [3, 2, 4, 3]
Generation 650:	Best = 4.018700e+02	FES = 65100	Diversity = 1.8583e-01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 4.017890e+02	FES = 66100	Diversity = 1.6202e-01	k_t = [2, 3, 4, 3]
Generation 670:	Best = 4.017137e+02	FES = 67100	Diversity = 1.9350e-01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 4.016257e+02	FES = 68100	Diversity = 1.7840e-01	k_t = [4, 4, 3, 4]
Generation 690:	Best = 4.015665e+02	FES = 69100	Diversity = 1.5958e-01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 4.015109e+02	FES = 70100	Diversity = 1.5001e-01	k_t = [4, 4, 2, 4]
Generation 710:	Best = 4.014516e+02	FES = 71100	Diversity = 1.3186e-01	k_t = [1, 1, 1, 1]
Generation 720:	Best = 4.013870e+02	FES = 72100	Diversity = 1.4452e-01	k_t = [4, 3, 4, 3]
Generation 730:	Best = 4.013388e+02	FES = 73100	Diversity = 1.1752e-01	k_t = [1, 1, 1, 1]
Generation 740:	Best = 4.012900e+02	FES = 74100	Diversity = 1.0969e-01	k_t = [3, 3, 4, 2]
Generation 750:	Best = 4.012477e+02	FES = 75100	Diversity = 1.0827e-01	k_t = [1, 1, 1, 1]
Generation 760:	Best = 4.011977e+02	FES = 76100	Diversity = 1.1765e-01	k_t = [3, 4, 3, 4]

Generation 770:	Best = 4.011322e+02	FES = 77100	Diversity = 1.4962e-01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 4.010496e+02	FES = 78100	Diversity = 1.8366e-01	k_t
= [3, 3, 4, 2]				
Generation 790:	Best = 4.009945e+02	FES = 79100	Diversity = 1.3065e-01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 4.009468e+02	FES = 80100	Diversity = 1.1232e-01	k_t
= [3, 4, 4, 4]				
Generation 810:	Best = 4.009085e+02	FES = 81100	Diversity = 1.0170e-01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 4.008790e+02	FES = 82100	Diversity = 8.9991e-02	k_t
= [3, 4, 3, 4]				
Generation 830:	Best = 4.008517e+02	FES = 83100	Diversity = 7.1610e-02	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 4.008292e+02	FES = 84100	Diversity = 9.2143e-02	k_t
= [4, 2, 2, 2]				
Generation 850:	Best = 4.008017e+02	FES = 85100	Diversity = 7.2672e-02	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 4.007850e+02	FES = 86100	Diversity = 5.4522e-02	k_t
= [3, 2, 2, 4]				
Generation 870:	Best = 4.007678e+02	FES = 87100	Diversity = 5.3193e-02	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 4.007515e+02	FES = 88100	Diversity = 5.9202e-02	k_t
= [2, 2, 2, 2]				
Generation 890:	Best = 4.007304e+02	FES = 89100	Diversity = 6.6856e-02	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 4.007131e+02	FES = 90100	Diversity = 6.0327e-02	k_t
= [2, 2, 4, 4]				
Generation 910:	Best = 4.007011e+02	FES = 91100	Diversity = 4.9268e-02	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 4.006838e+02	FES = 92100	Diversity = 5.8001e-02	k_t
= [4, 3, 3, 3]				
Generation 930:	Best = 4.006692e+02	FES = 93100	Diversity = 4.4608e-02	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 4.006533e+02	FES = 94100	Diversity = 5.4772e-02	k_t
= [2, 3, 4, 4]				
Generation 950:	Best = 4.006407e+02	FES = 95100	Diversity = 4.9297e-02	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 4.006304e+02	FES = 96100	Diversity = 4.6249e-02	k_t
= [4, 2, 4, 3]				
Generation 970:	Best = 4.006128e+02	FES = 97100	Diversity = 5.8472e-02	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 4.005888e+02	FES = 98100	Diversity = 7.8368e-02	k_t
= [4, 2, 3, 4]				
Generation 990:	Best = 4.005666e+02	FES = 99100	Diversity = 6.4844e-02	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 4.005515e+02

Total Function Evaluations = 100000

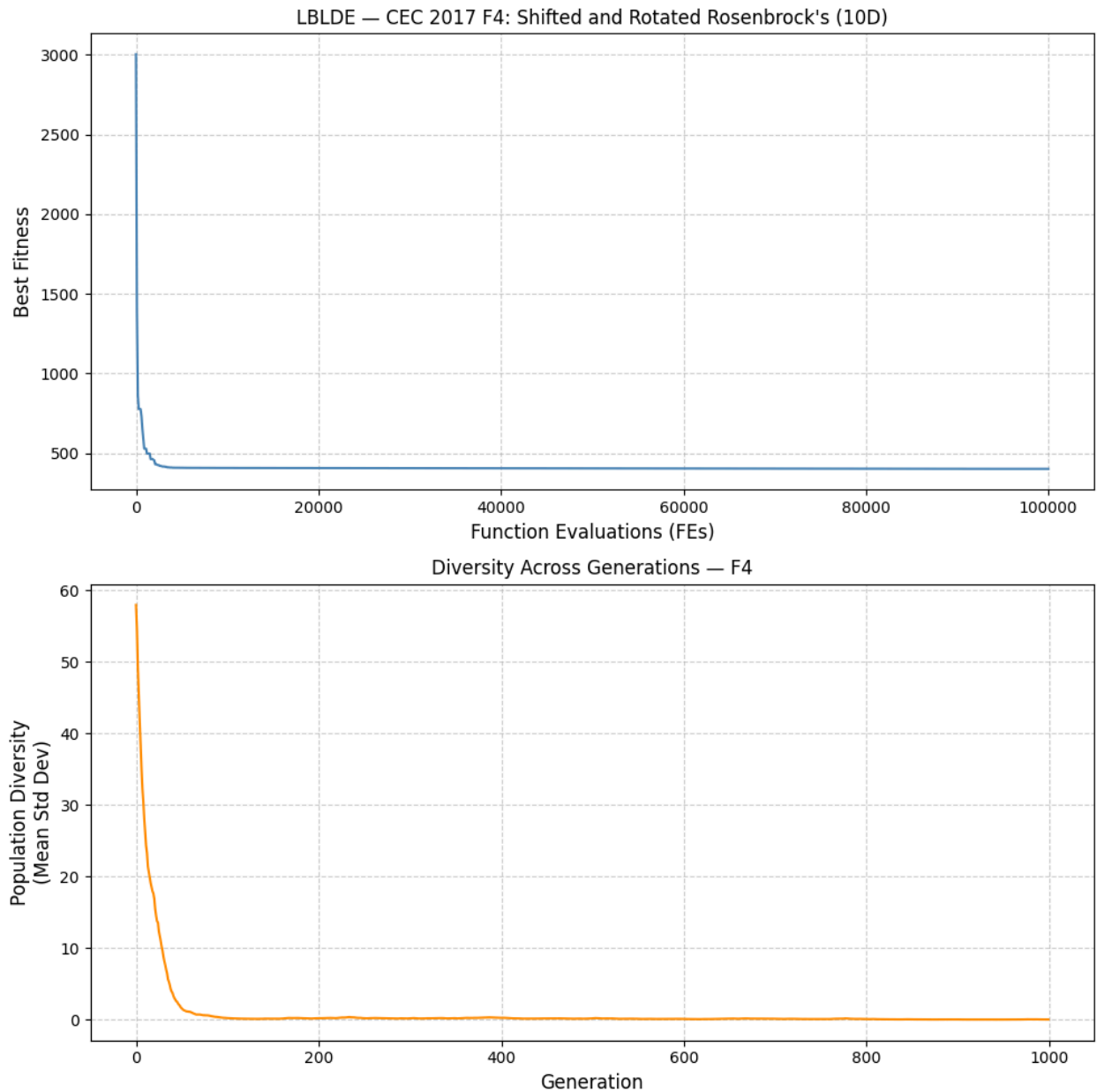
=====

Results:

```

Best Fitness      : 4.005515e+02
Global Optimum    : 4.000000e+02
Error (|best-optimum|) : 5.514667e-01
Best Solution (5 dim) : [ 49.038572    18.54244488 -65.09400628   64.31276458
25.07685381]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F4.png

```

In [21]: # =====
# F5 — Shifted and Rotated Rastrigin
# =====
run_and_plot('F5', make_wrapper(simple.f5), 500.0, "Shifted and Rotated Rastrigin")

```

```

=====
=
LBLDE_s on CEC 2017 F5 – Single Run
=====
=
Function      : Shifted and Rotated Rastrigin's (10D)
Bounds       : [-100, 100]
Global optimum : 500.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 6.377184e+02 | Diversity = 5.7928e+01
Generation 10: Best = 5.584894e+02 FES = 1100 Diversity = 3.0687e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 5.528411e+02 FES = 2100 Diversity = 2.4210e+01 k_t
= [4, 3, 4, 2]
Generation 30: Best = 5.375924e+02 FES = 3100 Diversity = 2.2218e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 5.373312e+02 FES = 4100 Diversity = 2.1276e+01 k_t
= [3, 3, 4, 4]
Generation 50: Best = 5.353656e+02 FES = 5100 Diversity = 2.0647e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 5.347053e+02 FES = 6100 Diversity = 2.0212e+01 k_t
= [2, 4, 3, 4]
Generation 70: Best = 5.347053e+02 FES = 7100 Diversity = 2.0070e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 5.339363e+02 FES = 8100 Diversity = 1.9983e+01 k_t
= [2, 2, 2, 2]
Generation 90: Best = 5.215939e+02 FES = 9100 Diversity = 1.9899e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 5.207014e+02 FES = 10100 Diversity = 1.9366e+01 k_t
= [3, 3, 2, 2]
Generation 110: Best = 5.177655e+02 FES = 11100 Diversity = 1.9238e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 5.177655e+02 FES = 12100 Diversity = 1.9242e+01 k_t
= [4, 3, 4, 4]
Generation 130: Best = 5.177655e+02 FES = 13100 Diversity = 1.8886e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 5.168209e+02 FES = 14100 Diversity = 1.8340e+01 k_t
= [3, 2, 4, 3]
Generation 150: Best = 5.162684e+02 FES = 15100 Diversity = 1.8117e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 5.115238e+02 FES = 16100 Diversity = 1.7668e+01 k_t
= [4, 3, 4, 2]
Generation 170: Best = 5.115238e+02 FES = 17100 Diversity = 1.7833e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 5.115238e+02 FES = 18100 Diversity = 1.7769e+01 k_t
= [3, 2, 4, 2]
Generation 190: Best = 5.115238e+02 FES = 19100 Diversity = 1.7758e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 5.115238e+02 FES = 20100 Diversity = 1.7508e+01 k_t
= [3, 2, 3, 4]
Generation 210: Best = 5.115238e+02 FES = 21100 Diversity = 1.7407e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 5.115238e+02 FES = 22100 Diversity = 1.7353e+01 k_t
= [2, 3, 3, 2]

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Generation 230:	Best = 5.115238e+02	FES = 23100	Diversity = 1.7331e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 5.107119e+02	FES = 24100	Diversity = 1.7277e+01	k_t = [4, 3, 2, 3]
Generation 250:	Best = 5.107119e+02	FES = 25100	Diversity = 1.7246e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 5.107119e+02	FES = 26100	Diversity = 1.7467e+01	k_t = [4, 3, 2, 2]
Generation 270:	Best = 5.107119e+02	FES = 27100	Diversity = 1.7443e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 5.107119e+02	FES = 28100	Diversity = 1.7259e+01	k_t = [4, 3, 3, 3]
Generation 290:	Best = 5.107119e+02	FES = 29100	Diversity = 1.7049e+01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 5.103513e+02	FES = 30100	Diversity = 1.6843e+01	k_t = [3, 3, 3, 3]
Generation 310:	Best = 5.103513e+02	FES = 31100	Diversity = 1.6701e+01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 5.103513e+02	FES = 32100	Diversity = 1.6310e+01	k_t = [2, 2, 2, 4]
Generation 330:	Best = 5.097565e+02	FES = 33100	Diversity = 1.6373e+01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 5.097565e+02	FES = 34100	Diversity = 1.6223e+01	k_t = [4, 4, 3, 3]
Generation 350:	Best = 5.091693e+02	FES = 35100	Diversity = 1.6014e+01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 5.086186e+02	FES = 36100	Diversity = 1.5517e+01	k_t = [2, 3, 3, 2]
Generation 370:	Best = 5.086186e+02	FES = 37100	Diversity = 1.5726e+01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 5.086186e+02	FES = 38100	Diversity = 1.5089e+01	k_t = [4, 3, 4, 4]
Generation 390:	Best = 5.086186e+02	FES = 39100	Diversity = 1.4876e+01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 5.086186e+02	FES = 40100	Diversity = 1.4631e+01	k_t = [2, 3, 4, 2]
Generation 410:	Best = 5.086186e+02	FES = 41100	Diversity = 1.4079e+01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 5.073085e+02	FES = 42100	Diversity = 1.3698e+01	k_t = [4, 3, 2, 4]
Generation 430:	Best = 5.067906e+02	FES = 43100	Diversity = 1.3460e+01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 5.048873e+02	FES = 44100	Diversity = 1.3240e+01	k_t = [3, 3, 3, 4]
Generation 450:	Best = 5.048873e+02	FES = 45100	Diversity = 1.2996e+01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 5.048873e+02	FES = 46100	Diversity = 1.2821e+01	k_t = [4, 4, 3, 3]
Generation 470:	Best = 5.044990e+02	FES = 47100	Diversity = 1.2727e+01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 5.032457e+02	FES = 48100	Diversity = 1.2270e+01	k_t = [3, 2, 2, 4]
Generation 490:	Best = 5.032457e+02	FES = 49100	Diversity = 1.2171e+01	k_t = [1, 1, 1, 1]

Generation 500:	Best = 5.032457e+02	FES = 50100	Diversity = 1.1564e+01	k_t = [2, 2, 3, 3]
Generation 510:	Best = 5.032457e+02	FES = 51100	Diversity = 1.0901e+01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 5.029060e+02	FES = 52100	Diversity = 1.0493e+01	k_t = [4, 4, 3, 3]
Generation 530:	Best = 5.024217e+02	FES = 53100	Diversity = 8.9669e+00	k_t = [1, 1, 1, 1]
Generation 540:	Best = 5.022445e+02	FES = 54100	Diversity = 6.9490e+00	k_t = [3, 2, 4, 3]
Generation 550:	Best = 5.021343e+02	FES = 55100	Diversity = 5.3570e+00	k_t = [1, 1, 1, 1]
Generation 560:	Best = 5.020025e+02	FES = 56100	Diversity = 4.4919e+00	k_t = [2, 3, 2, 3]
Generation 570:	Best = 5.013029e+02	FES = 57100	Diversity = 4.0833e+00	k_t = [1, 1, 1, 1]
Generation 580:	Best = 5.011101e+02	FES = 58100	Diversity = 3.3532e+00	k_t = [2, 3, 4, 4]
Generation 590:	Best = 5.010721e+02	FES = 59100	Diversity = 2.6176e+00	k_t = [1, 1, 1, 1]
Generation 600:	Best = 5.010291e+02	FES = 60100	Diversity = 8.6638e-01	k_t = [2, 3, 3, 3]
Generation 610:	Best = 5.010054e+02	FES = 61100	Diversity = 7.3478e-02	k_t = [1, 1, 1, 1]
Generation 620:	Best = 5.009972e+02	FES = 62100	Diversity = 3.5920e-02	k_t = [4, 3, 3, 2]
Generation 630:	Best = 5.009954e+02	FES = 63100	Diversity = 1.7002e-02	k_t = [1, 1, 1, 1]
Generation 640:	Best = 5.009951e+02	FES = 64100	Diversity = 8.4259e-03	k_t = [3, 3, 4, 2]
Generation 650:	Best = 5.009950e+02	FES = 65100	Diversity = 3.8313e-03	k_t = [1, 1, 1, 1]
Generation 660:	Best = 5.009950e+02	FES = 66100	Diversity = 1.7359e-03	k_t = [4, 4, 4, 2]
Generation 670:	Best = 5.009950e+02	FES = 67100	Diversity = 8.4872e-04	k_t = [1, 1, 1, 1]
Generation 680:	Best = 5.009950e+02	FES = 68100	Diversity = 3.6052e-04	k_t = [4, 3, 4, 4]
Generation 690:	Best = 5.009950e+02	FES = 69100	Diversity = 1.5919e-04	k_t = [1, 1, 1, 1]
Generation 700:	Best = 5.009950e+02	FES = 70100	Diversity = 7.8164e-05	k_t = [2, 4, 2, 2]
Generation 710:	Best = 5.009950e+02	FES = 71100	Diversity = 3.3300e-05	k_t = [1, 1, 1, 1]
Generation 720:	Best = 5.009950e+02	FES = 72100	Diversity = 1.5109e-05	k_t = [2, 4, 3, 2]
Generation 730:	Best = 5.009950e+02	FES = 73100	Diversity = 6.8920e-06	k_t = [1, 1, 1, 1]
Generation 740:	Best = 5.009950e+02	FES = 74100	Diversity = 3.1880e-06	k_t = [4, 3, 2, 2]
Generation 750:	Best = 5.009950e+02	FES = 75100	Diversity = 1.4873e-06	k_t = [1, 1, 1, 1]
Generation 760:	Best = 5.009950e+02	FES = 76100	Diversity = 6.7187e-07	k_t = [3, 4, 3, 4]

Generation 770:	Best = 5.009950e+02	FES = 77100	Diversity = 3.1732e-07	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 5.009950e+02	FES = 78100	Diversity = 1.5163e-07	k_t
= [3, 4, 2, 3]				
Generation 790:	Best = 5.009950e+02	FES = 79100	Diversity = 9.5892e-08	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 5.009950e+02	FES = 80100	Diversity = 6.5846e-08	k_t
= [3, 3, 4, 2]				
Generation 810:	Best = 5.009950e+02	FES = 81100	Diversity = 5.7524e-08	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 5.009950e+02	FES = 82100	Diversity = 5.6861e-08	k_t
= [4, 3, 2, 4]				
Generation 830:	Best = 5.009950e+02	FES = 83100	Diversity = 5.5797e-08	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 5.009950e+02	FES = 84100	Diversity = 5.9002e-08	k_t
= [3, 2, 4, 3]				
Generation 850:	Best = 5.009950e+02	FES = 85100	Diversity = 5.8631e-08	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 5.009950e+02	FES = 86100	Diversity = 6.0082e-08	k_t
= [3, 2, 3, 4]				
Generation 870:	Best = 5.009950e+02	FES = 87100	Diversity = 5.7101e-08	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 5.009950e+02	FES = 88100	Diversity = 5.5253e-08	k_t
= [3, 3, 3, 4]				
Generation 890:	Best = 5.009950e+02	FES = 89100	Diversity = 5.6375e-08	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 5.009950e+02	FES = 90100	Diversity = 6.0168e-08	k_t
= [4, 4, 3, 3]				
Generation 910:	Best = 5.009950e+02	FES = 91100	Diversity = 5.8966e-08	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 5.009950e+02	FES = 92100	Diversity = 5.9955e-08	k_t
= [4, 3, 4, 4]				
Generation 930:	Best = 5.009950e+02	FES = 93100	Diversity = 5.9357e-08	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 5.009950e+02	FES = 94100	Diversity = 6.1159e-08	k_t
= [3, 4, 4, 2]				
Generation 950:	Best = 5.009950e+02	FES = 95100	Diversity = 6.0327e-08	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 5.009950e+02	FES = 96100	Diversity = 5.8206e-08	k_t
= [2, 2, 4, 2]				
Generation 970:	Best = 5.009950e+02	FES = 97100	Diversity = 5.9782e-08	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 5.009950e+02	FES = 98100	Diversity = 6.1195e-08	k_t
= [2, 2, 4, 3]				
Generation 990:	Best = 5.009950e+02	FES = 99100	Diversity = 5.6640e-08	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 5.009950e+02

Total Function Evaluations = 100000

=====

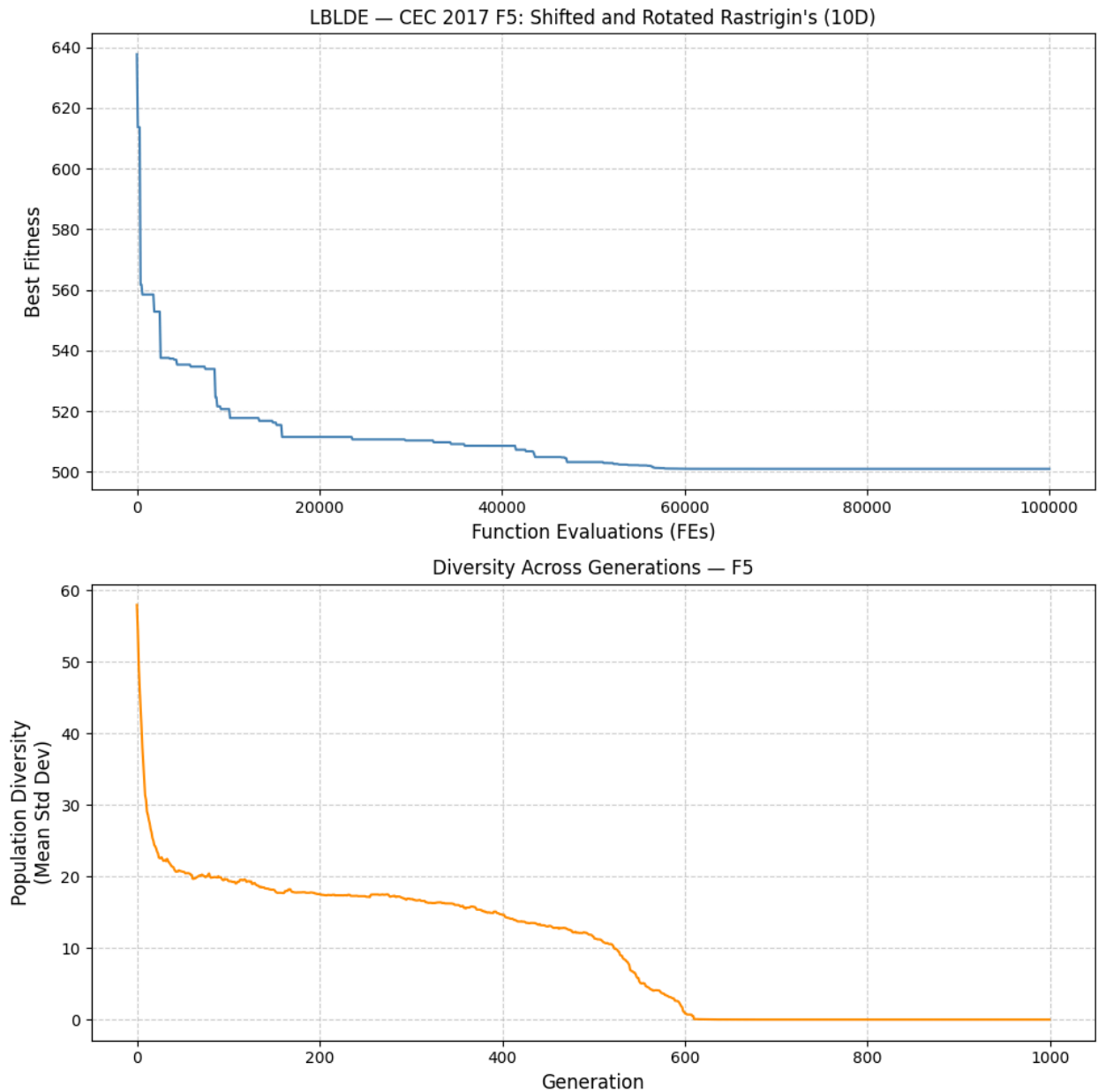
=

Results:

```

Best Fitness      : 5.009950e+02
Global Optimum    : 5.000000e+02
Error (|best-optimum|) : 9.949591e-01
Best Solution (5 dim) : [-17.4084504   56.17470328 -31.76157285 -42.55877148
14.25802259]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F5.png

```

In [22]: # =====
# F6 – Shifted and Rotated Schaffer's F7
# =====
run_and_plot('F6', make_wrapper(simple.f6), 600.0, "Shifted and Rotated Schaff

```

```

=====
=
LBLDE_s on CEC 2017 F6 – Single Run
=====
=
Function      : Shifted and Rotated Schaffer's F7 (10D)
Bounds       : [-100, 100]
Global optimum : 600.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 7.140452e+02 | Diversity = 5.7928e+01
Generation 10: Best = 6.348211e+02 FES = 1100 Diversity = 3.1966e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 6.155503e+02 FES = 2100 Diversity = 1.9129e+01 k_t
= [4, 3, 4, 2]
Generation 30: Best = 6.113481e+02 FES = 3100 Diversity = 1.0144e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 6.056095e+02 FES = 4100 Diversity = 5.9715e+00 k_t
= [3, 3, 4, 4]
Generation 50: Best = 6.020650e+02 FES = 5100 Diversity = 3.4664e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 6.013791e+02 FES = 6100 Diversity = 1.5808e+00 k_t
= [4, 2, 4, 4]
Generation 70: Best = 6.006072e+02 FES = 7100 Diversity = 1.0626e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 6.004720e+02 FES = 8100 Diversity = 5.5563e-01 k_t
= [4, 3, 4, 2]
Generation 90: Best = 6.002961e+02 FES = 9100 Diversity = 3.3977e-01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 6.001447e+02 FES = 10100 Diversity = 1.9322e-01 k_t
= [4, 4, 2, 4]
Generation 110: Best = 6.001180e+02 FES = 11100 Diversity = 1.1611e-01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 6.000521e+02 FES = 12100 Diversity = 6.5095e-02 k_t
= [2, 3, 3, 4]
Generation 130: Best = 6.000379e+02 FES = 13100 Diversity = 4.7714e-02 k_t
= [1, 1, 1, 1]
Generation 140: Best = 6.000314e+02 FES = 14100 Diversity = 3.5292e-02 k_t
= [2, 2, 3, 4]
Generation 150: Best = 6.000127e+02 FES = 15100 Diversity = 2.6363e-02 k_t
= [1, 1, 1, 1]
Generation 160: Best = 6.000094e+02 FES = 16100 Diversity = 1.3919e-02 k_t
= [3, 2, 2, 2]
Generation 170: Best = 6.000062e+02 FES = 17100 Diversity = 7.9286e-03 k_t
= [1, 1, 1, 1]
Generation 180: Best = 6.000036e+02 FES = 18100 Diversity = 5.0456e-03 k_t
= [4, 2, 3, 2]
Generation 190: Best = 6.000036e+02 FES = 19100 Diversity = 3.7263e-03 k_t
= [1, 1, 1, 1]
Generation 200: Best = 6.000020e+02 FES = 20100 Diversity = 2.2989e-03 k_t
= [3, 4, 4, 2]
Generation 210: Best = 6.000011e+02 FES = 21100 Diversity = 1.5071e-03 k_t
= [1, 1, 1, 1]
Generation 220: Best = 6.000005e+02 FES = 22100 Diversity = 9.1282e-04 k_t
= [4, 3, 4, 4]

```

Generation 230:	Best = 6.000005e+02	FES = 23100	Diversity = 5.9204e-04	k_t = [1, 1, 1, 1]
Generation 240:	Best = 6.000004e+02	FES = 24100	Diversity = 4.5656e-04	k_t = [2, 4, 3, 3]
Generation 250:	Best = 6.000002e+02	FES = 25100	Diversity = 3.0972e-04	k_t = [1, 1, 1, 1]
Generation 260:	Best = 6.000001e+02	FES = 26100	Diversity = 1.7950e-04	k_t = [2, 2, 3, 4]
Generation 270:	Best = 6.000001e+02	FES = 27100	Diversity = 1.1012e-04	k_t = [1, 1, 1, 1]
Generation 280:	Best = 6.000001e+02	FES = 28100	Diversity = 9.0711e-05	k_t = [4, 4, 2, 3]
Generation 290:	Best = 6.000001e+02	FES = 29100	Diversity = 7.4662e-05	k_t = [1, 1, 1, 1]
Generation 300:	Best = 6.000000e+02	FES = 30100	Diversity = 5.3362e-05	k_t = [2, 3, 4, 3]
Generation 310:	Best = 6.000000e+02	FES = 31100	Diversity = 2.3031e-05	k_t = [1, 1, 1, 1]
Generation 320:	Best = 6.000000e+02	FES = 32100	Diversity = 1.3660e-05	k_t = [4, 2, 3, 3]
Generation 330:	Best = 6.000000e+02	FES = 33100	Diversity = 1.2182e-05	k_t = [1, 1, 1, 1]
Generation 340:	Best = 6.000000e+02	FES = 34100	Diversity = 1.1942e-05	k_t = [4, 4, 3, 3]
Generation 350:	Best = 6.000000e+02	FES = 35100	Diversity = 1.0775e-05	k_t = [1, 1, 1, 1]
Generation 360:	Best = 6.000000e+02	FES = 36100	Diversity = 9.9614e-06	k_t = [4, 2, 2, 3]
Generation 370:	Best = 6.000000e+02	FES = 37100	Diversity = 9.3780e-06	k_t = [1, 1, 1, 1]
Generation 380:	Best = 6.000000e+02	FES = 38100	Diversity = 8.4625e-06	k_t = [3, 4, 3, 2]
Generation 390:	Best = 6.000000e+02	FES = 39100	Diversity = 7.2573e-06	k_t = [1, 1, 1, 1]
Generation 400:	Best = 6.000000e+02	FES = 40100	Diversity = 5.0974e-06	k_t = [3, 4, 4, 3]
Generation 410:	Best = 6.000000e+02	FES = 41100	Diversity = 2.5210e-06	k_t = [1, 1, 1, 1]
Generation 420:	Best = 6.000000e+02	FES = 42100	Diversity = 6.1929e-07	k_t = [2, 4, 4, 2]
Generation 430:	Best = 6.000000e+02	FES = 43100	Diversity = 3.4186e-07	k_t = [1, 1, 1, 1]
Generation 440:	Best = 6.000000e+02	FES = 44100	Diversity = 2.3210e-07	k_t = [4, 4, 4, 2]
Generation 450:	Best = 6.000000e+02	FES = 45100	Diversity = 1.5170e-07	k_t = [1, 1, 1, 1]
Generation 460:	Best = 6.000000e+02	FES = 46100	Diversity = 1.0108e-07	k_t = [3, 3, 2, 3]
Generation 470:	Best = 6.000000e+02	FES = 47100	Diversity = 5.5784e-08	k_t = [1, 1, 1, 1]
Generation 480:	Best = 6.000000e+02	FES = 48100	Diversity = 2.8255e-08	k_t = [3, 2, 4, 4]
Generation 490:	Best = 6.000000e+02	FES = 49100	Diversity = 1.4424e-08	k_t = [1, 1, 1, 1]

Generation 500:	Best = 6.000000e+02	FES = 50100	Diversity = 7.0347e-09	k_t = [3, 4, 3, 2]
Generation 510:	Best = 6.000000e+02	FES = 51100	Diversity = 3.6189e-09	k_t = [1, 1, 1, 1]
Generation 520:	Best = 6.000000e+02	FES = 52100	Diversity = 1.9834e-09	k_t = [2, 4, 2, 2]
Generation 530:	Best = 6.000000e+02	FES = 53100	Diversity = 1.0566e-09	k_t = [1, 1, 1, 1]
Generation 540:	Best = 6.000000e+02	FES = 54100	Diversity = 5.7868e-10	k_t = [3, 3, 3, 2]
Generation 550:	Best = 6.000000e+02	FES = 55100	Diversity = 2.7430e-10	k_t = [1, 1, 1, 1]
Generation 560:	Best = 6.000000e+02	FES = 56100	Diversity = 1.3809e-10	k_t = [3, 2, 3, 3]
Generation 570:	Best = 6.000000e+02	FES = 57100	Diversity = 6.3483e-11	k_t = [1, 1, 1, 1]
Generation 580:	Best = 6.000000e+02	FES = 58100	Diversity = 2.8975e-11	k_t = [3, 2, 4, 2]
Generation 590:	Best = 6.000000e+02	FES = 59100	Diversity = 1.3126e-11	k_t = [1, 1, 1, 1]
Generation 600:	Best = 6.000000e+02	FES = 60100	Diversity = 5.8041e-12	k_t = [2, 4, 3, 2]
Generation 610:	Best = 6.000000e+02	FES = 61100	Diversity = 2.6115e-12	k_t = [1, 1, 1, 1]
Generation 620:	Best = 6.000000e+02	FES = 62100	Diversity = 1.3536e-12	k_t = [2, 3, 3, 4]
Generation 630:	Best = 6.000000e+02	FES = 63100	Diversity = 7.1247e-13	k_t = [1, 1, 1, 1]
Generation 640:	Best = 6.000000e+02	FES = 64100	Diversity = 3.7175e-13	k_t = [3, 4, 4, 2]
Generation 650:	Best = 6.000000e+02	FES = 65100	Diversity = 2.0589e-13	k_t = [1, 1, 1, 1]
Generation 660:	Best = 6.000000e+02	FES = 66100	Diversity = 1.5646e-13	k_t = [2, 4, 3, 2]
Generation 670:	Best = 6.000000e+02	FES = 67100	Diversity = 1.3214e-13	k_t = [1, 1, 1, 1]
Generation 680:	Best = 6.000000e+02	FES = 68100	Diversity = 1.1847e-13	k_t = [2, 3, 2, 4]
Generation 690:	Best = 6.000000e+02	FES = 69100	Diversity = 1.0850e-13	k_t = [1, 1, 1, 1]
Generation 700:	Best = 6.000000e+02	FES = 70100	Diversity = 1.0804e-13	k_t = [2, 4, 4, 2]
Generation 710:	Best = 6.000000e+02	FES = 71100	Diversity = 9.7361e-14	k_t = [1, 1, 1, 1]
Generation 720:	Best = 6.000000e+02	FES = 72100	Diversity = 8.9819e-14	k_t = [2, 3, 2, 2]
Generation 730:	Best = 6.000000e+02	FES = 73100	Diversity = 7.4840e-14	k_t = [1, 1, 1, 1]
Generation 740:	Best = 6.000000e+02	FES = 74100	Diversity = 6.5497e-14	k_t = [3, 4, 4, 3]
Generation 750:	Best = 6.000000e+02	FES = 75100	Diversity = 6.4581e-14	k_t = [1, 1, 1, 1]
Generation 760:	Best = 6.000000e+02	FES = 76100	Diversity = 6.4346e-14	k_t = [3, 3, 2, 2]

Generation 770:	Best = 6.000000e+02	FES = 77100	Diversity = 6.3005e-14	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 6.000000e+02	FES = 78100	Diversity = 5.9640e-14	k_t
= [3, 3, 3, 3]				
Generation 790:	Best = 6.000000e+02	FES = 79100	Diversity = 5.9845e-14	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 6.000000e+02	FES = 80100	Diversity = 5.5761e-14	k_t
= [3, 4, 2, 3]				
Generation 810:	Best = 6.000000e+02	FES = 81100	Diversity = 6.0856e-14	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 6.000000e+02	FES = 82100	Diversity = 6.3883e-14	k_t
= [2, 4, 4, 2]				
Generation 830:	Best = 6.000000e+02	FES = 83100	Diversity = 6.8003e-14	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 6.000000e+02	FES = 84100	Diversity = 6.4149e-14	k_t
= [2, 4, 4, 4]				
Generation 850:	Best = 6.000000e+02	FES = 85100	Diversity = 6.1619e-14	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 6.000000e+02	FES = 86100	Diversity = 5.7399e-14	k_t
= [4, 2, 2, 4]				
Generation 870:	Best = 6.000000e+02	FES = 87100	Diversity = 6.2252e-14	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 6.000000e+02	FES = 88100	Diversity = 6.3652e-14	k_t
= [4, 2, 3, 4]				
Generation 890:	Best = 6.000000e+02	FES = 89100	Diversity = 6.2235e-14	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 6.000000e+02	FES = 90100	Diversity = 6.1959e-14	k_t
= [4, 2, 3, 4]				
Generation 910:	Best = 6.000000e+02	FES = 91100	Diversity = 6.1058e-14	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 6.000000e+02	FES = 92100	Diversity = 6.1598e-14	k_t
= [2, 4, 2, 3]				
Generation 930:	Best = 6.000000e+02	FES = 93100	Diversity = 5.9983e-14	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 6.000000e+02	FES = 94100	Diversity = 6.0889e-14	k_t
= [2, 4, 4, 4]				
Generation 950:	Best = 6.000000e+02	FES = 95100	Diversity = 6.1089e-14	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 6.000000e+02	FES = 96100	Diversity = 6.5368e-14	k_t
= [3, 4, 3, 4]				
Generation 970:	Best = 6.000000e+02	FES = 97100	Diversity = 6.6174e-14	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 6.000000e+02	FES = 98100	Diversity = 6.4389e-14	k_t
= [2, 4, 2, 4]				
Generation 990:	Best = 6.000000e+02	FES = 99100	Diversity = 6.1741e-14	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 6.000000e+02

Total Function Evaluations = 100000

=====

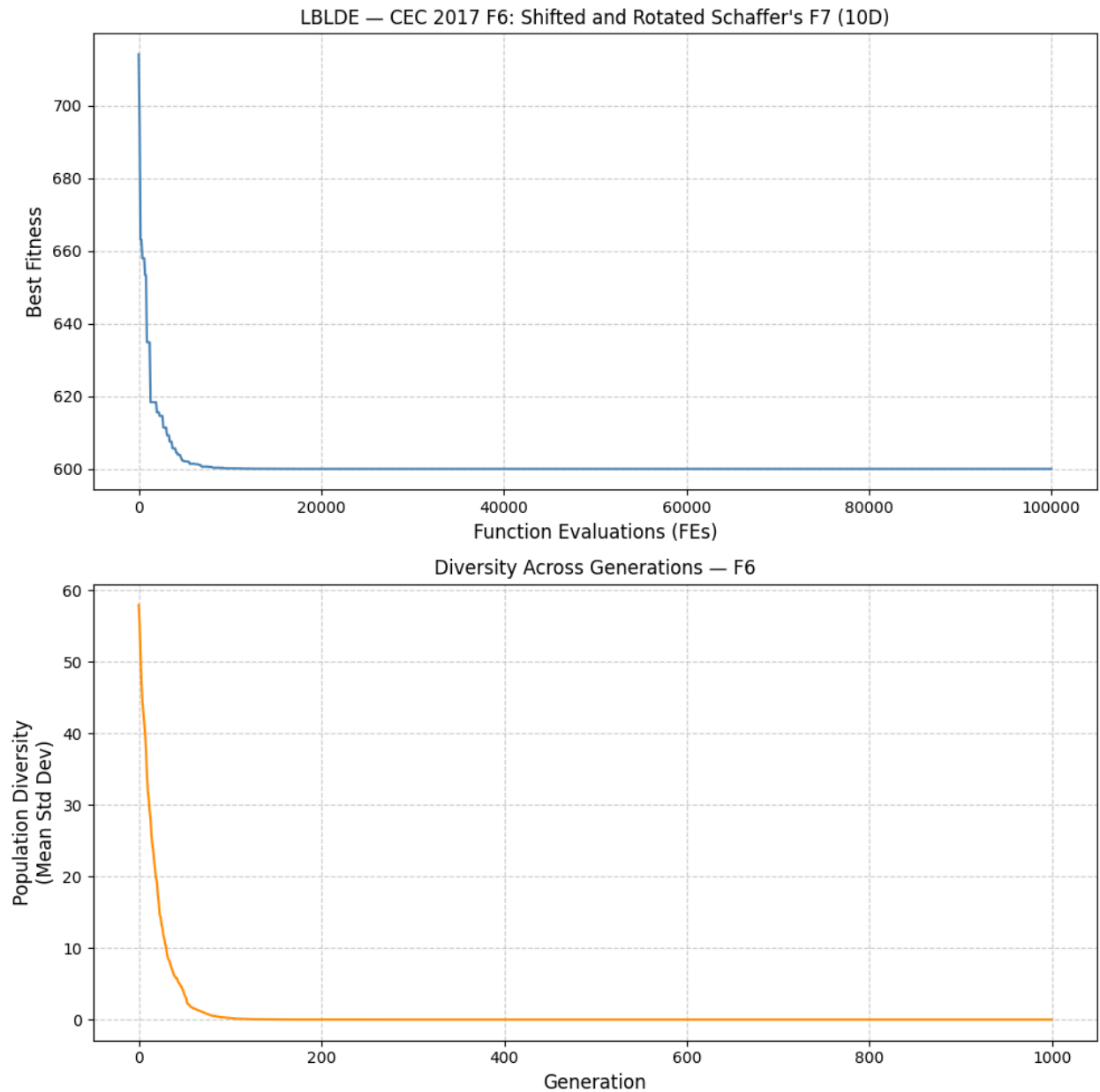
=

Results:

```

Best Fitness      : 6.000000e+02
Global Optimum    : 6.000000e+02
Error (|best-optimum|) : 0.000000e+00
Best Solution (5 dim) : [ 79.08939294 -24.57277765 -34.67392644 -44.68162088
75.99787965]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F6.png

```

In [23]: # =====
# F7 — Shifted and Rotated Lunacek Bi-Rastrigin
# =====
run_and_plot('F7', make_wrapper(simple.f7), 700.0, "Shifted and Rotated Lunacek Bi-Rastrigin")

```



```

=====
=
LBLDE_s on CEC 2017 F7 – Single Run
=====
=
Function      : Shifted and Rotated Lunacek Bi-Rastrigin's (10D)
Bounds       : [-100, 100]
Global optimum : 700.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 1.202264e+03 | Diversity = 5.7928e+01
Generation 10: Best = 8.078117e+02 FES = 1100 Diversity = 2.0412e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 7.719853e+02 FES = 2100 Diversity = 1.0872e+01 k_t
= [2, 4, 2, 3]
Generation 30: Best = 7.552852e+02 FES = 3100 Diversity = 9.1963e+00 k_t
= [1, 1, 1, 1]
Generation 40: Best = 7.552852e+02 FES = 4100 Diversity = 8.4676e+00 k_t
= [2, 2, 3, 2]
Generation 50: Best = 7.455100e+02 FES = 5100 Diversity = 8.0924e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 7.455100e+02 FES = 6100 Diversity = 7.7833e+00 k_t
= [2, 2, 2, 2]
Generation 70: Best = 7.429293e+02 FES = 7100 Diversity = 7.4200e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 7.404690e+02 FES = 8100 Diversity = 7.2616e+00 k_t
= [4, 4, 3, 3]
Generation 90: Best = 7.326571e+02 FES = 9100 Diversity = 7.1589e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 7.326571e+02 FES = 10100 Diversity = 7.0788e+00 k_t
= [3, 3, 3, 4]
Generation 110: Best = 7.326571e+02 FES = 11100 Diversity = 6.9922e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 7.326571e+02 FES = 12100 Diversity = 7.0790e+00 k_t
= [2, 4, 4, 3]
Generation 130: Best = 7.308913e+02 FES = 13100 Diversity = 6.9534e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 7.308913e+02 FES = 14100 Diversity = 6.9104e+00 k_t
= [2, 3, 2, 3]
Generation 150: Best = 7.238202e+02 FES = 15100 Diversity = 6.9770e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 7.238202e+02 FES = 16100 Diversity = 7.0535e+00 k_t
= [2, 2, 4, 3]
Generation 170: Best = 7.238202e+02 FES = 17100 Diversity = 6.8207e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 7.238202e+02 FES = 18100 Diversity = 6.6973e+00 k_t
= [4, 4, 2, 2]
Generation 190: Best = 7.238202e+02 FES = 19100 Diversity = 6.7315e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 7.238202e+02 FES = 20100 Diversity = 6.6929e+00 k_t
= [3, 4, 3, 2]
Generation 210: Best = 7.238202e+02 FES = 21100 Diversity = 6.6313e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 7.238202e+02 FES = 22100 Diversity = 6.5745e+00 k_t
= [4, 3, 2, 3]

```

Generation 230:	Best = 7.238202e+02	FES = 23100	Diversity = 6.5034e+00	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 7.224054e+02	FES = 24100	Diversity = 6.4689e+00	k_t
= [4, 3, 2, 2]				
Generation 250:	Best = 7.223738e+02	FES = 25100	Diversity = 6.4671e+00	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 7.223738e+02	FES = 26100	Diversity = 6.4507e+00	k_t
= [2, 2, 3, 2]				
Generation 270:	Best = 7.223738e+02	FES = 27100	Diversity = 6.4592e+00	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 7.223738e+02	FES = 28100	Diversity = 6.3602e+00	k_t
= [4, 3, 3, 4]				
Generation 290:	Best = 7.223738e+02	FES = 29100	Diversity = 6.2908e+00	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 7.223738e+02	FES = 30100	Diversity = 6.1463e+00	k_t
= [3, 3, 3, 2]				
Generation 310:	Best = 7.223738e+02	FES = 31100	Diversity = 6.0521e+00	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 7.223738e+02	FES = 32100	Diversity = 5.8677e+00	k_t
= [3, 3, 2, 4]				
Generation 330:	Best = 7.203808e+02	FES = 33100	Diversity = 5.6730e+00	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 7.203368e+02	FES = 34100	Diversity = 5.5835e+00	k_t
= [2, 4, 3, 2]				
Generation 350:	Best = 7.171721e+02	FES = 35100	Diversity = 5.3157e+00	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 7.171721e+02	FES = 36100	Diversity = 5.2146e+00	k_t
= [4, 3, 3, 3]				
Generation 370:	Best = 7.171721e+02	FES = 37100	Diversity = 5.1238e+00	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 7.171721e+02	FES = 38100	Diversity = 5.0987e+00	k_t
= [2, 4, 4, 2]				
Generation 390:	Best = 7.171721e+02	FES = 39100	Diversity = 4.9986e+00	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 7.171721e+02	FES = 40100	Diversity = 4.9155e+00	k_t
= [2, 3, 2, 4]				
Generation 410:	Best = 7.171721e+02	FES = 41100	Diversity = 4.8124e+00	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 7.159570e+02	FES = 42100	Diversity = 4.5987e+00	k_t
= [3, 4, 4, 2]				
Generation 430:	Best = 7.150988e+02	FES = 43100	Diversity = 4.3898e+00	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 7.138116e+02	FES = 44100	Diversity = 4.1091e+00	k_t
= [2, 3, 3, 4]				
Generation 450:	Best = 7.133389e+02	FES = 45100	Diversity = 3.9130e+00	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 7.131751e+02	FES = 46100	Diversity = 3.3845e+00	k_t
= [2, 2, 3, 4]				
Generation 470:	Best = 7.128432e+02	FES = 47100	Diversity = 2.2132e+00	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 7.128322e+02	FES = 48100	Diversity = 6.2183e-01	k_t
= [4, 2, 3, 2]				
Generation 490:	Best = 7.127816e+02	FES = 49100	Diversity = 5.3532e-01	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 7.127449e+02	FES = 50100	Diversity = 5.1728e-01	k_t = [4, 2, 2, 2]
Generation 510:	Best = 7.127318e+02	FES = 51100	Diversity = 4.8370e-01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 7.127311e+02	FES = 52100	Diversity = 2.7585e-01	k_t = [4, 3, 2, 4]
Generation 530:	Best = 7.127295e+02	FES = 53100	Diversity = 4.6760e-03	k_t = [1, 1, 1, 1]
Generation 540:	Best = 7.127292e+02	FES = 54100	Diversity = 2.0981e-03	k_t = [3, 3, 2, 2]
Generation 550:	Best = 7.127291e+02	FES = 55100	Diversity = 1.0700e-03	k_t = [1, 1, 1, 1]
Generation 560:	Best = 7.127291e+02	FES = 56100	Diversity = 5.2319e-04	k_t = [3, 4, 2, 4]
Generation 570:	Best = 7.127291e+02	FES = 57100	Diversity = 2.5012e-04	k_t = [1, 1, 1, 1]
Generation 580:	Best = 7.127291e+02	FES = 58100	Diversity = 9.9909e-05	k_t = [2, 2, 3, 4]
Generation 590:	Best = 7.127291e+02	FES = 59100	Diversity = 4.7191e-05	k_t = [1, 1, 1, 1]
Generation 600:	Best = 7.127291e+02	FES = 60100	Diversity = 2.2059e-05	k_t = [2, 3, 3, 3]
Generation 610:	Best = 7.127291e+02	FES = 61100	Diversity = 9.7317e-06	k_t = [1, 1, 1, 1]
Generation 620:	Best = 7.127291e+02	FES = 62100	Diversity = 4.1221e-06	k_t = [2, 2, 4, 4]
Generation 630:	Best = 7.127291e+02	FES = 63100	Diversity = 1.7574e-06	k_t = [1, 1, 1, 1]
Generation 640:	Best = 7.127291e+02	FES = 64100	Diversity = 6.4817e-07	k_t = [2, 3, 3, 3]
Generation 650:	Best = 7.127291e+02	FES = 65100	Diversity = 3.0115e-07	k_t = [1, 1, 1, 1]
Generation 660:	Best = 7.127291e+02	FES = 66100	Diversity = 1.3952e-07	k_t = [2, 2, 4, 2]
Generation 670:	Best = 7.127291e+02	FES = 67100	Diversity = 6.4160e-08	k_t = [1, 1, 1, 1]
Generation 680:	Best = 7.127291e+02	FES = 68100	Diversity = 3.2389e-08	k_t = [3, 4, 3, 2]
Generation 690:	Best = 7.127291e+02	FES = 69100	Diversity = 2.2979e-08	k_t = [1, 1, 1, 1]
Generation 700:	Best = 7.127291e+02	FES = 70100	Diversity = 1.9510e-08	k_t = [2, 3, 2, 3]
Generation 710:	Best = 7.127291e+02	FES = 71100	Diversity = 1.8863e-08	k_t = [1, 1, 1, 1]
Generation 720:	Best = 7.127291e+02	FES = 72100	Diversity = 1.8189e-08	k_t = [3, 3, 3, 3]
Generation 730:	Best = 7.127291e+02	FES = 73100	Diversity = 1.8041e-08	k_t = [1, 1, 1, 1]
Generation 740:	Best = 7.127291e+02	FES = 74100	Diversity = 1.8627e-08	k_t = [3, 2, 2, 4]
Generation 750:	Best = 7.127291e+02	FES = 75100	Diversity = 1.8772e-08	k_t = [1, 1, 1, 1]
Generation 760:	Best = 7.127291e+02	FES = 76100	Diversity = 1.9262e-08	k_t = [4, 2, 4, 2]

Generation 770:	Best = 7.127291e+02	FES = 77100	Diversity = 1.8724e-08	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 7.127291e+02	FES = 78100	Diversity = 1.8388e-08	k_t
= [4, 4, 2, 3]				
Generation 790:	Best = 7.127291e+02	FES = 79100	Diversity = 1.9122e-08	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 7.127291e+02	FES = 80100	Diversity = 1.8902e-08	k_t
= [4, 4, 4, 4]				
Generation 810:	Best = 7.127291e+02	FES = 81100	Diversity = 1.7790e-08	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 7.127291e+02	FES = 82100	Diversity = 1.7777e-08	k_t
= [3, 2, 2, 3]				
Generation 830:	Best = 7.127291e+02	FES = 83100	Diversity = 1.9289e-08	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 7.127291e+02	FES = 84100	Diversity = 1.8885e-08	k_t
= [4, 2, 3, 2]				
Generation 850:	Best = 7.127291e+02	FES = 85100	Diversity = 1.8407e-08	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 7.127291e+02	FES = 86100	Diversity = 1.9116e-08	k_t
= [3, 2, 3, 3]				
Generation 870:	Best = 7.127291e+02	FES = 87100	Diversity = 1.8241e-08	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 7.127291e+02	FES = 88100	Diversity = 1.8272e-08	k_t
= [3, 2, 4, 3]				
Generation 890:	Best = 7.127291e+02	FES = 89100	Diversity = 1.8757e-08	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 7.127291e+02	FES = 90100	Diversity = 1.9483e-08	k_t
= [4, 2, 3, 2]				
Generation 910:	Best = 7.127291e+02	FES = 91100	Diversity = 1.9404e-08	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 7.127291e+02	FES = 92100	Diversity = 1.8636e-08	k_t
= [4, 4, 2, 2]				
Generation 930:	Best = 7.127291e+02	FES = 93100	Diversity = 1.8897e-08	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 7.127291e+02	FES = 94100	Diversity = 1.8365e-08	k_t
= [3, 2, 2, 4]				
Generation 950:	Best = 7.127291e+02	FES = 95100	Diversity = 1.8272e-08	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 7.127291e+02	FES = 96100	Diversity = 1.8870e-08	k_t
= [4, 4, 3, 3]				
Generation 970:	Best = 7.127291e+02	FES = 97100	Diversity = 1.7652e-08	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 7.127291e+02	FES = 98100	Diversity = 1.7996e-08	k_t
= [2, 4, 4, 4]				
Generation 990:	Best = 7.127291e+02	FES = 99100	Diversity = 1.9141e-08	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 7.127291e+02

Total Function Evaluations = 100000

=====

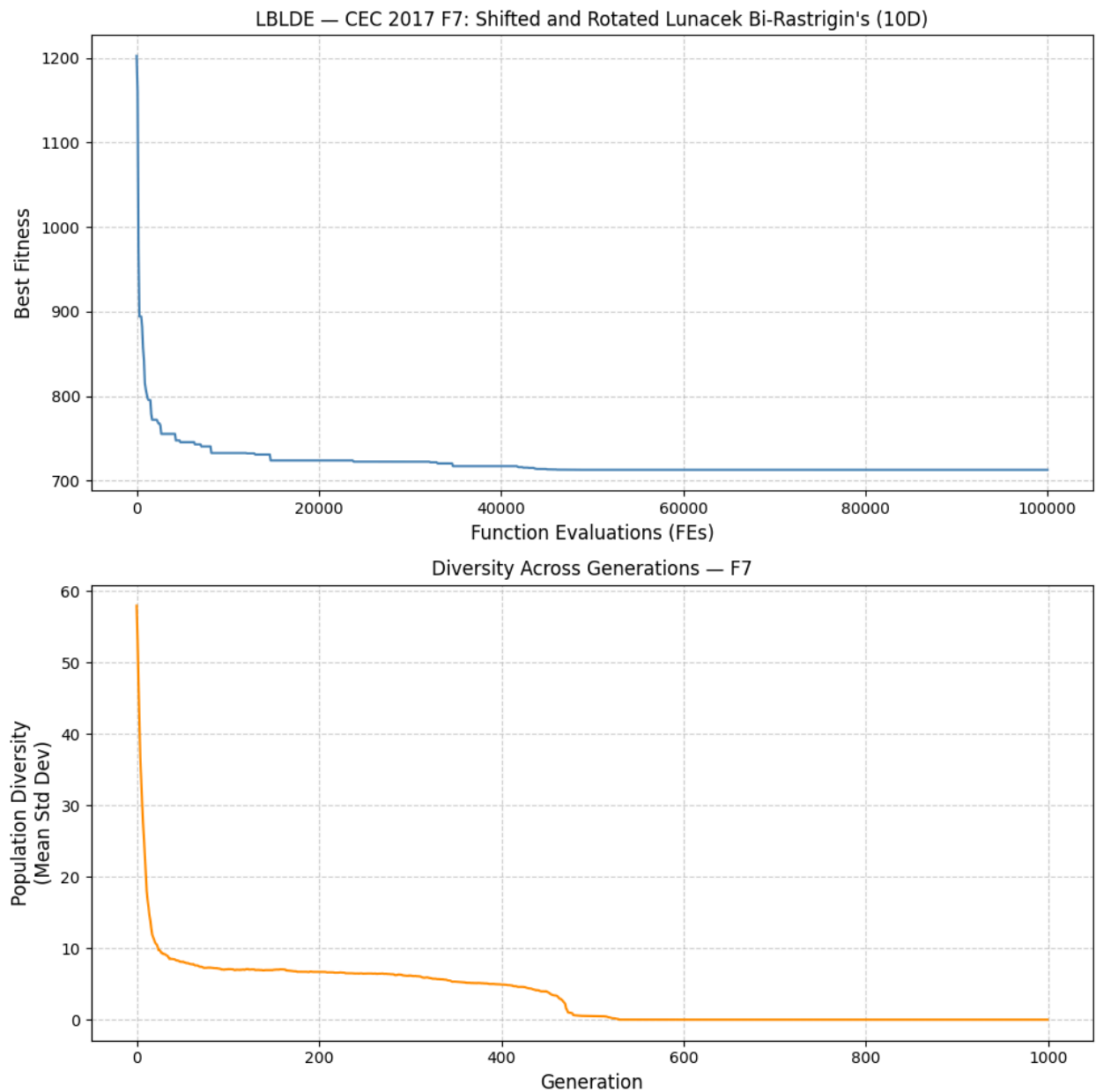
=

Results:

```

Best Fitness      : 7.127291e+02
Global Optimum    : 7.000000e+02
Error (|best-optimum|) : 1.272909e+01
Best Solution (5 dim) : [-22.33976368  21.80351398 -22.82990488  -7.30792362
15.67018647]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F7.png

```

In [24]: # =====
# F8 — Shifted and Rotated Non-Continuous Rastrigin
# =====
run_and_plot('F8', make_wrapper(simple.f8), 800.0, "Shifted and Rotated Non-Co

```

```

=====
=
LBLDE_s on CEC 2017 F8 – Single Run
=====
=
Function      : Shifted and Rotated Non-Continuous Rastrigin's (10D)
Bounds       : [-100, 100]
Global optimum : 800.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 9.505966e+02 | Diversity = 5.7928e+01
Generation 10: Best = 8.489533e+02 FES = 1100 Diversity = 3.0560e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 8.357564e+02 FES = 2100 Diversity = 2.4076e+01 k_t
= [2, 4, 2, 2]
Generation 30: Best = 8.357564e+02 FES = 3100 Diversity = 2.4254e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 8.342071e+02 FES = 4100 Diversity = 2.3374e+01 k_t
= [2, 2, 3, 2]
Generation 50: Best = 8.322486e+02 FES = 5100 Diversity = 2.2525e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 8.314113e+02 FES = 6100 Diversity = 2.1962e+01 k_t
= [2, 2, 3, 4]
Generation 70: Best = 8.314113e+02 FES = 7100 Diversity = 2.2039e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 8.314113e+02 FES = 8100 Diversity = 2.1480e+01 k_t
= [2, 2, 2, 2]
Generation 90: Best = 8.255044e+02 FES = 9100 Diversity = 2.0714e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 8.224576e+02 FES = 10100 Diversity = 2.1003e+01 k_t
= [2, 3, 2, 2]
Generation 110: Best = 8.223613e+02 FES = 11100 Diversity = 2.0649e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 8.223613e+02 FES = 12100 Diversity = 2.1162e+01 k_t
= [2, 2, 4, 4]
Generation 130: Best = 8.202198e+02 FES = 13100 Diversity = 2.0647e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 8.202198e+02 FES = 14100 Diversity = 2.0378e+01 k_t
= [3, 3, 4, 2]
Generation 150: Best = 8.179453e+02 FES = 15100 Diversity = 1.9666e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 8.174529e+02 FES = 16100 Diversity = 1.9725e+01 k_t
= [2, 2, 4, 2]
Generation 170: Best = 8.174529e+02 FES = 17100 Diversity = 1.9728e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 8.174529e+02 FES = 18100 Diversity = 1.9721e+01 k_t
= [2, 3, 4, 3]
Generation 190: Best = 8.159764e+02 FES = 19100 Diversity = 1.9363e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 8.159764e+02 FES = 20100 Diversity = 1.9275e+01 k_t
= [2, 3, 4, 2]
Generation 210: Best = 8.158035e+02 FES = 21100 Diversity = 1.9230e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 8.148004e+02 FES = 22100 Diversity = 1.9185e+01 k_t
= [3, 4, 3, 4]

```

Generation 230:	Best = 8.148004e+02	FES = 23100	Diversity = 1.9082e+01	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 8.147671e+02	FES = 24100	Diversity = 1.9285e+01	k_t
= [2, 3, 4, 4]				
Generation 250:	Best = 8.143549e+02	FES = 25100	Diversity = 1.9473e+01	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 8.141798e+02	FES = 26100	Diversity = 1.9413e+01	k_t
= [4, 2, 4, 3]				
Generation 270:	Best = 8.141798e+02	FES = 27100	Diversity = 1.9443e+01	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 8.141798e+02	FES = 28100	Diversity = 1.9271e+01	k_t
= [4, 4, 2, 4]				
Generation 290:	Best = 8.139909e+02	FES = 29100	Diversity = 1.9319e+01	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 8.128000e+02	FES = 30100	Diversity = 1.9218e+01	k_t
= [3, 2, 4, 3]				
Generation 310:	Best = 8.117323e+02	FES = 31100	Diversity = 1.9038e+01	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 8.111429e+02	FES = 32100	Diversity = 1.8656e+01	k_t
= [2, 2, 2, 2]				
Generation 330:	Best = 8.111429e+02	FES = 33100	Diversity = 1.8323e+01	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 8.111429e+02	FES = 34100	Diversity = 1.8035e+01	k_t
= [2, 3, 2, 3]				
Generation 350:	Best = 8.111429e+02	FES = 35100	Diversity = 1.7839e+01	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 8.109064e+02	FES = 36100	Diversity = 1.7699e+01	k_t
= [4, 4, 3, 2]				
Generation 370:	Best = 8.109064e+02	FES = 37100	Diversity = 1.7212e+01	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 8.109064e+02	FES = 38100	Diversity = 1.7163e+01	k_t
= [3, 2, 3, 3]				
Generation 390:	Best = 8.087421e+02	FES = 39100	Diversity = 1.7051e+01	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 8.087421e+02	FES = 40100	Diversity = 1.6837e+01	k_t
= [3, 4, 2, 3]				
Generation 410:	Best = 8.087421e+02	FES = 41100	Diversity = 1.6479e+01	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 8.087421e+02	FES = 42100	Diversity = 1.6484e+01	k_t
= [2, 2, 3, 4]				
Generation 430:	Best = 8.087381e+02	FES = 43100	Diversity = 1.6155e+01	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 8.073490e+02	FES = 44100	Diversity = 1.5497e+01	k_t
= [2, 2, 2, 4]				
Generation 450:	Best = 8.073490e+02	FES = 45100	Diversity = 1.5258e+01	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 8.073490e+02	FES = 46100	Diversity = 1.4575e+01	k_t
= [2, 4, 4, 2]				
Generation 470:	Best = 8.071072e+02	FES = 47100	Diversity = 1.4447e+01	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 8.071072e+02	FES = 48100	Diversity = 1.4060e+01	k_t
= [4, 2, 4, 3]				
Generation 490:	Best = 8.071057e+02	FES = 49100	Diversity = 1.3756e+01	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 8.066428e+02	FES = 50100	Diversity = 1.3953e+01	k_t = [4, 4, 2, 4]
Generation 510:	Best = 8.065222e+02	FES = 51100	Diversity = 1.3586e+01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 8.055411e+02	FES = 52100	Diversity = 1.3105e+01	k_t = [2, 4, 2, 2]
Generation 530:	Best = 8.049875e+02	FES = 53100	Diversity = 1.2596e+01	k_t = [1, 1, 1, 1]
Generation 540:	Best = 8.049875e+02	FES = 54100	Diversity = 1.2117e+01	k_t = [3, 4, 2, 3]
Generation 550:	Best = 8.043285e+02	FES = 55100	Diversity = 1.1034e+01	k_t = [1, 1, 1, 1]
Generation 560:	Best = 8.036936e+02	FES = 56100	Diversity = 9.8715e+00	k_t = [2, 2, 3, 2]
Generation 570:	Best = 8.030885e+02	FES = 57100	Diversity = 8.7421e+00	k_t = [1, 1, 1, 1]
Generation 580:	Best = 8.028011e+02	FES = 58100	Diversity = 8.0035e+00	k_t = [2, 4, 3, 2]
Generation 590:	Best = 8.018239e+02	FES = 59100	Diversity = 7.5679e+00	k_t = [1, 1, 1, 1]
Generation 600:	Best = 8.016205e+02	FES = 60100	Diversity = 6.7676e+00	k_t = [4, 2, 2, 3]
Generation 610:	Best = 8.013140e+02	FES = 61100	Diversity = 5.6893e+00	k_t = [1, 1, 1, 1]
Generation 620:	Best = 8.013140e+02	FES = 62100	Diversity = 4.8340e+00	k_t = [4, 4, 4, 3]
Generation 630:	Best = 8.010582e+02	FES = 63100	Diversity = 3.7587e+00	k_t = [1, 1, 1, 1]
Generation 640:	Best = 8.010582e+02	FES = 64100	Diversity = 3.4711e+00	k_t = [2, 3, 2, 4]
Generation 650:	Best = 8.010395e+02	FES = 65100	Diversity = 2.3382e+00	k_t = [1, 1, 1, 1]
Generation 660:	Best = 8.002909e+02	FES = 66100	Diversity = 1.9257e+00	k_t = [4, 3, 2, 4]
Generation 670:	Best = 8.000811e+02	FES = 67100	Diversity = 1.4305e+00	k_t = [1, 1, 1, 1]
Generation 680:	Best = 8.000318e+02	FES = 68100	Diversity = 1.2751e+00	k_t = [4, 2, 4, 3]
Generation 690:	Best = 8.000117e+02	FES = 69100	Diversity = 7.3899e-01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 8.000056e+02	FES = 70100	Diversity = 6.9557e-02	k_t = [4, 2, 4, 3]
Generation 710:	Best = 8.000011e+02	FES = 71100	Diversity = 3.5068e-02	k_t = [1, 1, 1, 1]
Generation 720:	Best = 8.000004e+02	FES = 72100	Diversity = 1.6287e-02	k_t = [3, 4, 3, 4]
Generation 730:	Best = 8.000001e+02	FES = 73100	Diversity = 7.2655e-03	k_t = [1, 1, 1, 1]
Generation 740:	Best = 8.000000e+02	FES = 74100	Diversity = 3.2778e-03	k_t = [4, 2, 2, 3]
Generation 750:	Best = 8.000000e+02	FES = 75100	Diversity = 1.5534e-03	k_t = [1, 1, 1, 1]
Generation 760:	Best = 8.000000e+02	FES = 76100	Diversity = 6.9556e-04	k_t = [2, 4, 2, 3]

Generation 770:	Best = 8.000000e+02	FES = 77100	Diversity = 2.8201e-04	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 8.000000e+02	FES = 78100	Diversity = 1.2457e-04	k_t
= [3, 3, 2, 2]				
Generation 790:	Best = 8.000000e+02	FES = 79100	Diversity = 5.3125e-05	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 8.000000e+02	FES = 80100	Diversity = 2.4528e-05	k_t
= [3, 2, 4, 4]				
Generation 810:	Best = 8.000000e+02	FES = 81100	Diversity = 1.1485e-05	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 8.000000e+02	FES = 82100	Diversity = 4.8662e-06	k_t
= [4, 4, 2, 4]				
Generation 830:	Best = 8.000000e+02	FES = 83100	Diversity = 2.1170e-06	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 8.000000e+02	FES = 84100	Diversity = 9.6079e-07	k_t
= [4, 2, 3, 3]				
Generation 850:	Best = 8.000000e+02	FES = 85100	Diversity = 4.1298e-07	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 8.000000e+02	FES = 86100	Diversity = 1.6926e-07	k_t
= [3, 2, 2, 4]				
Generation 870:	Best = 8.000000e+02	FES = 87100	Diversity = 1.1740e-07	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 8.000000e+02	FES = 88100	Diversity = 8.4339e-08	k_t
= [3, 2, 3, 4]				
Generation 890:	Best = 8.000000e+02	FES = 89100	Diversity = 7.7428e-08	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 8.000000e+02	FES = 90100	Diversity = 7.6637e-08	k_t
= [4, 2, 2, 3]				
Generation 910:	Best = 8.000000e+02	FES = 91100	Diversity = 7.5599e-08	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 8.000000e+02	FES = 92100	Diversity = 7.7404e-08	k_t
= [4, 3, 2, 2]				
Generation 930:	Best = 8.000000e+02	FES = 93100	Diversity = 7.5158e-08	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 8.000000e+02	FES = 94100	Diversity = 7.4542e-08	k_t
= [3, 2, 4, 3]				
Generation 950:	Best = 8.000000e+02	FES = 95100	Diversity = 7.0591e-08	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 8.000000e+02	FES = 96100	Diversity = 7.0451e-08	k_t
= [2, 3, 2, 2]				
Generation 970:	Best = 8.000000e+02	FES = 97100	Diversity = 7.1480e-08	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 8.000000e+02	FES = 98100	Diversity = 7.2673e-08	k_t
= [4, 3, 3, 4]				
Generation 990:	Best = 8.000000e+02	FES = 99100	Diversity = 7.6315e-08	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 8.000000e+02

Total Function Evaluations = 100000

=====

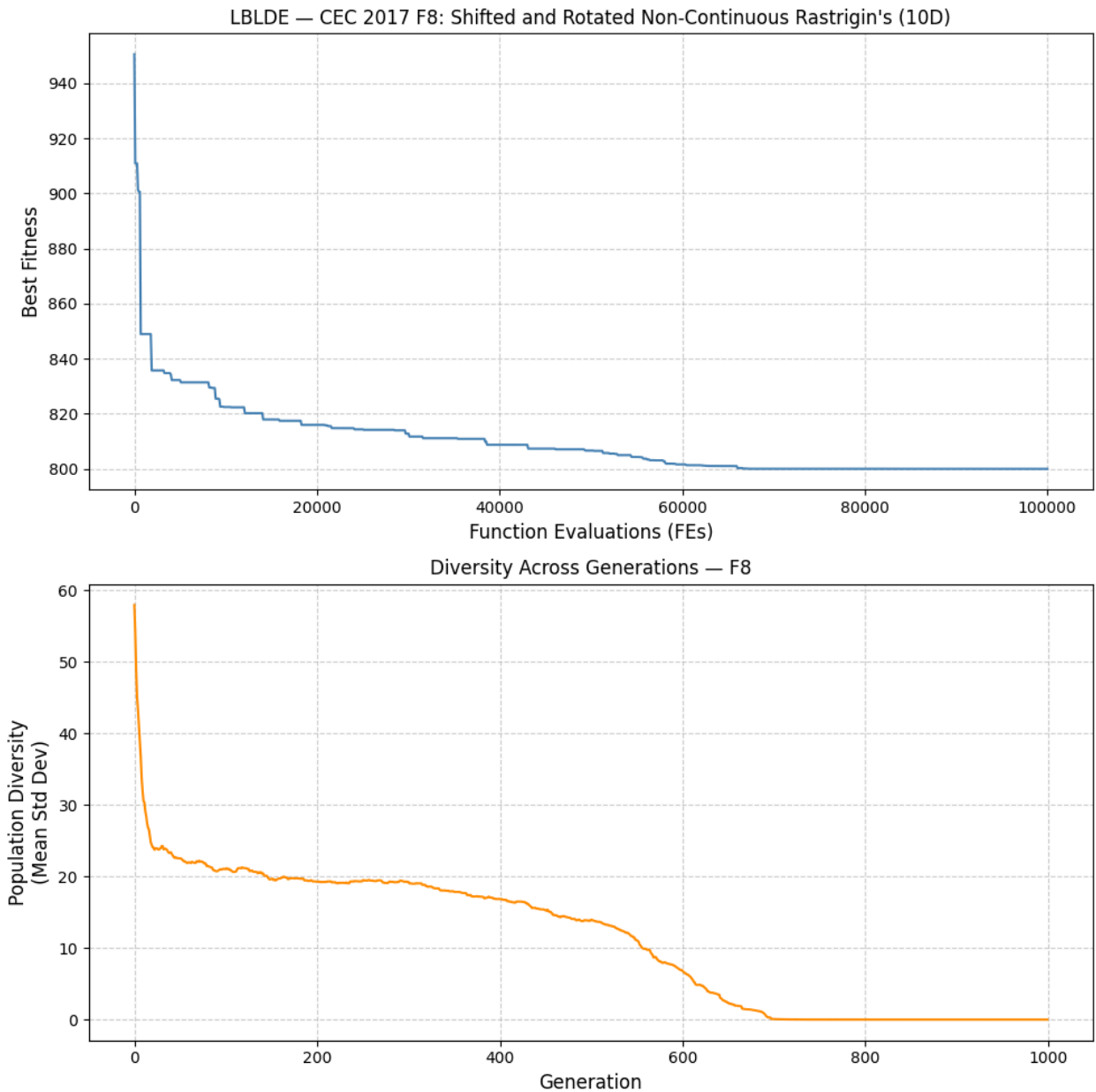
=

Results:

```

Best Fitness      : 8.000000e+02
Global Optimum    : 8.000000e+02
Error (|best-optimum|) : 0.000000e+00
Best Solution (5 dim) : [ 32.16421947 -55.965014      36.01071261 -47.95352988
-34.97768523]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F8.png

```

In [25]: # =====
# F9 – Shifted and Rotated Levy
# =====
run_and_plot('F9', make_wrapper(simple.f9), 900.0, 'Shifted and Rotated Levy')

```

```

=====
=
LBLDE_s on CEC 2017 F9 – Single Run
=====
=
Function      : Shifted and Rotated Levy (10D)
Bounds       : [-100, 100]
Global optimum : 900.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 6.112557e+03 | Diversity = 5.7928e+01
Generation 10: Best = 1.468838e+03 FES = 1100 Diversity = 3.3710e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.014235e+03 FES = 2100 Diversity = 1.4519e+01 k_t
= [2, 4, 2, 2]
Generation 30: Best = 9.298416e+02 FES = 3100 Diversity = 7.1596e+00 k_t
= [1, 1, 1, 1]
Generation 40: Best = 9.035500e+02 FES = 4100 Diversity = 3.9198e+00 k_t
= [2, 3, 4, 4]
Generation 50: Best = 9.013425e+02 FES = 5100 Diversity = 1.8765e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 9.005268e+02 FES = 6100 Diversity = 7.9229e-01 k_t
= [2, 2, 2, 2]
Generation 70: Best = 9.001479e+02 FES = 7100 Diversity = 4.3089e-01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 9.000502e+02 FES = 8100 Diversity = 2.2594e-01 k_t
= [3, 3, 2, 3]
Generation 90: Best = 9.000113e+02 FES = 9100 Diversity = 9.9050e-02 k_t
= [1, 1, 1, 1]
Generation 100: Best = 9.000020e+02 FES = 10100 Diversity = 3.9665e-02 k_t
= [4, 3, 4, 4]
Generation 110: Best = 9.000004e+02 FES = 11100 Diversity = 1.6369e-02 k_t
= [1, 1, 1, 1]
Generation 120: Best = 9.000000e+02 FES = 12100 Diversity = 7.2415e-03 k_t
= [4, 3, 2, 3]
Generation 130: Best = 9.000000e+02 FES = 13100 Diversity = 3.1206e-03 k_t
= [1, 1, 1, 1]
Generation 140: Best = 9.000000e+02 FES = 14100 Diversity = 1.2387e-03 k_t
= [3, 2, 2, 3]
Generation 150: Best = 9.000000e+02 FES = 15100 Diversity = 5.2351e-04 k_t
= [1, 1, 1, 1]
Generation 160: Best = 9.000000e+02 FES = 16100 Diversity = 2.2798e-04 k_t
= [2, 3, 3, 4]
Generation 170: Best = 9.000000e+02 FES = 17100 Diversity = 9.7452e-05 k_t
= [1, 1, 1, 1]
Generation 180: Best = 9.000000e+02 FES = 18100 Diversity = 4.1922e-05 k_t
= [3, 4, 2, 4]
Generation 190: Best = 9.000000e+02 FES = 19100 Diversity = 1.6958e-05 k_t
= [1, 1, 1, 1]
Generation 200: Best = 9.000000e+02 FES = 20100 Diversity = 7.3845e-06 k_t
= [4, 2, 2, 2]
Generation 210: Best = 9.000000e+02 FES = 21100 Diversity = 2.8292e-06 k_t
= [1, 1, 1, 1]
Generation 220: Best = 9.000000e+02 FES = 22100 Diversity = 1.2074e-06 k_t
= [2, 4, 2, 2]

```

Generation 230:	Best = 9.000000e+02	FES = 23100	Diversity = 5.2371e-07	k_t = [1, 1, 1, 1]
Generation 240:	Best = 9.000000e+02	FES = 24100	Diversity = 2.2466e-07	k_t = [3, 4, 2, 3]
Generation 250:	Best = 9.000000e+02	FES = 25100	Diversity = 1.2818e-07	k_t = [1, 1, 1, 1]
Generation 260:	Best = 9.000000e+02	FES = 26100	Diversity = 8.4186e-08	k_t = [2, 2, 3, 4]
Generation 270:	Best = 9.000000e+02	FES = 27100	Diversity = 8.2096e-08	k_t = [1, 1, 1, 1]
Generation 280:	Best = 9.000000e+02	FES = 28100	Diversity = 8.7626e-08	k_t = [3, 4, 3, 2]
Generation 290:	Best = 9.000000e+02	FES = 29100	Diversity = 8.4583e-08	k_t = [1, 1, 1, 1]
Generation 300:	Best = 9.000000e+02	FES = 30100	Diversity = 8.6518e-08	k_t = [2, 4, 2, 2]
Generation 310:	Best = 9.000000e+02	FES = 31100	Diversity = 8.3151e-08	k_t = [1, 1, 1, 1]
Generation 320:	Best = 9.000000e+02	FES = 32100	Diversity = 7.7314e-08	k_t = [2, 4, 3, 2]
Generation 330:	Best = 9.000000e+02	FES = 33100	Diversity = 7.5049e-08	k_t = [1, 1, 1, 1]
Generation 340:	Best = 9.000000e+02	FES = 34100	Diversity = 7.9535e-08	k_t = [4, 2, 3, 2]
Generation 350:	Best = 9.000000e+02	FES = 35100	Diversity = 8.2152e-08	k_t = [1, 1, 1, 1]
Generation 360:	Best = 9.000000e+02	FES = 36100	Diversity = 8.3440e-08	k_t = [2, 4, 4, 4]
Generation 370:	Best = 9.000000e+02	FES = 37100	Diversity = 8.3587e-08	k_t = [1, 1, 1, 1]
Generation 380:	Best = 9.000000e+02	FES = 38100	Diversity = 8.5140e-08	k_t = [4, 3, 2, 4]
Generation 390:	Best = 9.000000e+02	FES = 39100	Diversity = 8.2235e-08	k_t = [1, 1, 1, 1]
Generation 400:	Best = 9.000000e+02	FES = 40100	Diversity = 8.1379e-08	k_t = [2, 2, 2, 4]
Generation 410:	Best = 9.000000e+02	FES = 41100	Diversity = 7.8232e-08	k_t = [1, 1, 1, 1]
Generation 420:	Best = 9.000000e+02	FES = 42100	Diversity = 8.2987e-08	k_t = [3, 4, 4, 4]
Generation 430:	Best = 9.000000e+02	FES = 43100	Diversity = 8.0529e-08	k_t = [1, 1, 1, 1]
Generation 440:	Best = 9.000000e+02	FES = 44100	Diversity = 8.2420e-08	k_t = [4, 4, 3, 4]
Generation 450:	Best = 9.000000e+02	FES = 45100	Diversity = 8.9894e-08	k_t = [1, 1, 1, 1]
Generation 460:	Best = 9.000000e+02	FES = 46100	Diversity = 9.1051e-08	k_t = [4, 2, 3, 3]
Generation 470:	Best = 9.000000e+02	FES = 47100	Diversity = 8.8652e-08	k_t = [1, 1, 1, 1]
Generation 480:	Best = 9.000000e+02	FES = 48100	Diversity = 8.3783e-08	k_t = [3, 3, 4, 2]
Generation 490:	Best = 9.000000e+02	FES = 49100	Diversity = 8.3841e-08	k_t = [1, 1, 1, 1]

Generation 500:	Best = 9.000000e+02	FES = 50100	Diversity = 8.6457e-08	k_t = [2, 3, 2, 4]
Generation 510:	Best = 9.000000e+02	FES = 51100	Diversity = 8.3434e-08	k_t = [1, 1, 1, 1]
Generation 520:	Best = 9.000000e+02	FES = 52100	Diversity = 8.4497e-08	k_t = [3, 2, 3, 4]
Generation 530:	Best = 9.000000e+02	FES = 53100	Diversity = 8.5447e-08	k_t = [1, 1, 1, 1]
Generation 540:	Best = 9.000000e+02	FES = 54100	Diversity = 8.7421e-08	k_t = [4, 4, 4, 4]
Generation 550:	Best = 9.000000e+02	FES = 55100	Diversity = 8.8613e-08	k_t = [1, 1, 1, 1]
Generation 560:	Best = 9.000000e+02	FES = 56100	Diversity = 8.8545e-08	k_t = [2, 2, 3, 3]
Generation 570:	Best = 9.000000e+02	FES = 57100	Diversity = 8.9713e-08	k_t = [1, 1, 1, 1]
Generation 580:	Best = 9.000000e+02	FES = 58100	Diversity = 8.4123e-08	k_t = [2, 3, 3, 2]
Generation 590:	Best = 9.000000e+02	FES = 59100	Diversity = 8.4070e-08	k_t = [1, 1, 1, 1]
Generation 600:	Best = 9.000000e+02	FES = 60100	Diversity = 8.2696e-08	k_t = [3, 2, 4, 4]
Generation 610:	Best = 9.000000e+02	FES = 61100	Diversity = 8.2964e-08	k_t = [1, 1, 1, 1]
Generation 620:	Best = 9.000000e+02	FES = 62100	Diversity = 8.0274e-08	k_t = [4, 3, 4, 3]
Generation 630:	Best = 9.000000e+02	FES = 63100	Diversity = 9.1551e-08	k_t = [1, 1, 1, 1]
Generation 640:	Best = 9.000000e+02	FES = 64100	Diversity = 9.5295e-08	k_t = [3, 4, 4, 3]
Generation 650:	Best = 9.000000e+02	FES = 65100	Diversity = 9.4334e-08	k_t = [1, 1, 1, 1]
Generation 660:	Best = 9.000000e+02	FES = 66100	Diversity = 9.0982e-08	k_t = [2, 4, 2, 3]
Generation 670:	Best = 9.000000e+02	FES = 67100	Diversity = 8.7560e-08	k_t = [1, 1, 1, 1]
Generation 680:	Best = 9.000000e+02	FES = 68100	Diversity = 8.6915e-08	k_t = [3, 4, 2, 4]
Generation 690:	Best = 9.000000e+02	FES = 69100	Diversity = 9.0348e-08	k_t = [1, 1, 1, 1]
Generation 700:	Best = 9.000000e+02	FES = 70100	Diversity = 8.5333e-08	k_t = [4, 4, 3, 4]
Generation 710:	Best = 9.000000e+02	FES = 71100	Diversity = 8.5456e-08	k_t = [1, 1, 1, 1]
Generation 720:	Best = 9.000000e+02	FES = 72100	Diversity = 8.2645e-08	k_t = [4, 3, 4, 2]
Generation 730:	Best = 9.000000e+02	FES = 73100	Diversity = 8.2001e-08	k_t = [1, 1, 1, 1]
Generation 740:	Best = 9.000000e+02	FES = 74100	Diversity = 7.9633e-08	k_t = [3, 3, 2, 4]
Generation 750:	Best = 9.000000e+02	FES = 75100	Diversity = 8.1756e-08	k_t = [1, 1, 1, 1]
Generation 760:	Best = 9.000000e+02	FES = 76100	Diversity = 8.1639e-08	k_t = [3, 4, 2, 2]

Generation 770:	Best = 9.000000e+02	FES = 77100	Diversity = 8.0783e-08	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 9.000000e+02	FES = 78100	Diversity = 8.2485e-08	k_t
= [3, 2, 2, 2]				
Generation 790:	Best = 9.000000e+02	FES = 79100	Diversity = 8.5924e-08	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 9.000000e+02	FES = 80100	Diversity = 8.4321e-08	k_t
= [3, 4, 4, 2]				
Generation 810:	Best = 9.000000e+02	FES = 81100	Diversity = 8.7901e-08	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 9.000000e+02	FES = 82100	Diversity = 8.0551e-08	k_t
= [4, 4, 2, 4]				
Generation 830:	Best = 9.000000e+02	FES = 83100	Diversity = 8.5314e-08	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 9.000000e+02	FES = 84100	Diversity = 8.6185e-08	k_t
= [3, 3, 2, 4]				
Generation 850:	Best = 9.000000e+02	FES = 85100	Diversity = 8.9758e-08	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 9.000000e+02	FES = 86100	Diversity = 8.8350e-08	k_t
= [3, 4, 4, 4]				
Generation 870:	Best = 9.000000e+02	FES = 87100	Diversity = 8.7901e-08	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 9.000000e+02	FES = 88100	Diversity = 8.3337e-08	k_t
= [2, 4, 2, 2]				
Generation 890:	Best = 9.000000e+02	FES = 89100	Diversity = 8.0076e-08	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 9.000000e+02	FES = 90100	Diversity = 8.5442e-08	k_t
= [4, 2, 2, 3]				
Generation 910:	Best = 9.000000e+02	FES = 91100	Diversity = 8.6811e-08	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 9.000000e+02	FES = 92100	Diversity = 8.4452e-08	k_t
= [4, 2, 3, 4]				
Generation 930:	Best = 9.000000e+02	FES = 93100	Diversity = 8.1261e-08	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 9.000000e+02	FES = 94100	Diversity = 8.0973e-08	k_t
= [2, 2, 2, 4]				
Generation 950:	Best = 9.000000e+02	FES = 95100	Diversity = 8.1431e-08	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 9.000000e+02	FES = 96100	Diversity = 7.7350e-08	k_t
= [2, 4, 3, 4]				
Generation 970:	Best = 9.000000e+02	FES = 97100	Diversity = 8.3784e-08	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 9.000000e+02	FES = 98100	Diversity = 8.1081e-08	k_t
= [2, 2, 2, 3]				
Generation 990:	Best = 9.000000e+02	FES = 99100	Diversity = 7.9982e-08	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 9.000000e+02

Total Function Evaluations = 100000

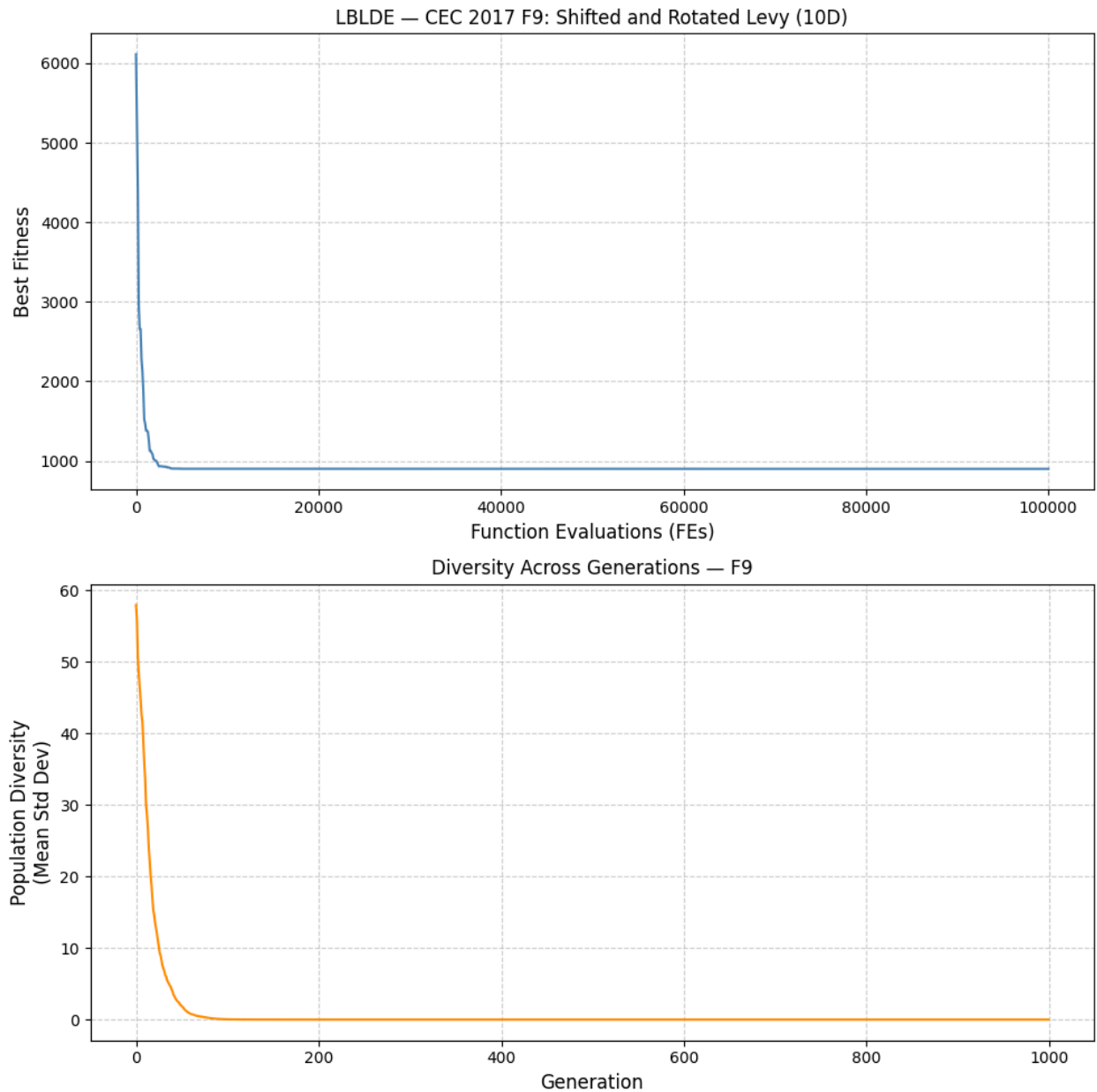
=====

Results:

```

Best Fitness      : 9.000000e+02
Global Optimum    : 9.000000e+02
Error (|best-optimum|) : 0.000000e+00
Best Solution (5 dim) : [-23.5931402    0.12032413   20.83264082   52.54992609
-18.2278721 ]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F9.png

```

In [26]: # =====
# F10 — Shifted and Rotated Schwefel
# =====
run_and_plot('F10', make_wrapper(simple.f10), 1000.0, "Shifted and Rotated Sch

```

```

=====
=
LBLDE_s on CEC 2017 F10 – Single Run
=====
=
Function      : Shifted and Rotated Schwefel's (10D)
Bounds       : [-100, 100]
Global optimum : 1000.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 3.749402e+03 | Diversity = 5.7928e+01
Generation 10: Best = 2.469815e+03 FES = 1100 Diversity = 5.3521e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.261264e+03 FES = 2100 Diversity = 5.2305e+01 k_t
= [2, 4, 3, 2]
Generation 30: Best = 2.261264e+03 FES = 3100 Diversity = 5.1845e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.261264e+03 FES = 4100 Diversity = 5.2483e+01 k_t
= [2, 2, 2, 3]
Generation 50: Best = 2.261264e+03 FES = 5100 Diversity = 5.3470e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.261264e+03 FES = 6100 Diversity = 5.2902e+01 k_t
= [4, 2, 2, 4]
Generation 70: Best = 2.261264e+03 FES = 7100 Diversity = 5.2606e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.261264e+03 FES = 8100 Diversity = 5.1955e+01 k_t
= [3, 4, 2, 4]
Generation 90: Best = 2.261264e+03 FES = 9100 Diversity = 5.1908e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.261264e+03 FES = 10100 Diversity = 5.1280e+01 k_t
= [3, 3, 3, 4]
Generation 110: Best = 2.059748e+03 FES = 11100 Diversity = 5.1657e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.059748e+03 FES = 12100 Diversity = 5.1620e+01 k_t
= [4, 2, 3, 2]
Generation 130: Best = 2.059748e+03 FES = 13100 Diversity = 5.1537e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.897117e+03 FES = 14100 Diversity = 5.1923e+01 k_t
= [2, 4, 4, 4]
Generation 150: Best = 1.739222e+03 FES = 15100 Diversity = 5.0968e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.739222e+03 FES = 16100 Diversity = 5.1228e+01 k_t
= [2, 3, 4, 2]
Generation 170: Best = 1.739222e+03 FES = 17100 Diversity = 5.1210e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.739222e+03 FES = 18100 Diversity = 5.1525e+01 k_t
= [3, 4, 2, 3]
Generation 190: Best = 1.739222e+03 FES = 19100 Diversity = 5.1899e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.728543e+03 FES = 20100 Diversity = 5.1757e+01 k_t
= [2, 4, 4, 3]
Generation 210: Best = 1.683596e+03 FES = 21100 Diversity = 5.1494e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.522940e+03 FES = 22100 Diversity = 5.1470e+01 k_t
= [3, 3, 4, 4]

```


Generation 230:	Best = 1.522940e+03	FES = 23100	Diversity = 5.1701e+01	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 1.495255e+03	FES = 24100	Diversity = 5.1912e+01	k_t
= [2, 3, 2, 4]				
Generation 250:	Best = 1.495255e+03	FES = 25100	Diversity = 5.1775e+01	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 1.495255e+03	FES = 26100	Diversity = 5.2015e+01	k_t
= [3, 4, 2, 3]				
Generation 270:	Best = 1.495255e+03	FES = 27100	Diversity = 5.1892e+01	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 1.495255e+03	FES = 28100	Diversity = 5.2489e+01	k_t
= [2, 4, 3, 4]				
Generation 290:	Best = 1.495255e+03	FES = 29100	Diversity = 5.2375e+01	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 1.495255e+03	FES = 30100	Diversity = 5.2186e+01	k_t
= [2, 2, 4, 3]				
Generation 310:	Best = 1.440190e+03	FES = 31100	Diversity = 5.1790e+01	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 1.440190e+03	FES = 32100	Diversity = 5.1326e+01	k_t
= [3, 2, 3, 2]				
Generation 330:	Best = 1.440190e+03	FES = 33100	Diversity = 5.1687e+01	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 1.440190e+03	FES = 34100	Diversity = 5.0527e+01	k_t
= [2, 3, 4, 3]				
Generation 350:	Best = 1.440190e+03	FES = 35100	Diversity = 5.0401e+01	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 1.440190e+03	FES = 36100	Diversity = 5.0360e+01	k_t
= [4, 2, 3, 3]				
Generation 370:	Best = 1.344164e+03	FES = 37100	Diversity = 5.0052e+01	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 1.344164e+03	FES = 38100	Diversity = 5.0190e+01	k_t
= [4, 3, 4, 3]				
Generation 390:	Best = 1.341666e+03	FES = 39100	Diversity = 4.9487e+01	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 1.340493e+03	FES = 40100	Diversity = 4.8690e+01	k_t
= [2, 2, 3, 3]				
Generation 410:	Best = 1.297963e+03	FES = 41100	Diversity = 4.8822e+01	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 1.297963e+03	FES = 42100	Diversity = 4.8831e+01	k_t
= [2, 4, 4, 2]				
Generation 430:	Best = 1.291431e+03	FES = 43100	Diversity = 4.8839e+01	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 1.285457e+03	FES = 44100	Diversity = 4.8250e+01	k_t
= [2, 3, 4, 2]				
Generation 450:	Best = 1.285457e+03	FES = 45100	Diversity = 4.7679e+01	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 1.285457e+03	FES = 46100	Diversity = 4.7562e+01	k_t
= [4, 4, 4, 4]				
Generation 470:	Best = 1.285457e+03	FES = 47100	Diversity = 4.7285e+01	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 1.285457e+03	FES = 48100	Diversity = 4.7324e+01	k_t
= [4, 4, 2, 2]				
Generation 490:	Best = 1.285457e+03	FES = 49100	Diversity = 4.6632e+01	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 1.221540e+03	FES = 50100	Diversity = 4.6800e+01	k_t = [3, 4, 3, 2]
Generation 510:	Best = 1.219876e+03	FES = 51100	Diversity = 4.6488e+01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 1.213999e+03	FES = 52100	Diversity = 4.5095e+01	k_t = [3, 3, 3, 4]
Generation 530:	Best = 1.213999e+03	FES = 53100	Diversity = 4.4601e+01	k_t = [1, 1, 1, 1]
Generation 540:	Best = 1.213999e+03	FES = 54100	Diversity = 4.4721e+01	k_t = [3, 4, 4, 3]
Generation 550:	Best = 1.114471e+03	FES = 55100	Diversity = 4.4859e+01	k_t = [1, 1, 1, 1]
Generation 560:	Best = 1.114471e+03	FES = 56100	Diversity = 4.4135e+01	k_t = [2, 2, 3, 2]
Generation 570:	Best = 1.109704e+03	FES = 57100	Diversity = 4.3235e+01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 1.109704e+03	FES = 58100	Diversity = 4.1018e+01	k_t = [3, 4, 2, 4]
Generation 590:	Best = 1.095570e+03	FES = 59100	Diversity = 3.9174e+01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 1.085821e+03	FES = 60100	Diversity = 3.7945e+01	k_t = [4, 3, 3, 3]
Generation 610:	Best = 1.070762e+03	FES = 61100	Diversity = 3.5519e+01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 1.054940e+03	FES = 62100	Diversity = 3.1571e+01	k_t = [2, 2, 4, 2]
Generation 630:	Best = 1.047625e+03	FES = 63100	Diversity = 2.8789e+01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 1.047625e+03	FES = 64100	Diversity = 2.3404e+01	k_t = [2, 4, 3, 2]
Generation 650:	Best = 1.037668e+03	FES = 65100	Diversity = 1.3875e+01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 1.027724e+03	FES = 66100	Diversity = 9.0163e+00	k_t = [4, 2, 4, 4]
Generation 670:	Best = 1.023324e+03	FES = 67100	Diversity = 1.0657e+01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 1.019398e+03	FES = 68100	Diversity = 9.2394e+00	k_t = [4, 4, 3, 3]
Generation 690:	Best = 1.018783e+03	FES = 69100	Diversity = 3.6973e+00	k_t = [1, 1, 1, 1]
Generation 700:	Best = 1.018537e+03	FES = 70100	Diversity = 4.2109e-02	k_t = [2, 4, 4, 2]
Generation 710:	Best = 1.018491e+03	FES = 71100	Diversity = 1.9898e-02	k_t = [1, 1, 1, 1]
Generation 720:	Best = 1.018475e+03	FES = 72100	Diversity = 1.0028e-02	k_t = [3, 2, 3, 4]
Generation 730:	Best = 1.018473e+03	FES = 73100	Diversity = 4.6008e-03	k_t = [1, 1, 1, 1]
Generation 740:	Best = 1.018472e+03	FES = 74100	Diversity = 2.0301e-03	k_t = [3, 3, 2, 4]
Generation 750:	Best = 1.018472e+03	FES = 75100	Diversity = 9.1661e-04	k_t = [1, 1, 1, 1]
Generation 760:	Best = 1.018472e+03	FES = 76100	Diversity = 3.8406e-04	k_t = [2, 4, 4, 3]

Generation 770:	Best = 1.018472e+03	FES = 77100	Diversity = 1.6064e-04	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.018472e+03	FES = 78100	Diversity = 6.0753e-05	k_t
= [3, 3, 2, 4]				
Generation 790:	Best = 1.018472e+03	FES = 79100	Diversity = 2.6467e-05	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.018472e+03	FES = 80100	Diversity = 1.1631e-05	k_t
= [3, 2, 4, 2]				
Generation 810:	Best = 1.018472e+03	FES = 81100	Diversity = 4.8577e-06	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.018472e+03	FES = 82100	Diversity = 2.0728e-06	k_t
= [4, 2, 3, 3]				
Generation 830:	Best = 1.018472e+03	FES = 83100	Diversity = 7.7397e-07	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.018472e+03	FES = 84100	Diversity = 3.2091e-07	k_t
= [4, 2, 2, 2]				
Generation 850:	Best = 1.018472e+03	FES = 85100	Diversity = 1.4835e-07	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.018472e+03	FES = 86100	Diversity = 7.9809e-08	k_t
= [3, 4, 3, 2]				
Generation 870:	Best = 1.018472e+03	FES = 87100	Diversity = 6.2120e-08	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.018472e+03	FES = 88100	Diversity = 5.7982e-08	k_t
= [4, 4, 3, 4]				
Generation 890:	Best = 1.018472e+03	FES = 89100	Diversity = 5.6893e-08	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.018472e+03	FES = 90100	Diversity = 5.7881e-08	k_t
= [3, 4, 2, 2]				
Generation 910:	Best = 1.018472e+03	FES = 91100	Diversity = 6.0191e-08	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.018472e+03	FES = 92100	Diversity = 6.1258e-08	k_t
= [2, 3, 3, 4]				
Generation 930:	Best = 1.018472e+03	FES = 93100	Diversity = 5.9408e-08	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.018472e+03	FES = 94100	Diversity = 5.8894e-08	k_t
= [3, 2, 3, 2]				
Generation 950:	Best = 1.018472e+03	FES = 95100	Diversity = 5.9714e-08	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.018472e+03	FES = 96100	Diversity = 6.0395e-08	k_t
= [2, 2, 3, 4]				
Generation 970:	Best = 1.018472e+03	FES = 97100	Diversity = 6.0641e-08	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.018472e+03	FES = 98100	Diversity = 5.7311e-08	k_t
= [4, 2, 3, 2]				
Generation 990:	Best = 1.018472e+03	FES = 99100	Diversity = 5.6493e-08	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 1.018472e+03

Total Function Evaluations = 100000

=====

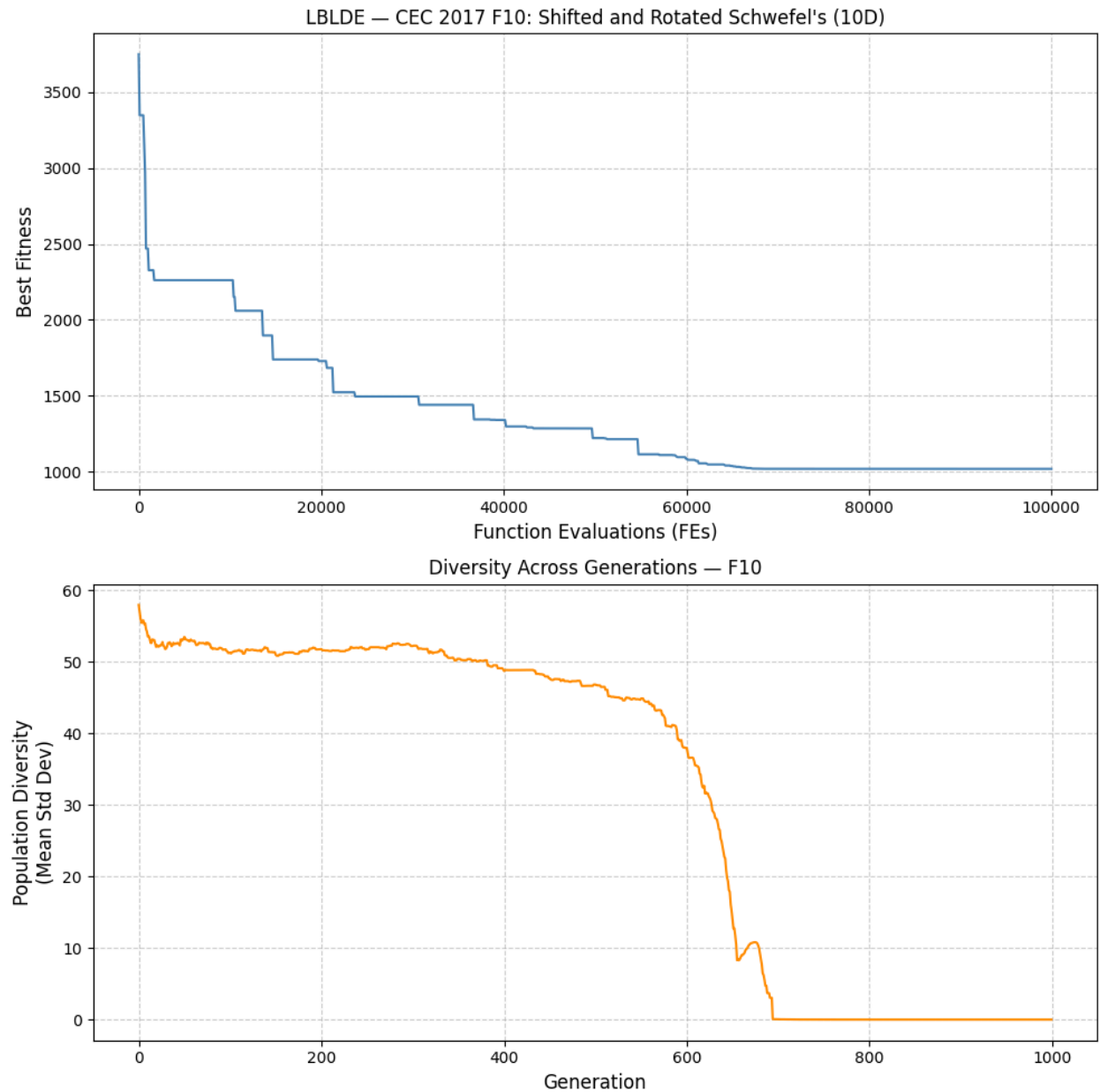
=

Results:

```

Best Fitness          : 1.018472e+03
Global Optimum        : 1.000000e+03
Error (|best-optimum|) : 1.847221e+01
Best Solution (5 dim)  : [-2.3246776  43.5531457  99.611919  74.36411535  4
2.11118 ]
D^(0) (init. diversity): 5.7928e+01
=====

```



Graph saved as convergence_F10.png

6.3 Hybrid Functions (F11-F20)

```

In [27]: # =====
# F11 – Hybrid Function 1 (Zakharov; Rosenbrock; Rastrigin)

```

```
# =====  
run_and_plot('F11', make_wrapper(hybrid.f11), 1100.0, 'Hybrid: Zakharov; Rosen
```

```

=====
=
LBLDE_s on CEC 2017 F11 – Single Run
=====
=
Function      : Hybrid: Zakharov; Rosenbrock; Rastrigin (10D)
Bounds       : [-100, 100]
Global optimum : 1100.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 2.292232e+04 | Diversity = 5.7928e+01
Generation 10: Best = 1.428521e+03 FES = 1100 Diversity = 4.2695e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.199543e+03 FES = 2100 Diversity = 2.6967e+01 k_t
= [4, 3, 2, 3]
Generation 30: Best = 1.147298e+03 FES = 3100 Diversity = 1.7597e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.114240e+03 FES = 4100 Diversity = 1.1833e+01 k_t
= [3, 2, 3, 3]
Generation 50: Best = 1.111977e+03 FES = 5100 Diversity = 9.7056e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.111124e+03 FES = 6100 Diversity = 8.3440e+00 k_t
= [3, 2, 2, 2]
Generation 70: Best = 1.106225e+03 FES = 7100 Diversity = 8.1165e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.106225e+03 FES = 8100 Diversity = 7.4137e+00 k_t
= [3, 3, 2, 3]
Generation 90: Best = 1.105535e+03 FES = 9100 Diversity = 7.2215e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.105535e+03 FES = 10100 Diversity = 6.8732e+00 k_t
= [4, 3, 4, 4]
Generation 110: Best = 1.105535e+03 FES = 11100 Diversity = 6.6982e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.105444e+03 FES = 12100 Diversity = 6.6394e+00 k_t
= [2, 3, 3, 4]
Generation 130: Best = 1.105110e+03 FES = 13100 Diversity = 6.6504e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.105110e+03 FES = 14100 Diversity = 6.5876e+00 k_t
= [3, 3, 4, 4]
Generation 150: Best = 1.103929e+03 FES = 15100 Diversity = 6.4302e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.103929e+03 FES = 16100 Diversity = 6.3112e+00 k_t
= [4, 4, 4, 2]
Generation 170: Best = 1.103906e+03 FES = 17100 Diversity = 6.1494e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.103906e+03 FES = 18100 Diversity = 5.7751e+00 k_t
= [4, 2, 2, 3]
Generation 190: Best = 1.103881e+03 FES = 19100 Diversity = 5.5700e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.103881e+03 FES = 20100 Diversity = 5.2757e+00 k_t
= [4, 4, 3, 3]
Generation 210: Best = 1.103815e+03 FES = 21100 Diversity = 5.1882e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.103815e+03 FES = 22100 Diversity = 5.1200e+00 k_t
= [3, 2, 4, 2]

```

Generation 230:	Best = 1.103815e+03	FES = 23100	Diversity = 4.9636e+00	k_t = [1, 1, 1, 1]
Generation 240:	Best = 1.103346e+03	FES = 24100	Diversity = 4.9377e+00	k_t = [2, 3, 2, 3]
Generation 250:	Best = 1.102040e+03	FES = 25100	Diversity = 4.7585e+00	k_t = [1, 1, 1, 1]
Generation 260:	Best = 1.101963e+03	FES = 26100	Diversity = 4.5954e+00	k_t = [2, 4, 2, 3]
Generation 270:	Best = 1.101963e+03	FES = 27100	Diversity = 4.5139e+00	k_t = [1, 1, 1, 1]
Generation 280:	Best = 1.101963e+03	FES = 28100	Diversity = 4.5200e+00	k_t = [2, 2, 2, 3]
Generation 290:	Best = 1.101754e+03	FES = 29100	Diversity = 4.2378e+00	k_t = [1, 1, 1, 1]
Generation 300:	Best = 1.101709e+03	FES = 30100	Diversity = 4.0026e+00	k_t = [4, 2, 3, 2]
Generation 310:	Best = 1.101497e+03	FES = 31100	Diversity = 3.3805e+00	k_t = [1, 1, 1, 1]
Generation 320:	Best = 1.100928e+03	FES = 32100	Diversity = 2.7982e+00	k_t = [4, 3, 2, 2]
Generation 330:	Best = 1.100918e+03	FES = 33100	Diversity = 1.6569e+00	k_t = [1, 1, 1, 1]
Generation 340:	Best = 1.100489e+03	FES = 34100	Diversity = 1.0642e+00	k_t = [3, 2, 3, 3]
Generation 350:	Best = 1.100424e+03	FES = 35100	Diversity = 6.7377e-01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 1.100308e+03	FES = 36100	Diversity = 5.2832e-01	k_t = [2, 4, 3, 4]
Generation 370:	Best = 1.100237e+03	FES = 37100	Diversity = 5.3906e-01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 1.100191e+03	FES = 38100	Diversity = 4.6772e-01	k_t = [3, 2, 4, 3]
Generation 390:	Best = 1.100123e+03	FES = 39100	Diversity = 3.8448e-01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 1.100094e+03	FES = 40100	Diversity = 3.4963e-01	k_t = [4, 2, 4, 2]
Generation 410:	Best = 1.100051e+03	FES = 41100	Diversity = 4.7709e-01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 1.100015e+03	FES = 42100	Diversity = 4.0265e-01	k_t = [2, 3, 3, 3]
Generation 430:	Best = 1.100012e+03	FES = 43100	Diversity = 3.3770e-01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 1.100009e+03	FES = 44100	Diversity = 2.2391e-01	k_t = [3, 3, 3, 4]
Generation 450:	Best = 1.100003e+03	FES = 45100	Diversity = 1.8362e-01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 1.100001e+03	FES = 46100	Diversity = 1.4712e-01	k_t = [2, 3, 2, 2]
Generation 470:	Best = 1.100000e+03	FES = 47100	Diversity = 1.0872e-01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 1.100000e+03	FES = 48100	Diversity = 8.3419e-02	k_t = [2, 3, 2, 4]
Generation 490:	Best = 1.100000e+03	FES = 49100	Diversity = 7.9207e-02	k_t = [1, 1, 1, 1]

Generation 500:	Best = 1.100000e+03	FES = 50100	Diversity = 6.2155e-02	k_t
= [3, 3, 3, 4]				
Generation 510:	Best = 1.100000e+03	FES = 51100	Diversity = 3.7974e-02	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 1.100000e+03	FES = 52100	Diversity = 2.2094e-02	k_t
= [3, 4, 3, 3]				
Generation 530:	Best = 1.100000e+03	FES = 53100	Diversity = 1.3403e-02	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 1.100000e+03	FES = 54100	Diversity = 8.6148e-03	k_t
= [3, 2, 3, 3]				
Generation 550:	Best = 1.100000e+03	FES = 55100	Diversity = 6.3916e-03	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 1.100000e+03	FES = 56100	Diversity = 4.2488e-03	k_t
= [4, 2, 2, 4]				
Generation 570:	Best = 1.100000e+03	FES = 57100	Diversity = 2.7247e-03	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 1.100000e+03	FES = 58100	Diversity = 1.7542e-03	k_t
= [2, 3, 2, 2]				
Generation 590:	Best = 1.100000e+03	FES = 59100	Diversity = 1.1972e-03	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 1.100000e+03	FES = 60100	Diversity = 7.0671e-04	k_t
= [3, 3, 2, 3]				
Generation 610:	Best = 1.100000e+03	FES = 61100	Diversity = 4.7015e-04	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 1.100000e+03	FES = 62100	Diversity = 2.7419e-04	k_t
= [4, 4, 3, 4]				
Generation 630:	Best = 1.100000e+03	FES = 63100	Diversity = 2.0180e-04	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 1.100000e+03	FES = 64100	Diversity = 1.3133e-04	k_t
= [4, 2, 4, 4]				
Generation 650:	Best = 1.100000e+03	FES = 65100	Diversity = 8.5242e-05	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 1.100000e+03	FES = 66100	Diversity = 5.0130e-05	k_t
= [3, 2, 2, 3]				
Generation 670:	Best = 1.100000e+03	FES = 67100	Diversity = 3.7116e-05	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 1.100000e+03	FES = 68100	Diversity = 2.4715e-05	k_t
= [4, 2, 2, 3]				
Generation 690:	Best = 1.100000e+03	FES = 69100	Diversity = 1.5776e-05	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 1.100000e+03	FES = 70100	Diversity = 1.0870e-05	k_t
= [4, 4, 4, 3]				
Generation 710:	Best = 1.100000e+03	FES = 71100	Diversity = 7.9101e-06	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 1.100000e+03	FES = 72100	Diversity = 5.4470e-06	k_t
= [4, 4, 2, 4]				
Generation 730:	Best = 1.100000e+03	FES = 73100	Diversity = 3.6704e-06	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 1.100000e+03	FES = 74100	Diversity = 2.4824e-06	k_t
= [4, 4, 3, 4]				
Generation 750:	Best = 1.100000e+03	FES = 75100	Diversity = 1.6379e-06	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 1.100000e+03	FES = 76100	Diversity = 1.3948e-06	k_t
= [2, 4, 3, 4]				

Generation 770:	Best = 1.100000e+03	FES = 77100	Diversity = 1.4837e-06	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.100000e+03	FES = 78100	Diversity = 9.8854e-07	k_t
= [3, 4, 4, 4]				
Generation 790:	Best = 1.100000e+03	FES = 79100	Diversity = 9.8023e-07	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.100000e+03	FES = 80100	Diversity = 9.8525e-07	k_t
= [4, 2, 2, 3]				
Generation 810:	Best = 1.100000e+03	FES = 81100	Diversity = 9.8767e-07	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.100000e+03	FES = 82100	Diversity = 9.5753e-07	k_t
= [3, 4, 3, 3]				
Generation 830:	Best = 1.100000e+03	FES = 83100	Diversity = 9.8067e-07	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.100000e+03	FES = 84100	Diversity = 9.8445e-07	k_t
= [3, 4, 4, 4]				
Generation 850:	Best = 1.100000e+03	FES = 85100	Diversity = 9.5566e-07	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.100000e+03	FES = 86100	Diversity = 9.8482e-07	k_t
= [3, 4, 3, 3]				
Generation 870:	Best = 1.100000e+03	FES = 87100	Diversity = 9.4746e-07	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.100000e+03	FES = 88100	Diversity = 9.7378e-07	k_t
= [4, 2, 4, 4]				
Generation 890:	Best = 1.100000e+03	FES = 89100	Diversity = 9.5932e-07	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.100000e+03	FES = 90100	Diversity = 9.9810e-07	k_t
= [3, 4, 3, 2]				
Generation 910:	Best = 1.100000e+03	FES = 91100	Diversity = 9.0764e-07	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.100000e+03	FES = 92100	Diversity = 9.8763e-07	k_t
= [2, 4, 3, 2]				
Generation 930:	Best = 1.100000e+03	FES = 93100	Diversity = 9.4753e-07	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.100000e+03	FES = 94100	Diversity = 9.6843e-07	k_t
= [2, 4, 2, 2]				
Generation 950:	Best = 1.100000e+03	FES = 95100	Diversity = 9.3918e-07	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.100000e+03	FES = 96100	Diversity = 9.4350e-07	k_t
= [3, 2, 2, 4]				
Generation 970:	Best = 1.100000e+03	FES = 97100	Diversity = 9.1265e-07	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.100000e+03	FES = 98100	Diversity = 9.4455e-07	k_t
= [3, 4, 2, 4]				
Generation 990:	Best = 1.100000e+03	FES = 99100	Diversity = 9.3442e-07	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 1.100000e+03

Total Function Evaluations = 100000

=====

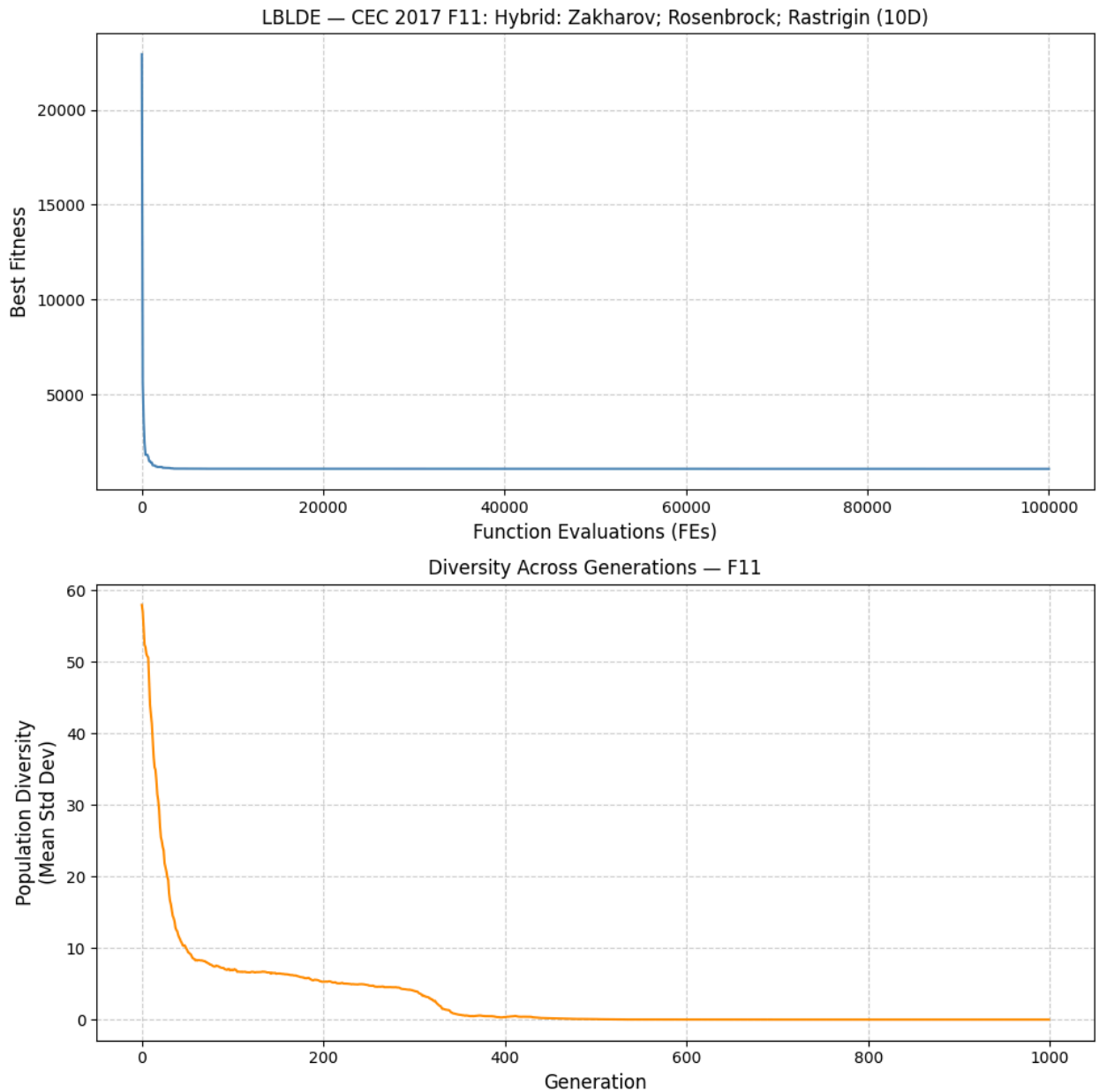
=

Results:

```

Best Fitness      : 1.100000e+03
Global Optimum    : 1.100000e+03
Error (|best-optimum|) : 0.000000e+00
Best Solution (5 dim) : [-24.75259024 -16.78697179 -58.62265228  9.08309141
19.51774125]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F11.png

```

In [28]: # =====
# F12 — Hybrid Function 2 (Elliptic; Schwefel; Bent Cigar)
# =====
run_and_plot('F12', make_wrapper(hybrid.f12), 1200.0, 'Hybrid: Elliptic; Schwefel; Bent Cigar')

```

```

=====
=
LBLDE_s on CEC 2017 F12 – Single Run
=====
=
Function      : Hybrid: Elliptic; Schwefel; Bent Cigar (10D)
Bounds       : [-100, 100]
Global optimum : 1200.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 7.052719e+09 | Diversity = 5.7928e+01
Generation 10: Best = 1.059919e+08 FES = 1100 Diversity = 4.1516e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.683205e+07 FES = 2100 Diversity = 3.7975e+01 k_t
= [4, 3, 4, 2]
Generation 30: Best = 6.874830e+06 FES = 3100 Diversity = 3.6144e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.754730e+06 FES = 4100 Diversity = 3.2893e+01 k_t
= [3, 3, 4, 4]
Generation 50: Best = 7.043446e+05 FES = 5100 Diversity = 3.1587e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 3.428052e+05 FES = 6100 Diversity = 3.2298e+01 k_t
= [2, 3, 4, 3]
Generation 70: Best = 9.383564e+04 FES = 7100 Diversity = 3.1674e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 3.682086e+04 FES = 8100 Diversity = 3.0731e+01 k_t
= [2, 2, 2, 2]
Generation 90: Best = 1.912260e+04 FES = 9100 Diversity = 3.0959e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.154781e+04 FES = 10100 Diversity = 2.9088e+01 k_t
= [2, 3, 4, 2]
Generation 110: Best = 5.361282e+03 FES = 11100 Diversity = 2.7515e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 3.967029e+03 FES = 12100 Diversity = 2.7949e+01 k_t
= [4, 2, 2, 4]
Generation 130: Best = 3.140811e+03 FES = 13100 Diversity = 2.1970e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.499198e+03 FES = 14100 Diversity = 2.1175e+01 k_t
= [3, 4, 2, 2]
Generation 150: Best = 1.706467e+03 FES = 15100 Diversity = 1.9978e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.706467e+03 FES = 16100 Diversity = 1.7973e+01 k_t
= [2, 3, 3, 3]
Generation 170: Best = 1.706467e+03 FES = 17100 Diversity = 1.5884e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.662399e+03 FES = 18100 Diversity = 1.5728e+01 k_t
= [4, 3, 4, 2]
Generation 190: Best = 1.636197e+03 FES = 19100 Diversity = 1.4246e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.394264e+03 FES = 20100 Diversity = 1.3538e+01 k_t
= [4, 3, 4, 3]
Generation 210: Best = 1.394264e+03 FES = 21100 Diversity = 1.3158e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.394264e+03 FES = 22100 Diversity = 1.2158e+01 k_t
= [4, 3, 4, 4]

```

Generation 230:	Best = 1.319344e+03	FES = 23100	Diversity = 1.1987e+01	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 1.319344e+03	FES = 24100	Diversity = 1.1549e+01	k_t
= [4, 2, 2, 2]				
Generation 250:	Best = 1.287113e+03	FES = 25100	Diversity = 1.1231e+01	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 1.287113e+03	FES = 26100	Diversity = 1.0954e+01	k_t
= [2, 3, 4, 3]				
Generation 270:	Best = 1.287113e+03	FES = 27100	Diversity = 1.0166e+01	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 1.273452e+03	FES = 28100	Diversity = 9.3342e+00	k_t
= [3, 4, 2, 3]				
Generation 290:	Best = 1.258929e+03	FES = 29100	Diversity = 8.7479e+00	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 1.257218e+03	FES = 30100	Diversity = 8.6053e+00	k_t
= [2, 3, 2, 4]				
Generation 310:	Best = 1.257218e+03	FES = 31100	Diversity = 8.3157e+00	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 1.257218e+03	FES = 32100	Diversity = 8.4177e+00	k_t
= [4, 4, 2, 3]				
Generation 330:	Best = 1.253288e+03	FES = 33100	Diversity = 7.9424e+00	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 1.253288e+03	FES = 34100	Diversity = 7.7176e+00	k_t
= [3, 4, 3, 2]				
Generation 350:	Best = 1.253288e+03	FES = 35100	Diversity = 7.6573e+00	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 1.253288e+03	FES = 36100	Diversity = 7.1865e+00	k_t
= [4, 3, 4, 3]				
Generation 370:	Best = 1.253288e+03	FES = 37100	Diversity = 7.0254e+00	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 1.253288e+03	FES = 38100	Diversity = 6.1970e+00	k_t
= [4, 3, 4, 3]				
Generation 390:	Best = 1.251120e+03	FES = 39100	Diversity = 5.3675e+00	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 1.251120e+03	FES = 40100	Diversity = 5.0237e+00	k_t
= [3, 3, 4, 2]				
Generation 410:	Best = 1.248631e+03	FES = 41100	Diversity = 5.0919e+00	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 1.248631e+03	FES = 42100	Diversity = 5.2265e+00	k_t
= [4, 2, 4, 2]				
Generation 430:	Best = 1.248631e+03	FES = 43100	Diversity = 5.0384e+00	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 1.242470e+03	FES = 44100	Diversity = 5.0854e+00	k_t
= [4, 2, 2, 2]				
Generation 450:	Best = 1.242470e+03	FES = 45100	Diversity = 4.9416e+00	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 1.242470e+03	FES = 46100	Diversity = 4.7090e+00	k_t
= [3, 3, 2, 4]				
Generation 470:	Best = 1.219996e+03	FES = 47100	Diversity = 4.3903e+00	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 1.219996e+03	FES = 48100	Diversity = 3.8451e+00	k_t
= [3, 4, 4, 4]				
Generation 490:	Best = 1.219996e+03	FES = 49100	Diversity = 3.3185e+00	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 1.219996e+03	FES = 50100	Diversity = 3.0169e+00	k_t = [3, 4, 3, 3]
Generation 510:	Best = 1.219996e+03	FES = 51100	Diversity = 2.7900e+00	k_t = [1, 1, 1, 1]
Generation 520:	Best = 1.219996e+03	FES = 52100	Diversity = 2.7827e+00	k_t = [3, 2, 2, 2]
Generation 530:	Best = 1.219996e+03	FES = 53100	Diversity = 2.7896e+00	k_t = [1, 1, 1, 1]
Generation 540:	Best = 1.219185e+03	FES = 54100	Diversity = 2.7482e+00	k_t = [2, 2, 4, 4]
Generation 550:	Best = 1.219185e+03	FES = 55100	Diversity = 2.7749e+00	k_t = [1, 1, 1, 1]
Generation 560:	Best = 1.216946e+03	FES = 56100	Diversity = 2.7735e+00	k_t = [2, 4, 3, 4]
Generation 570:	Best = 1.216946e+03	FES = 57100	Diversity = 2.8270e+00	k_t = [1, 1, 1, 1]
Generation 580:	Best = 1.214050e+03	FES = 58100	Diversity = 2.8169e+00	k_t = [2, 2, 4, 2]
Generation 590:	Best = 1.214050e+03	FES = 59100	Diversity = 2.8162e+00	k_t = [1, 1, 1, 1]
Generation 600:	Best = 1.208807e+03	FES = 60100	Diversity = 2.7944e+00	k_t = [3, 3, 2, 2]
Generation 610:	Best = 1.208807e+03	FES = 61100	Diversity = 2.7581e+00	k_t = [1, 1, 1, 1]
Generation 620:	Best = 1.208807e+03	FES = 62100	Diversity = 2.7279e+00	k_t = [4, 4, 3, 4]
Generation 630:	Best = 1.206857e+03	FES = 63100	Diversity = 2.7074e+00	k_t = [1, 1, 1, 1]
Generation 640:	Best = 1.206857e+03	FES = 64100	Diversity = 2.6955e+00	k_t = [2, 3, 4, 2]
Generation 650:	Best = 1.206857e+03	FES = 65100	Diversity = 2.6794e+00	k_t = [1, 1, 1, 1]
Generation 660:	Best = 1.201184e+03	FES = 66100	Diversity = 2.6619e+00	k_t = [2, 2, 3, 2]
Generation 670:	Best = 1.201184e+03	FES = 67100	Diversity = 2.6548e+00	k_t = [1, 1, 1, 1]
Generation 680:	Best = 1.200981e+03	FES = 68100	Diversity = 2.6047e+00	k_t = [3, 3, 3, 3]
Generation 690:	Best = 1.200981e+03	FES = 69100	Diversity = 2.5752e+00	k_t = [1, 1, 1, 1]
Generation 700:	Best = 1.200981e+03	FES = 70100	Diversity = 2.4778e+00	k_t = [3, 4, 4, 2]
Generation 710:	Best = 1.200973e+03	FES = 71100	Diversity = 2.3503e+00	k_t = [1, 1, 1, 1]
Generation 720:	Best = 1.200973e+03	FES = 72100	Diversity = 2.2342e+00	k_t = [3, 2, 3, 4]
Generation 730:	Best = 1.200973e+03	FES = 73100	Diversity = 1.9991e+00	k_t = [1, 1, 1, 1]
Generation 740:	Best = 1.200973e+03	FES = 74100	Diversity = 1.9150e+00	k_t = [4, 4, 2, 2]
Generation 750:	Best = 1.200965e+03	FES = 75100	Diversity = 1.7634e+00	k_t = [1, 1, 1, 1]
Generation 760:	Best = 1.200440e+03	FES = 76100	Diversity = 1.5164e+00	k_t = [2, 2, 2, 2]

Generation 770:	Best = 1.200345e+03	FES = 77100	Diversity = 9.4946e-01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.200315e+03	FES = 78100	Diversity = 1.3093e-01	k_t
= [4, 3, 2, 3]				
Generation 790:	Best = 1.200198e+03	FES = 79100	Diversity = 9.9325e-02	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.200079e+03	FES = 80100	Diversity = 7.0629e-02	k_t
= [4, 4, 2, 4]				
Generation 810:	Best = 1.200042e+03	FES = 81100	Diversity = 5.4900e-02	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.200026e+03	FES = 82100	Diversity = 3.6981e-02	k_t
= [3, 2, 2, 4]				
Generation 830:	Best = 1.200014e+03	FES = 83100	Diversity = 2.7773e-02	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.200008e+03	FES = 84100	Diversity = 1.9925e-02	k_t
= [4, 4, 4, 3]				
Generation 850:	Best = 1.200004e+03	FES = 85100	Diversity = 1.5639e-02	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.200002e+03	FES = 86100	Diversity = 1.1028e-02	k_t
= [4, 2, 3, 2]				
Generation 870:	Best = 1.200001e+03	FES = 87100	Diversity = 6.9218e-03	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.200001e+03	FES = 88100	Diversity = 5.3712e-03	k_t
= [2, 3, 4, 4]				
Generation 890:	Best = 1.200000e+03	FES = 89100	Diversity = 3.5752e-03	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.200000e+03	FES = 90100	Diversity = 2.6293e-03	k_t
= [3, 4, 3, 4]				
Generation 910:	Best = 1.200000e+03	FES = 91100	Diversity = 1.9007e-03	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.200000e+03	FES = 92100	Diversity = 1.3458e-03	k_t
= [3, 3, 2, 4]				
Generation 930:	Best = 1.200000e+03	FES = 93100	Diversity = 9.5694e-04	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.200000e+03	FES = 94100	Diversity = 6.3852e-04	k_t
= [3, 3, 2, 3]				
Generation 950:	Best = 1.200000e+03	FES = 95100	Diversity = 4.9373e-04	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.200000e+03	FES = 96100	Diversity = 3.1743e-04	k_t
= [2, 4, 2, 4]				
Generation 970:	Best = 1.200000e+03	FES = 97100	Diversity = 2.3005e-04	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.200000e+03	FES = 98100	Diversity = 1.6702e-04	k_t
= [3, 2, 4, 4]				
Generation 990:	Best = 1.200000e+03	FES = 99100	Diversity = 1.0923e-04	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 1.200000e+03

Total Function Evaluations = 100000

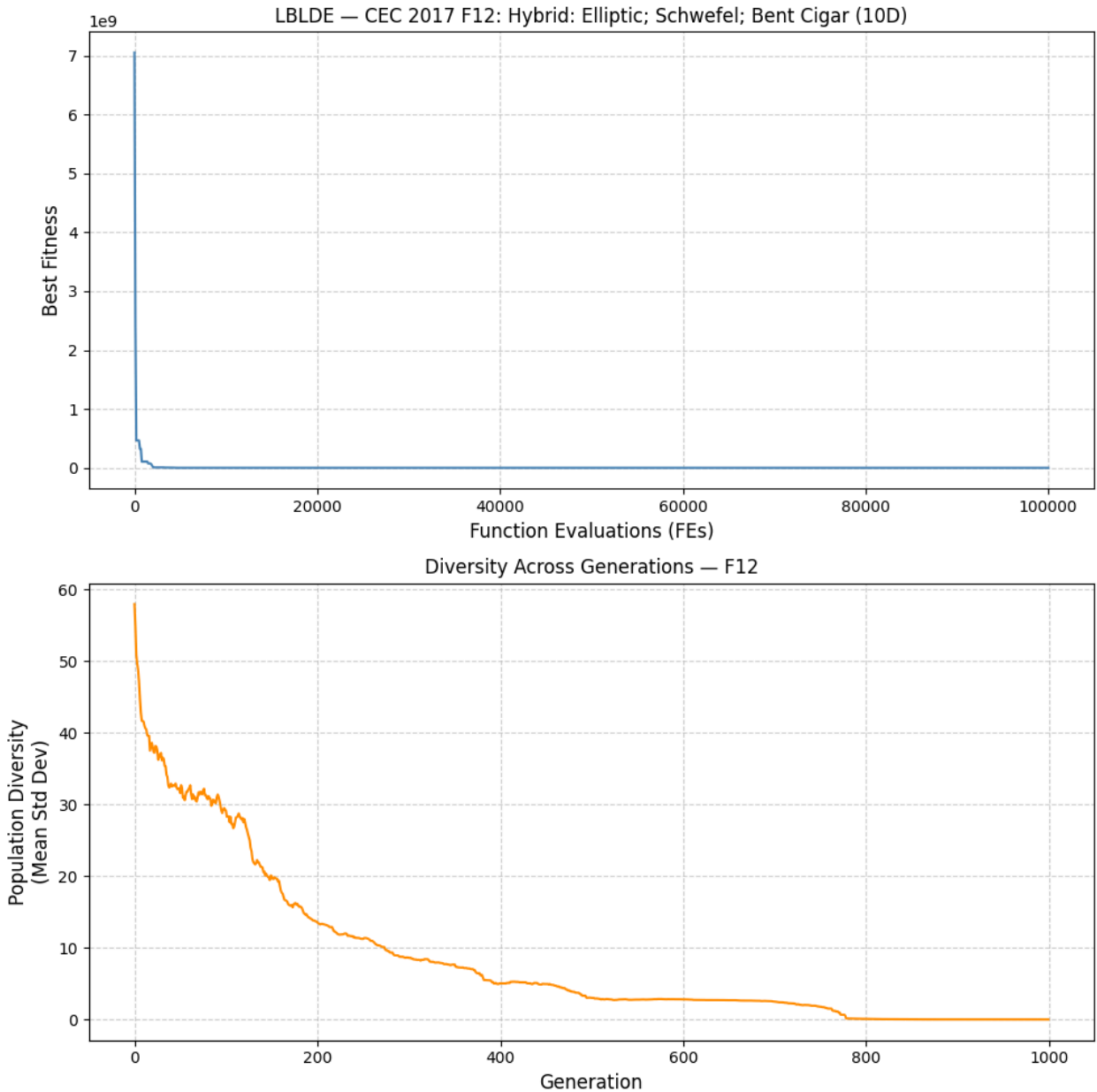
=====

Results:

```

Best Fitness      : 1.200000e+03
Global Optimum    : 1.200000e+03
Error (|best-optimum|) : 3.825973e-05
Best Solution (5 dim) : [ 65.25881846   5.20268055 -78.04181781  31.47055842
-34.47153964]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F12.png

```

In [29]: # =====
# F13 – Hybrid Function 3 (Bent Cigar; Rosenbrock; Bi-Rastrigin)
# =====
run_and_plot('F13', make_wrapper(hybrid.f13), 1300.0, 'Hybrid: Bent Cigar; Ros

```

```

=====
=
LBLDE_s on CEC 2017 F13 – Single Run
=====
=
Function      : Hybrid: Bent Cigar; Rosenbrock; Bi-Rastrigin (10D)
Bounds       : [-100, 100]
Global optimum : 1300.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 1.212051e+09 | Diversity = 5.7928e+01
Generation 10: Best = 1.925106e+06 FES = 1100 Diversity = 4.7486e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 5.175112e+04 FES = 2100 Diversity = 4.7320e+01 k_t
= [4, 3, 2, 3]
Generation 30: Best = 5.418305e+03 FES = 3100 Diversity = 4.5013e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.041340e+03 FES = 4100 Diversity = 4.2569e+01 k_t
= [3, 4, 3, 2]
Generation 50: Best = 1.658190e+03 FES = 5100 Diversity = 3.4083e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.440907e+03 FES = 6100 Diversity = 2.5644e+01 k_t
= [2, 4, 4, 3]
Generation 70: Best = 1.396807e+03 FES = 7100 Diversity = 1.9548e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.335469e+03 FES = 8100 Diversity = 1.5598e+01 k_t
= [4, 2, 2, 4]
Generation 90: Best = 1.328998e+03 FES = 9100 Diversity = 1.2390e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.328998e+03 FES = 10100 Diversity = 1.0510e+01 k_t
= [2, 4, 3, 3]
Generation 110: Best = 1.323350e+03 FES = 11100 Diversity = 9.8206e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.323350e+03 FES = 12100 Diversity = 9.4883e+00 k_t
= [2, 3, 4, 4]
Generation 130: Best = 1.323350e+03 FES = 13100 Diversity = 9.5272e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.318421e+03 FES = 14100 Diversity = 9.5545e+00 k_t
= [2, 2, 4, 2]
Generation 150: Best = 1.318421e+03 FES = 15100 Diversity = 9.4190e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.316406e+03 FES = 16100 Diversity = 9.3409e+00 k_t
= [4, 3, 4, 4]
Generation 170: Best = 1.312858e+03 FES = 17100 Diversity = 9.0721e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.312858e+03 FES = 18100 Diversity = 9.1384e+00 k_t
= [3, 4, 4, 2]
Generation 190: Best = 1.311859e+03 FES = 19100 Diversity = 9.0515e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.311594e+03 FES = 20100 Diversity = 8.8267e+00 k_t
= [3, 4, 3, 3]
Generation 210: Best = 1.311594e+03 FES = 21100 Diversity = 8.6897e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.311579e+03 FES = 22100 Diversity = 8.7196e+00 k_t
= [4, 3, 3, 4]

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Generation 230:	Best = 1.311476e+03	FES = 23100	Diversity = 8.6651e+00	k_t = [1, 1, 1, 1]
Generation 240:	Best = 1.309855e+03	FES = 24100	Diversity = 9.0617e+00	k_t = [4, 3, 4, 4]
Generation 250:	Best = 1.305814e+03	FES = 25100	Diversity = 9.1398e+00	k_t = [1, 1, 1, 1]
Generation 260:	Best = 1.305814e+03	FES = 26100	Diversity = 9.1973e+00	k_t = [3, 2, 2, 3]
Generation 270:	Best = 1.305814e+03	FES = 27100	Diversity = 8.7725e+00	k_t = [1, 1, 1, 1]
Generation 280:	Best = 1.305010e+03	FES = 28100	Diversity = 8.5036e+00	k_t = [3, 3, 2, 4]
Generation 290:	Best = 1.305010e+03	FES = 29100	Diversity = 8.3209e+00	k_t = [1, 1, 1, 1]
Generation 300:	Best = 1.304780e+03	FES = 30100	Diversity = 8.0467e+00	k_t = [2, 4, 2, 4]
Generation 310:	Best = 1.304521e+03	FES = 31100	Diversity = 7.9439e+00	k_t = [1, 1, 1, 1]
Generation 320:	Best = 1.304521e+03	FES = 32100	Diversity = 8.0381e+00	k_t = [4, 3, 3, 4]
Generation 330:	Best = 1.303798e+03	FES = 33100	Diversity = 7.7603e+00	k_t = [1, 1, 1, 1]
Generation 340:	Best = 1.303552e+03	FES = 34100	Diversity = 7.6676e+00	k_t = [3, 2, 4, 2]
Generation 350:	Best = 1.303552e+03	FES = 35100	Diversity = 7.7278e+00	k_t = [1, 1, 1, 1]
Generation 360:	Best = 1.303552e+03	FES = 36100	Diversity = 7.6189e+00	k_t = [4, 3, 2, 3]
Generation 370:	Best = 1.303552e+03	FES = 37100	Diversity = 7.0759e+00	k_t = [1, 1, 1, 1]
Generation 380:	Best = 1.303552e+03	FES = 38100	Diversity = 7.0672e+00	k_t = [2, 2, 3, 2]
Generation 390:	Best = 1.303505e+03	FES = 39100	Diversity = 7.0064e+00	k_t = [1, 1, 1, 1]
Generation 400:	Best = 1.303505e+03	FES = 40100	Diversity = 6.9939e+00	k_t = [2, 2, 4, 4]
Generation 410:	Best = 1.303505e+03	FES = 41100	Diversity = 6.8496e+00	k_t = [1, 1, 1, 1]
Generation 420:	Best = 1.303505e+03	FES = 42100	Diversity = 6.6612e+00	k_t = [2, 3, 2, 4]
Generation 430:	Best = 1.303494e+03	FES = 43100	Diversity = 6.5777e+00	k_t = [1, 1, 1, 1]
Generation 440:	Best = 1.303494e+03	FES = 44100	Diversity = 6.1991e+00	k_t = [2, 3, 2, 2]
Generation 450:	Best = 1.303494e+03	FES = 45100	Diversity = 6.1448e+00	k_t = [1, 1, 1, 1]
Generation 460:	Best = 1.303218e+03	FES = 46100	Diversity = 5.9108e+00	k_t = [4, 2, 2, 2]
Generation 470:	Best = 1.303218e+03	FES = 47100	Diversity = 5.5197e+00	k_t = [1, 1, 1, 1]
Generation 480:	Best = 1.302960e+03	FES = 48100	Diversity = 4.6544e+00	k_t = [3, 2, 3, 2]
Generation 490:	Best = 1.302423e+03	FES = 49100	Diversity = 4.2181e+00	k_t = [1, 1, 1, 1]

Generation 500:	Best = 1.302368e+03	FES = 50100	Diversity = 3.9445e+00	k_t = [2, 4, 2, 3]
Generation 510:	Best = 1.302328e+03	FES = 51100	Diversity = 3.6436e+00	k_t = [1, 1, 1, 1]
Generation 520:	Best = 1.302328e+03	FES = 52100	Diversity = 3.1256e+00	k_t = [3, 3, 3, 2]
Generation 530:	Best = 1.302327e+03	FES = 53100	Diversity = 2.6361e+00	k_t = [1, 1, 1, 1]
Generation 540:	Best = 1.302142e+03	FES = 54100	Diversity = 2.4481e+00	k_t = [3, 3, 3, 3]
Generation 550:	Best = 1.302037e+03	FES = 55100	Diversity = 2.1130e+00	k_t = [1, 1, 1, 1]
Generation 560:	Best = 1.301668e+03	FES = 56100	Diversity = 1.8753e+00	k_t = [3, 3, 3, 3]
Generation 570:	Best = 1.301656e+03	FES = 57100	Diversity = 1.7281e+00	k_t = [1, 1, 1, 1]
Generation 580:	Best = 1.301613e+03	FES = 58100	Diversity = 1.6290e+00	k_t = [4, 3, 2, 3]
Generation 590:	Best = 1.301348e+03	FES = 59100	Diversity = 1.4885e+00	k_t = [1, 1, 1, 1]
Generation 600:	Best = 1.301053e+03	FES = 60100	Diversity = 1.4819e+00	k_t = [4, 3, 3, 2]
Generation 610:	Best = 1.300967e+03	FES = 61100	Diversity = 1.5430e+00	k_t = [1, 1, 1, 1]
Generation 620:	Best = 1.300962e+03	FES = 62100	Diversity = 1.4855e+00	k_t = [4, 3, 3, 4]
Generation 630:	Best = 1.300723e+03	FES = 63100	Diversity = 1.3946e+00	k_t = [1, 1, 1, 1]
Generation 640:	Best = 1.300514e+03	FES = 64100	Diversity = 1.3641e+00	k_t = [3, 4, 3, 3]
Generation 650:	Best = 1.300446e+03	FES = 65100	Diversity = 1.3836e+00	k_t = [1, 1, 1, 1]
Generation 660:	Best = 1.300332e+03	FES = 66100	Diversity = 1.3725e+00	k_t = [4, 2, 2, 2]
Generation 670:	Best = 1.300332e+03	FES = 67100	Diversity = 1.2306e+00	k_t = [1, 1, 1, 1]
Generation 680:	Best = 1.300310e+03	FES = 68100	Diversity = 8.2917e-01	k_t = [4, 3, 4, 2]
Generation 690:	Best = 1.300212e+03	FES = 69100	Diversity = 7.9224e-01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 1.300095e+03	FES = 70100	Diversity = 9.5457e-01	k_t = [2, 4, 2, 3]
Generation 710:	Best = 1.300085e+03	FES = 71100	Diversity = 8.0412e-01	k_t = [1, 1, 1, 1]
Generation 720:	Best = 1.300050e+03	FES = 72100	Diversity = 5.3352e-01	k_t = [4, 2, 4, 3]
Generation 730:	Best = 1.300031e+03	FES = 73100	Diversity = 6.6910e-01	k_t = [1, 1, 1, 1]
Generation 740:	Best = 1.300013e+03	FES = 74100	Diversity = 6.8963e-01	k_t = [3, 2, 3, 2]
Generation 750:	Best = 1.300004e+03	FES = 75100	Diversity = 5.4726e-01	k_t = [1, 1, 1, 1]
Generation 760:	Best = 1.300003e+03	FES = 76100	Diversity = 4.2576e-01	k_t = [4, 2, 4, 4]

Generation 770:	Best = 1.300002e+03	FES = 77100	Diversity = 3.3393e-01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.300001e+03	FES = 78100	Diversity = 2.4113e-01	k_t
= [2, 3, 4, 3]				
Generation 790:	Best = 1.300000e+03	FES = 79100	Diversity = 1.8173e-01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.300000e+03	FES = 80100	Diversity = 1.2657e-01	k_t
= [3, 3, 2, 3]				
Generation 810:	Best = 1.300000e+03	FES = 81100	Diversity = 9.5268e-02	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.300000e+03	FES = 82100	Diversity = 7.9825e-02	k_t
= [4, 4, 2, 3]				
Generation 830:	Best = 1.300000e+03	FES = 83100	Diversity = 5.3536e-02	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.300000e+03	FES = 84100	Diversity = 3.8460e-02	k_t
= [2, 2, 4, 3]				
Generation 850:	Best = 1.300000e+03	FES = 85100	Diversity = 2.6789e-02	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.300000e+03	FES = 86100	Diversity = 2.0728e-02	k_t
= [2, 2, 2, 3]				
Generation 870:	Best = 1.300000e+03	FES = 87100	Diversity = 1.2220e-02	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.300000e+03	FES = 88100	Diversity = 7.9696e-03	k_t
= [2, 3, 2, 4]				
Generation 890:	Best = 1.300000e+03	FES = 89100	Diversity = 5.3852e-03	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.300000e+03	FES = 90100	Diversity = 4.4331e-03	k_t
= [3, 4, 2, 2]				
Generation 910:	Best = 1.300000e+03	FES = 91100	Diversity = 3.2446e-03	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.300000e+03	FES = 92100	Diversity = 2.5357e-03	k_t
= [3, 3, 2, 2]				
Generation 930:	Best = 1.300000e+03	FES = 93100	Diversity = 1.3615e-03	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.300000e+03	FES = 94100	Diversity = 8.6227e-04	k_t
= [4, 2, 3, 3]				
Generation 950:	Best = 1.300000e+03	FES = 95100	Diversity = 5.2624e-04	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.300000e+03	FES = 96100	Diversity = 3.4279e-04	k_t
= [3, 2, 2, 3]				
Generation 970:	Best = 1.300000e+03	FES = 97100	Diversity = 2.8411e-04	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.300000e+03	FES = 98100	Diversity = 2.0169e-04	k_t
= [4, 4, 2, 4]				
Generation 990:	Best = 1.300000e+03	FES = 99100	Diversity = 1.1771e-04	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 1.300000e+03

Total Function Evaluations = 100000

=====

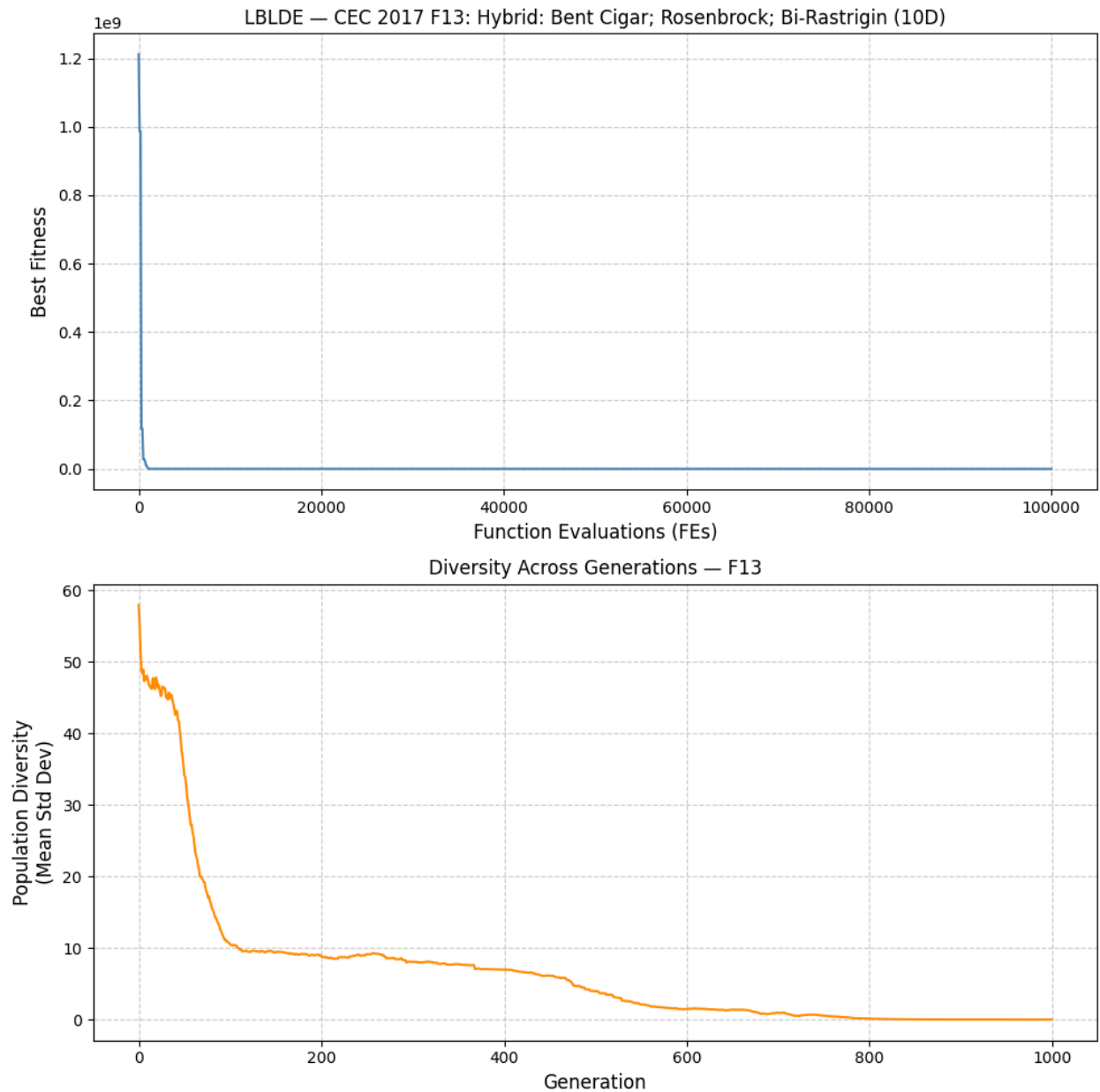
=

Results:

```

Best Fitness          : 1.300000e+03
Global Optimum        : 1.300000e+03
Error (|best-optimum|) : 8.890311e-11
Best Solution (5 dim)  : [ 4.574813   -71.12311738  -6.24587961 -14.50696723
25.61587896]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F13.png

```

In [30]: # =====
# F14 — Hybrid Function 4 (Elliptic; Ackley; Schaffer; Rastrigin)
# =====
run_and_plot('F14', make_wrapper(hybrid.f14), 1400.0, 'Hybrid: Elliptic; Ackle

```

```

=====
=
LBLDE_s on CEC 2017 F14 – Single Run
=====
=
Function      : Hybrid: Elliptic; Ackley; Schaffer; Rastrigin (10D)
Bounds       : [-100, 100]
Global optimum : 1400.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 2.917787e+04 | Diversity = 5.7928e+01
Generation 10: Best = 2.099243e+03 FES = 1100 Diversity = 4.9720e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.616899e+03 FES = 2100 Diversity = 4.8220e+01 k_t
= [4, 3, 2, 3]
Generation 30: Best = 1.526426e+03 FES = 3100 Diversity = 4.4809e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.473168e+03 FES = 4100 Diversity = 3.5950e+01 k_t
= [4, 3, 4, 4]
Generation 50: Best = 1.457257e+03 FES = 5100 Diversity = 3.3286e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.446679e+03 FES = 6100 Diversity = 2.8648e+01 k_t
= [2, 2, 4, 2]
Generation 70: Best = 1.444576e+03 FES = 7100 Diversity = 2.6101e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.430370e+03 FES = 8100 Diversity = 2.5072e+01 k_t
= [2, 3, 2, 4]
Generation 90: Best = 1.430370e+03 FES = 9100 Diversity = 2.4297e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.429947e+03 FES = 10100 Diversity = 2.3104e+01 k_t
= [3, 3, 2, 3]
Generation 110: Best = 1.429947e+03 FES = 11100 Diversity = 2.3073e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.428375e+03 FES = 12100 Diversity = 2.2369e+01 k_t
= [3, 2, 4, 2]
Generation 130: Best = 1.428375e+03 FES = 13100 Diversity = 2.1811e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.428375e+03 FES = 14100 Diversity = 2.2497e+01 k_t
= [3, 2, 3, 3]
Generation 150: Best = 1.426111e+03 FES = 15100 Diversity = 2.1944e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.426111e+03 FES = 16100 Diversity = 2.1864e+01 k_t
= [4, 2, 2, 3]
Generation 170: Best = 1.422266e+03 FES = 17100 Diversity = 2.0652e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.419382e+03 FES = 18100 Diversity = 2.0298e+01 k_t
= [2, 4, 3, 3]
Generation 190: Best = 1.419382e+03 FES = 19100 Diversity = 1.9089e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.419382e+03 FES = 20100 Diversity = 1.8732e+01 k_t
= [2, 2, 4, 4]
Generation 210: Best = 1.418599e+03 FES = 21100 Diversity = 1.8238e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.415454e+03 FES = 22100 Diversity = 1.8203e+01 k_t
= [4, 3, 2, 3]

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Generation 230:	Best = 1.411228e+03	FES = 23100	Diversity = 1.7389e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 1.410088e+03	FES = 24100	Diversity = 1.7076e+01	k_t = [2, 3, 2, 4]
Generation 250:	Best = 1.410088e+03	FES = 25100	Diversity = 1.6127e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 1.406839e+03	FES = 26100	Diversity = 1.4416e+01	k_t = [2, 3, 4, 4]
Generation 270:	Best = 1.406839e+03	FES = 27100	Diversity = 1.1900e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 1.406557e+03	FES = 28100	Diversity = 9.7695e+00	k_t = [3, 2, 4, 2]
Generation 290:	Best = 1.406557e+03	FES = 29100	Diversity = 7.3269e+00	k_t = [1, 1, 1, 1]
Generation 300:	Best = 1.405585e+03	FES = 30100	Diversity = 6.8416e+00	k_t = [2, 3, 2, 3]
Generation 310:	Best = 1.405585e+03	FES = 31100	Diversity = 6.4407e+00	k_t = [1, 1, 1, 1]
Generation 320:	Best = 1.404409e+03	FES = 32100	Diversity = 6.2200e+00	k_t = [4, 3, 3, 3]
Generation 330:	Best = 1.404007e+03	FES = 33100	Diversity = 5.8744e+00	k_t = [1, 1, 1, 1]
Generation 340:	Best = 1.402219e+03	FES = 34100	Diversity = 5.5000e+00	k_t = [4, 2, 2, 2]
Generation 350:	Best = 1.401458e+03	FES = 35100	Diversity = 5.3971e+00	k_t = [1, 1, 1, 1]
Generation 360:	Best = 1.401458e+03	FES = 36100	Diversity = 5.1463e+00	k_t = [3, 4, 4, 2]
Generation 370:	Best = 1.401427e+03	FES = 37100	Diversity = 4.8678e+00	k_t = [1, 1, 1, 1]
Generation 380:	Best = 1.401311e+03	FES = 38100	Diversity = 4.4973e+00	k_t = [2, 2, 4, 2]
Generation 390:	Best = 1.401084e+03	FES = 39100	Diversity = 4.2366e+00	k_t = [1, 1, 1, 1]
Generation 400:	Best = 1.400353e+03	FES = 40100	Diversity = 3.9198e+00	k_t = [3, 4, 2, 4]
Generation 410:	Best = 1.400353e+03	FES = 41100	Diversity = 3.2641e+00	k_t = [1, 1, 1, 1]
Generation 420:	Best = 1.400189e+03	FES = 42100	Diversity = 2.2208e+00	k_t = [2, 3, 3, 3]
Generation 430:	Best = 1.400168e+03	FES = 43100	Diversity = 1.2084e+00	k_t = [1, 1, 1, 1]
Generation 440:	Best = 1.400155e+03	FES = 44100	Diversity = 1.3276e-01	k_t = [2, 3, 3, 3]
Generation 450:	Best = 1.400100e+03	FES = 45100	Diversity = 9.4756e-02	k_t = [1, 1, 1, 1]
Generation 460:	Best = 1.400059e+03	FES = 46100	Diversity = 6.6577e-02	k_t = [2, 3, 4, 4]
Generation 470:	Best = 1.400022e+03	FES = 47100	Diversity = 4.6909e-02	k_t = [1, 1, 1, 1]
Generation 480:	Best = 1.400012e+03	FES = 48100	Diversity = 4.0534e-02	k_t = [4, 2, 4, 2]
Generation 490:	Best = 1.400010e+03	FES = 49100	Diversity = 2.8164e-02	k_t = [1, 1, 1, 1]

Generation 500:	Best = 1.400005e+03	FES = 50100	Diversity = 1.8378e-02	k_t = [2, 2, 4, 2]
Generation 510:	Best = 1.400002e+03	FES = 51100	Diversity = 1.3550e-02	k_t = [1, 1, 1, 1]
Generation 520:	Best = 1.400002e+03	FES = 52100	Diversity = 1.0006e-02	k_t = [3, 3, 4, 4]
Generation 530:	Best = 1.400001e+03	FES = 53100	Diversity = 7.5018e-03	k_t = [1, 1, 1, 1]
Generation 540:	Best = 1.400000e+03	FES = 54100	Diversity = 5.6271e-03	k_t = [4, 2, 2, 4]
Generation 550:	Best = 1.400000e+03	FES = 55100	Diversity = 3.6885e-03	k_t = [1, 1, 1, 1]
Generation 560:	Best = 1.400000e+03	FES = 56100	Diversity = 2.7393e-03	k_t = [4, 3, 2, 4]
Generation 570:	Best = 1.400000e+03	FES = 57100	Diversity = 2.2282e-03	k_t = [1, 1, 1, 1]
Generation 580:	Best = 1.400000e+03	FES = 58100	Diversity = 1.3824e-03	k_t = [2, 2, 2, 2]
Generation 590:	Best = 1.400000e+03	FES = 59100	Diversity = 9.9106e-04	k_t = [1, 1, 1, 1]
Generation 600:	Best = 1.400000e+03	FES = 60100	Diversity = 7.5538e-04	k_t = [2, 2, 4, 2]
Generation 610:	Best = 1.400000e+03	FES = 61100	Diversity = 5.6578e-04	k_t = [1, 1, 1, 1]
Generation 620:	Best = 1.400000e+03	FES = 62100	Diversity = 3.2919e-04	k_t = [4, 4, 3, 4]
Generation 630:	Best = 1.400000e+03	FES = 63100	Diversity = 2.2831e-04	k_t = [1, 1, 1, 1]
Generation 640:	Best = 1.400000e+03	FES = 64100	Diversity = 1.6187e-04	k_t = [4, 4, 4, 3]
Generation 650:	Best = 1.400000e+03	FES = 65100	Diversity = 1.2063e-04	k_t = [1, 1, 1, 1]
Generation 660:	Best = 1.400000e+03	FES = 66100	Diversity = 9.7168e-05	k_t = [2, 4, 2, 2]
Generation 670:	Best = 1.400000e+03	FES = 67100	Diversity = 6.5119e-05	k_t = [1, 1, 1, 1]
Generation 680:	Best = 1.400000e+03	FES = 68100	Diversity = 4.4036e-05	k_t = [3, 2, 2, 3]
Generation 690:	Best = 1.400000e+03	FES = 69100	Diversity = 2.8475e-05	k_t = [1, 1, 1, 1]
Generation 700:	Best = 1.400000e+03	FES = 70100	Diversity = 1.9303e-05	k_t = [2, 3, 4, 3]
Generation 710:	Best = 1.400000e+03	FES = 71100	Diversity = 1.2454e-05	k_t = [1, 1, 1, 1]
Generation 720:	Best = 1.400000e+03	FES = 72100	Diversity = 8.5017e-06	k_t = [3, 4, 4, 2]
Generation 730:	Best = 1.400000e+03	FES = 73100	Diversity = 5.3926e-06	k_t = [1, 1, 1, 1]
Generation 740:	Best = 1.400000e+03	FES = 74100	Diversity = 3.4121e-06	k_t = [4, 2, 4, 4]
Generation 750:	Best = 1.400000e+03	FES = 75100	Diversity = 2.0557e-06	k_t = [1, 1, 1, 1]
Generation 760:	Best = 1.400000e+03	FES = 76100	Diversity = 1.3848e-06	k_t = [3, 2, 2, 4]

Generation 770:	Best = 1.400000e+03	FES = 77100	Diversity = 9.8072e-07	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.400000e+03	FES = 78100	Diversity = 6.2739e-07	k_t
= [4, 3, 4, 4]				
Generation 790:	Best = 1.400000e+03	FES = 79100	Diversity = 4.1042e-07	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.400000e+03	FES = 80100	Diversity = 2.5757e-07	k_t
= [3, 3, 3, 2]				
Generation 810:	Best = 1.400000e+03	FES = 81100	Diversity = 1.7582e-07	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.400000e+03	FES = 82100	Diversity = 1.3841e-07	k_t
= [4, 4, 4, 3]				
Generation 830:	Best = 1.400000e+03	FES = 83100	Diversity = 1.0265e-07	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.400000e+03	FES = 84100	Diversity = 9.3416e-08	k_t
= [3, 3, 2, 4]				
Generation 850:	Best = 1.400000e+03	FES = 85100	Diversity = 9.3295e-08	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.400000e+03	FES = 86100	Diversity = 8.1964e-08	k_t
= [2, 3, 3, 2]				
Generation 870:	Best = 1.400000e+03	FES = 87100	Diversity = 6.1644e-08	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.400000e+03	FES = 88100	Diversity = 4.5299e-08	k_t
= [3, 4, 2, 2]				
Generation 890:	Best = 1.400000e+03	FES = 89100	Diversity = 4.3825e-08	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.400000e+03	FES = 90100	Diversity = 4.4444e-08	k_t
= [2, 2, 4, 3]				
Generation 910:	Best = 1.400000e+03	FES = 91100	Diversity = 4.5114e-08	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.400000e+03	FES = 92100	Diversity = 5.0102e-08	k_t
= [3, 3, 3, 3]				
Generation 930:	Best = 1.400000e+03	FES = 93100	Diversity = 5.3624e-08	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.400000e+03	FES = 94100	Diversity = 5.0405e-08	k_t
= [3, 3, 2, 2]				
Generation 950:	Best = 1.400000e+03	FES = 95100	Diversity = 4.9347e-08	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.400000e+03	FES = 96100	Diversity = 4.7363e-08	k_t
= [4, 4, 3, 2]				
Generation 970:	Best = 1.400000e+03	FES = 97100	Diversity = 4.7622e-08	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.400000e+03	FES = 98100	Diversity = 4.6703e-08	k_t
= [4, 3, 4, 3]				
Generation 990:	Best = 1.400000e+03	FES = 99100	Diversity = 4.4290e-08	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 1.400000e+03

Total Function Evaluations = 100000

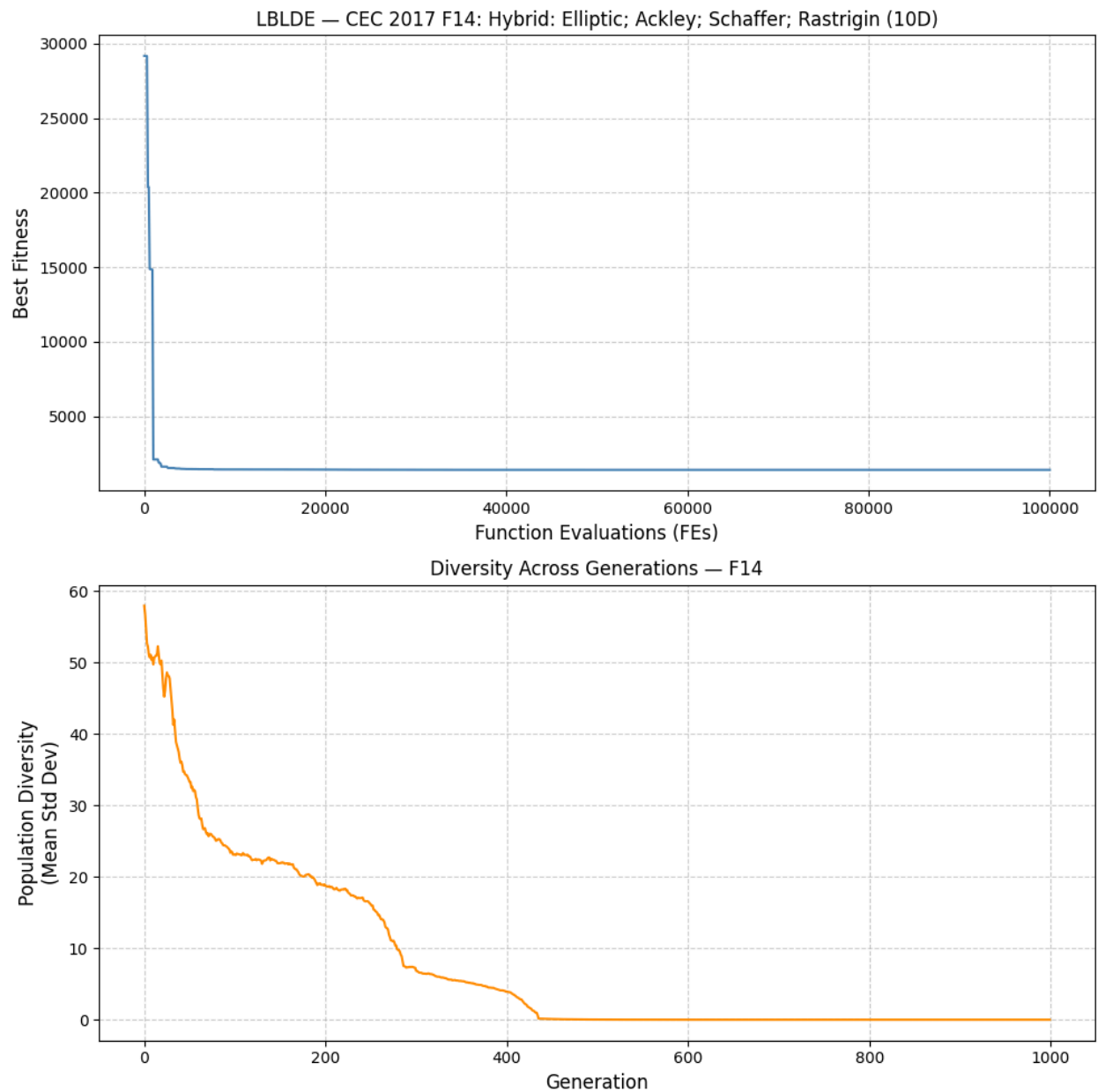
=====

Results:


```

Best Fitness      : 1.400000e+03
Global Optimum    : 1.400000e+03
Error (|best-optimum|) : 0.000000e+00
Best Solution (5 dim) : [-36.61752711 -70.77608488 66.34140998 -39.92255194
61.66827386]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F14.png

```

In [31]: # =====
# F15 – Hybrid Function 5 (Bent Cigar; HGBat; Rastrigin; Rosenbrock)
# =====
run_and_plot('F15', make_wrapper(hybrid.f15), 1500.0, 'Hybrid: Bent Cigar; HGBat; Rastrigin; Rosenbrock')

```

```

=====
=
LBLDE_s on CEC 2017 F15 – Single Run
=====
=
Function      : Hybrid: Bent Cigar; HGBat; Rastrigin; Rosenbrock (10D)
Bounds       : [-100, 100]
Global optimum : 1500.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 3.114909e+07 | Diversity = 5.7928e+01
Generation 10: Best = 1.135625e+04 FES = 1100 Diversity = 4.9175e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.851904e+03 FES = 2100 Diversity = 4.8004e+01 k_t
= [4, 3, 2, 3]
Generation 30: Best = 1.671210e+03 FES = 3100 Diversity = 3.6787e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.558833e+03 FES = 4100 Diversity = 2.5295e+01 k_t
= [3, 4, 3, 2]
Generation 50: Best = 1.523567e+03 FES = 5100 Diversity = 1.9133e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.515247e+03 FES = 6100 Diversity = 1.6586e+01 k_t
= [2, 4, 4, 3]
Generation 70: Best = 1.513387e+03 FES = 7100 Diversity = 1.4090e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.508925e+03 FES = 8100 Diversity = 1.3049e+01 k_t
= [2, 4, 4, 2]
Generation 90: Best = 1.508925e+03 FES = 9100 Diversity = 1.1253e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.507956e+03 FES = 10100 Diversity = 1.0515e+01 k_t
= [4, 4, 4, 4]
Generation 110: Best = 1.505745e+03 FES = 11100 Diversity = 1.0138e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.505745e+03 FES = 12100 Diversity = 9.9147e+00 k_t
= [2, 4, 4, 2]
Generation 130: Best = 1.504823e+03 FES = 13100 Diversity = 9.7452e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.503378e+03 FES = 14100 Diversity = 9.4539e+00 k_t
= [2, 3, 4, 2]
Generation 150: Best = 1.503025e+03 FES = 15100 Diversity = 9.0621e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.502966e+03 FES = 16100 Diversity = 9.0828e+00 k_t
= [2, 4, 2, 4]
Generation 170: Best = 1.502896e+03 FES = 17100 Diversity = 8.7036e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.502896e+03 FES = 18100 Diversity = 8.5013e+00 k_t
= [2, 2, 3, 4]
Generation 190: Best = 1.502697e+03 FES = 19100 Diversity = 8.2159e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.502697e+03 FES = 20100 Diversity = 7.9469e+00 k_t
= [2, 2, 3, 3]
Generation 210: Best = 1.502541e+03 FES = 21100 Diversity = 7.9120e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.502541e+03 FES = 22100 Diversity = 7.6273e+00 k_t
= [3, 4, 3, 4]

```

Generation 230:	Best = 1.502541e+03	FES = 23100	Diversity = 7.6342e+00	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 1.502541e+03	FES = 24100	Diversity = 7.3626e+00	k_t
= [3, 4, 3, 3]				
Generation 250:	Best = 1.502541e+03	FES = 25100	Diversity = 7.2509e+00	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 1.501771e+03	FES = 26100	Diversity = 7.0016e+00	k_t
= [4, 3, 3, 2]				
Generation 270:	Best = 1.501771e+03	FES = 27100	Diversity = 7.0398e+00	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 1.501647e+03	FES = 28100	Diversity = 6.8392e+00	k_t
= [2, 4, 4, 2]				
Generation 290:	Best = 1.501290e+03	FES = 29100	Diversity = 6.6011e+00	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 1.501290e+03	FES = 30100	Diversity = 6.4978e+00	k_t
= [4, 2, 4, 2]				
Generation 310:	Best = 1.501290e+03	FES = 31100	Diversity = 6.4592e+00	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 1.501290e+03	FES = 32100	Diversity = 6.3726e+00	k_t
= [2, 3, 4, 4]				
Generation 330:	Best = 1.500869e+03	FES = 33100	Diversity = 6.3404e+00	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 1.500869e+03	FES = 34100	Diversity = 6.2196e+00	k_t
= [3, 4, 2, 3]				
Generation 350:	Best = 1.500764e+03	FES = 35100	Diversity = 5.8760e+00	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 1.500741e+03	FES = 36100	Diversity = 5.7371e+00	k_t
= [3, 3, 3, 4]				
Generation 370:	Best = 1.500736e+03	FES = 37100	Diversity = 5.5869e+00	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 1.500564e+03	FES = 38100	Diversity = 5.2878e+00	k_t
= [3, 3, 3, 2]				
Generation 390:	Best = 1.500524e+03	FES = 39100	Diversity = 4.7400e+00	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 1.500357e+03	FES = 40100	Diversity = 4.5161e+00	k_t
= [2, 4, 4, 2]				
Generation 410:	Best = 1.500353e+03	FES = 41100	Diversity = 3.9177e+00	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 1.500308e+03	FES = 42100	Diversity = 3.3995e+00	k_t
= [2, 3, 2, 4]				
Generation 430:	Best = 1.500292e+03	FES = 43100	Diversity = 2.8285e+00	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 1.500227e+03	FES = 44100	Diversity = 2.7841e+00	k_t
= [2, 2, 2, 4]				
Generation 450:	Best = 1.500195e+03	FES = 45100	Diversity = 2.4636e+00	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 1.500151e+03	FES = 46100	Diversity = 2.3560e+00	k_t
= [2, 4, 2, 2]				
Generation 470:	Best = 1.500151e+03	FES = 47100	Diversity = 2.1478e+00	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 1.500151e+03	FES = 48100	Diversity = 2.0946e+00	k_t
= [2, 4, 4, 4]				
Generation 490:	Best = 1.500151e+03	FES = 49100	Diversity = 2.0581e+00	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 1.500145e+03	FES = 50100	Diversity = 1.9635e+00	k_t
= [2, 4, 3, 2]				
Generation 510:	Best = 1.500123e+03	FES = 51100	Diversity = 1.8717e+00	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 1.500103e+03	FES = 52100	Diversity = 1.6662e+00	k_t
= [4, 3, 4, 3]				
Generation 530:	Best = 1.500071e+03	FES = 53100	Diversity = 1.5532e+00	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 1.500070e+03	FES = 54100	Diversity = 1.5538e+00	k_t
= [4, 3, 3, 3]				
Generation 550:	Best = 1.500070e+03	FES = 55100	Diversity = 1.5273e+00	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 1.500069e+03	FES = 56100	Diversity = 1.3602e+00	k_t
= [2, 4, 2, 4]				
Generation 570:	Best = 1.500069e+03	FES = 57100	Diversity = 1.2867e+00	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 1.500069e+03	FES = 58100	Diversity = 1.1549e+00	k_t
= [4, 4, 4, 2]				
Generation 590:	Best = 1.500053e+03	FES = 59100	Diversity = 1.1031e+00	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 1.500053e+03	FES = 60100	Diversity = 1.1021e+00	k_t
= [2, 2, 3, 4]				
Generation 610:	Best = 1.500053e+03	FES = 61100	Diversity = 1.1207e+00	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 1.500052e+03	FES = 62100	Diversity = 1.0628e+00	k_t
= [3, 2, 4, 4]				
Generation 630:	Best = 1.500045e+03	FES = 63100	Diversity = 1.0106e+00	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 1.500043e+03	FES = 64100	Diversity = 9.2599e-01	k_t
= [4, 4, 4, 2]				
Generation 650:	Best = 1.500035e+03	FES = 65100	Diversity = 9.5637e-01	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 1.500034e+03	FES = 66100	Diversity = 9.5256e-01	k_t
= [3, 2, 4, 3]				
Generation 670:	Best = 1.500034e+03	FES = 67100	Diversity = 9.2826e-01	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 1.500024e+03	FES = 68100	Diversity = 8.8563e-01	k_t
= [3, 2, 3, 4]				
Generation 690:	Best = 1.500011e+03	FES = 69100	Diversity = 8.6868e-01	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 1.500011e+03	FES = 70100	Diversity = 8.2886e-01	k_t
= [4, 4, 3, 2]				
Generation 710:	Best = 1.500011e+03	FES = 71100	Diversity = 8.0860e-01	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 1.500011e+03	FES = 72100	Diversity = 7.5392e-01	k_t
= [4, 2, 2, 3]				
Generation 730:	Best = 1.500011e+03	FES = 73100	Diversity = 7.3516e-01	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 1.500011e+03	FES = 74100	Diversity = 7.3064e-01	k_t
= [2, 3, 2, 3]				
Generation 750:	Best = 1.500011e+03	FES = 75100	Diversity = 6.3414e-01	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 1.500011e+03	FES = 76100	Diversity = 6.1110e-01	k_t
= [4, 2, 4, 3]				

Generation 770:	Best = 1.500011e+03	FES = 77100	Diversity = 6.0418e-01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.500010e+03	FES = 78100	Diversity = 5.6924e-01	k_t
= [3, 3, 3, 3]				
Generation 790:	Best = 1.500010e+03	FES = 79100	Diversity = 5.5781e-01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.500009e+03	FES = 80100	Diversity = 5.2197e-01	k_t
= [3, 2, 2, 2]				
Generation 810:	Best = 1.500008e+03	FES = 81100	Diversity = 4.9386e-01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.500008e+03	FES = 82100	Diversity = 4.8155e-01	k_t
= [3, 2, 2, 3]				
Generation 830:	Best = 1.500008e+03	FES = 83100	Diversity = 4.7803e-01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.500008e+03	FES = 84100	Diversity = 4.5438e-01	k_t
= [3, 4, 4, 3]				
Generation 850:	Best = 1.500005e+03	FES = 85100	Diversity = 4.5287e-01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.500005e+03	FES = 86100	Diversity = 4.4310e-01	k_t
= [2, 4, 3, 3]				
Generation 870:	Best = 1.500005e+03	FES = 87100	Diversity = 4.2107e-01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.500005e+03	FES = 88100	Diversity = 4.1934e-01	k_t
= [2, 2, 3, 2]				
Generation 890:	Best = 1.500003e+03	FES = 89100	Diversity = 3.7677e-01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.500003e+03	FES = 90100	Diversity = 3.7271e-01	k_t
= [4, 2, 2, 3]				
Generation 910:	Best = 1.500003e+03	FES = 91100	Diversity = 3.8084e-01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.500003e+03	FES = 92100	Diversity = 3.5089e-01	k_t
= [3, 3, 2, 3]				
Generation 930:	Best = 1.500003e+03	FES = 93100	Diversity = 3.3939e-01	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.500003e+03	FES = 94100	Diversity = 3.2825e-01	k_t
= [4, 3, 2, 4]				
Generation 950:	Best = 1.500003e+03	FES = 95100	Diversity = 3.2196e-01	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.500002e+03	FES = 96100	Diversity = 3.0006e-01	k_t
= [4, 4, 4, 4]				
Generation 970:	Best = 1.500002e+03	FES = 97100	Diversity = 2.9995e-01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.500002e+03	FES = 98100	Diversity = 2.8791e-01	k_t
= [3, 3, 3, 3]				
Generation 990:	Best = 1.500002e+03	FES = 99100	Diversity = 2.3303e-01	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 1.500002e+03

Total Function Evaluations = 100000

=====

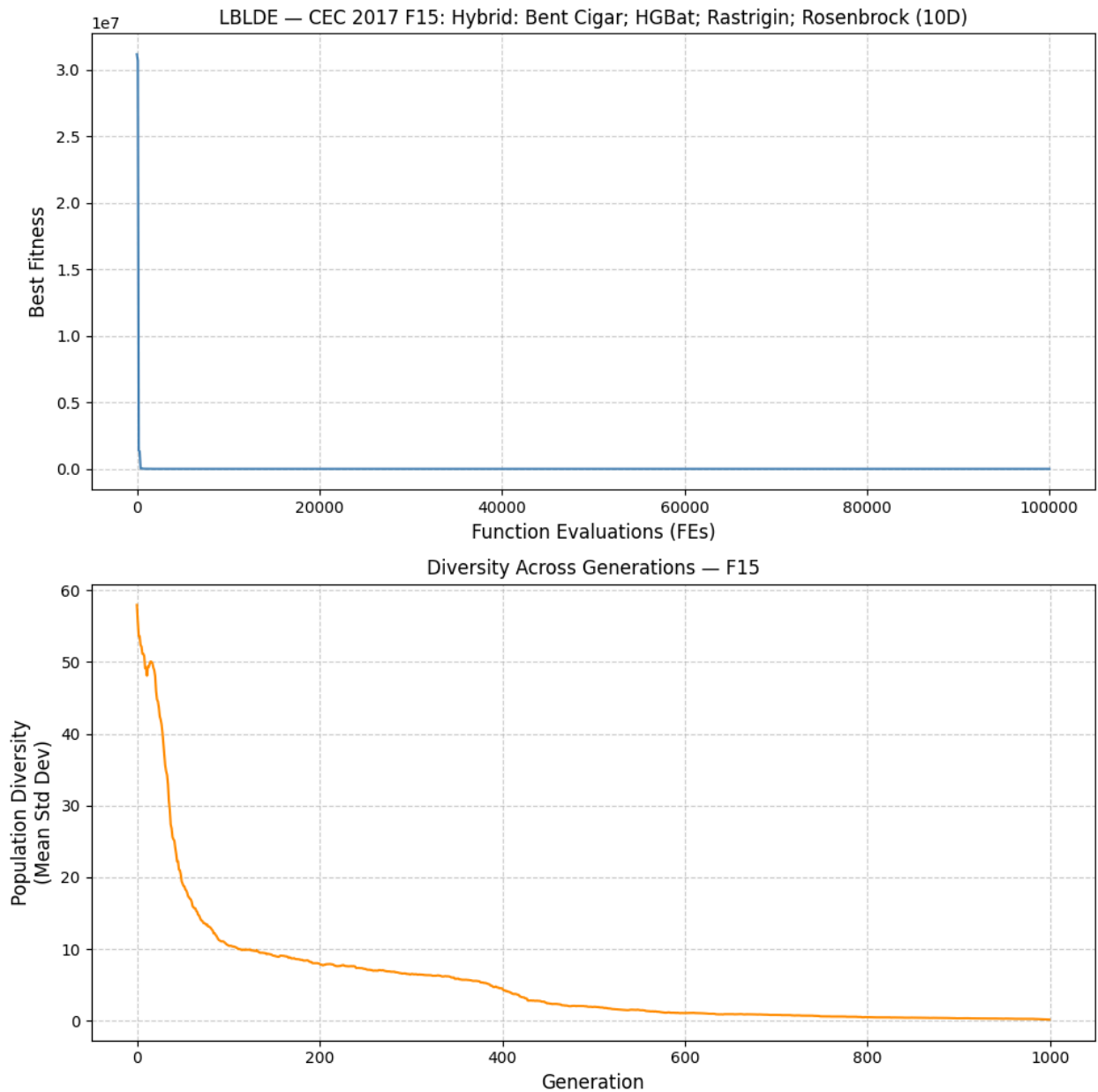
=

Results:

```

Best Fitness      : 1.500002e+03
Global Optimum    : 1.500000e+03
Error (|best-optimum|) : 1.624597e-03
Best Solution (5 dim) : [ 5.47591182  1.97056796  65.49392237 -42.02455798
72.40713809]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F15.png

```

In [32]: # =====
# F16 – Hybrid Function 6 (Schaffer; HGBat; Rosenbrock; Schwefel)
# =====
run_and_plot('F16', make_wrapper(hybrid.f16), 1600.0, 'Hybrid: Schaffer; HGBat

```

```

=====
=
LBLDE_s on CEC 2017 F16 – Single Run
=====
=
Function      : Hybrid: Schaffer; HGBat; Rosenbrock; Schwefel (10D)
Bounds       : [-100, 100]
Global optimum : 1600.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 2.512498e+03 | Diversity = 5.7928e+01
Generation 10: Best = 2.081722e+03 FES = 1100 Diversity = 4.4706e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.042823e+03 FES = 2100 Diversity = 4.1494e+01 k_t
= [4, 3, 2, 3]
Generation 30: Best = 1.767255e+03 FES = 3100 Diversity = 3.9022e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.668924e+03 FES = 4100 Diversity = 3.7690e+01 k_t
= [3, 3, 4, 3]
Generation 50: Best = 1.667174e+03 FES = 5100 Diversity = 3.4074e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.667174e+03 FES = 6100 Diversity = 3.1939e+01 k_t
= [2, 2, 2, 2]
Generation 70: Best = 1.661917e+03 FES = 7100 Diversity = 3.1354e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.621191e+03 FES = 8100 Diversity = 3.0120e+01 k_t
= [4, 3, 4, 2]
Generation 90: Best = 1.621191e+03 FES = 9100 Diversity = 2.8958e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.620927e+03 FES = 10100 Diversity = 2.9293e+01 k_t
= [4, 4, 4, 4]
Generation 110: Best = 1.620137e+03 FES = 11100 Diversity = 2.8058e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.620137e+03 FES = 12100 Diversity = 2.6509e+01 k_t
= [2, 3, 2, 3]
Generation 130: Best = 1.618131e+03 FES = 13100 Diversity = 2.4787e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.617313e+03 FES = 14100 Diversity = 2.3671e+01 k_t
= [3, 2, 2, 2]
Generation 150: Best = 1.610806e+03 FES = 15100 Diversity = 2.1752e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.608188e+03 FES = 16100 Diversity = 2.1797e+01 k_t
= [2, 4, 2, 2]
Generation 170: Best = 1.606839e+03 FES = 17100 Diversity = 2.1614e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.604551e+03 FES = 18100 Diversity = 2.1455e+01 k_t
= [3, 2, 3, 3]
Generation 190: Best = 1.603336e+03 FES = 19100 Diversity = 2.1438e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.603336e+03 FES = 20100 Diversity = 2.2052e+01 k_t
= [3, 3, 2, 3]
Generation 210: Best = 1.601856e+03 FES = 21100 Diversity = 2.2153e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.601856e+03 FES = 22100 Diversity = 2.2695e+01 k_t
= [2, 2, 4, 2]

```

Generation 230:	Best = 1.601856e+03	FES = 23100	Diversity = 2.2960e+01	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 1.601759e+03	FES = 24100	Diversity = 2.2327e+01	k_t
= [4, 3, 4, 3]				
Generation 250:	Best = 1.601560e+03	FES = 25100	Diversity = 2.1163e+01	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 1.601560e+03	FES = 26100	Diversity = 1.9577e+01	k_t
= [4, 4, 4, 3]				
Generation 270:	Best = 1.601433e+03	FES = 27100	Diversity = 1.7834e+01	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 1.601433e+03	FES = 28100	Diversity = 1.8172e+01	k_t
= [3, 3, 4, 2]				
Generation 290:	Best = 1.601354e+03	FES = 29100	Diversity = 1.7139e+01	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 1.601172e+03	FES = 30100	Diversity = 1.6175e+01	k_t
= [3, 2, 4, 2]				
Generation 310:	Best = 1.601165e+03	FES = 31100	Diversity = 1.6828e+01	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 1.601108e+03	FES = 32100	Diversity = 1.7258e+01	k_t
= [4, 2, 2, 2]				
Generation 330:	Best = 1.601081e+03	FES = 33100	Diversity = 1.6319e+01	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 1.601050e+03	FES = 34100	Diversity = 1.7074e+01	k_t
= [2, 2, 2, 4]				
Generation 350:	Best = 1.601050e+03	FES = 35100	Diversity = 1.6632e+01	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 1.601018e+03	FES = 36100	Diversity = 1.6454e+01	k_t
= [2, 4, 3, 2]				
Generation 370:	Best = 1.600979e+03	FES = 37100	Diversity = 1.5501e+01	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 1.600938e+03	FES = 38100	Diversity = 1.5390e+01	k_t
= [4, 4, 4, 4]				
Generation 390:	Best = 1.600752e+03	FES = 39100	Diversity = 1.4579e+01	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 1.600678e+03	FES = 40100	Diversity = 1.4331e+01	k_t
= [4, 4, 2, 2]				
Generation 410:	Best = 1.600673e+03	FES = 41100	Diversity = 1.2994e+01	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 1.600673e+03	FES = 42100	Diversity = 1.0615e+01	k_t
= [3, 4, 4, 4]				
Generation 430:	Best = 1.600604e+03	FES = 43100	Diversity = 9.9559e+00	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 1.600574e+03	FES = 44100	Diversity = 8.9256e+00	k_t
= [3, 4, 4, 3]				
Generation 450:	Best = 1.600574e+03	FES = 45100	Diversity = 7.7124e+00	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 1.600518e+03	FES = 46100	Diversity = 6.8148e+00	k_t
= [2, 4, 3, 2]				
Generation 470:	Best = 1.600518e+03	FES = 47100	Diversity = 5.4985e+00	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 1.600518e+03	FES = 48100	Diversity = 4.5453e+00	k_t
= [2, 2, 4, 3]				
Generation 490:	Best = 1.600472e+03	FES = 49100	Diversity = 3.6427e+00	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 1.600472e+03	FES = 50100	Diversity = 2.7920e+00	k_t
= [4, 2, 2, 2]				
Generation 510:	Best = 1.600472e+03	FES = 51100	Diversity = 2.6929e+00	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 1.600472e+03	FES = 52100	Diversity = 2.6618e+00	k_t
= [2, 3, 4, 4]				
Generation 530:	Best = 1.600472e+03	FES = 53100	Diversity = 2.3952e+00	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 1.600470e+03	FES = 54100	Diversity = 2.3216e+00	k_t
= [3, 3, 3, 2]				
Generation 550:	Best = 1.600470e+03	FES = 55100	Diversity = 2.2382e+00	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 1.600465e+03	FES = 56100	Diversity = 2.1159e+00	k_t
= [4, 4, 2, 2]				
Generation 570:	Best = 1.600463e+03	FES = 57100	Diversity = 2.1016e+00	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 1.600460e+03	FES = 58100	Diversity = 2.0589e+00	k_t
= [3, 4, 3, 2]				
Generation 590:	Best = 1.600460e+03	FES = 59100	Diversity = 2.0197e+00	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 1.600460e+03	FES = 60100	Diversity = 1.9474e+00	k_t
= [2, 4, 4, 4]				
Generation 610:	Best = 1.600459e+03	FES = 61100	Diversity = 1.9283e+00	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 1.600459e+03	FES = 62100	Diversity = 1.8520e+00	k_t
= [2, 4, 4, 3]				
Generation 630:	Best = 1.600459e+03	FES = 63100	Diversity = 1.8514e+00	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 1.600454e+03	FES = 64100	Diversity = 1.7892e+00	k_t
= [4, 4, 3, 4]				
Generation 650:	Best = 1.600454e+03	FES = 65100	Diversity = 1.7093e+00	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 1.600454e+03	FES = 66100	Diversity = 1.6488e+00	k_t
= [4, 4, 4, 2]				
Generation 670:	Best = 1.600454e+03	FES = 67100	Diversity = 1.6074e+00	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 1.600454e+03	FES = 68100	Diversity = 1.5601e+00	k_t
= [3, 3, 3, 3]				
Generation 690:	Best = 1.600439e+03	FES = 69100	Diversity = 1.4939e+00	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 1.600439e+03	FES = 70100	Diversity = 1.4624e+00	k_t
= [3, 4, 2, 4]				
Generation 710:	Best = 1.600439e+03	FES = 71100	Diversity = 1.4746e+00	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 1.600439e+03	FES = 72100	Diversity = 1.4789e+00	k_t
= [4, 4, 4, 3]				
Generation 730:	Best = 1.600439e+03	FES = 73100	Diversity = 1.4679e+00	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 1.600439e+03	FES = 74100	Diversity = 1.4453e+00	k_t
= [4, 2, 2, 4]				
Generation 750:	Best = 1.600439e+03	FES = 75100	Diversity = 1.4011e+00	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 1.600439e+03	FES = 76100	Diversity = 1.3440e+00	k_t
= [3, 3, 4, 3]				

Generation 770:	Best = 1.600439e+03	FES = 77100	Diversity = 1.2920e+00	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.600439e+03	FES = 78100	Diversity = 1.2479e+00	k_t
= [4, 2, 3, 2]				
Generation 790:	Best = 1.600437e+03	FES = 79100	Diversity = 1.2079e+00	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.600437e+03	FES = 80100	Diversity = 1.2049e+00	k_t
= [4, 2, 2, 4]				
Generation 810:	Best = 1.600437e+03	FES = 81100	Diversity = 1.1844e+00	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.600433e+03	FES = 82100	Diversity = 1.1761e+00	k_t
= [3, 3, 4, 4]				
Generation 830:	Best = 1.600433e+03	FES = 83100	Diversity = 1.1484e+00	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.600433e+03	FES = 84100	Diversity = 1.1540e+00	k_t
= [2, 3, 4, 2]				
Generation 850:	Best = 1.600433e+03	FES = 85100	Diversity = 1.1372e+00	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.600433e+03	FES = 86100	Diversity = 1.1316e+00	k_t
= [2, 4, 4, 3]				
Generation 870:	Best = 1.600432e+03	FES = 87100	Diversity = 1.1228e+00	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.600432e+03	FES = 88100	Diversity = 1.0983e+00	k_t
= [4, 2, 3, 3]				
Generation 890:	Best = 1.600432e+03	FES = 89100	Diversity = 1.1024e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.600432e+03	FES = 90100	Diversity = 1.0866e+00	k_t
= [4, 4, 3, 2]				
Generation 910:	Best = 1.600432e+03	FES = 91100	Diversity = 1.0847e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.600432e+03	FES = 92100	Diversity = 1.0861e+00	k_t
= [2, 2, 3, 4]				
Generation 930:	Best = 1.600432e+03	FES = 93100	Diversity = 1.0767e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.600432e+03	FES = 94100	Diversity = 1.0507e+00	k_t
= [2, 3, 4, 4]				
Generation 950:	Best = 1.600432e+03	FES = 95100	Diversity = 1.0006e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.600432e+03	FES = 96100	Diversity = 9.6118e-01	k_t
= [3, 3, 4, 2]				
Generation 970:	Best = 1.600432e+03	FES = 97100	Diversity = 9.5130e-01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.600432e+03	FES = 98100	Diversity = 9.3535e-01	k_t
= [4, 4, 2, 2]				
Generation 990:	Best = 1.600432e+03	FES = 99100	Diversity = 9.0562e-01	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 1.600430e+03

Total Function Evaluations = 100000

=====

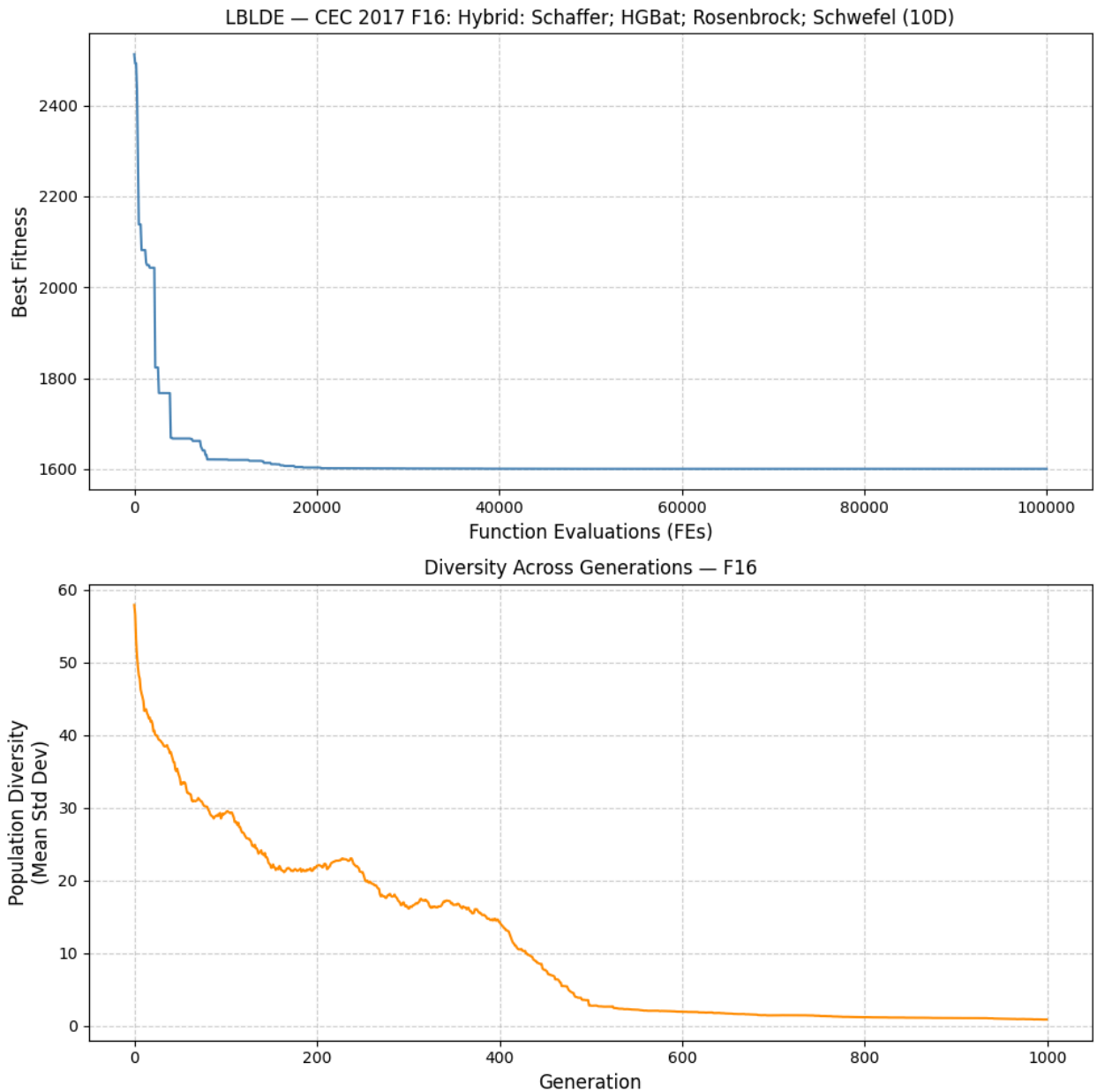
=

Results:

```

Best Fitness      : 1.600430e+03
Global Optimum    : 1.600000e+03
Error (|best-optimum|) : 4.301387e-01
Best Solution (5 dim) : [ 35.68204007  57.8864251  62.76964369 -67.31630948
-37.50693655]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F16.png

```

In [33]: # =====
# F17 — Hybrid Function 7 (Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Ra
# =====
run_and_plot('F17', make_wrapper(hybrid.f17), 1700.0, 'Hybrid: Katsuura; Ackle

```

```

=====
=
LBLDE_s on CEC 2017 F17 – Single Run
=====
=
Function      : Hybrid: Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastr
igin (10D)
Bounds       : [-100, 100]
Global optimum : 1700.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 2.471803e+03 | Diversity = 5.7928e+01
Generation 10: Best = 1.951175e+03 FES = 1100 Diversity = 4.3729e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.814513e+03 FES = 2100 Diversity = 4.0786e+01 k_t
= [4, 3, 2, 3]
Generation 30: Best = 1.791424e+03 FES = 3100 Diversity = 3.9140e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.785678e+03 FES = 4100 Diversity = 3.8602e+01 k_t
= [3, 4, 3, 2]
Generation 50: Best = 1.785622e+03 FES = 5100 Diversity = 3.8272e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.756723e+03 FES = 6100 Diversity = 3.7291e+01 k_t
= [2, 2, 4, 2]
Generation 70: Best = 1.756723e+03 FES = 7100 Diversity = 3.6120e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.746833e+03 FES = 8100 Diversity = 3.5649e+01 k_t
= [4, 3, 4, 2]
Generation 90: Best = 1.745887e+03 FES = 9100 Diversity = 3.4550e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.745887e+03 FES = 10100 Diversity = 3.3819e+01 k_t
= [4, 4, 4, 4]
Generation 110: Best = 1.739623e+03 FES = 11100 Diversity = 3.3044e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.735034e+03 FES = 12100 Diversity = 3.2644e+01 k_t
= [2, 3, 2, 3]
Generation 130: Best = 1.726691e+03 FES = 13100 Diversity = 3.2046e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.726691e+03 FES = 14100 Diversity = 3.1285e+01 k_t
= [2, 3, 2, 3]
Generation 150: Best = 1.726691e+03 FES = 15100 Diversity = 3.0686e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.726691e+03 FES = 16100 Diversity = 2.9019e+01 k_t
= [4, 4, 4, 2]
Generation 170: Best = 1.725747e+03 FES = 17100 Diversity = 2.8117e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.724630e+03 FES = 18100 Diversity = 2.6809e+01 k_t
= [3, 2, 3, 3]
Generation 190: Best = 1.721065e+03 FES = 19100 Diversity = 2.6129e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.721065e+03 FES = 20100 Diversity = 2.3908e+01 k_t
= [3, 4, 3, 3]
Generation 210: Best = 1.716851e+03 FES = 21100 Diversity = 2.2414e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.712953e+03 FES = 22100 Diversity = 2.1210e+01 k_t

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= [2, 4, 2, 3]
Generation 230: Best = 1.710861e+03 FES = 23100 Diversity = 1.9783e+01 k_t
= [1, 1, 1, 1]
Generation 240: Best = 1.710861e+03 FES = 24100 Diversity = 1.8143e+01 k_t
= [4, 4, 4, 3]
Generation 250: Best = 1.709174e+03 FES = 25100 Diversity = 1.7382e+01 k_t
= [1, 1, 1, 1]
Generation 260: Best = 1.708366e+03 FES = 26100 Diversity = 1.6064e+01 k_t
= [4, 3, 2, 3]
Generation 270: Best = 1.705640e+03 FES = 27100 Diversity = 1.4969e+01 k_t
= [1, 1, 1, 1]
Generation 280: Best = 1.704864e+03 FES = 28100 Diversity = 1.4513e+01 k_t
= [3, 4, 4, 3]
Generation 290: Best = 1.704864e+03 FES = 29100 Diversity = 1.3213e+01 k_t
= [1, 1, 1, 1]
Generation 300: Best = 1.704573e+03 FES = 30100 Diversity = 1.2963e+01 k_t
= [2, 4, 2, 4]
Generation 310: Best = 1.704573e+03 FES = 31100 Diversity = 1.2780e+01 k_t
= [1, 1, 1, 1]
Generation 320: Best = 1.703526e+03 FES = 32100 Diversity = 1.2421e+01 k_t
= [2, 3, 3, 4]
Generation 330: Best = 1.703513e+03 FES = 33100 Diversity = 1.2152e+01 k_t
= [1, 1, 1, 1]
Generation 340: Best = 1.703513e+03 FES = 34100 Diversity = 1.2404e+01 k_t
= [3, 4, 4, 2]
Generation 350: Best = 1.703069e+03 FES = 35100 Diversity = 1.2122e+01 k_t
= [1, 1, 1, 1]
Generation 360: Best = 1.703069e+03 FES = 36100 Diversity = 1.1750e+01 k_t
= [2, 4, 2, 3]
Generation 370: Best = 1.702208e+03 FES = 37100 Diversity = 1.1192e+01 k_t
= [1, 1, 1, 1]
Generation 380: Best = 1.701979e+03 FES = 38100 Diversity = 1.1174e+01 k_t
= [2, 3, 3, 4]
Generation 390: Best = 1.701978e+03 FES = 39100 Diversity = 1.1022e+01 k_t
= [1, 1, 1, 1]
Generation 400: Best = 1.701745e+03 FES = 40100 Diversity = 1.0866e+01 k_t
= [3, 2, 4, 2]
Generation 410: Best = 1.701745e+03 FES = 41100 Diversity = 1.0704e+01 k_t
= [1, 1, 1, 1]
Generation 420: Best = 1.701711e+03 FES = 42100 Diversity = 1.1094e+01 k_t
= [4, 3, 4, 2]
Generation 430: Best = 1.701614e+03 FES = 43100 Diversity = 1.1033e+01 k_t
= [1, 1, 1, 1]
Generation 440: Best = 1.701613e+03 FES = 44100 Diversity = 1.1003e+01 k_t
= [4, 4, 3, 4]
Generation 450: Best = 1.700687e+03 FES = 45100 Diversity = 1.1084e+01 k_t
= [1, 1, 1, 1]
Generation 460: Best = 1.700679e+03 FES = 46100 Diversity = 1.0950e+01 k_t
= [4, 4, 3, 4]
Generation 470: Best = 1.700679e+03 FES = 47100 Diversity = 1.1218e+01 k_t
= [1, 1, 1, 1]
Generation 480: Best = 1.700679e+03 FES = 48100 Diversity = 1.1255e+01 k_t
= [2, 3, 4, 2]
Generation 490: Best = 1.700679e+03 FES = 49100 Diversity = 1.1175e+01 k_t

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= [1, 1, 1, 1]
Generation 500: Best = 1.700582e+03 FES = 50100 Diversity = 1.0788e+01 k_t
= [2, 3, 4, 3]
Generation 510: Best = 1.700560e+03 FES = 51100 Diversity = 1.1033e+01 k_t
= [1, 1, 1, 1]
Generation 520: Best = 1.700560e+03 FES = 52100 Diversity = 1.0872e+01 k_t
= [4, 2, 2, 3]
Generation 530: Best = 1.700525e+03 FES = 53100 Diversity = 1.0546e+01 k_t
= [1, 1, 1, 1]
Generation 540: Best = 1.700524e+03 FES = 54100 Diversity = 1.0627e+01 k_t
= [3, 3, 4, 3]
Generation 550: Best = 1.700453e+03 FES = 55100 Diversity = 1.0168e+01 k_t
= [1, 1, 1, 1]
Generation 560: Best = 1.700453e+03 FES = 56100 Diversity = 1.0166e+01 k_t
= [3, 4, 4, 2]
Generation 570: Best = 1.700386e+03 FES = 57100 Diversity = 1.0179e+01 k_t
= [1, 1, 1, 1]
Generation 580: Best = 1.700386e+03 FES = 58100 Diversity = 1.0039e+01 k_t
= [4, 3, 4, 4]
Generation 590: Best = 1.700378e+03 FES = 59100 Diversity = 9.8034e+00 k_t
= [1, 1, 1, 1]
Generation 600: Best = 1.700377e+03 FES = 60100 Diversity = 9.2400e+00 k_t
= [4, 4, 3, 3]
Generation 610: Best = 1.700377e+03 FES = 61100 Diversity = 8.5928e+00 k_t
= [1, 1, 1, 1]
Generation 620: Best = 1.700366e+03 FES = 62100 Diversity = 8.5096e+00 k_t
= [4, 4, 4, 3]
Generation 630: Best = 1.700354e+03 FES = 63100 Diversity = 8.0184e+00 k_t
= [1, 1, 1, 1]
Generation 640: Best = 1.700346e+03 FES = 64100 Diversity = 7.8693e+00 k_t
= [4, 2, 3, 4]
Generation 650: Best = 1.700206e+03 FES = 65100 Diversity = 7.3668e+00 k_t
= [1, 1, 1, 1]
Generation 660: Best = 1.700140e+03 FES = 66100 Diversity = 7.2039e+00 k_t
= [3, 2, 4, 3]
Generation 670: Best = 1.700096e+03 FES = 67100 Diversity = 6.9623e+00 k_t
= [1, 1, 1, 1]
Generation 680: Best = 1.700086e+03 FES = 68100 Diversity = 6.9759e+00 k_t
= [4, 4, 4, 3]
Generation 690: Best = 1.700082e+03 FES = 69100 Diversity = 6.8784e+00 k_t
= [1, 1, 1, 1]
Generation 700: Best = 1.700039e+03 FES = 70100 Diversity = 6.8006e+00 k_t
= [3, 2, 4, 4]
Generation 710: Best = 1.700036e+03 FES = 71100 Diversity = 6.9666e+00 k_t
= [1, 1, 1, 1]
Generation 720: Best = 1.700018e+03 FES = 72100 Diversity = 7.0134e+00 k_t
= [3, 3, 4, 2]
Generation 730: Best = 1.700015e+03 FES = 73100 Diversity = 7.0338e+00 k_t
= [1, 1, 1, 1]
Generation 740: Best = 1.700013e+03 FES = 74100 Diversity = 6.6702e+00 k_t
= [4, 4, 3, 3]
Generation 750: Best = 1.700007e+03 FES = 75100 Diversity = 6.5779e+00 k_t
= [1, 1, 1, 1]
Generation 760: Best = 1.700006e+03 FES = 76100 Diversity = 6.2772e+00 k_t

```

```

= [4, 3, 2, 4]
Generation 770: Best = 1.700004e+03  FES = 77100  Diversity = 5.4097e+00  k_t
= [1, 1, 1, 1]
Generation 780: Best = 1.700002e+03  FES = 78100  Diversity = 5.1714e+00  k_t
= [4, 2, 3, 4]
Generation 790: Best = 1.700001e+03  FES = 79100  Diversity = 5.0915e+00  k_t
= [1, 1, 1, 1]
Generation 800: Best = 1.700000e+03  FES = 80100  Diversity = 5.0640e+00  k_t
= [4, 3, 3, 3]
Generation 810: Best = 1.700000e+03  FES = 81100  Diversity = 4.9839e+00  k_t
= [1, 1, 1, 1]
Generation 820: Best = 1.700000e+03  FES = 82100  Diversity = 5.0258e+00  k_t
= [2, 3, 4, 2]
Generation 830: Best = 1.700000e+03  FES = 83100  Diversity = 5.0574e+00  k_t
= [1, 1, 1, 1]
Generation 840: Best = 1.700000e+03  FES = 84100  Diversity = 5.3296e+00  k_t
= [2, 2, 4, 3]
Generation 850: Best = 1.700000e+03  FES = 85100  Diversity = 5.2797e+00  k_t
= [1, 1, 1, 1]
Generation 860: Best = 1.700000e+03  FES = 86100  Diversity = 5.2913e+00  k_t
= [4, 2, 3, 4]
Generation 870: Best = 1.700000e+03  FES = 87100  Diversity = 5.1235e+00  k_t
= [1, 1, 1, 1]
Generation 880: Best = 1.700000e+03  FES = 88100  Diversity = 5.4166e+00  k_t
= [3, 4, 3, 3]
Generation 890: Best = 1.700000e+03  FES = 89100  Diversity = 5.5374e+00  k_t
= [1, 1, 1, 1]
Generation 900: Best = 1.700000e+03  FES = 90100  Diversity = 5.4580e+00  k_t
= [2, 3, 3, 3]
Generation 910: Best = 1.700000e+03  FES = 91100  Diversity = 4.7947e+00  k_t
= [1, 1, 1, 1]
Generation 920: Best = 1.700000e+03  FES = 92100  Diversity = 4.7250e+00  k_t
= [3, 4, 3, 2]
Generation 930: Best = 1.700000e+03  FES = 93100  Diversity = 4.6381e+00  k_t
= [1, 1, 1, 1]
Generation 940: Best = 1.700000e+03  FES = 94100  Diversity = 5.1107e+00  k_t
= [2, 2, 2, 2]
Generation 950: Best = 1.700000e+03  FES = 95100  Diversity = 5.1387e+00  k_t
= [1, 1, 1, 1]
Generation 960: Best = 1.700000e+03  FES = 96100  Diversity = 5.2081e+00  k_t
= [2, 4, 3, 3]
Generation 970: Best = 1.700000e+03  FES = 97100  Diversity = 5.3452e+00  k_t
= [1, 1, 1, 1]
Generation 980: Best = 1.700000e+03  FES = 98100  Diversity = 5.1714e+00  k_t
= [4, 3, 4, 4]
Generation 990: Best = 1.700000e+03  FES = 99100  Diversity = 4.8612e+00  k_t
= [1, 1, 1, 1]

```

Optimization completed!

Final Best Fitness = 1.700000e+03

Total Function Evaluations = 100000

=====

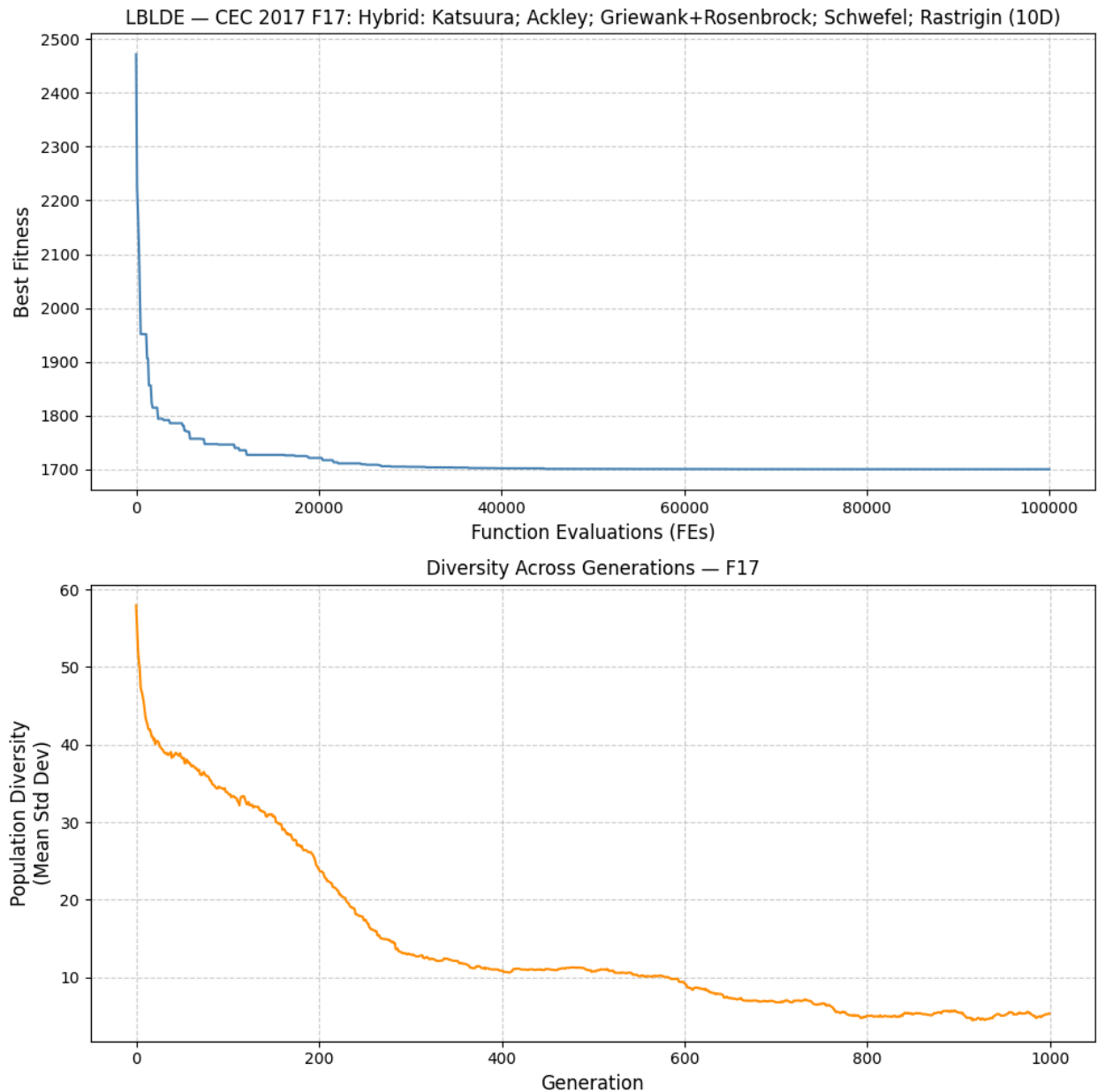
=

Results:

```
Best Fitness          : 1.700000e+03
Global Optimum        : 1.700000e+03
Error (|best-optimum|) : 2.545672e-05
Best Solution (5 dim)  : [-38.84138634 -7.1602438 -22.21764003 28.25058987
-53.05441624]
D^(0) (init. diversity): 5.7928e+01
```

=====

==



Graph saved as convergence_F17.png

```
In [34]: # =====
# F18 — Hybrid Function 8 (Elliptic; Ackley; Rastrigin; HGBat; Discus)
# =====
run_and_plot('F18', make_wrapper(hybrid.f18), 1800.0, 'Hybrid: Elliptic; Ackle
```



```

=====
=
LBLDE_s on CEC 2017 F18 – Single Run
=====
=
Function      : Hybrid: Elliptic; Ackley; Rastrigin; HGBat; Discus (10D)
Bounds       : [-100, 100]
Global optimum : 1800.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 6.135676e+08 | Diversity = 5.7928e+01
Generation 10: Best = 3.581854e+05 FES = 1100 Diversity = 4.6477e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.959011e+04 FES = 2100 Diversity = 4.5327e+01 k_t
= [4, 3, 2, 3]
Generation 30: Best = 1.491932e+04 FES = 3100 Diversity = 4.3970e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 4.065514e+03 FES = 4100 Diversity = 4.1559e+01 k_t
= [4, 3, 4, 3]
Generation 50: Best = 2.344200e+03 FES = 5100 Diversity = 4.0202e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.003112e+03 FES = 6100 Diversity = 3.6825e+01 k_t
= [3, 3, 2, 4]
Generation 70: Best = 1.889674e+03 FES = 7100 Diversity = 3.0032e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.849402e+03 FES = 8100 Diversity = 2.6209e+01 k_t
= [2, 4, 3, 2]
Generation 90: Best = 1.831818e+03 FES = 9100 Diversity = 2.2694e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.827375e+03 FES = 10100 Diversity = 2.1263e+01 k_t
= [3, 3, 2, 3]
Generation 110: Best = 1.825698e+03 FES = 11100 Diversity = 1.9112e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.825698e+03 FES = 12100 Diversity = 1.8621e+01 k_t
= [2, 3, 3, 2]
Generation 130: Best = 1.824084e+03 FES = 13100 Diversity = 1.7929e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.824084e+03 FES = 14100 Diversity = 1.7081e+01 k_t
= [3, 3, 4, 4]
Generation 150: Best = 1.822796e+03 FES = 15100 Diversity = 1.7012e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.822244e+03 FES = 16100 Diversity = 1.6043e+01 k_t
= [2, 3, 4, 2]
Generation 170: Best = 1.822040e+03 FES = 17100 Diversity = 1.5466e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.820152e+03 FES = 18100 Diversity = 1.5511e+01 k_t
= [3, 4, 2, 2]
Generation 190: Best = 1.812530e+03 FES = 19100 Diversity = 1.5378e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.812295e+03 FES = 20100 Diversity = 1.4347e+01 k_t
= [3, 2, 2, 4]
Generation 210: Best = 1.812295e+03 FES = 21100 Diversity = 1.3614e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.812107e+03 FES = 22100 Diversity = 1.3213e+01 k_t
= [3, 2, 3, 3]

```

Generation 230:	Best = 1.809017e+03	FES = 23100	Diversity = 1.1779e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 1.807523e+03	FES = 24100	Diversity = 1.0200e+01	k_t = [2, 3, 2, 3]
Generation 250:	Best = 1.805910e+03	FES = 25100	Diversity = 8.3307e+00	k_t = [1, 1, 1, 1]
Generation 260:	Best = 1.802762e+03	FES = 26100	Diversity = 6.7235e+00	k_t = [3, 4, 2, 4]
Generation 270:	Best = 1.802762e+03	FES = 27100	Diversity = 3.9606e+00	k_t = [1, 1, 1, 1]
Generation 280:	Best = 1.802068e+03	FES = 28100	Diversity = 3.2943e+00	k_t = [3, 3, 3, 4]
Generation 290:	Best = 1.801948e+03	FES = 29100	Diversity = 2.5508e+00	k_t = [1, 1, 1, 1]
Generation 300:	Best = 1.801834e+03	FES = 30100	Diversity = 2.3655e+00	k_t = [3, 2, 4, 2]
Generation 310:	Best = 1.801668e+03	FES = 31100	Diversity = 1.8729e+00	k_t = [1, 1, 1, 1]
Generation 320:	Best = 1.800935e+03	FES = 32100	Diversity = 1.1436e+00	k_t = [4, 2, 2, 2]
Generation 330:	Best = 1.800794e+03	FES = 33100	Diversity = 9.8939e-01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 1.800600e+03	FES = 34100	Diversity = 6.5136e-01	k_t = [3, 2, 4, 4]
Generation 350:	Best = 1.800585e+03	FES = 35100	Diversity = 6.8457e-01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 1.800523e+03	FES = 36100	Diversity = 7.2361e-01	k_t = [3, 3, 4, 4]
Generation 370:	Best = 1.800509e+03	FES = 37100	Diversity = 7.6303e-01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 1.800484e+03	FES = 38100	Diversity = 7.1691e-01	k_t = [2, 3, 3, 4]
Generation 390:	Best = 1.800428e+03	FES = 39100	Diversity = 7.7194e-01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 1.800428e+03	FES = 40100	Diversity = 8.6634e-01	k_t = [4, 4, 2, 4]
Generation 410:	Best = 1.800428e+03	FES = 41100	Diversity = 9.1908e-01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 1.800299e+03	FES = 42100	Diversity = 9.7679e-01	k_t = [2, 2, 4, 3]
Generation 430:	Best = 1.800234e+03	FES = 43100	Diversity = 1.0876e+00	k_t = [1, 1, 1, 1]
Generation 440:	Best = 1.800232e+03	FES = 44100	Diversity = 1.1601e+00	k_t = [2, 3, 3, 4]
Generation 450:	Best = 1.800053e+03	FES = 45100	Diversity = 1.3121e+00	k_t = [1, 1, 1, 1]
Generation 460:	Best = 1.800053e+03	FES = 46100	Diversity = 1.4338e+00	k_t = [3, 3, 2, 3]
Generation 470:	Best = 1.800053e+03	FES = 47100	Diversity = 1.4655e+00	k_t = [1, 1, 1, 1]
Generation 480:	Best = 1.800053e+03	FES = 48100	Diversity = 1.4907e+00	k_t = [4, 3, 2, 3]
Generation 490:	Best = 1.800053e+03	FES = 49100	Diversity = 1.4917e+00	k_t = [1, 1, 1, 1]

Generation 500:	Best = 1.800049e+03	FES = 50100	Diversity = 1.4728e+00	k_t = [4, 2, 4, 2]
Generation 510:	Best = 1.800046e+03	FES = 51100	Diversity = 1.4202e+00	k_t = [1, 1, 1, 1]
Generation 520:	Best = 1.800046e+03	FES = 52100	Diversity = 1.3957e+00	k_t = [3, 4, 4, 3]
Generation 530:	Best = 1.800046e+03	FES = 53100	Diversity = 1.3668e+00	k_t = [1, 1, 1, 1]
Generation 540:	Best = 1.800046e+03	FES = 54100	Diversity = 1.2603e+00	k_t = [3, 3, 2, 4]
Generation 550:	Best = 1.800046e+03	FES = 55100	Diversity = 1.1721e+00	k_t = [1, 1, 1, 1]
Generation 560:	Best = 1.800046e+03	FES = 56100	Diversity = 1.0653e+00	k_t = [4, 3, 4, 3]
Generation 570:	Best = 1.800027e+03	FES = 57100	Diversity = 8.4342e-01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 1.800027e+03	FES = 58100	Diversity = 6.7662e-01	k_t = [4, 3, 4, 4]
Generation 590:	Best = 1.800027e+03	FES = 59100	Diversity = 5.9999e-01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 1.800027e+03	FES = 60100	Diversity = 5.3959e-01	k_t = [3, 3, 4, 4]
Generation 610:	Best = 1.800026e+03	FES = 61100	Diversity = 3.8772e-01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 1.800026e+03	FES = 62100	Diversity = 3.4169e-01	k_t = [3, 3, 2, 3]
Generation 630:	Best = 1.800026e+03	FES = 63100	Diversity = 3.2860e-01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 1.800024e+03	FES = 64100	Diversity = 2.9551e-01	k_t = [2, 2, 3, 2]
Generation 650:	Best = 1.800022e+03	FES = 65100	Diversity = 2.9137e-01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 1.800021e+03	FES = 66100	Diversity = 2.6685e-01	k_t = [3, 4, 2, 3]
Generation 670:	Best = 1.800018e+03	FES = 67100	Diversity = 2.6679e-01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 1.800017e+03	FES = 68100	Diversity = 2.5772e-01	k_t = [4, 2, 3, 4]
Generation 690:	Best = 1.800017e+03	FES = 69100	Diversity = 2.4719e-01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 1.800017e+03	FES = 70100	Diversity = 2.3976e-01	k_t = [4, 4, 3, 2]
Generation 710:	Best = 1.800016e+03	FES = 71100	Diversity = 2.3851e-01	k_t = [1, 1, 1, 1]
Generation 720:	Best = 1.800015e+03	FES = 72100	Diversity = 2.2194e-01	k_t = [2, 3, 3, 4]
Generation 730:	Best = 1.800015e+03	FES = 73100	Diversity = 2.2301e-01	k_t = [1, 1, 1, 1]
Generation 740:	Best = 1.800014e+03	FES = 74100	Diversity = 2.2106e-01	k_t = [4, 2, 4, 4]
Generation 750:	Best = 1.800013e+03	FES = 75100	Diversity = 2.1475e-01	k_t = [1, 1, 1, 1]
Generation 760:	Best = 1.800012e+03	FES = 76100	Diversity = 2.0933e-01	k_t = [4, 4, 2, 4]

Generation 770:	Best = 1.800012e+03	FES = 77100	Diversity = 2.0014e-01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.800012e+03	FES = 78100	Diversity = 1.8540e-01	k_t
= [2, 2, 3, 2]				
Generation 790:	Best = 1.800012e+03	FES = 79100	Diversity = 1.7901e-01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.800012e+03	FES = 80100	Diversity = 1.6674e-01	k_t
= [2, 2, 3, 2]				
Generation 810:	Best = 1.800012e+03	FES = 81100	Diversity = 1.6978e-01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.800011e+03	FES = 82100	Diversity = 1.6740e-01	k_t
= [3, 4, 2, 3]				
Generation 830:	Best = 1.800010e+03	FES = 83100	Diversity = 1.4674e-01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.800010e+03	FES = 84100	Diversity = 1.4400e-01	k_t
= [4, 4, 3, 3]				
Generation 850:	Best = 1.800009e+03	FES = 85100	Diversity = 1.3767e-01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.800007e+03	FES = 86100	Diversity = 1.2757e-01	k_t
= [2, 3, 2, 4]				
Generation 870:	Best = 1.800006e+03	FES = 87100	Diversity = 1.1726e-01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.800006e+03	FES = 88100	Diversity = 1.1682e-01	k_t
= [3, 4, 4, 3]				
Generation 890:	Best = 1.800004e+03	FES = 89100	Diversity = 1.0118e-01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.800004e+03	FES = 90100	Diversity = 8.8297e-02	k_t
= [4, 2, 3, 2]				
Generation 910:	Best = 1.800004e+03	FES = 91100	Diversity = 7.3363e-02	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.800004e+03	FES = 92100	Diversity = 5.4439e-02	k_t
= [3, 3, 2, 2]				
Generation 930:	Best = 1.800004e+03	FES = 93100	Diversity = 4.2634e-02	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.800004e+03	FES = 94100	Diversity = 3.3458e-02	k_t
= [2, 2, 4, 4]				
Generation 950:	Best = 1.800004e+03	FES = 95100	Diversity = 2.9609e-02	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.800003e+03	FES = 96100	Diversity = 1.5959e-02	k_t
= [2, 2, 4, 2]				
Generation 970:	Best = 1.800003e+03	FES = 97100	Diversity = 1.4883e-02	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.800003e+03	FES = 98100	Diversity = 1.2201e-02	k_t
= [4, 3, 3, 4]				
Generation 990:	Best = 1.800003e+03	FES = 99100	Diversity = 1.1494e-02	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 1.800003e+03

Total Function Evaluations = 100000

=====

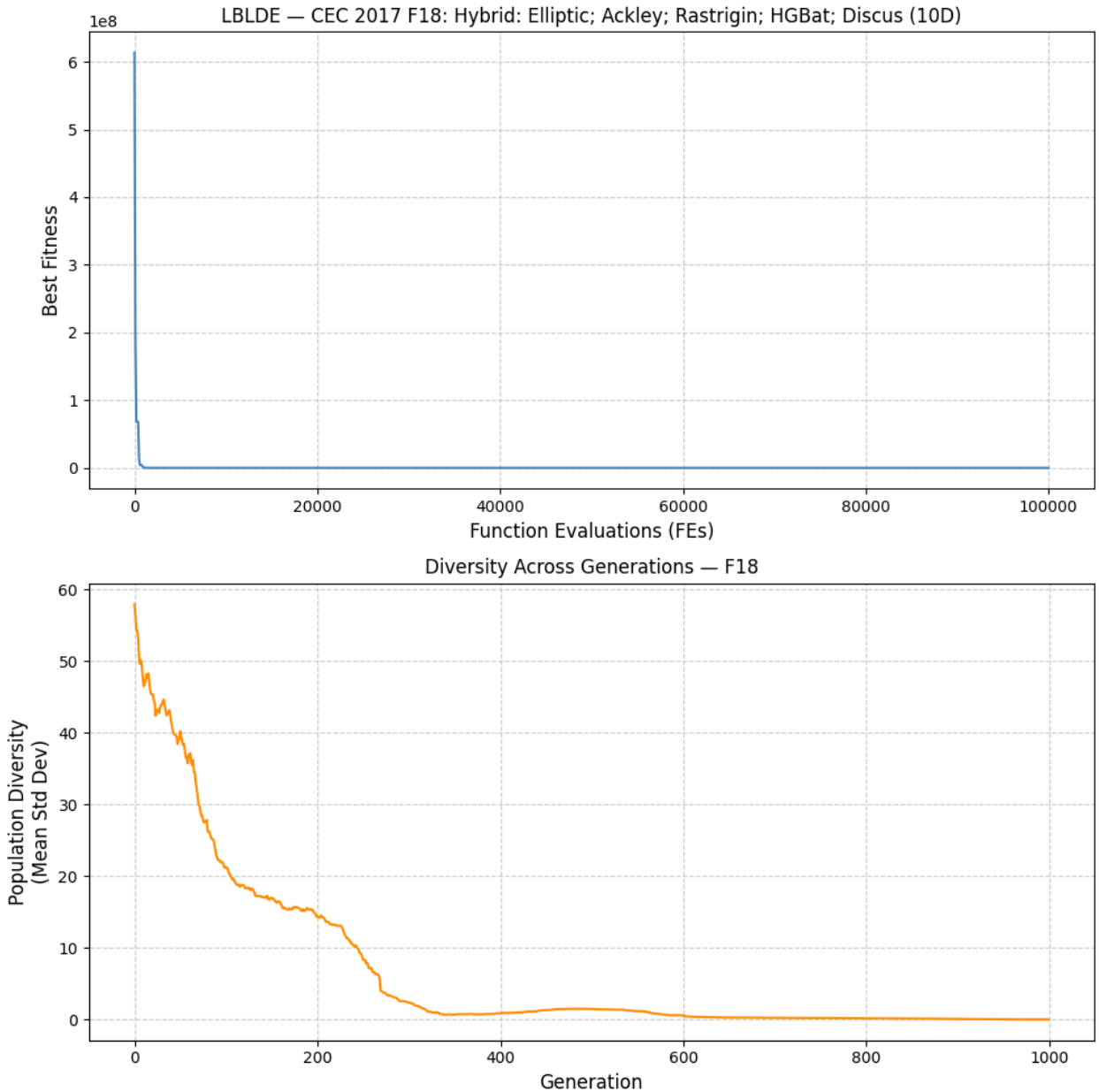
=

Results:

```

Best Fitness      : 1.800003e+03
Global Optimum    : 1.800000e+03
Error (|best-optimum|) : 2.647263e-03
Best Solution (5 dim) : [ 18.15619908   1.84792009  17.95103819 -48.08214574
-11.41760006]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F18.png

```

In [35]: # =====
# F19 – Hybrid Function 9 (Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierst
# =====
run_and_plot('F19', make_wrapper(hybrid.f19), 1900.0, 'Hybrid: Bent Cigar; Ras

```

```

=====
=
LBLDE_s on CEC 2017 F19 – Single Run
=====
=
Function      : Hybrid: Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstras
s; Schaffer (10D)
Bounds       : [-100, 100]
Global optimum : 1900.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 2.530389e+08 | Diversity = 5.7928e+01
Generation 10: Best = 1.707848e+04 FES = 1100 Diversity = 4.3769e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.144040e+03 FES = 2100 Diversity = 4.2135e+01 k_t
= [4, 3, 2, 3]
Generation 30: Best = 1.947849e+03 FES = 3100 Diversity = 3.4475e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.918177e+03 FES = 4100 Diversity = 3.0506e+01 k_t
= [4, 3, 4, 4]
Generation 50: Best = 1.912163e+03 FES = 5100 Diversity = 2.5216e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.909120e+03 FES = 6100 Diversity = 2.4515e+01 k_t
= [4, 4, 2, 3]
Generation 70: Best = 1.905809e+03 FES = 7100 Diversity = 2.3809e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.905725e+03 FES = 8100 Diversity = 2.3200e+01 k_t
= [2, 3, 2, 4]
Generation 90: Best = 1.904477e+03 FES = 9100 Diversity = 2.1641e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.904071e+03 FES = 10100 Diversity = 2.1450e+01 k_t
= [4, 2, 3, 2]
Generation 110: Best = 1.903574e+03 FES = 11100 Diversity = 2.2561e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.903308e+03 FES = 12100 Diversity = 2.2104e+01 k_t
= [4, 3, 3, 4]
Generation 130: Best = 1.902482e+03 FES = 13100 Diversity = 2.2208e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.902482e+03 FES = 14100 Diversity = 2.1946e+01 k_t
= [2, 2, 4, 2]
Generation 150: Best = 1.902126e+03 FES = 15100 Diversity = 2.0702e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.901658e+03 FES = 16100 Diversity = 2.0105e+01 k_t
= [2, 4, 3, 2]
Generation 170: Best = 1.901628e+03 FES = 17100 Diversity = 1.8994e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.901345e+03 FES = 18100 Diversity = 1.9340e+01 k_t
= [3, 2, 3, 2]
Generation 190: Best = 1.901296e+03 FES = 19100 Diversity = 1.8165e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.901105e+03 FES = 20100 Diversity = 1.6350e+01 k_t
= [3, 4, 4, 4]
Generation 210: Best = 1.900950e+03 FES = 21100 Diversity = 1.4189e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.900636e+03 FES = 22100 Diversity = 1.3242e+01 k_t

```

```

= [2, 4, 2, 4]
Generation 230: Best = 1.900636e+03 FES = 23100 Diversity = 1.2670e+01 k_t
= [1, 1, 1, 1]
Generation 240: Best = 1.900635e+03 FES = 24100 Diversity = 1.2350e+01 k_t
= [2, 4, 4, 2]
Generation 250: Best = 1.900506e+03 FES = 25100 Diversity = 1.2568e+01 k_t
= [1, 1, 1, 1]
Generation 260: Best = 1.900419e+03 FES = 26100 Diversity = 1.0736e+01 k_t
= [2, 3, 2, 4]
Generation 270: Best = 1.900210e+03 FES = 27100 Diversity = 1.0195e+01 k_t
= [1, 1, 1, 1]
Generation 280: Best = 1.900190e+03 FES = 28100 Diversity = 9.8196e+00 k_t
= [2, 3, 2, 3]
Generation 290: Best = 1.900185e+03 FES = 29100 Diversity = 8.0502e+00 k_t
= [1, 1, 1, 1]
Generation 300: Best = 1.900141e+03 FES = 30100 Diversity = 4.9645e+00 k_t
= [2, 4, 4, 3]
Generation 310: Best = 1.900124e+03 FES = 31100 Diversity = 2.1808e+00 k_t
= [1, 1, 1, 1]
Generation 320: Best = 1.900078e+03 FES = 32100 Diversity = 1.5465e+00 k_t
= [4, 2, 3, 2]
Generation 330: Best = 1.900053e+03 FES = 33100 Diversity = 1.2442e+00 k_t
= [1, 1, 1, 1]
Generation 340: Best = 1.900051e+03 FES = 34100 Diversity = 1.0424e+00 k_t
= [2, 4, 3, 4]
Generation 350: Best = 1.900029e+03 FES = 35100 Diversity = 9.2098e-01 k_t
= [1, 1, 1, 1]
Generation 360: Best = 1.900029e+03 FES = 36100 Diversity = 8.2194e-01 k_t
= [3, 3, 4, 3]
Generation 370: Best = 1.900025e+03 FES = 37100 Diversity = 6.9433e-01 k_t
= [1, 1, 1, 1]
Generation 380: Best = 1.900018e+03 FES = 38100 Diversity = 6.6505e-01 k_t
= [2, 2, 3, 3]
Generation 390: Best = 1.900016e+03 FES = 39100 Diversity = 4.8866e-01 k_t
= [1, 1, 1, 1]
Generation 400: Best = 1.900015e+03 FES = 40100 Diversity = 4.4353e-01 k_t
= [4, 4, 3, 4]
Generation 410: Best = 1.900014e+03 FES = 41100 Diversity = 3.8709e-01 k_t
= [1, 1, 1, 1]
Generation 420: Best = 1.900011e+03 FES = 42100 Diversity = 3.7719e-01 k_t
= [4, 4, 3, 4]
Generation 430: Best = 1.900011e+03 FES = 43100 Diversity = 3.4754e-01 k_t
= [1, 1, 1, 1]
Generation 440: Best = 1.900011e+03 FES = 44100 Diversity = 3.3001e-01 k_t
= [4, 4, 4, 4]
Generation 450: Best = 1.900011e+03 FES = 45100 Diversity = 3.2198e-01 k_t
= [1, 1, 1, 1]
Generation 460: Best = 1.900011e+03 FES = 46100 Diversity = 3.2239e-01 k_t
= [4, 2, 2, 4]
Generation 470: Best = 1.900010e+03 FES = 47100 Diversity = 3.1048e-01 k_t
= [1, 1, 1, 1]
Generation 480: Best = 1.900010e+03 FES = 48100 Diversity = 3.1633e-01 k_t
= [3, 2, 3, 4]
Generation 490: Best = 1.900010e+03 FES = 49100 Diversity = 3.1204e-01 k_t

```

```

= [1, 1, 1, 1]
Generation 500: Best = 1.900010e+03 FES = 50100 Diversity = 3.0457e-01 k_t
= [2, 4, 2, 4]
Generation 510: Best = 1.900010e+03 FES = 51100 Diversity = 3.0365e-01 k_t
= [1, 1, 1, 1]
Generation 520: Best = 1.900010e+03 FES = 52100 Diversity = 3.0650e-01 k_t
= [4, 3, 4, 4]
Generation 530: Best = 1.900010e+03 FES = 53100 Diversity = 3.0451e-01 k_t
= [1, 1, 1, 1]
Generation 540: Best = 1.900010e+03 FES = 54100 Diversity = 2.9534e-01 k_t
= [2, 2, 2, 4]
Generation 550: Best = 1.900010e+03 FES = 55100 Diversity = 2.9142e-01 k_t
= [1, 1, 1, 1]
Generation 560: Best = 1.900010e+03 FES = 56100 Diversity = 2.8564e-01 k_t
= [4, 2, 3, 2]
Generation 570: Best = 1.900010e+03 FES = 57100 Diversity = 2.7584e-01 k_t
= [1, 1, 1, 1]
Generation 580: Best = 1.900010e+03 FES = 58100 Diversity = 2.7361e-01 k_t
= [2, 4, 2, 4]
Generation 590: Best = 1.900010e+03 FES = 59100 Diversity = 2.7436e-01 k_t
= [1, 1, 1, 1]
Generation 600: Best = 1.900010e+03 FES = 60100 Diversity = 2.7038e-01 k_t
= [3, 2, 4, 4]
Generation 610: Best = 1.900010e+03 FES = 61100 Diversity = 2.6702e-01 k_t
= [1, 1, 1, 1]
Generation 620: Best = 1.900010e+03 FES = 62100 Diversity = 2.6590e-01 k_t
= [2, 4, 4, 2]
Generation 630: Best = 1.900010e+03 FES = 63100 Diversity = 2.6259e-01 k_t
= [1, 1, 1, 1]
Generation 640: Best = 1.900010e+03 FES = 64100 Diversity = 2.6077e-01 k_t
= [2, 4, 4, 2]
Generation 650: Best = 1.900010e+03 FES = 65100 Diversity = 2.6444e-01 k_t
= [1, 1, 1, 1]
Generation 660: Best = 1.900010e+03 FES = 66100 Diversity = 2.6563e-01 k_t
= [2, 4, 2, 2]
Generation 670: Best = 1.900010e+03 FES = 67100 Diversity = 2.6294e-01 k_t
= [1, 1, 1, 1]
Generation 680: Best = 1.900010e+03 FES = 68100 Diversity = 2.6511e-01 k_t
= [4, 4, 2, 3]
Generation 690: Best = 1.900010e+03 FES = 69100 Diversity = 2.5902e-01 k_t
= [1, 1, 1, 1]
Generation 700: Best = 1.900010e+03 FES = 70100 Diversity = 2.5688e-01 k_t
= [3, 2, 3, 4]
Generation 710: Best = 1.900010e+03 FES = 71100 Diversity = 2.4597e-01 k_t
= [1, 1, 1, 1]
Generation 720: Best = 1.900010e+03 FES = 72100 Diversity = 2.4241e-01 k_t
= [2, 3, 2, 3]
Generation 730: Best = 1.900010e+03 FES = 73100 Diversity = 2.1814e-01 k_t
= [1, 1, 1, 1]
Generation 740: Best = 1.900010e+03 FES = 74100 Diversity = 2.1230e-01 k_t
= [3, 4, 2, 3]
Generation 750: Best = 1.900010e+03 FES = 75100 Diversity = 2.0310e-01 k_t
= [1, 1, 1, 1]
Generation 760: Best = 1.900010e+03 FES = 76100 Diversity = 2.0161e-01 k_t

```



```

= [3, 4, 3, 3]
Generation 770: Best = 1.900010e+03 FES = 77100 Diversity = 1.9896e-01 k_t
= [1, 1, 1, 1]
Generation 780: Best = 1.900010e+03 FES = 78100 Diversity = 1.8847e-01 k_t
= [2, 3, 4, 4]
Generation 790: Best = 1.900010e+03 FES = 79100 Diversity = 1.6860e-01 k_t
= [1, 1, 1, 1]
Generation 800: Best = 1.900010e+03 FES = 80100 Diversity = 1.4337e-01 k_t
= [2, 2, 2, 2]
Generation 810: Best = 1.900010e+03 FES = 81100 Diversity = 9.8429e-02 k_t
= [1, 1, 1, 1]
Generation 820: Best = 1.900010e+03 FES = 82100 Diversity = 7.6982e-02 k_t
= [4, 4, 4, 3]
Generation 830: Best = 1.900010e+03 FES = 83100 Diversity = 4.3323e-02 k_t
= [1, 1, 1, 1]
Generation 840: Best = 1.900010e+03 FES = 84100 Diversity = 3.1257e-02 k_t
= [3, 2, 3, 2]
Generation 850: Best = 1.900010e+03 FES = 85100 Diversity = 2.3194e-02 k_t
= [1, 1, 1, 1]
Generation 860: Best = 1.900010e+03 FES = 86100 Diversity = 2.3062e-02 k_t
= [2, 2, 4, 3]
Generation 870: Best = 1.900010e+03 FES = 87100 Diversity = 2.2985e-02 k_t
= [1, 1, 1, 1]
Generation 880: Best = 1.900010e+03 FES = 88100 Diversity = 3.2919e-03 k_t
= [4, 2, 3, 3]
Generation 890: Best = 1.900010e+03 FES = 89100 Diversity = 2.8666e-03 k_t
= [1, 1, 1, 1]
Generation 900: Best = 1.900010e+03 FES = 90100 Diversity = 2.5432e-03 k_t
= [2, 2, 2, 2]
Generation 910: Best = 1.900010e+03 FES = 91100 Diversity = 2.1400e-03 k_t
= [1, 1, 1, 1]
Generation 920: Best = 1.900010e+03 FES = 92100 Diversity = 1.7588e-03 k_t
= [3, 2, 3, 2]
Generation 930: Best = 1.900010e+03 FES = 93100 Diversity = 1.5658e-03 k_t
= [1, 1, 1, 1]
Generation 940: Best = 1.900010e+03 FES = 94100 Diversity = 1.4072e-03 k_t
= [4, 2, 4, 2]
Generation 950: Best = 1.900010e+03 FES = 95100 Diversity = 1.2823e-03 k_t
= [1, 1, 1, 1]
Generation 960: Best = 1.900010e+03 FES = 96100 Diversity = 1.1403e-03 k_t
= [3, 4, 2, 3]
Generation 970: Best = 1.900010e+03 FES = 97100 Diversity = 1.0254e-03 k_t
= [1, 1, 1, 1]
Generation 980: Best = 1.900010e+03 FES = 98100 Diversity = 1.0043e-03 k_t
= [3, 2, 2, 3]
Generation 990: Best = 1.900010e+03 FES = 99100 Diversity = 9.2385e-04 k_t
= [1, 1, 1, 1]

```

Optimization completed!

Final Best Fitness = 1.900010e+03

Total Function Evaluations = 100000

=====

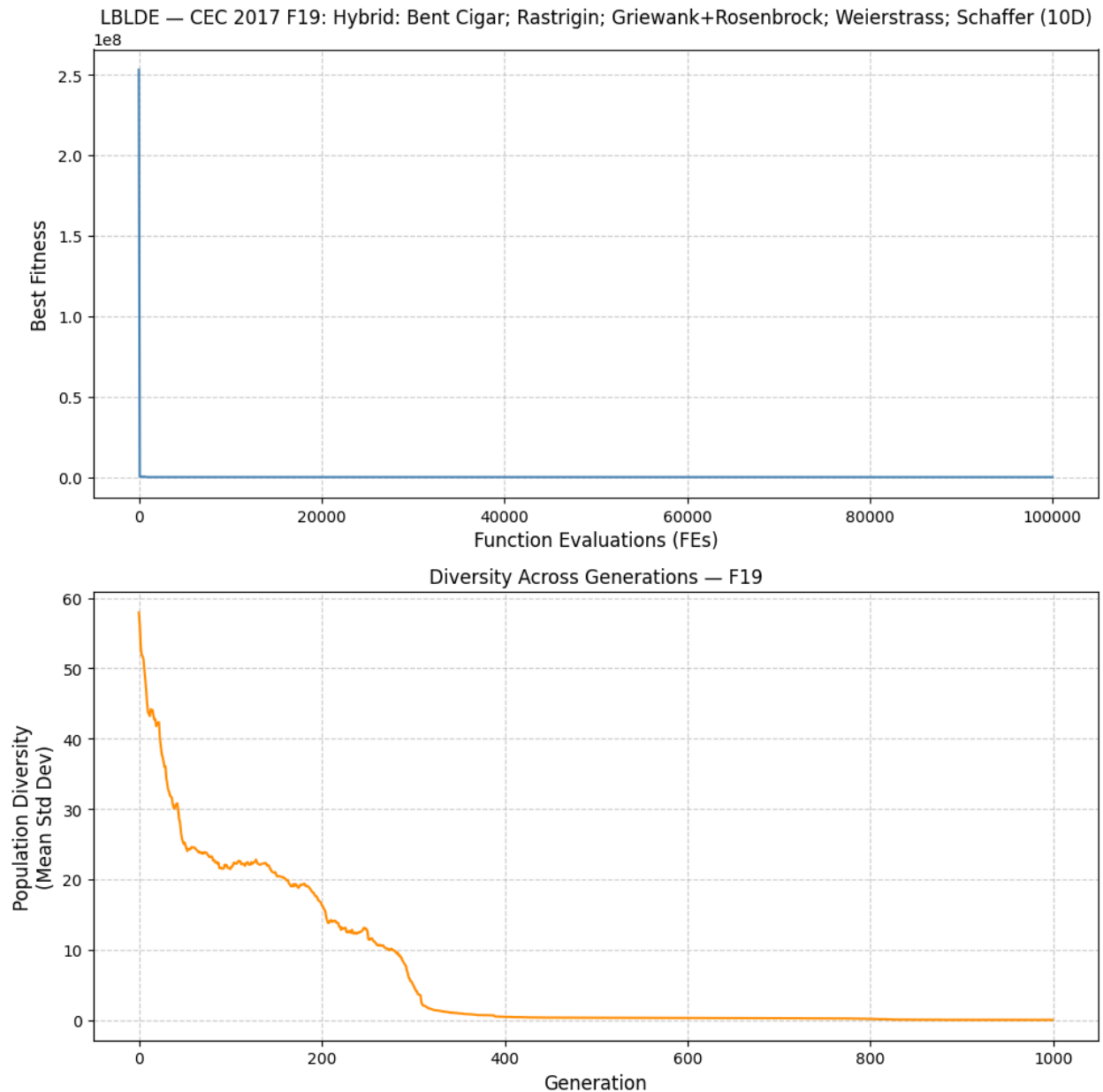
=

Results:

```
Best Fitness      : 1.900010e+03
Global Optimum    : 1.900000e+03
Error (|best-optimum|) : 9.715910e-03
Best Solution (5 dim) : [ 79.29118526   6.9474003  -56.25302276   35.23447592
24.21108062]
D^(0) (init. diversity): 5.7928e+01
```

=====

=



Graph saved as convergence_F19.png

```
In [36]: # =====
# F20 — Hybrid Function 10 (HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; S
# =====
run_and_plot('F20', make_wrapper(hybrid.f20), 2000.0, 'Hybrid: HappyCat; Katsu
```

```

=====
=
LBLDE_s on CEC 2017 F20 – Single Run
=====
=
Function      : Hybrid: HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaf
fer (10D)
Bounds       : [-100, 100]
Global optimum : 2000.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 2.391561e+03 | Diversity = 5.7928e+01
Generation 10: Best = 2.213162e+03 FES = 1100 Diversity = 5.0755e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.143433e+03 FES = 2100 Diversity = 4.8190e+01 k_t
= [4, 3, 2, 3]
Generation 30: Best = 2.079213e+03 FES = 3100 Diversity = 4.5220e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.071589e+03 FES = 4100 Diversity = 3.8425e+01 k_t
= [4, 2, 3, 2]
Generation 50: Best = 2.043684e+03 FES = 5100 Diversity = 3.5963e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.043684e+03 FES = 6100 Diversity = 3.2786e+01 k_t
= [2, 2, 2, 2]
Generation 70: Best = 2.039913e+03 FES = 7100 Diversity = 3.0682e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.035027e+03 FES = 8100 Diversity = 2.9625e+01 k_t
= [2, 3, 2, 4]
Generation 90: Best = 2.028167e+03 FES = 9100 Diversity = 2.8843e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.024130e+03 FES = 10100 Diversity = 2.8284e+01 k_t
= [2, 4, 3, 3]
Generation 110: Best = 2.023621e+03 FES = 11100 Diversity = 2.7260e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.023620e+03 FES = 12100 Diversity = 2.6143e+01 k_t
= [4, 2, 2, 3]
Generation 130: Best = 2.023620e+03 FES = 13100 Diversity = 2.6211e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.017482e+03 FES = 14100 Diversity = 2.5468e+01 k_t
= [2, 2, 4, 3]
Generation 150: Best = 2.013960e+03 FES = 15100 Diversity = 2.4202e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.013717e+03 FES = 16100 Diversity = 2.3359e+01 k_t
= [4, 3, 3, 2]
Generation 170: Best = 2.008824e+03 FES = 17100 Diversity = 2.2418e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.006979e+03 FES = 18100 Diversity = 2.0799e+01 k_t
= [3, 2, 3, 2]
Generation 190: Best = 2.005629e+03 FES = 19100 Diversity = 1.6996e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.004814e+03 FES = 20100 Diversity = 1.2665e+01 k_t
= [3, 4, 3, 2]
Generation 210: Best = 2.003421e+03 FES = 21100 Diversity = 1.0817e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.002205e+03 FES = 22100 Diversity = 9.3485e+00 k_t

```

```

= [4, 2, 4, 4]
Generation 230: Best = 2.001509e+03 FES = 23100 Diversity = 8.4660e+00 k_t
= [1, 1, 1, 1]
Generation 240: Best = 2.001018e+03 FES = 24100 Diversity = 7.7296e+00 k_t
= [2, 3, 4, 3]
Generation 250: Best = 2.000810e+03 FES = 25100 Diversity = 6.6451e+00 k_t
= [1, 1, 1, 1]
Generation 260: Best = 2.000569e+03 FES = 26100 Diversity = 5.5970e+00 k_t
= [3, 3, 2, 2]
Generation 270: Best = 2.000567e+03 FES = 27100 Diversity = 5.0113e+00 k_t
= [1, 1, 1, 1]
Generation 280: Best = 2.000282e+03 FES = 28100 Diversity = 5.2347e+00 k_t
= [3, 3, 2, 3]
Generation 290: Best = 2.000205e+03 FES = 29100 Diversity = 5.5349e+00 k_t
= [1, 1, 1, 1]
Generation 300: Best = 2.000173e+03 FES = 30100 Diversity = 5.3954e+00 k_t
= [4, 2, 3, 2]
Generation 310: Best = 2.000152e+03 FES = 31100 Diversity = 5.3616e+00 k_t
= [1, 1, 1, 1]
Generation 320: Best = 2.000089e+03 FES = 32100 Diversity = 5.6414e+00 k_t
= [2, 4, 3, 3]
Generation 330: Best = 2.000086e+03 FES = 33100 Diversity = 5.4545e+00 k_t
= [1, 1, 1, 1]
Generation 340: Best = 2.000052e+03 FES = 34100 Diversity = 5.4207e+00 k_t
= [3, 2, 2, 4]
Generation 350: Best = 2.000043e+03 FES = 35100 Diversity = 5.4114e+00 k_t
= [1, 1, 1, 1]
Generation 360: Best = 2.000027e+03 FES = 36100 Diversity = 5.4804e+00 k_t
= [3, 4, 3, 2]
Generation 370: Best = 2.000022e+03 FES = 37100 Diversity = 5.4033e+00 k_t
= [1, 1, 1, 1]
Generation 380: Best = 2.000012e+03 FES = 38100 Diversity = 5.5428e+00 k_t
= [3, 3, 4, 3]
Generation 390: Best = 2.000010e+03 FES = 39100 Diversity = 5.1904e+00 k_t
= [1, 1, 1, 1]
Generation 400: Best = 2.000006e+03 FES = 40100 Diversity = 5.3919e+00 k_t
= [3, 4, 4, 4]
Generation 410: Best = 2.000005e+03 FES = 41100 Diversity = 5.4821e+00 k_t
= [1, 1, 1, 1]
Generation 420: Best = 2.000003e+03 FES = 42100 Diversity = 5.1996e+00 k_t
= [2, 4, 2, 2]
Generation 430: Best = 2.000002e+03 FES = 43100 Diversity = 5.2394e+00 k_t
= [1, 1, 1, 1]
Generation 440: Best = 2.000002e+03 FES = 44100 Diversity = 5.3611e+00 k_t
= [2, 3, 3, 3]
Generation 450: Best = 2.000001e+03 FES = 45100 Diversity = 5.2987e+00 k_t
= [1, 1, 1, 1]
Generation 460: Best = 2.000001e+03 FES = 46100 Diversity = 5.4118e+00 k_t
= [3, 4, 4, 3]
Generation 470: Best = 2.000000e+03 FES = 47100 Diversity = 5.1015e+00 k_t
= [1, 1, 1, 1]
Generation 480: Best = 2.000000e+03 FES = 48100 Diversity = 4.8682e+00 k_t
= [3, 3, 2, 4]
Generation 490: Best = 2.000000e+03 FES = 49100 Diversity = 5.2415e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 500: Best = 2.000000e+03 FES = 50100 Diversity = 5.1382e+00 k_t
= [4, 4, 3, 3]
Generation 510: Best = 2.000000e+03 FES = 51100 Diversity = 4.7993e+00 k_t
= [1, 1, 1, 1]
Generation 520: Best = 2.000000e+03 FES = 52100 Diversity = 4.9694e+00 k_t
= [4, 4, 2, 4]
Generation 530: Best = 2.000000e+03 FES = 53100 Diversity = 4.9326e+00 k_t
= [1, 1, 1, 1]
Generation 540: Best = 2.000000e+03 FES = 54100 Diversity = 4.8371e+00 k_t
= [2, 4, 3, 2]
Generation 550: Best = 2.000000e+03 FES = 55100 Diversity = 4.4803e+00 k_t
= [1, 1, 1, 1]
Generation 560: Best = 2.000000e+03 FES = 56100 Diversity = 4.1646e+00 k_t
= [3, 4, 4, 2]
Generation 570: Best = 2.000000e+03 FES = 57100 Diversity = 4.7310e+00 k_t
= [1, 1, 1, 1]
Generation 580: Best = 2.000000e+03 FES = 58100 Diversity = 5.2385e+00 k_t
= [3, 2, 4, 4]
Generation 590: Best = 2.000000e+03 FES = 59100 Diversity = 5.2774e+00 k_t
= [1, 1, 1, 1]
Generation 600: Best = 2.000000e+03 FES = 60100 Diversity = 5.7361e+00 k_t
= [3, 3, 3, 2]
Generation 610: Best = 2.000000e+03 FES = 61100 Diversity = 5.7172e+00 k_t
= [1, 1, 1, 1]
Generation 620: Best = 2.000000e+03 FES = 62100 Diversity = 5.9271e+00 k_t
= [3, 4, 4, 4]
Generation 630: Best = 2.000000e+03 FES = 63100 Diversity = 5.2914e+00 k_t
= [1, 1, 1, 1]
Generation 640: Best = 2.000000e+03 FES = 64100 Diversity = 4.9794e+00 k_t
= [3, 4, 3, 4]
Generation 650: Best = 2.000000e+03 FES = 65100 Diversity = 4.7083e+00 k_t
= [1, 1, 1, 1]
Generation 660: Best = 2.000000e+03 FES = 66100 Diversity = 4.7300e+00 k_t
= [2, 4, 3, 3]
Generation 670: Best = 2.000000e+03 FES = 67100 Diversity = 4.1626e+00 k_t
= [1, 1, 1, 1]
Generation 680: Best = 2.000000e+03 FES = 68100 Diversity = 4.6548e+00 k_t
= [4, 3, 2, 4]
Generation 690: Best = 2.000000e+03 FES = 69100 Diversity = 4.6979e+00 k_t
= [1, 1, 1, 1]
Generation 700: Best = 2.000000e+03 FES = 70100 Diversity = 4.4605e+00 k_t
= [4, 2, 2, 4]
Generation 710: Best = 2.000000e+03 FES = 71100 Diversity = 4.7789e+00 k_t
= [1, 1, 1, 1]
Generation 720: Best = 2.000000e+03 FES = 72100 Diversity = 5.2257e+00 k_t
= [4, 4, 2, 3]
Generation 730: Best = 2.000000e+03 FES = 73100 Diversity = 5.4275e+00 k_t
= [1, 1, 1, 1]
Generation 740: Best = 2.000000e+03 FES = 74100 Diversity = 5.4953e+00 k_t
= [2, 3, 2, 3]
Generation 750: Best = 2.000000e+03 FES = 75100 Diversity = 5.4560e+00 k_t
= [1, 1, 1, 1]
Generation 760: Best = 2.000000e+03 FES = 76100 Diversity = 5.2980e+00 k_t

```

```

= [2, 4, 4, 2]
Generation 770: Best = 2.000000e+03 FES = 77100 Diversity = 5.2360e+00 k_t
= [1, 1, 1, 1]
Generation 780: Best = 2.000000e+03 FES = 78100 Diversity = 5.4800e+00 k_t
= [2, 3, 3, 2]
Generation 790: Best = 2.000000e+03 FES = 79100 Diversity = 5.2278e+00 k_t
= [1, 1, 1, 1]
Generation 800: Best = 2.000000e+03 FES = 80100 Diversity = 5.4645e+00 k_t
= [3, 3, 3, 2]
Generation 810: Best = 2.000000e+03 FES = 81100 Diversity = 5.6707e+00 k_t
= [1, 1, 1, 1]
Generation 820: Best = 2.000000e+03 FES = 82100 Diversity = 5.5453e+00 k_t
= [4, 2, 4, 2]
Generation 830: Best = 2.000000e+03 FES = 83100 Diversity = 5.4605e+00 k_t
= [1, 1, 1, 1]
Generation 840: Best = 2.000000e+03 FES = 84100 Diversity = 5.4172e+00 k_t
= [3, 2, 3, 4]
Generation 850: Best = 2.000000e+03 FES = 85100 Diversity = 5.2669e+00 k_t
= [1, 1, 1, 1]
Generation 860: Best = 2.000000e+03 FES = 86100 Diversity = 5.1962e+00 k_t
= [4, 2, 2, 2]
Generation 870: Best = 2.000000e+03 FES = 87100 Diversity = 5.1312e+00 k_t
= [1, 1, 1, 1]
Generation 880: Best = 2.000000e+03 FES = 88100 Diversity = 5.1832e+00 k_t
= [2, 4, 4, 3]
Generation 890: Best = 2.000000e+03 FES = 89100 Diversity = 5.5783e+00 k_t
= [1, 1, 1, 1]
Generation 900: Best = 2.000000e+03 FES = 90100 Diversity = 5.5610e+00 k_t
= [3, 2, 4, 4]
Generation 910: Best = 2.000000e+03 FES = 91100 Diversity = 5.6551e+00 k_t
= [1, 1, 1, 1]
Generation 920: Best = 2.000000e+03 FES = 92100 Diversity = 5.7830e+00 k_t
= [3, 3, 4, 4]
Generation 930: Best = 2.000000e+03 FES = 93100 Diversity = 6.0476e+00 k_t
= [1, 1, 1, 1]
Generation 940: Best = 2.000000e+03 FES = 94100 Diversity = 5.8731e+00 k_t
= [4, 2, 3, 3]
Generation 950: Best = 2.000000e+03 FES = 95100 Diversity = 5.9355e+00 k_t
= [1, 1, 1, 1]
Generation 960: Best = 2.000000e+03 FES = 96100 Diversity = 5.7345e+00 k_t
= [2, 4, 3, 3]
Generation 970: Best = 2.000000e+03 FES = 97100 Diversity = 5.5719e+00 k_t
= [1, 1, 1, 1]
Generation 980: Best = 2.000000e+03 FES = 98100 Diversity = 5.3732e+00 k_t
= [3, 3, 2, 3]
Generation 990: Best = 2.000000e+03 FES = 99100 Diversity = 5.5174e+00 k_t
= [1, 1, 1, 1]

```

Optimization completed!

Final Best Fitness = 2.000000e+03

Total Function Evaluations = 100000

=====

=

Results:

Best Fitness : 2.000000e+03

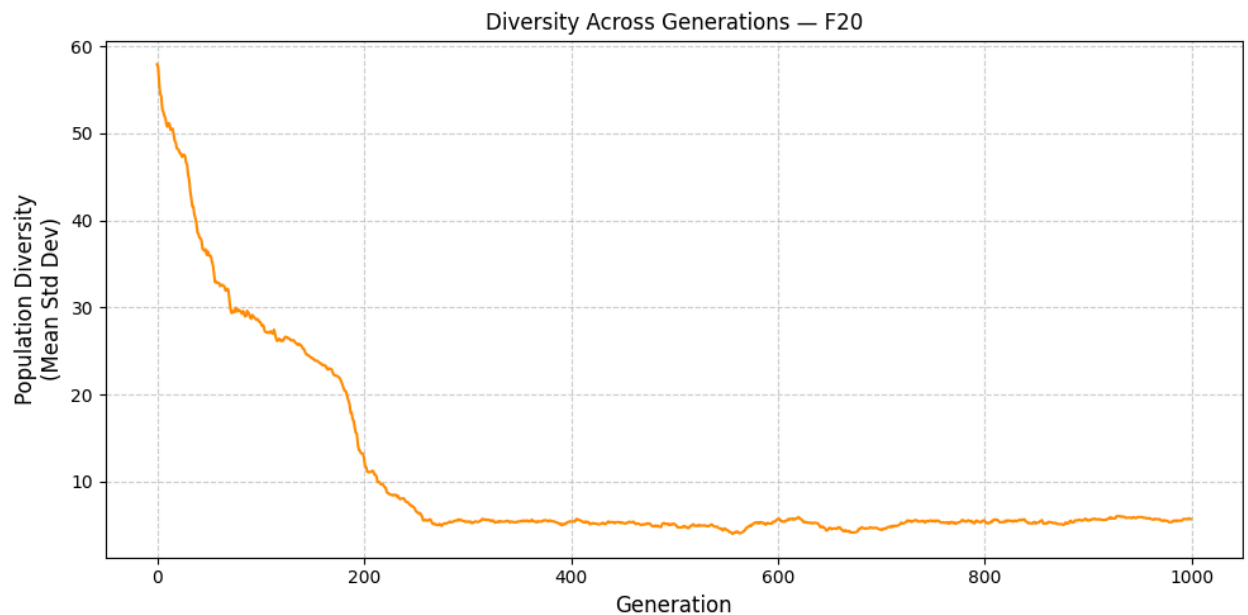
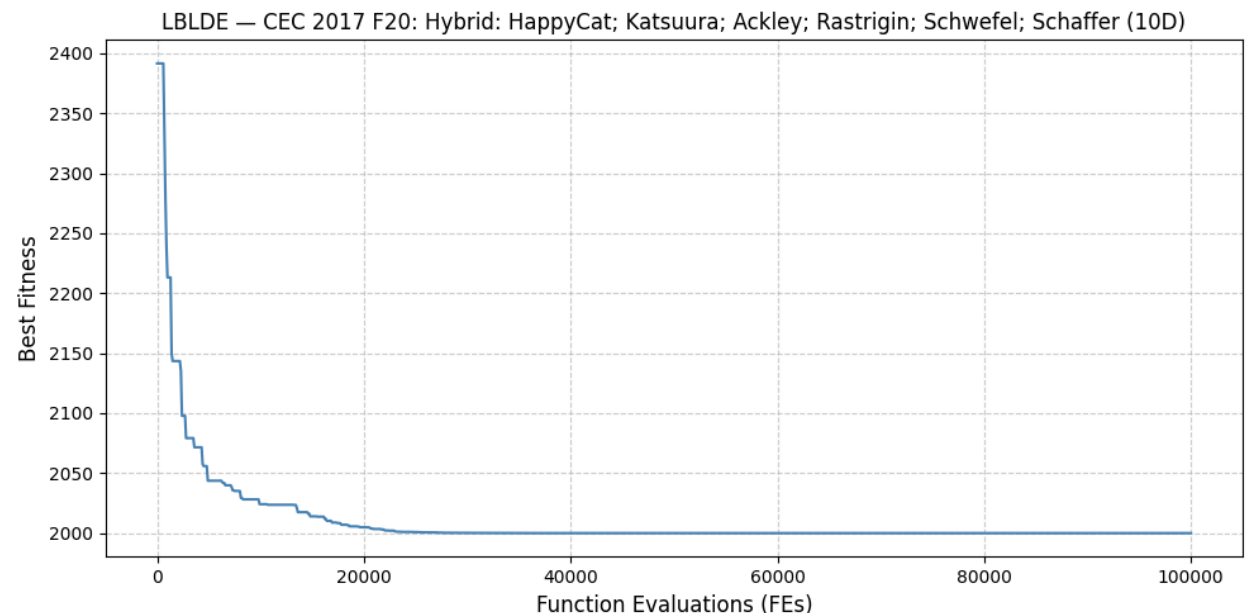
Global Optimum : 2.000000e+03

Error (|best-optimum|) : 2.545513e-05

Best Solution (5 dim) : [-30.59057464 -55.78803921 -29.57451967 61.04539565
69.96466593]

$D^*(0)$ (init. diversity): 5.7928e+01

=====



Graph saved as convergence_F20.png

6.4 Composite Functions (F21-F30)

In [37]: # =====

```
# F21 – Composition Function 1 (Rosenbrock; Elliptic; Rastrigin)
# =====
run_and_plot('F21', make_wrapper(composition.f21), 2100.0, 'Composition: Rosenbrock; Elliptic; Rastrigin')
```



```

=====
=
LBLDE_s on CEC 2017 F21 – Single Run
=====
=
Function      : Composition: Rosenbrock; Elliptic; Rastrigin (10D)
Bounds       : [-100, 100]
Global optimum : 2100.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 2.446690e+03 | Diversity = 5.7928e+01
Generation 10: Best = 2.380196e+03 FES = 1100 Diversity = 3.6841e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.343862e+03 FES = 2100 Diversity = 2.9371e+01 k_t
= [4, 3, 2, 3]
Generation 30: Best = 2.343862e+03 FES = 3100 Diversity = 2.5738e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.342142e+03 FES = 4100 Diversity = 2.4591e+01 k_t
= [4, 3, 4, 3]
Generation 50: Best = 2.339877e+03 FES = 5100 Diversity = 2.3005e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.326885e+03 FES = 6100 Diversity = 2.3038e+01 k_t
= [2, 4, 4, 2]
Generation 70: Best = 2.326885e+03 FES = 7100 Diversity = 2.2884e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.326885e+03 FES = 8100 Diversity = 2.2494e+01 k_t
= [2, 4, 3, 3]
Generation 90: Best = 2.326885e+03 FES = 9100 Diversity = 2.2595e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.322681e+03 FES = 10100 Diversity = 2.2357e+01 k_t
= [3, 4, 3, 4]
Generation 110: Best = 2.322681e+03 FES = 11100 Diversity = 2.2255e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.322681e+03 FES = 12100 Diversity = 2.2259e+01 k_t
= [2, 2, 4, 4]
Generation 130: Best = 2.322681e+03 FES = 13100 Diversity = 2.2434e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.322681e+03 FES = 14100 Diversity = 2.2700e+01 k_t
= [3, 4, 4, 4]
Generation 150: Best = 2.319747e+03 FES = 15100 Diversity = 2.2574e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.319747e+03 FES = 16100 Diversity = 2.2841e+01 k_t
= [2, 2, 4, 2]
Generation 170: Best = 2.319747e+03 FES = 17100 Diversity = 2.2790e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.319340e+03 FES = 18100 Diversity = 2.2363e+01 k_t
= [2, 2, 2, 3]
Generation 190: Best = 2.319340e+03 FES = 19100 Diversity = 2.2282e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.319340e+03 FES = 20100 Diversity = 2.2294e+01 k_t
= [3, 3, 4, 4]
Generation 210: Best = 2.317911e+03 FES = 21100 Diversity = 2.2033e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.317911e+03 FES = 22100 Diversity = 2.1688e+01 k_t
= [2, 2, 2, 2]

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Generation 230:	Best = 2.317911e+03	FES = 23100	Diversity = 2.1558e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 2.317911e+03	FES = 24100	Diversity = 2.1458e+01	k_t = [3, 2, 4, 4]
Generation 250:	Best = 2.317911e+03	FES = 25100	Diversity = 2.0984e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 2.317911e+03	FES = 26100	Diversity = 2.0982e+01	k_t = [2, 3, 4, 3]
Generation 270:	Best = 2.317911e+03	FES = 27100	Diversity = 2.0762e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 2.314393e+03	FES = 28100	Diversity = 2.0745e+01	k_t = [4, 4, 3, 4]
Generation 290:	Best = 2.313359e+03	FES = 29100	Diversity = 2.0599e+01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 2.313220e+03	FES = 30100	Diversity = 2.0260e+01	k_t = [3, 3, 4, 2]
Generation 310:	Best = 2.312294e+03	FES = 31100	Diversity = 1.9555e+01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 2.309092e+03	FES = 32100	Diversity = 1.9050e+01	k_t = [2, 3, 4, 3]
Generation 330:	Best = 2.309059e+03	FES = 33100	Diversity = 1.7979e+01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 2.308277e+03	FES = 34100	Diversity = 1.7250e+01	k_t = [3, 4, 3, 3]
Generation 350:	Best = 2.307693e+03	FES = 35100	Diversity = 1.6632e+01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 2.307693e+03	FES = 36100	Diversity = 1.6524e+01	k_t = [4, 2, 3, 3]
Generation 370:	Best = 2.306332e+03	FES = 37100	Diversity = 1.5973e+01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 2.306332e+03	FES = 38100	Diversity = 1.5057e+01	k_t = [2, 4, 2, 4]
Generation 390:	Best = 2.306332e+03	FES = 39100	Diversity = 1.5102e+01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 2.305471e+03	FES = 40100	Diversity = 1.4542e+01	k_t = [4, 2, 2, 2]
Generation 410:	Best = 2.305273e+03	FES = 41100	Diversity = 1.4026e+01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 2.305058e+03	FES = 42100	Diversity = 1.2340e+01	k_t = [2, 4, 2, 2]
Generation 430:	Best = 2.304928e+03	FES = 43100	Diversity = 5.5735e+00	k_t = [1, 1, 1, 1]
Generation 440:	Best = 2.304732e+03	FES = 44100	Diversity = 2.0564e-01	k_t = [4, 2, 4, 3]
Generation 450:	Best = 2.304714e+03	FES = 45100	Diversity = 9.3729e-02	k_t = [1, 1, 1, 1]
Generation 460:	Best = 2.304702e+03	FES = 46100	Diversity = 4.6151e-02	k_t = [3, 4, 4, 3]
Generation 470:	Best = 2.304701e+03	FES = 47100	Diversity = 2.1604e-02	k_t = [1, 1, 1, 1]
Generation 480:	Best = 2.304700e+03	FES = 48100	Diversity = 1.1094e-02	k_t = [2, 3, 4, 4]
Generation 490:	Best = 2.304700e+03	FES = 49100	Diversity = 5.4569e-03	k_t = [1, 1, 1, 1]

Generation 500:	Best = 2.304700e+03	FES = 50100	Diversity = 2.3866e-03	k_t = [4, 2, 4, 4]
Generation 510:	Best = 2.304700e+03	FES = 51100	Diversity = 1.0302e-03	k_t = [1, 1, 1, 1]
Generation 520:	Best = 2.304700e+03	FES = 52100	Diversity = 4.8827e-04	k_t = [2, 3, 2, 3]
Generation 530:	Best = 2.304700e+03	FES = 53100	Diversity = 2.0632e-04	k_t = [1, 1, 1, 1]
Generation 540:	Best = 2.304700e+03	FES = 54100	Diversity = 9.0987e-05	k_t = [3, 2, 2, 2]
Generation 550:	Best = 2.304700e+03	FES = 55100	Diversity = 3.8935e-05	k_t = [1, 1, 1, 1]
Generation 560:	Best = 2.304700e+03	FES = 56100	Diversity = 1.7655e-05	k_t = [3, 2, 3, 2]
Generation 570:	Best = 2.304700e+03	FES = 57100	Diversity = 8.0601e-06	k_t = [1, 1, 1, 1]
Generation 580:	Best = 2.304700e+03	FES = 58100	Diversity = 3.5159e-06	k_t = [2, 2, 4, 4]
Generation 590:	Best = 2.304700e+03	FES = 59100	Diversity = 1.4541e-06	k_t = [1, 1, 1, 1]
Generation 600:	Best = 2.304700e+03	FES = 60100	Diversity = 6.6877e-07	k_t = [3, 4, 4, 3]
Generation 610:	Best = 2.304700e+03	FES = 61100	Diversity = 3.7540e-07	k_t = [1, 1, 1, 1]
Generation 620:	Best = 2.304700e+03	FES = 62100	Diversity = 2.6897e-07	k_t = [2, 4, 3, 2]
Generation 630:	Best = 2.304700e+03	FES = 63100	Diversity = 1.9104e-07	k_t = [1, 1, 1, 1]
Generation 640:	Best = 2.304700e+03	FES = 64100	Diversity = 1.7752e-07	k_t = [4, 4, 4, 4]
Generation 650:	Best = 2.304700e+03	FES = 65100	Diversity = 1.8186e-07	k_t = [1, 1, 1, 1]
Generation 660:	Best = 2.304700e+03	FES = 66100	Diversity = 1.7832e-07	k_t = [3, 3, 3, 4]
Generation 670:	Best = 2.304700e+03	FES = 67100	Diversity = 1.7680e-07	k_t = [1, 1, 1, 1]
Generation 680:	Best = 2.304700e+03	FES = 68100	Diversity = 1.7809e-07	k_t = [3, 4, 4, 3]
Generation 690:	Best = 2.304700e+03	FES = 69100	Diversity = 1.8488e-07	k_t = [1, 1, 1, 1]
Generation 700:	Best = 2.304700e+03	FES = 70100	Diversity = 1.7748e-07	k_t = [2, 4, 3, 3]
Generation 710:	Best = 2.304700e+03	FES = 71100	Diversity = 1.8692e-07	k_t = [1, 1, 1, 1]
Generation 720:	Best = 2.304700e+03	FES = 72100	Diversity = 1.7886e-07	k_t = [2, 2, 3, 3]
Generation 730:	Best = 2.304700e+03	FES = 73100	Diversity = 1.8822e-07	k_t = [1, 1, 1, 1]
Generation 740:	Best = 2.304700e+03	FES = 74100	Diversity = 1.8435e-07	k_t = [3, 2, 4, 2]
Generation 750:	Best = 2.304700e+03	FES = 75100	Diversity = 1.7469e-07	k_t = [1, 1, 1, 1]
Generation 760:	Best = 2.304700e+03	FES = 76100	Diversity = 1.7404e-07	k_t = [4, 2, 4, 2]

Generation 770:	Best = 2.304700e+03	FES = 77100	Diversity = 1.7698e-07	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 2.304700e+03	FES = 78100	Diversity = 1.7740e-07	k_t
= [4, 4, 2, 3]				
Generation 790:	Best = 2.304700e+03	FES = 79100	Diversity = 1.7649e-07	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 2.304700e+03	FES = 80100	Diversity = 1.7718e-07	k_t
= [4, 4, 2, 2]				
Generation 810:	Best = 2.304700e+03	FES = 81100	Diversity = 1.7885e-07	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 2.304700e+03	FES = 82100	Diversity = 1.7726e-07	k_t
= [4, 4, 3, 2]				
Generation 830:	Best = 2.304700e+03	FES = 83100	Diversity = 1.8273e-07	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 2.304700e+03	FES = 84100	Diversity = 1.8206e-07	k_t
= [4, 4, 3, 3]				
Generation 850:	Best = 2.304700e+03	FES = 85100	Diversity = 1.7043e-07	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 2.304700e+03	FES = 86100	Diversity = 1.6282e-07	k_t
= [3, 2, 4, 2]				
Generation 870:	Best = 2.304700e+03	FES = 87100	Diversity = 1.5727e-07	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 2.304700e+03	FES = 88100	Diversity = 1.6900e-07	k_t
= [2, 2, 3, 3]				
Generation 890:	Best = 2.304700e+03	FES = 89100	Diversity = 1.6379e-07	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 2.304700e+03	FES = 90100	Diversity = 1.7228e-07	k_t
= [2, 2, 4, 3]				
Generation 910:	Best = 2.304700e+03	FES = 91100	Diversity = 1.7270e-07	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 2.304700e+03	FES = 92100	Diversity = 1.8389e-07	k_t
= [3, 3, 3, 2]				
Generation 930:	Best = 2.304700e+03	FES = 93100	Diversity = 1.8067e-07	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 2.304700e+03	FES = 94100	Diversity = 1.7981e-07	k_t
= [2, 4, 2, 2]				
Generation 950:	Best = 2.304700e+03	FES = 95100	Diversity = 1.8032e-07	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 2.304700e+03	FES = 96100	Diversity = 1.8799e-07	k_t
= [3, 2, 2, 3]				
Generation 970:	Best = 2.304700e+03	FES = 97100	Diversity = 1.9111e-07	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 2.304700e+03	FES = 98100	Diversity = 1.9230e-07	k_t
= [4, 4, 4, 3]				
Generation 990:	Best = 2.304700e+03	FES = 99100	Diversity = 1.8984e-07	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 2.304700e+03

Total Function Evaluations = 100000

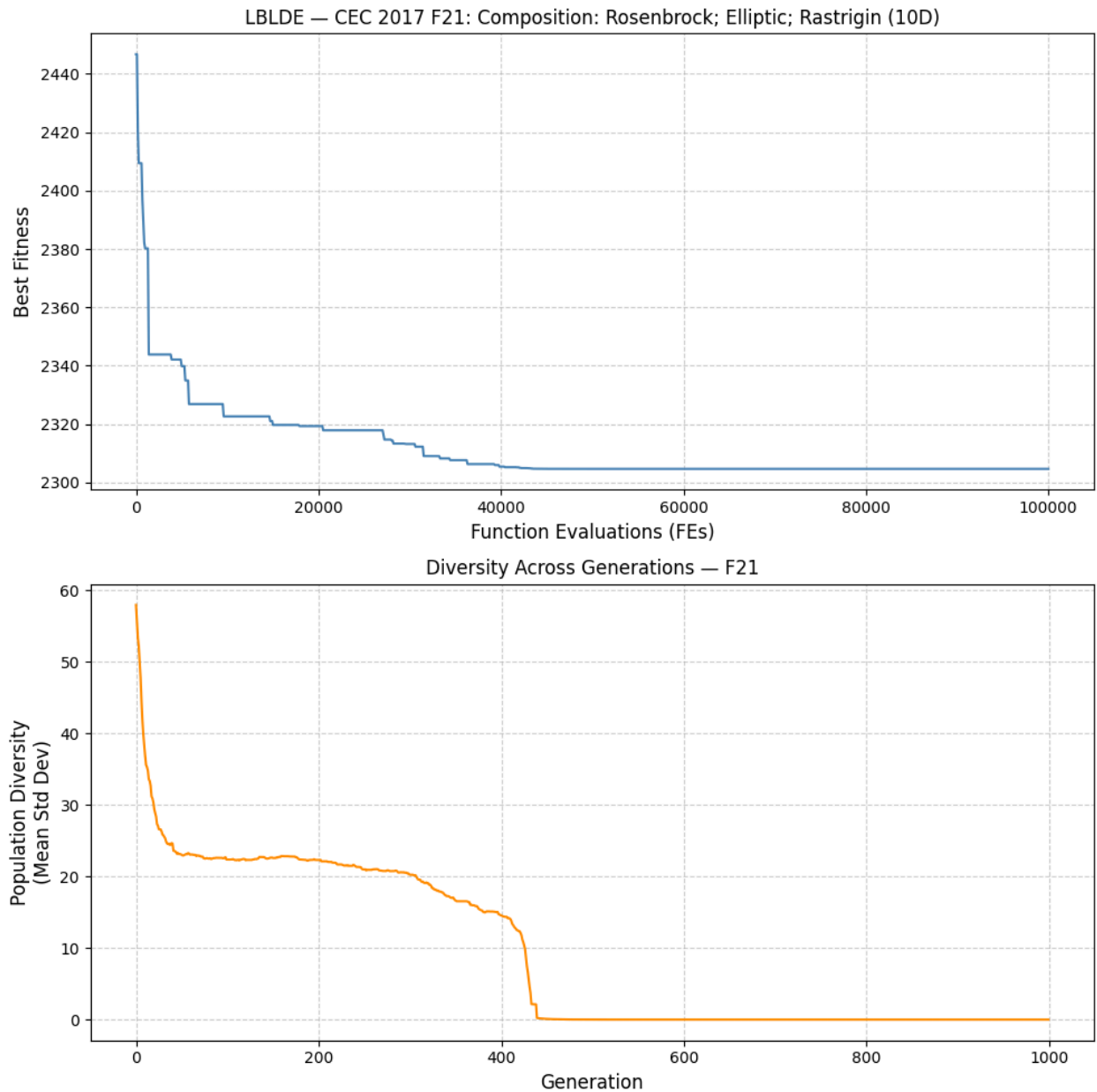
=====

Results:

```

Best Fitness      : 2.304700e+03
Global Optimum    : 2.100000e+03
Error (|best-optimum|) : 2.046999e+02
Best Solution (5 dim) : [ 8.31995652 -39.46687931  43.4165755  52.06127369
63.01453199]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F21.png

```

In [38]: # =====
# F22 — Composition Function 2 (Rastrigin; Griewank; Schwefel)
# =====
run_and_plot('F22', make_wrapper(composition.f22), 2200.0, 'Composition: Rastr

```

```

=====
=
LBLDE_s on CEC 2017 F22 – Single Run
=====
=
Function      : Composition: Rastrigin; Griewank; Schwefel (10D)
Bounds       : [-100, 100]
Global optimum : 2200.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 3.528003e+03 | Diversity = 5.7928e+01
Generation 10: Best = 2.428352e+03 FES = 1100 Diversity = 2.2510e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.323160e+03 FES = 2100 Diversity = 7.6191e+00 k_t
= [4, 3, 4, 2]
Generation 30: Best = 2.311955e+03 FES = 3100 Diversity = 2.6963e+00 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.307363e+03 FES = 4100 Diversity = 1.1191e+00 k_t
= [3, 3, 4, 4]
Generation 50: Best = 2.303687e+03 FES = 5100 Diversity = 8.2678e-01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.303604e+03 FES = 6100 Diversity = 7.6497e-01 k_t
= [2, 3, 4, 3]
Generation 70: Best = 2.303203e+03 FES = 7100 Diversity = 7.3409e-01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.302054e+03 FES = 8100 Diversity = 7.0764e-01 k_t
= [2, 2, 3, 2]
Generation 90: Best = 2.301897e+03 FES = 9100 Diversity = 6.8353e-01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.301376e+03 FES = 10100 Diversity = 6.3379e-01 k_t
= [4, 2, 2, 4]
Generation 110: Best = 2.301220e+03 FES = 11100 Diversity = 5.7410e-01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.301039e+03 FES = 12100 Diversity = 5.2231e-01 k_t
= [2, 4, 2, 3]
Generation 130: Best = 2.300769e+03 FES = 13100 Diversity = 4.7949e-01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.300580e+03 FES = 14100 Diversity = 4.4682e-01 k_t
= [4, 2, 4, 4]
Generation 150: Best = 2.300325e+03 FES = 15100 Diversity = 3.9069e-01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.300299e+03 FES = 16100 Diversity = 3.4270e-01 k_t
= [2, 2, 4, 2]
Generation 170: Best = 2.300195e+03 FES = 17100 Diversity = 2.8460e-01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.300190e+03 FES = 18100 Diversity = 2.5553e-01 k_t
= [2, 2, 2, 4]
Generation 190: Best = 2.300119e+03 FES = 19100 Diversity = 2.0769e-01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.300076e+03 FES = 20100 Diversity = 1.5275e-01 k_t
= [4, 2, 4, 2]
Generation 210: Best = 2.300062e+03 FES = 21100 Diversity = 9.0202e-02 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.300030e+03 FES = 22100 Diversity = 3.1403e-02 k_t
= [4, 3, 2, 4]

```

Generation 230:	Best = 2.300008e+03	FES = 23100	Diversity = 1.9898e-02	k_t = [1, 1, 1, 1]
Generation 240:	Best = 2.300005e+03	FES = 24100	Diversity = 1.0447e-02	k_t = [3, 3, 3, 3]
Generation 250:	Best = 2.300004e+03	FES = 25100	Diversity = 5.5950e-03	k_t = [1, 1, 1, 1]
Generation 260:	Best = 2.300001e+03	FES = 26100	Diversity = 2.8998e-03	k_t = [2, 3, 3, 2]
Generation 270:	Best = 2.300000e+03	FES = 27100	Diversity = 1.3024e-03	k_t = [1, 1, 1, 1]
Generation 280:	Best = 2.300000e+03	FES = 28100	Diversity = 5.6925e-04	k_t = [3, 2, 2, 4]
Generation 290:	Best = 2.300000e+03	FES = 29100	Diversity = 2.6440e-04	k_t = [1, 1, 1, 1]
Generation 300:	Best = 2.300000e+03	FES = 30100	Diversity = 1.1274e-04	k_t = [3, 4, 3, 3]
Generation 310:	Best = 2.300000e+03	FES = 31100	Diversity = 4.4846e-05	k_t = [1, 1, 1, 1]
Generation 320:	Best = 2.300000e+03	FES = 32100	Diversity = 2.0204e-05	k_t = [2, 4, 4, 2]
Generation 330:	Best = 2.300000e+03	FES = 33100	Diversity = 8.9197e-06	k_t = [1, 1, 1, 1]
Generation 340:	Best = 2.300000e+03	FES = 34100	Diversity = 3.7734e-06	k_t = [4, 2, 3, 2]
Generation 350:	Best = 2.300000e+03	FES = 35100	Diversity = 1.7579e-06	k_t = [1, 1, 1, 1]
Generation 360:	Best = 2.300000e+03	FES = 36100	Diversity = 7.0546e-07	k_t = [2, 4, 3, 2]
Generation 370:	Best = 2.300000e+03	FES = 37100	Diversity = 2.9805e-07	k_t = [1, 1, 1, 1]
Generation 380:	Best = 2.300000e+03	FES = 38100	Diversity = 1.2232e-07	k_t = [4, 4, 4, 4]
Generation 390:	Best = 2.300000e+03	FES = 39100	Diversity = 5.4088e-08	k_t = [1, 1, 1, 1]
Generation 400:	Best = 2.300000e+03	FES = 40100	Diversity = 1.9326e-08	k_t = [4, 4, 2, 2]
Generation 410:	Best = 2.300000e+03	FES = 41100	Diversity = 6.7803e-09	k_t = [1, 1, 1, 1]
Generation 420:	Best = 2.300000e+03	FES = 42100	Diversity = 2.6060e-09	k_t = [4, 4, 2, 4]
Generation 430:	Best = 2.300000e+03	FES = 43100	Diversity = 1.0296e-09	k_t = [1, 1, 1, 1]
Generation 440:	Best = 2.300000e+03	FES = 44100	Diversity = 4.3564e-10	k_t = [4, 2, 2, 4]
Generation 450:	Best = 2.300000e+03	FES = 45100	Diversity = 1.6529e-10	k_t = [1, 1, 1, 1]
Generation 460:	Best = 2.300000e+03	FES = 46100	Diversity = 5.9851e-11	k_t = [4, 4, 4, 3]
Generation 470:	Best = 2.300000e+03	FES = 47100	Diversity = 2.3829e-11	k_t = [1, 1, 1, 1]
Generation 480:	Best = 2.300000e+03	FES = 48100	Diversity = 9.5019e-12	k_t = [3, 3, 3, 4]
Generation 490:	Best = 2.300000e+03	FES = 49100	Diversity = 3.8709e-12	k_t = [1, 1, 1, 1]

Generation 500:	Best = 2.300000e+03	FES = 50100	Diversity = 1.8765e-12	k_t = [2, 4, 4, 3]
Generation 510:	Best = 2.300000e+03	FES = 51100	Diversity = 1.1200e-12	k_t = [1, 1, 1, 1]
Generation 520:	Best = 2.300000e+03	FES = 52100	Diversity = 7.9928e-13	k_t = [2, 3, 2, 3]
Generation 530:	Best = 2.300000e+03	FES = 53100	Diversity = 7.2888e-13	k_t = [1, 1, 1, 1]
Generation 540:	Best = 2.300000e+03	FES = 54100	Diversity = 7.4207e-13	k_t = [4, 4, 3, 4]
Generation 550:	Best = 2.300000e+03	FES = 55100	Diversity = 7.4387e-13	k_t = [1, 1, 1, 1]
Generation 560:	Best = 2.300000e+03	FES = 56100	Diversity = 7.2890e-13	k_t = [3, 3, 4, 3]
Generation 570:	Best = 2.300000e+03	FES = 57100	Diversity = 7.4133e-13	k_t = [1, 1, 1, 1]
Generation 580:	Best = 2.300000e+03	FES = 58100	Diversity = 7.2522e-13	k_t = [2, 4, 3, 4]
Generation 590:	Best = 2.300000e+03	FES = 59100	Diversity = 7.2204e-13	k_t = [1, 1, 1, 1]
Generation 600:	Best = 2.300000e+03	FES = 60100	Diversity = 7.1123e-13	k_t = [3, 3, 3, 2]
Generation 610:	Best = 2.300000e+03	FES = 61100	Diversity = 6.9115e-13	k_t = [1, 1, 1, 1]
Generation 620:	Best = 2.300000e+03	FES = 62100	Diversity = 6.8156e-13	k_t = [4, 4, 4, 2]
Generation 630:	Best = 2.300000e+03	FES = 63100	Diversity = 6.1350e-13	k_t = [1, 1, 1, 1]
Generation 640:	Best = 2.300000e+03	FES = 64100	Diversity = 5.5685e-13	k_t = [4, 4, 2, 2]
Generation 650:	Best = 2.300000e+03	FES = 65100	Diversity = 3.8165e-13	k_t = [1, 1, 1, 1]
Generation 660:	Best = 2.300000e+03	FES = 66100	Diversity = 2.6227e-13	k_t = [3, 3, 3, 3]
Generation 670:	Best = 2.300000e+03	FES = 67100	Diversity = 2.3629e-13	k_t = [1, 1, 1, 1]
Generation 680:	Best = 2.300000e+03	FES = 68100	Diversity = 2.5330e-13	k_t = [4, 2, 2, 2]
Generation 690:	Best = 2.300000e+03	FES = 69100	Diversity = 2.4838e-13	k_t = [1, 1, 1, 1]
Generation 700:	Best = 2.300000e+03	FES = 70100	Diversity = 2.6195e-13	k_t = [2, 4, 4, 2]
Generation 710:	Best = 2.300000e+03	FES = 71100	Diversity = 2.4473e-13	k_t = [1, 1, 1, 1]
Generation 720:	Best = 2.300000e+03	FES = 72100	Diversity = 2.4406e-13	k_t = [2, 3, 2, 2]
Generation 730:	Best = 2.300000e+03	FES = 73100	Diversity = 2.3041e-13	k_t = [1, 1, 1, 1]
Generation 740:	Best = 2.300000e+03	FES = 74100	Diversity = 2.2658e-13	k_t = [4, 3, 4, 4]
Generation 750:	Best = 2.300000e+03	FES = 75100	Diversity = 2.3311e-13	k_t = [1, 1, 1, 1]
Generation 760:	Best = 2.300000e+03	FES = 76100	Diversity = 2.4445e-13	k_t = [4, 3, 2, 3]

Generation 770:	Best = 2.300000e+03	FES = 77100	Diversity = 2.3011e-13	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 2.300000e+03	FES = 78100	Diversity = 2.3296e-13	k_t
= [3, 4, 3, 4]				
Generation 790:	Best = 2.300000e+03	FES = 79100	Diversity = 2.4264e-13	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 2.300000e+03	FES = 80100	Diversity = 2.4425e-13	k_t
= [4, 3, 4, 2]				
Generation 810:	Best = 2.300000e+03	FES = 81100	Diversity = 2.4557e-13	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 2.300000e+03	FES = 82100	Diversity = 2.4585e-13	k_t
= [4, 3, 3, 2]				
Generation 830:	Best = 2.300000e+03	FES = 83100	Diversity = 2.3632e-13	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 2.300000e+03	FES = 84100	Diversity = 2.4269e-13	k_t
= [4, 2, 2, 2]				
Generation 850:	Best = 2.300000e+03	FES = 85100	Diversity = 2.3074e-13	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 2.300000e+03	FES = 86100	Diversity = 2.3388e-13	k_t
= [2, 2, 3, 3]				
Generation 870:	Best = 2.300000e+03	FES = 87100	Diversity = 2.4535e-13	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 2.300000e+03	FES = 88100	Diversity = 2.4700e-13	k_t
= [4, 4, 2, 3]				
Generation 890:	Best = 2.300000e+03	FES = 89100	Diversity = 2.3841e-13	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 2.300000e+03	FES = 90100	Diversity = 2.4400e-13	k_t
= [3, 3, 4, 3]				
Generation 910:	Best = 2.300000e+03	FES = 91100	Diversity = 2.4264e-13	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 2.300000e+03	FES = 92100	Diversity = 2.4517e-13	k_t
= [4, 4, 2, 4]				
Generation 930:	Best = 2.300000e+03	FES = 93100	Diversity = 2.3555e-13	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 2.300000e+03	FES = 94100	Diversity = 2.4903e-13	k_t
= [4, 2, 2, 2]				
Generation 950:	Best = 2.300000e+03	FES = 95100	Diversity = 2.4290e-13	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 2.300000e+03	FES = 96100	Diversity = 2.5707e-13	k_t
= [4, 4, 4, 3]				
Generation 970:	Best = 2.300000e+03	FES = 97100	Diversity = 2.5045e-13	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 2.300000e+03	FES = 98100	Diversity = 2.4415e-13	k_t
= [2, 2, 4, 2]				
Generation 990:	Best = 2.300000e+03	FES = 99100	Diversity = 2.4277e-13	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 2.300000e+03

Total Function Evaluations = 100000

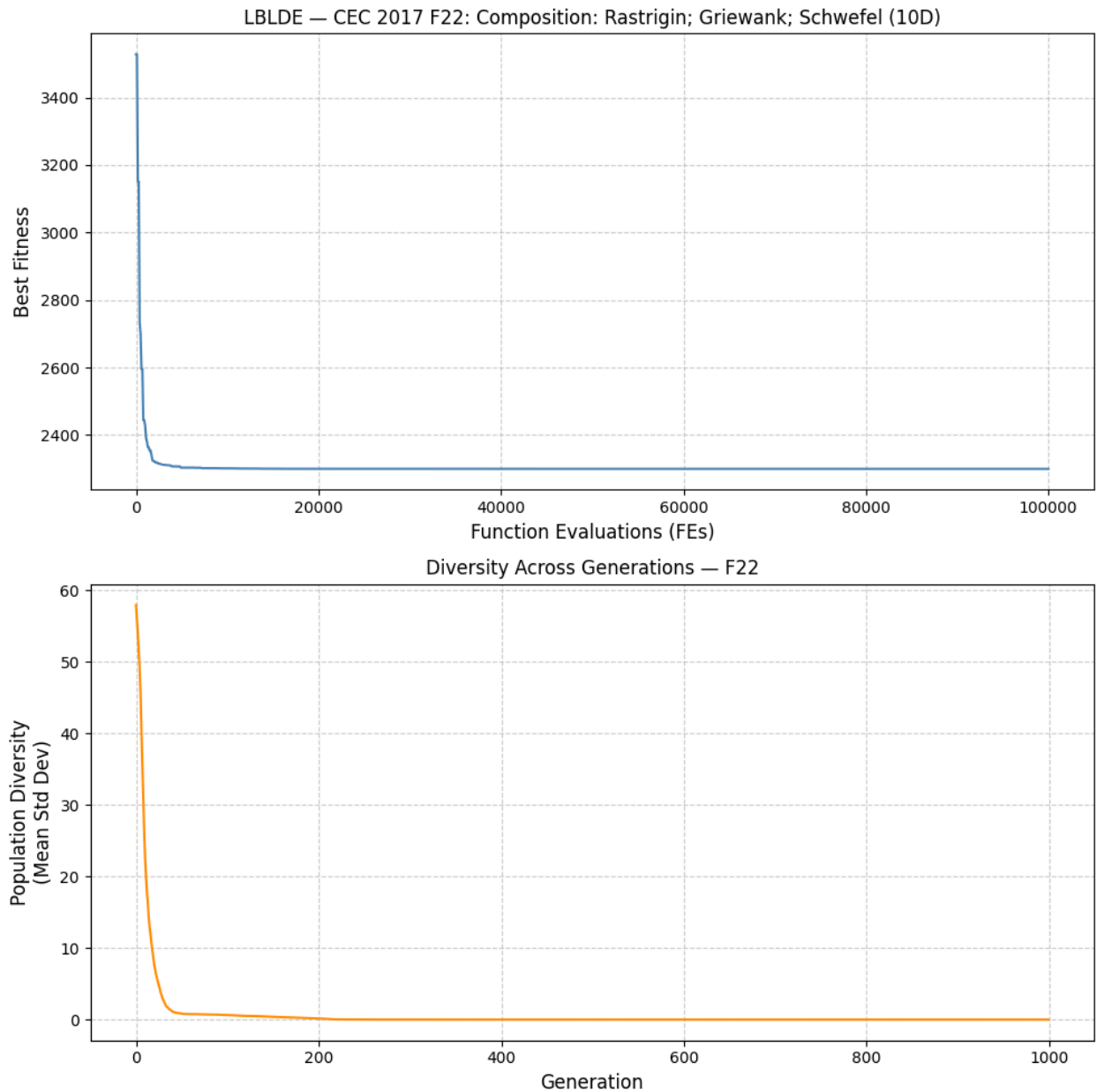
=====

Results:

```

Best Fitness      : 2.300000e+03
Global Optimum    : 2.200000e+03
Error (|best-optimum|) : 1.000000e+02
Best Solution (5 dim) : [-52.41266063  11.16831479  23.05456506  35.08347775
30.48935237]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F22.png

```

In [39]: # =====
# F23 — Composition Function 3 (Rosenbrock; Ackley; Schwefel; Rastrigin)
# =====
run_and_plot('F23', make_wrapper(composition.f23), 2300.0, 'Composition: Roser

```

```

=====
=
LBLDE_s on CEC 2017 F23 – Single Run
=====
=
Function      : Composition: Rosenbrock; Ackley; Schwefel; Rastrigin (10D)
Bounds       : [-100, 100]
Global optimum : 2300.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 2.773921e+03 | Diversity = 5.7928e+01
Generation 10: Best = 2.657895e+03 FES = 1100 Diversity = 2.9956e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.654510e+03 FES = 2100 Diversity = 2.5035e+01 k_t
= [4, 3, 2, 3]
Generation 30: Best = 2.644554e+03 FES = 3100 Diversity = 2.4497e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.642748e+03 FES = 4100 Diversity = 2.3047e+01 k_t
= [3, 4, 3, 2]
Generation 50: Best = 2.633870e+03 FES = 5100 Diversity = 2.2619e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.633870e+03 FES = 6100 Diversity = 2.2197e+01 k_t
= [2, 2, 2, 2]
Generation 70: Best = 2.633870e+03 FES = 7100 Diversity = 2.1676e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.633139e+03 FES = 8100 Diversity = 2.0757e+01 k_t
= [4, 2, 2, 2]
Generation 90: Best = 2.627876e+03 FES = 9100 Diversity = 2.0087e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.623507e+03 FES = 10100 Diversity = 1.9809e+01 k_t
= [2, 3, 3, 4]
Generation 110: Best = 2.623507e+03 FES = 11100 Diversity = 1.9667e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.623051e+03 FES = 12100 Diversity = 1.9146e+01 k_t
= [3, 3, 4, 4]
Generation 130: Best = 2.621046e+03 FES = 13100 Diversity = 1.8754e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.620938e+03 FES = 14100 Diversity = 1.8666e+01 k_t
= [3, 4, 3, 3]
Generation 150: Best = 2.620938e+03 FES = 15100 Diversity = 1.8755e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.620938e+03 FES = 16100 Diversity = 1.8518e+01 k_t
= [4, 4, 2, 3]
Generation 170: Best = 2.619143e+03 FES = 17100 Diversity = 1.8443e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.619143e+03 FES = 18100 Diversity = 1.8003e+01 k_t
= [2, 4, 3, 3]
Generation 190: Best = 2.619143e+03 FES = 19100 Diversity = 1.7552e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.616166e+03 FES = 20100 Diversity = 1.7528e+01 k_t
= [2, 3, 2, 4]
Generation 210: Best = 2.615805e+03 FES = 21100 Diversity = 1.7671e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.615805e+03 FES = 22100 Diversity = 1.7391e+01 k_t
= [3, 2, 4, 3]

```

Generation 230:	Best = 2.614725e+03	FES = 23100	Diversity = 1.7397e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 2.611991e+03	FES = 24100	Diversity = 1.7151e+01	k_t = [4, 3, 2, 3]
Generation 250:	Best = 2.611991e+03	FES = 25100	Diversity = 1.7174e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 2.611362e+03	FES = 26100	Diversity = 1.6470e+01	k_t = [3, 4, 3, 4]
Generation 270:	Best = 2.611362e+03	FES = 27100	Diversity = 1.6369e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 2.610507e+03	FES = 28100	Diversity = 1.5890e+01	k_t = [2, 4, 4, 3]
Generation 290:	Best = 2.608492e+03	FES = 29100	Diversity = 1.5647e+01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 2.608492e+03	FES = 30100	Diversity = 1.5543e+01	k_t = [2, 4, 4, 2]
Generation 310:	Best = 2.608492e+03	FES = 31100	Diversity = 1.4959e+01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 2.607368e+03	FES = 32100	Diversity = 1.4673e+01	k_t = [2, 4, 3, 2]
Generation 330:	Best = 2.607368e+03	FES = 33100	Diversity = 1.4430e+01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 2.607368e+03	FES = 34100	Diversity = 1.3575e+01	k_t = [4, 3, 4, 4]
Generation 350:	Best = 2.606009e+03	FES = 35100	Diversity = 1.2916e+01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 2.605816e+03	FES = 36100	Diversity = 1.1297e+01	k_t = [4, 2, 4, 4]
Generation 370:	Best = 2.604844e+03	FES = 37100	Diversity = 8.7987e+00	k_t = [1, 1, 1, 1]
Generation 380:	Best = 2.604844e+03	FES = 38100	Diversity = 3.6201e+00	k_t = [3, 2, 4, 4]
Generation 390:	Best = 2.604363e+03	FES = 39100	Diversity = 2.4759e+00	k_t = [1, 1, 1, 1]
Generation 400:	Best = 2.604280e+03	FES = 40100	Diversity = 1.6486e+00	k_t = [4, 3, 3, 2]
Generation 410:	Best = 2.604237e+03	FES = 41100	Diversity = 9.2431e-01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 2.604197e+03	FES = 42100	Diversity = 5.0287e-01	k_t = [2, 3, 4, 2]
Generation 430:	Best = 2.604171e+03	FES = 43100	Diversity = 4.0875e-01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 2.604170e+03	FES = 44100	Diversity = 3.5545e-01	k_t = [2, 4, 4, 3]
Generation 450:	Best = 2.604159e+03	FES = 45100	Diversity = 3.2844e-01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 2.604156e+03	FES = 46100	Diversity = 3.2309e-01	k_t = [2, 4, 3, 3]
Generation 470:	Best = 2.604153e+03	FES = 47100	Diversity = 3.4092e-01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 2.604151e+03	FES = 48100	Diversity = 3.4217e-01	k_t = [4, 3, 3, 2]
Generation 490:	Best = 2.604151e+03	FES = 49100	Diversity = 3.2442e-01	k_t = [1, 1, 1, 1]

Generation 500:	Best = 2.604150e+03	FES = 50100	Diversity = 2.5280e-01	k_t
= [2, 4, 2, 3]				
Generation 510:	Best = 2.604150e+03	FES = 51100	Diversity = 1.3957e-01	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 2.604150e+03	FES = 52100	Diversity = 6.6138e-02	k_t
= [3, 2, 4, 2]				
Generation 530:	Best = 2.604150e+03	FES = 53100	Diversity = 4.8811e-03	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 2.604150e+03	FES = 54100	Diversity = 3.4003e-03	k_t
= [2, 2, 3, 4]				
Generation 550:	Best = 2.604150e+03	FES = 55100	Diversity = 2.3748e-03	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 2.604150e+03	FES = 56100	Diversity = 1.5652e-03	k_t
= [3, 4, 2, 4]				
Generation 570:	Best = 2.604150e+03	FES = 57100	Diversity = 1.2106e-03	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 2.604150e+03	FES = 58100	Diversity = 8.8285e-04	k_t
= [4, 3, 4, 4]				
Generation 590:	Best = 2.604150e+03	FES = 59100	Diversity = 6.1902e-04	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 2.604150e+03	FES = 60100	Diversity = 4.8422e-04	k_t
= [4, 4, 4, 3]				
Generation 610:	Best = 2.604150e+03	FES = 61100	Diversity = 3.8450e-04	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 2.604150e+03	FES = 62100	Diversity = 2.9835e-04	k_t
= [2, 2, 3, 4]				
Generation 630:	Best = 2.604150e+03	FES = 63100	Diversity = 2.2125e-04	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 2.604150e+03	FES = 64100	Diversity = 1.6567e-04	k_t
= [3, 2, 3, 4]				
Generation 650:	Best = 2.604150e+03	FES = 65100	Diversity = 1.1665e-04	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 2.604150e+03	FES = 66100	Diversity = 8.1326e-05	k_t
= [2, 4, 2, 2]				
Generation 670:	Best = 2.604150e+03	FES = 67100	Diversity = 6.5592e-05	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 2.604150e+03	FES = 68100	Diversity = 5.1396e-05	k_t
= [4, 2, 4, 4]				
Generation 690:	Best = 2.604150e+03	FES = 69100	Diversity = 4.4029e-05	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 2.604150e+03	FES = 70100	Diversity = 3.3288e-05	k_t
= [2, 2, 4, 3]				
Generation 710:	Best = 2.604150e+03	FES = 71100	Diversity = 2.5465e-05	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 2.604150e+03	FES = 72100	Diversity = 2.1710e-05	k_t
= [4, 4, 4, 3]				
Generation 730:	Best = 2.604150e+03	FES = 73100	Diversity = 1.7620e-05	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 2.604150e+03	FES = 74100	Diversity = 1.3760e-05	k_t
= [4, 3, 3, 2]				
Generation 750:	Best = 2.604150e+03	FES = 75100	Diversity = 1.1126e-05	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 2.604150e+03	FES = 76100	Diversity = 9.9849e-06	k_t
= [2, 2, 2, 4]				

Generation 770:	Best = 2.604150e+03	FES = 77100	Diversity = 8.4201e-06	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 2.604150e+03	FES = 78100	Diversity = 6.3935e-06	k_t
= [3, 3, 4, 3]				
Generation 790:	Best = 2.604150e+03	FES = 79100	Diversity = 5.1634e-06	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 2.604150e+03	FES = 80100	Diversity = 3.9611e-06	k_t
= [4, 3, 4, 2]				
Generation 810:	Best = 2.604150e+03	FES = 81100	Diversity = 3.3035e-06	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 2.604150e+03	FES = 82100	Diversity = 2.5575e-06	k_t
= [4, 3, 4, 4]				
Generation 830:	Best = 2.604150e+03	FES = 83100	Diversity = 2.0719e-06	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 2.604150e+03	FES = 84100	Diversity = 1.7661e-06	k_t
= [3, 2, 3, 2]				
Generation 850:	Best = 2.604150e+03	FES = 85100	Diversity = 1.4788e-06	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 2.604150e+03	FES = 86100	Diversity = 1.2040e-06	k_t
= [3, 4, 3, 2]				
Generation 870:	Best = 2.604150e+03	FES = 87100	Diversity = 9.2385e-07	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 2.604150e+03	FES = 88100	Diversity = 7.0908e-07	k_t
= [4, 2, 3, 4]				
Generation 890:	Best = 2.604150e+03	FES = 89100	Diversity = 6.0552e-07	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 2.604150e+03	FES = 90100	Diversity = 5.1574e-07	k_t
= [3, 4, 3, 3]				
Generation 910:	Best = 2.604150e+03	FES = 91100	Diversity = 4.5323e-07	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 2.604150e+03	FES = 92100	Diversity = 3.6616e-07	k_t
= [4, 3, 4, 4]				
Generation 930:	Best = 2.604150e+03	FES = 93100	Diversity = 3.0325e-07	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 2.604150e+03	FES = 94100	Diversity = 2.3821e-07	k_t
= [2, 2, 4, 4]				
Generation 950:	Best = 2.604150e+03	FES = 95100	Diversity = 2.3091e-07	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 2.604150e+03	FES = 96100	Diversity = 2.0762e-07	k_t
= [3, 2, 2, 3]				
Generation 970:	Best = 2.604150e+03	FES = 97100	Diversity = 2.0922e-07	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 2.604150e+03	FES = 98100	Diversity = 2.0908e-07	k_t
= [3, 2, 2, 3]				
Generation 990:	Best = 2.604150e+03	FES = 99100	Diversity = 2.0581e-07	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 2.604150e+03

Total Function Evaluations = 100000

=====

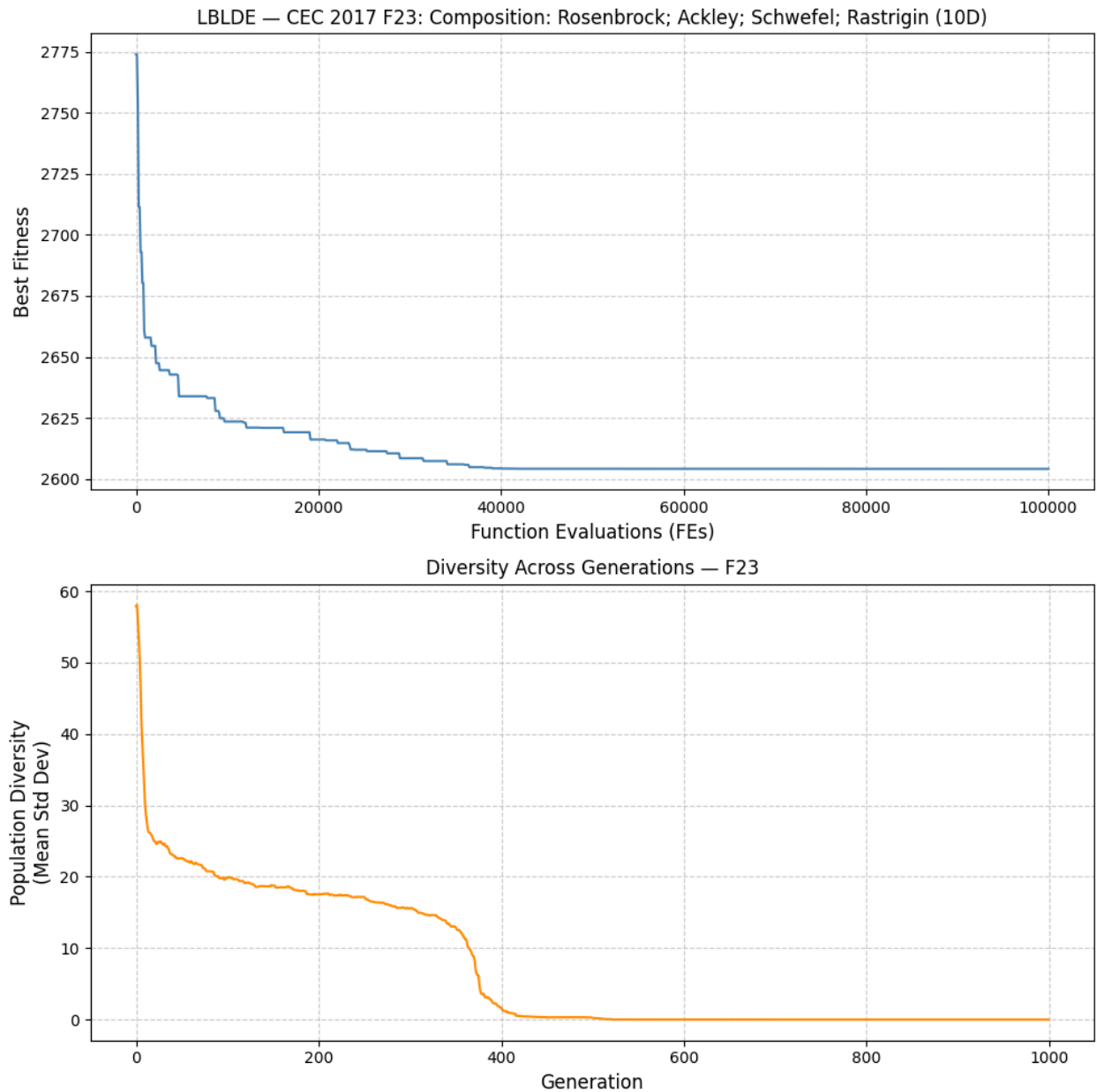
=

Results:

```

Best Fitness      : 2.604150e+03
Global Optimum    : 2.300000e+03
Error (|best-optimum|) : 3.041496e+02
Best Solution (5 dim) : [-77.79270967 -72.60183167 -63.36040657  65.81980581
-68.87589106]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F23.png

```

In [40]: # =====
# F24 — Composition Function 4 (Ackley; Elliptic; Griewank; Rastrigin)
# =====
run_and_plot('F24', make_wrapper(composition.f24), 2400.0, 'Composition: Ackle

```

```

=====
=
LBLDE_s on CEC 2017 F24 – Single Run
=====
=
Function      : Composition: Ackley; Elliptic; Griewank; Rastrigin (10D)
Bounds       : [-100, 100]
Global optimum : 2400.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 2.997998e+03 | Diversity = 5.7928e+01
Generation 10: Best = 2.811216e+03 FES = 1100 Diversity = 3.0557e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.783314e+03 FES = 2100 Diversity = 2.3179e+01 k_t
= [4, 3, 2, 3]
Generation 30: Best = 2.779605e+03 FES = 3100 Diversity = 2.1504e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.760086e+03 FES = 4100 Diversity = 2.0129e+01 k_t
= [3, 4, 3, 2]
Generation 50: Best = 2.760086e+03 FES = 5100 Diversity = 1.9663e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.760086e+03 FES = 6100 Diversity = 1.9215e+01 k_t
= [2, 2, 2, 2]
Generation 70: Best = 2.759986e+03 FES = 7100 Diversity = 1.8667e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.753754e+03 FES = 8100 Diversity = 1.8443e+01 k_t
= [3, 4, 3, 4]
Generation 90: Best = 2.753754e+03 FES = 9100 Diversity = 1.8150e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.749566e+03 FES = 10100 Diversity = 1.7632e+01 k_t
= [4, 3, 4, 4]
Generation 110: Best = 2.749566e+03 FES = 11100 Diversity = 1.7228e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.749566e+03 FES = 12100 Diversity = 1.7047e+01 k_t
= [2, 4, 2, 2]
Generation 130: Best = 2.749566e+03 FES = 13100 Diversity = 1.6887e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.749219e+03 FES = 14100 Diversity = 1.6772e+01 k_t
= [3, 4, 4, 4]
Generation 150: Best = 2.749219e+03 FES = 15100 Diversity = 1.6460e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.748271e+03 FES = 16100 Diversity = 1.6302e+01 k_t
= [3, 3, 2, 3]
Generation 170: Best = 2.748237e+03 FES = 17100 Diversity = 1.6168e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.747600e+03 FES = 18100 Diversity = 1.6019e+01 k_t
= [3, 2, 3, 3]
Generation 190: Best = 2.746168e+03 FES = 19100 Diversity = 1.5939e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.745281e+03 FES = 20100 Diversity = 1.5387e+01 k_t
= [3, 3, 4, 4]
Generation 210: Best = 2.745281e+03 FES = 21100 Diversity = 1.5339e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.745281e+03 FES = 22100 Diversity = 1.5256e+01 k_t
= [2, 3, 4, 3]

```


Generation 230: Best = 2.745194e+03	FES = 23100	Diversity = 1.5212e+01	k_t
= [1, 1, 1, 1]			
Generation 240: Best = 2.745194e+03	FES = 24100	Diversity = 1.5128e+01	k_t
= [3, 2, 4, 4]			
Generation 250: Best = 2.745194e+03	FES = 25100	Diversity = 1.5017e+01	k_t
= [1, 1, 1, 1]			
Generation 260: Best = 2.745194e+03	FES = 26100	Diversity = 1.4872e+01	k_t
= [2, 4, 2, 3]			
Generation 270: Best = 2.744164e+03	FES = 27100	Diversity = 1.4823e+01	k_t
= [1, 1, 1, 1]			
Generation 280: Best = 2.743823e+03	FES = 28100	Diversity = 1.4462e+01	k_t
= [2, 4, 4, 3]			
Generation 290: Best = 2.743759e+03	FES = 29100	Diversity = 1.4397e+01	k_t
= [1, 1, 1, 1]			
Generation 300: Best = 2.740408e+03	FES = 30100	Diversity = 1.4389e+01	k_t
= [2, 2, 2, 2]			
Generation 310: Best = 2.740284e+03	FES = 31100	Diversity = 1.4264e+01	k_t
= [1, 1, 1, 1]			
Generation 320: Best = 2.740034e+03	FES = 32100	Diversity = 1.4080e+01	k_t
= [2, 4, 3, 2]			
Generation 330: Best = 2.739984e+03	FES = 33100	Diversity = 1.3777e+01	k_t
= [1, 1, 1, 1]			
Generation 340: Best = 2.739143e+03	FES = 34100	Diversity = 1.3762e+01	k_t
= [2, 2, 3, 3]			
Generation 350: Best = 2.739143e+03	FES = 35100	Diversity = 1.3367e+01	k_t
= [1, 1, 1, 1]			
Generation 360: Best = 2.739143e+03	FES = 36100	Diversity = 1.2924e+01	k_t
= [4, 2, 2, 3]			
Generation 370: Best = 2.737206e+03	FES = 37100	Diversity = 1.2573e+01	k_t
= [1, 1, 1, 1]			
Generation 380: Best = 2.736494e+03	FES = 38100	Diversity = 1.2552e+01	k_t
= [3, 3, 4, 2]			
Generation 390: Best = 2.736271e+03	FES = 39100	Diversity = 1.2231e+01	k_t
= [1, 1, 1, 1]			
Generation 400: Best = 2.735388e+03	FES = 40100	Diversity = 1.2064e+01	k_t
= [2, 3, 4, 2]			
Generation 410: Best = 2.734990e+03	FES = 41100	Diversity = 1.1556e+01	k_t
= [1, 1, 1, 1]			
Generation 420: Best = 2.734494e+03	FES = 42100	Diversity = 9.7506e+00	k_t
= [3, 4, 4, 2]			
Generation 430: Best = 2.734435e+03	FES = 43100	Diversity = 3.9209e+00	k_t
= [1, 1, 1, 1]			
Generation 440: Best = 2.734355e+03	FES = 44100	Diversity = 1.4264e-01	k_t
= [3, 3, 3, 4]			
Generation 450: Best = 2.734345e+03	FES = 45100	Diversity = 7.5052e-02	k_t
= [1, 1, 1, 1]			
Generation 460: Best = 2.734340e+03	FES = 46100	Diversity = 3.8591e-02	k_t
= [2, 4, 3, 3]			
Generation 470: Best = 2.734338e+03	FES = 47100	Diversity = 1.9522e-02	k_t
= [1, 1, 1, 1]			
Generation 480: Best = 2.734337e+03	FES = 48100	Diversity = 8.8178e-03	k_t
= [2, 3, 4, 4]			
Generation 490: Best = 2.734337e+03	FES = 49100	Diversity = 4.0911e-03	k_t
= [1, 1, 1, 1]			

Generation 500:	Best = 2.734337e+03	FES = 50100	Diversity = 1.6179e-03	k_t = [3, 4, 4, 4]
Generation 510:	Best = 2.734337e+03	FES = 51100	Diversity = 6.5937e-04	k_t = [1, 1, 1, 1]
Generation 520:	Best = 2.734337e+03	FES = 52100	Diversity = 2.8458e-04	k_t = [2, 2, 4, 4]
Generation 530:	Best = 2.734337e+03	FES = 53100	Diversity = 1.2718e-04	k_t = [1, 1, 1, 1]
Generation 540:	Best = 2.734337e+03	FES = 54100	Diversity = 6.0457e-05	k_t = [4, 2, 4, 2]
Generation 550:	Best = 2.734337e+03	FES = 55100	Diversity = 2.6856e-05	k_t = [1, 1, 1, 1]
Generation 560:	Best = 2.734337e+03	FES = 56100	Diversity = 1.2329e-05	k_t = [4, 2, 4, 4]
Generation 570:	Best = 2.734337e+03	FES = 57100	Diversity = 5.7388e-06	k_t = [1, 1, 1, 1]
Generation 580:	Best = 2.734337e+03	FES = 58100	Diversity = 2.2904e-06	k_t = [2, 4, 3, 4]
Generation 590:	Best = 2.734337e+03	FES = 59100	Diversity = 9.6598e-07	k_t = [1, 1, 1, 1]
Generation 600:	Best = 2.734337e+03	FES = 60100	Diversity = 4.6862e-07	k_t = [2, 2, 2, 3]
Generation 610:	Best = 2.734337e+03	FES = 61100	Diversity = 2.9950e-07	k_t = [1, 1, 1, 1]
Generation 620:	Best = 2.734337e+03	FES = 62100	Diversity = 2.0709e-07	k_t = [3, 4, 2, 4]
Generation 630:	Best = 2.734337e+03	FES = 63100	Diversity = 2.0427e-07	k_t = [1, 1, 1, 1]
Generation 640:	Best = 2.734337e+03	FES = 64100	Diversity = 2.0522e-07	k_t = [3, 3, 4, 2]
Generation 650:	Best = 2.734337e+03	FES = 65100	Diversity = 1.9564e-07	k_t = [1, 1, 1, 1]
Generation 660:	Best = 2.734337e+03	FES = 66100	Diversity = 1.9638e-07	k_t = [2, 3, 3, 4]
Generation 670:	Best = 2.734337e+03	FES = 67100	Diversity = 2.1224e-07	k_t = [1, 1, 1, 1]
Generation 680:	Best = 2.734337e+03	FES = 68100	Diversity = 2.0120e-07	k_t = [2, 3, 3, 4]
Generation 690:	Best = 2.734337e+03	FES = 69100	Diversity = 1.9358e-07	k_t = [1, 1, 1, 1]
Generation 700:	Best = 2.734337e+03	FES = 70100	Diversity = 2.0995e-07	k_t = [3, 3, 3, 4]
Generation 710:	Best = 2.734337e+03	FES = 71100	Diversity = 2.0437e-07	k_t = [1, 1, 1, 1]
Generation 720:	Best = 2.734337e+03	FES = 72100	Diversity = 2.0152e-07	k_t = [4, 3, 4, 2]
Generation 730:	Best = 2.734337e+03	FES = 73100	Diversity = 1.9809e-07	k_t = [1, 1, 1, 1]
Generation 740:	Best = 2.734337e+03	FES = 74100	Diversity = 1.9097e-07	k_t = [2, 4, 4, 3]
Generation 750:	Best = 2.734337e+03	FES = 75100	Diversity = 1.9516e-07	k_t = [1, 1, 1, 1]
Generation 760:	Best = 2.734337e+03	FES = 76100	Diversity = 2.0193e-07	k_t = [3, 2, 3, 3]

Generation 770:	Best = 2.734337e+03	FES = 77100	Diversity = 1.9788e-07	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 2.734337e+03	FES = 78100	Diversity = 1.9972e-07	k_t
= [3, 4, 3, 3]				
Generation 790:	Best = 2.734337e+03	FES = 79100	Diversity = 2.0337e-07	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 2.734337e+03	FES = 80100	Diversity = 2.0226e-07	k_t
= [2, 3, 4, 4]				
Generation 810:	Best = 2.734337e+03	FES = 81100	Diversity = 2.0296e-07	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 2.734337e+03	FES = 82100	Diversity = 2.0262e-07	k_t
= [3, 4, 4, 3]				
Generation 830:	Best = 2.734337e+03	FES = 83100	Diversity = 2.0849e-07	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 2.734337e+03	FES = 84100	Diversity = 2.0420e-07	k_t
= [4, 3, 4, 4]				
Generation 850:	Best = 2.734337e+03	FES = 85100	Diversity = 2.0220e-07	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 2.734337e+03	FES = 86100	Diversity = 2.0377e-07	k_t
= [2, 2, 3, 2]				
Generation 870:	Best = 2.734337e+03	FES = 87100	Diversity = 2.0242e-07	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 2.734337e+03	FES = 88100	Diversity = 2.1253e-07	k_t
= [4, 3, 3, 4]				
Generation 890:	Best = 2.734337e+03	FES = 89100	Diversity = 1.9085e-07	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 2.734337e+03	FES = 90100	Diversity = 1.8707e-07	k_t
= [3, 2, 2, 3]				
Generation 910:	Best = 2.734337e+03	FES = 91100	Diversity = 1.8725e-07	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 2.734337e+03	FES = 92100	Diversity = 1.9789e-07	k_t
= [2, 4, 4, 3]				
Generation 930:	Best = 2.734337e+03	FES = 93100	Diversity = 2.0087e-07	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 2.734337e+03	FES = 94100	Diversity = 2.0564e-07	k_t
= [4, 3, 4, 3]				
Generation 950:	Best = 2.734337e+03	FES = 95100	Diversity = 2.0284e-07	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 2.734337e+03	FES = 96100	Diversity = 1.9567e-07	k_t
= [2, 2, 4, 4]				
Generation 970:	Best = 2.734337e+03	FES = 97100	Diversity = 2.0064e-07	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 2.734337e+03	FES = 98100	Diversity = 1.9732e-07	k_t
= [3, 4, 2, 4]				
Generation 990:	Best = 2.734337e+03	FES = 99100	Diversity = 1.9595e-07	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 2.734337e+03

Total Function Evaluations = 100000

=====

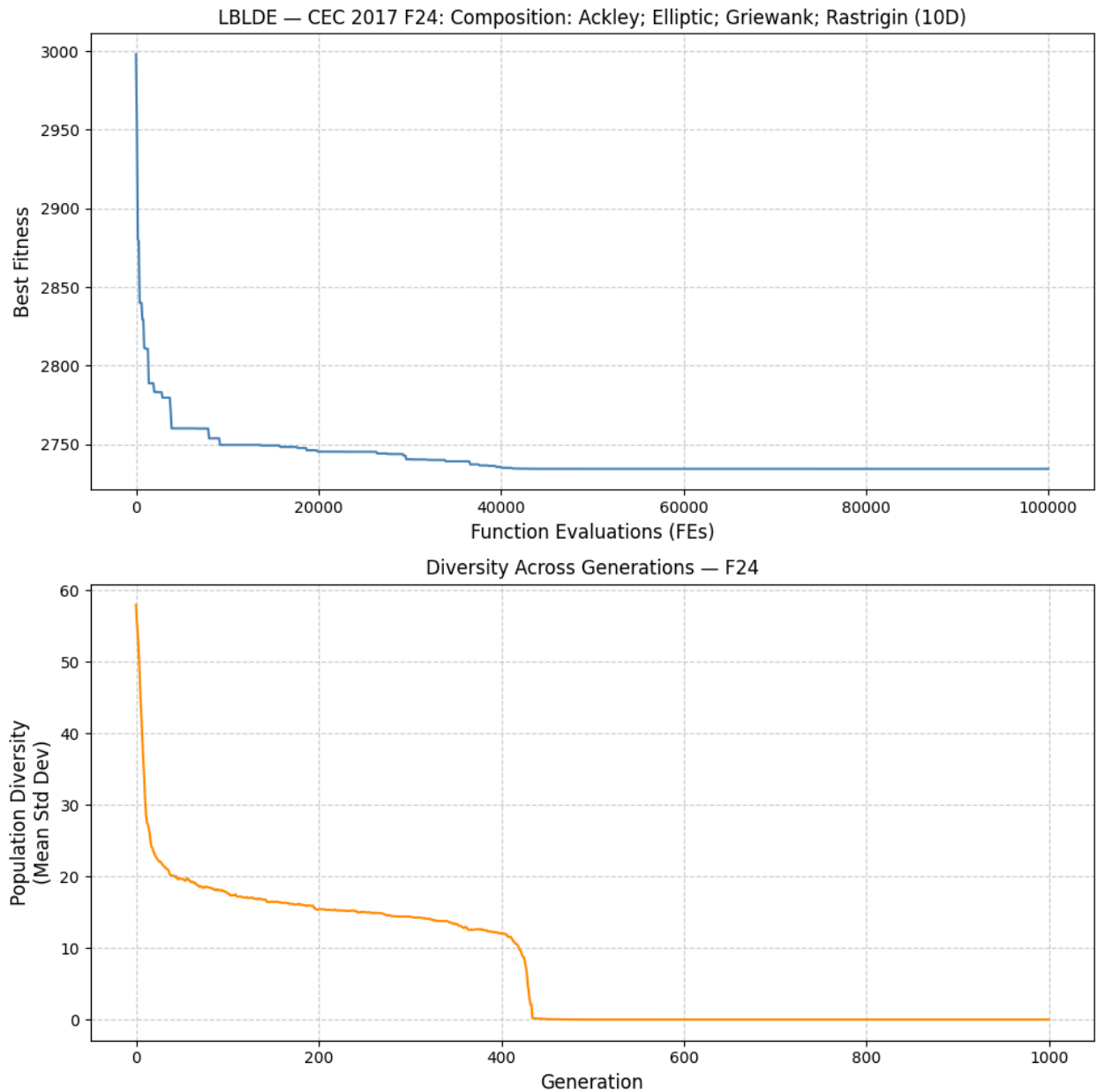
=

Results:

```

Best Fitness      : 2.734337e+03
Global Optimum    : 2.400000e+03
Error (|best-optimum|) : 3.343373e+02
Best Solution (5 dim) : [ 37.78667357  86.03168547 -67.89787924 -80.3959674
65.83343001]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F24.png

```

In [41]: # =====
# F25 — Composition Function 5 (Rastrigin; HappyCat; Ackley; Discus; Rosenbrock)
# =====
run_and_plot('F25', make_wrapper(composition.f25), 2500.0, 'Composition: Rastr

```

```

=====
=
LBLDE_s on CEC 2017 F25 – Single Run
=====
=
Function      : Composition: Rastrigin; HappyCat; Ackley; Discus; Rosenbrock
(10D)
Bounds        : [-100, 100]
Global optimum : 2500.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 4.081798e+03 | Diversity = 5.7928e+01
Generation 10: Best = 2.967089e+03 FES = 1100 Diversity = 2.3982e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.955683e+03 FES = 2100 Diversity = 1.2363e+01 k_t
= [2, 4, 2, 3]
Generation 30: Best = 2.940181e+03 FES = 3100 Diversity = 9.6102e+00 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.910353e+03 FES = 4100 Diversity = 8.1222e+00 k_t
= [3, 4, 3, 2]
Generation 50: Best = 2.903064e+03 FES = 5100 Diversity = 3.0088e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.900499e+03 FES = 6100 Diversity = 1.4128e+00 k_t
= [2, 2, 2, 2]
Generation 70: Best = 2.899797e+03 FES = 7100 Diversity = 7.7198e-01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.899646e+03 FES = 8100 Diversity = 4.2203e-01 k_t
= [3, 3, 2, 3]
Generation 90: Best = 2.899594e+03 FES = 9100 Diversity = 2.1286e-01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.899588e+03 FES = 10100 Diversity = 5.9109e-02 k_t
= [4, 3, 4, 4]
Generation 110: Best = 2.899585e+03 FES = 11100 Diversity = 2.2387e-02 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.899585e+03 FES = 12100 Diversity = 8.1128e-03 k_t
= [2, 4, 4, 2]
Generation 130: Best = 2.899585e+03 FES = 13100 Diversity = 3.3186e-03 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.899585e+03 FES = 14100 Diversity = 1.2193e-03 k_t
= [2, 3, 2, 4]
Generation 150: Best = 2.899585e+03 FES = 15100 Diversity = 4.5381e-04 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.899585e+03 FES = 16100 Diversity = 1.8102e-04 k_t
= [4, 2, 3, 2]
Generation 170: Best = 2.899585e+03 FES = 17100 Diversity = 6.3204e-05 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.899585e+03 FES = 18100 Diversity = 2.4298e-05 k_t
= [3, 4, 2, 4]
Generation 190: Best = 2.899585e+03 FES = 19100 Diversity = 9.3215e-06 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.899585e+03 FES = 20100 Diversity = 3.5478e-06 k_t
= [3, 2, 2, 4]
Generation 210: Best = 2.899585e+03 FES = 21100 Diversity = 1.2122e-06 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.899585e+03 FES = 22100 Diversity = 5.1966e-07 k_t

```

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= [2, 2, 4, 4]
Generation 230: Best = 2.899585e+03 FES = 23100 Diversity = 3.0252e-07 k_t
= [1, 1, 1, 1]
Generation 240: Best = 2.899585e+03 FES = 24100 Diversity = 2.8063e-07 k_t
= [4, 4, 2, 4]
Generation 250: Best = 2.899585e+03 FES = 25100 Diversity = 2.0720e-07 k_t
= [1, 1, 1, 1]
Generation 260: Best = 2.899585e+03 FES = 26100 Diversity = 1.7890e-07 k_t
= [3, 4, 3, 4]
Generation 270: Best = 2.899585e+03 FES = 27100 Diversity = 1.5624e-07 k_t
= [1, 1, 1, 1]
Generation 280: Best = 2.899585e+03 FES = 28100 Diversity = 1.4966e-07 k_t
= [2, 2, 3, 3]
Generation 290: Best = 2.899585e+03 FES = 29100 Diversity = 1.4556e-07 k_t
= [1, 1, 1, 1]
Generation 300: Best = 2.899585e+03 FES = 30100 Diversity = 1.4476e-07 k_t
= [2, 4, 3, 4]
Generation 310: Best = 2.899585e+03 FES = 31100 Diversity = 1.3984e-07 k_t
= [1, 1, 1, 1]
Generation 320: Best = 2.899585e+03 FES = 32100 Diversity = 1.4122e-07 k_t
= [2, 4, 4, 3]
Generation 330: Best = 2.899585e+03 FES = 33100 Diversity = 1.4326e-07 k_t
= [1, 1, 1, 1]
Generation 340: Best = 2.899585e+03 FES = 34100 Diversity = 1.4026e-07 k_t
= [2, 2, 4, 3]
Generation 350: Best = 2.899585e+03 FES = 35100 Diversity = 1.5134e-07 k_t
= [1, 1, 1, 1]
Generation 360: Best = 2.899585e+03 FES = 36100 Diversity = 1.4343e-07 k_t
= [4, 4, 4, 3]
Generation 370: Best = 2.899585e+03 FES = 37100 Diversity = 1.4811e-07 k_t
= [1, 1, 1, 1]
Generation 380: Best = 2.899585e+03 FES = 38100 Diversity = 1.4212e-07 k_t
= [3, 2, 4, 3]
Generation 390: Best = 2.899585e+03 FES = 39100 Diversity = 1.4123e-07 k_t
= [1, 1, 1, 1]
Generation 400: Best = 2.899585e+03 FES = 40100 Diversity = 1.4016e-07 k_t
= [2, 4, 4, 3]
Generation 410: Best = 2.899585e+03 FES = 41100 Diversity = 1.4009e-07 k_t
= [1, 1, 1, 1]
Generation 420: Best = 2.899585e+03 FES = 42100 Diversity = 1.3939e-07 k_t
= [2, 4, 2, 4]
Generation 430: Best = 2.899585e+03 FES = 43100 Diversity = 1.4406e-07 k_t
= [1, 1, 1, 1]
Generation 440: Best = 2.899585e+03 FES = 44100 Diversity = 1.4197e-07 k_t
= [3, 3, 3, 4]
Generation 450: Best = 2.899585e+03 FES = 45100 Diversity = 1.4137e-07 k_t
= [1, 1, 1, 1]
Generation 460: Best = 2.899585e+03 FES = 46100 Diversity = 1.4224e-07 k_t
= [2, 2, 2, 2]
Generation 470: Best = 2.899585e+03 FES = 47100 Diversity = 1.3743e-07 k_t
= [1, 1, 1, 1]
Generation 480: Best = 2.899585e+03 FES = 48100 Diversity = 1.3308e-07 k_t
= [4, 4, 4, 2]
Generation 490: Best = 2.899585e+03 FES = 49100 Diversity = 1.3673e-07 k_t

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= [1, 1, 1, 1]
Generation 500: Best = 2.899585e+03 FES = 50100 Diversity = 1.3508e-07 k_t
= [3, 4, 3, 2]
Generation 510: Best = 2.899585e+03 FES = 51100 Diversity = 1.4109e-07 k_t
= [1, 1, 1, 1]
Generation 520: Best = 2.899585e+03 FES = 52100 Diversity = 1.4375e-07 k_t
= [4, 2, 4, 2]
Generation 530: Best = 2.899585e+03 FES = 53100 Diversity = 1.3863e-07 k_t
= [1, 1, 1, 1]
Generation 540: Best = 2.899585e+03 FES = 54100 Diversity = 1.3890e-07 k_t
= [2, 4, 4, 4]
Generation 550: Best = 2.899585e+03 FES = 55100 Diversity = 1.4707e-07 k_t
= [1, 1, 1, 1]
Generation 560: Best = 2.899585e+03 FES = 56100 Diversity = 1.4570e-07 k_t
= [2, 3, 4, 4]
Generation 570: Best = 2.899585e+03 FES = 57100 Diversity = 1.4638e-07 k_t
= [1, 1, 1, 1]
Generation 580: Best = 2.899585e+03 FES = 58100 Diversity = 1.4203e-07 k_t
= [4, 4, 2, 2]
Generation 590: Best = 2.899585e+03 FES = 59100 Diversity = 1.5034e-07 k_t
= [1, 1, 1, 1]
Generation 600: Best = 2.899585e+03 FES = 60100 Diversity = 1.5308e-07 k_t
= [2, 2, 3, 2]
Generation 610: Best = 2.899585e+03 FES = 61100 Diversity = 1.5572e-07 k_t
= [1, 1, 1, 1]
Generation 620: Best = 2.899585e+03 FES = 62100 Diversity = 1.5006e-07 k_t
= [4, 4, 3, 2]
Generation 630: Best = 2.899585e+03 FES = 63100 Diversity = 1.4968e-07 k_t
= [1, 1, 1, 1]
Generation 640: Best = 2.899585e+03 FES = 64100 Diversity = 1.4294e-07 k_t
= [4, 2, 4, 2]
Generation 650: Best = 2.899585e+03 FES = 65100 Diversity = 1.4902e-07 k_t
= [1, 1, 1, 1]
Generation 660: Best = 2.899585e+03 FES = 66100 Diversity = 1.4810e-07 k_t
= [4, 4, 4, 3]
Generation 670: Best = 2.899585e+03 FES = 67100 Diversity = 1.4265e-07 k_t
= [1, 1, 1, 1]
Generation 680: Best = 2.899585e+03 FES = 68100 Diversity = 1.4100e-07 k_t
= [4, 2, 3, 3]
Generation 690: Best = 2.899585e+03 FES = 69100 Diversity = 1.4585e-07 k_t
= [1, 1, 1, 1]
Generation 700: Best = 2.899585e+03 FES = 70100 Diversity = 1.4361e-07 k_t
= [2, 2, 2, 3]
Generation 710: Best = 2.899585e+03 FES = 71100 Diversity = 1.4994e-07 k_t
= [1, 1, 1, 1]
Generation 720: Best = 2.899585e+03 FES = 72100 Diversity = 1.5657e-07 k_t
= [4, 2, 2, 2]
Generation 730: Best = 2.899585e+03 FES = 73100 Diversity = 1.6102e-07 k_t
= [1, 1, 1, 1]
Generation 740: Best = 2.899585e+03 FES = 74100 Diversity = 1.5024e-07 k_t
= [4, 3, 2, 4]
Generation 750: Best = 2.899585e+03 FES = 75100 Diversity = 1.5420e-07 k_t
= [1, 1, 1, 1]
Generation 760: Best = 2.899585e+03 FES = 76100 Diversity = 1.4719e-07 k_t

```

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= [4, 4, 2, 4]
Generation 770: Best = 2.899585e+03  FES = 77100  Diversity = 1.4725e-07  k_t
= [1, 1, 1, 1]
Generation 780: Best = 2.899585e+03  FES = 78100  Diversity = 1.4922e-07  k_t
= [3, 3, 4, 2]
Generation 790: Best = 2.899585e+03  FES = 79100  Diversity = 1.4696e-07  k_t
= [1, 1, 1, 1]
Generation 800: Best = 2.899585e+03  FES = 80100  Diversity = 1.4520e-07  k_t
= [2, 4, 3, 2]
Generation 810: Best = 2.899585e+03  FES = 81100  Diversity = 1.3913e-07  k_t
= [1, 1, 1, 1]
Generation 820: Best = 2.899585e+03  FES = 82100  Diversity = 1.3759e-07  k_t
= [3, 2, 4, 4]
Generation 830: Best = 2.899585e+03  FES = 83100  Diversity = 1.4174e-07  k_t
= [1, 1, 1, 1]
Generation 840: Best = 2.899585e+03  FES = 84100  Diversity = 1.3806e-07  k_t
= [3, 2, 2, 2]
Generation 850: Best = 2.899585e+03  FES = 85100  Diversity = 1.4784e-07  k_t
= [1, 1, 1, 1]
Generation 860: Best = 2.899585e+03  FES = 86100  Diversity = 1.4830e-07  k_t
= [4, 4, 3, 2]
Generation 870: Best = 2.899585e+03  FES = 87100  Diversity = 1.4832e-07  k_t
= [1, 1, 1, 1]
Generation 880: Best = 2.899585e+03  FES = 88100  Diversity = 1.4982e-07  k_t
= [3, 4, 3, 3]
Generation 890: Best = 2.899585e+03  FES = 89100  Diversity = 1.4708e-07  k_t
= [1, 1, 1, 1]
Generation 900: Best = 2.899585e+03  FES = 90100  Diversity = 1.4181e-07  k_t
= [4, 4, 4, 2]
Generation 910: Best = 2.899585e+03  FES = 91100  Diversity = 1.4355e-07  k_t
= [1, 1, 1, 1]
Generation 920: Best = 2.899585e+03  FES = 92100  Diversity = 1.5134e-07  k_t
= [3, 3, 3, 4]
Generation 930: Best = 2.899585e+03  FES = 93100  Diversity = 1.5647e-07  k_t
= [1, 1, 1, 1]
Generation 940: Best = 2.899585e+03  FES = 94100  Diversity = 1.4945e-07  k_t
= [2, 2, 2, 3]
Generation 950: Best = 2.899585e+03  FES = 95100  Diversity = 1.4979e-07  k_t
= [1, 1, 1, 1]
Generation 960: Best = 2.899585e+03  FES = 96100  Diversity = 1.5400e-07  k_t
= [2, 3, 2, 3]
Generation 970: Best = 2.899585e+03  FES = 97100  Diversity = 1.5609e-07  k_t
= [1, 1, 1, 1]
Generation 980: Best = 2.899585e+03  FES = 98100  Diversity = 1.5058e-07  k_t
= [2, 3, 4, 2]
Generation 990: Best = 2.899585e+03  FES = 99100  Diversity = 1.5285e-07  k_t
= [1, 1, 1, 1]

```

Optimization completed!

Final Best Fitness = 2.899585e+03

Total Function Evaluations = 100000

```

=====
=

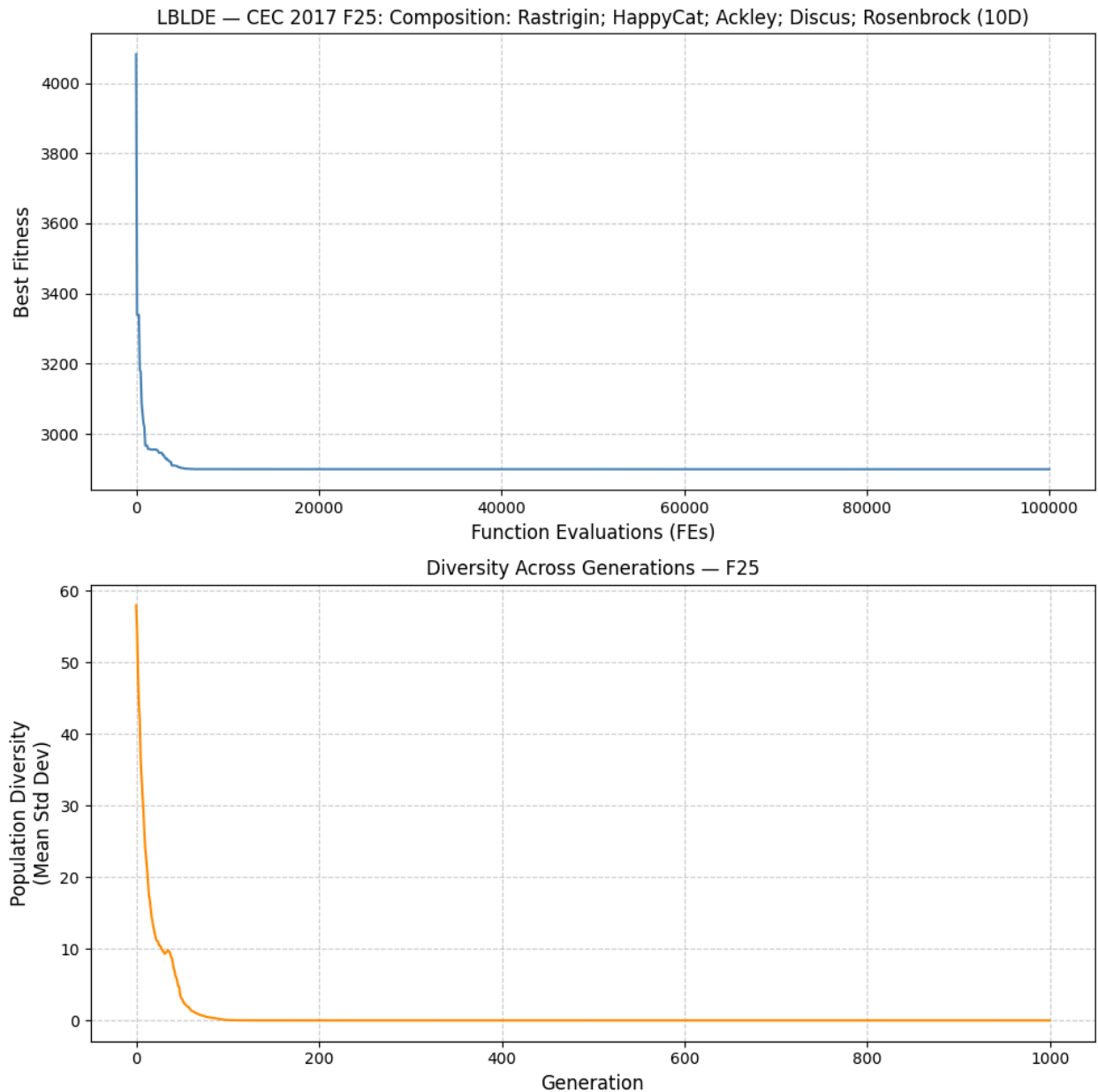
```


Results:

```
Best Fitness      : 2.899585e+03
Global Optimum    : 2.500000e+03
Error (|best-optimum|) : 3.995850e+02
Best Solution (5 dim) : [-9.6131142  46.49640673 12.3841954  1.33320085 1
1.74913676]
D^(0) (init. diversity): 5.7928e+01
```

=====

==



Graph saved as convergence_F25.png

```
In [42]: # =====
# F26 — Composition Function 6 (Schaffer; Schwefel; Griewank; Rosenbrock; Rast
# =====
run_and_plot('F26', make_wrapper(composition.f26), 2600.0, 'Composition: Schaf
```

```

=====
=
LBLDE_s on CEC 2017 F26 – Single Run
=====
=
Function      : Composition: Schaffer; Schwefel; Griewank; Rosenbrock; Rastrig
in (10D)
Bounds       : [-100, 100]
Global optimum : 2600.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 4.061684e+03 | Diversity = 5.7928e+01
Generation 10: Best = 3.186835e+03 FES = 1100 Diversity = 3.0325e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.043319e+03 FES = 2100 Diversity = 1.5138e+01 k_t
= [4, 3, 2, 3]
Generation 30: Best = 2.967395e+03 FES = 3100 Diversity = 9.1971e+00 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.925129e+03 FES = 4100 Diversity = 5.0020e+00 k_t
= [4, 3, 4, 3]
Generation 50: Best = 2.908144e+03 FES = 5100 Diversity = 1.8627e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.902292e+03 FES = 6100 Diversity = 6.6481e-01 k_t
= [3, 3, 2, 4]
Generation 70: Best = 2.900911e+03 FES = 7100 Diversity = 2.4142e-01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.900260e+03 FES = 8100 Diversity = 8.1501e-02 k_t
= [2, 4, 4, 4]
Generation 90: Best = 2.900066e+03 FES = 9100 Diversity = 2.8099e-02 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.900024e+03 FES = 10100 Diversity = 9.0353e-03 k_t
= [2, 2, 3, 3]
Generation 110: Best = 2.900011e+03 FES = 11100 Diversity = 2.8765e-03 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.900004e+03 FES = 12100 Diversity = 1.0836e-03 k_t
= [2, 3, 2, 4]
Generation 130: Best = 2.900001e+03 FES = 13100 Diversity = 4.2072e-04 k_t
= [1, 1, 1, 1]
Generation 140: Best = 2.900001e+03 FES = 14100 Diversity = 1.5908e-04 k_t
= [2, 3, 2, 4]
Generation 150: Best = 2.900000e+03 FES = 15100 Diversity = 5.5080e-05 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.900000e+03 FES = 16100 Diversity = 1.9209e-05 k_t
= [4, 2, 3, 2]
Generation 170: Best = 2.900000e+03 FES = 17100 Diversity = 7.4764e-06 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.900000e+03 FES = 18100 Diversity = 2.3307e-06 k_t
= [3, 4, 2, 4]
Generation 190: Best = 2.900000e+03 FES = 19100 Diversity = 8.0958e-07 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.900000e+03 FES = 20100 Diversity = 2.9479e-07 k_t
= [3, 2, 2, 4]
Generation 210: Best = 2.900000e+03 FES = 21100 Diversity = 1.1500e-07 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.900000e+03 FES = 22100 Diversity = 4.9679e-08 k_t

```

```

= [2, 2, 4, 4]
Generation 230: Best = 2.900000e+03 FES = 23100 Diversity = 1.9710e-08 k_t
= [1, 1, 1, 1]
Generation 240: Best = 2.900000e+03 FES = 24100 Diversity = 7.2450e-09 k_t
= [4, 4, 2, 4]
Generation 250: Best = 2.900000e+03 FES = 25100 Diversity = 2.5104e-09 k_t
= [1, 1, 1, 1]
Generation 260: Best = 2.900000e+03 FES = 26100 Diversity = 9.3517e-10 k_t
= [3, 4, 3, 4]
Generation 270: Best = 2.900000e+03 FES = 27100 Diversity = 3.4836e-10 k_t
= [1, 1, 1, 1]
Generation 280: Best = 2.900000e+03 FES = 28100 Diversity = 1.1813e-10 k_t
= [2, 2, 3, 3]
Generation 290: Best = 2.900000e+03 FES = 29100 Diversity = 4.5059e-11 k_t
= [1, 1, 1, 1]
Generation 300: Best = 2.900000e+03 FES = 30100 Diversity = 1.6662e-11 k_t
= [2, 4, 3, 4]
Generation 310: Best = 2.900000e+03 FES = 31100 Diversity = 6.7058e-12 k_t
= [1, 1, 1, 1]
Generation 320: Best = 2.900000e+03 FES = 32100 Diversity = 2.5330e-12 k_t
= [3, 4, 3, 4]
Generation 330: Best = 2.900000e+03 FES = 33100 Diversity = 9.0263e-13 k_t
= [1, 1, 1, 1]
Generation 340: Best = 2.900000e+03 FES = 34100 Diversity = 3.8351e-13 k_t
= [2, 2, 4, 3]
Generation 350: Best = 2.900000e+03 FES = 35100 Diversity = 1.8978e-13 k_t
= [1, 1, 1, 1]
Generation 360: Best = 2.900000e+03 FES = 36100 Diversity = 1.2305e-13 k_t
= [3, 4, 4, 4]
Generation 370: Best = 2.900000e+03 FES = 37100 Diversity = 9.0986e-14 k_t
= [1, 1, 1, 1]
Generation 380: Best = 2.900000e+03 FES = 38100 Diversity = 8.9621e-14 k_t
= [2, 2, 2, 2]
Generation 390: Best = 2.900000e+03 FES = 39100 Diversity = 7.2080e-14 k_t
= [1, 1, 1, 1]
Generation 400: Best = 2.900000e+03 FES = 40100 Diversity = 7.1845e-14 k_t
= [3, 3, 3, 3]
Generation 410: Best = 2.900000e+03 FES = 41100 Diversity = 6.7768e-14 k_t
= [1, 1, 1, 1]
Generation 420: Best = 2.900000e+03 FES = 42100 Diversity = 6.8759e-14 k_t
= [4, 3, 3, 4]
Generation 430: Best = 2.900000e+03 FES = 43100 Diversity = 6.6857e-14 k_t
= [1, 1, 1, 1]
Generation 440: Best = 2.900000e+03 FES = 44100 Diversity = 7.1619e-14 k_t
= [2, 3, 4, 3]
Generation 450: Best = 2.900000e+03 FES = 45100 Diversity = 6.8631e-14 k_t
= [1, 1, 1, 1]
Generation 460: Best = 2.900000e+03 FES = 46100 Diversity = 7.4170e-14 k_t
= [2, 3, 4, 3]
Generation 470: Best = 2.900000e+03 FES = 47100 Diversity = 6.2908e-14 k_t
= [1, 1, 1, 1]
Generation 480: Best = 2.900000e+03 FES = 48100 Diversity = 7.0482e-14 k_t
= [2, 3, 3, 2]
Generation 490: Best = 2.900000e+03 FES = 49100 Diversity = 6.3402e-14 k_t

```

```

= [1, 1, 1, 1]
Generation 500: Best = 2.900000e+03 FES = 50100 Diversity = 6.7338e-14 k_t
= [3, 3, 3, 4]
Generation 510: Best = 2.900000e+03 FES = 51100 Diversity = 6.6362e-14 k_t
= [1, 1, 1, 1]
Generation 520: Best = 2.900000e+03 FES = 52100 Diversity = 6.9762e-14 k_t
= [2, 3, 2, 3]
Generation 530: Best = 2.900000e+03 FES = 53100 Diversity = 6.3745e-14 k_t
= [1, 1, 1, 1]
Generation 540: Best = 2.900000e+03 FES = 54100 Diversity = 6.5585e-14 k_t
= [2, 2, 3, 2]
Generation 550: Best = 2.900000e+03 FES = 55100 Diversity = 6.3883e-14 k_t
= [1, 1, 1, 1]
Generation 560: Best = 2.900000e+03 FES = 56100 Diversity = 6.9957e-14 k_t
= [3, 4, 3, 2]
Generation 570: Best = 2.900000e+03 FES = 57100 Diversity = 7.0230e-14 k_t
= [1, 1, 1, 1]
Generation 580: Best = 2.900000e+03 FES = 58100 Diversity = 6.7305e-14 k_t
= [3, 4, 3, 4]
Generation 590: Best = 2.900000e+03 FES = 59100 Diversity = 6.7323e-14 k_t
= [1, 1, 1, 1]
Generation 600: Best = 2.900000e+03 FES = 60100 Diversity = 7.3973e-14 k_t
= [3, 3, 4, 3]
Generation 610: Best = 2.900000e+03 FES = 61100 Diversity = 6.9656e-14 k_t
= [1, 1, 1, 1]
Generation 620: Best = 2.900000e+03 FES = 62100 Diversity = 6.3907e-14 k_t
= [2, 4, 3, 2]
Generation 630: Best = 2.900000e+03 FES = 63100 Diversity = 6.7653e-14 k_t
= [1, 1, 1, 1]
Generation 640: Best = 2.900000e+03 FES = 64100 Diversity = 6.9201e-14 k_t
= [3, 4, 4, 2]
Generation 650: Best = 2.900000e+03 FES = 65100 Diversity = 7.1451e-14 k_t
= [1, 1, 1, 1]
Generation 660: Best = 2.900000e+03 FES = 66100 Diversity = 7.2571e-14 k_t
= [4, 4, 3, 4]
Generation 670: Best = 2.900000e+03 FES = 67100 Diversity = 7.1098e-14 k_t
= [1, 1, 1, 1]
Generation 680: Best = 2.900000e+03 FES = 68100 Diversity = 6.8815e-14 k_t
= [3, 3, 2, 4]
Generation 690: Best = 2.900000e+03 FES = 69100 Diversity = 6.8958e-14 k_t
= [1, 1, 1, 1]
Generation 700: Best = 2.900000e+03 FES = 70100 Diversity = 6.7212e-14 k_t
= [4, 2, 2, 2]
Generation 710: Best = 2.900000e+03 FES = 71100 Diversity = 6.6754e-14 k_t
= [1, 1, 1, 1]
Generation 720: Best = 2.900000e+03 FES = 72100 Diversity = 7.0889e-14 k_t
= [2, 2, 3, 4]
Generation 730: Best = 2.900000e+03 FES = 73100 Diversity = 6.7521e-14 k_t
= [1, 1, 1, 1]
Generation 740: Best = 2.900000e+03 FES = 74100 Diversity = 7.1136e-14 k_t
= [4, 2, 3, 3]
Generation 750: Best = 2.900000e+03 FES = 75100 Diversity = 7.0049e-14 k_t
= [1, 1, 1, 1]
Generation 760: Best = 2.900000e+03 FES = 76100 Diversity = 7.1366e-14 k_t

```

```

= [2, 2, 3, 3]
Generation 770: Best = 2.900000e+03 FES = 77100 Diversity = 6.8274e-14 k_t
= [1, 1, 1, 1]
Generation 780: Best = 2.900000e+03 FES = 78100 Diversity = 6.4566e-14 k_t
= [2, 3, 2, 3]
Generation 790: Best = 2.900000e+03 FES = 79100 Diversity = 7.0217e-14 k_t
= [1, 1, 1, 1]
Generation 800: Best = 2.900000e+03 FES = 80100 Diversity = 6.7198e-14 k_t
= [2, 2, 4, 2]
Generation 810: Best = 2.900000e+03 FES = 81100 Diversity = 6.9644e-14 k_t
= [1, 1, 1, 1]
Generation 820: Best = 2.900000e+03 FES = 82100 Diversity = 7.0330e-14 k_t
= [2, 3, 4, 2]
Generation 830: Best = 2.900000e+03 FES = 83100 Diversity = 7.1525e-14 k_t
= [1, 1, 1, 1]
Generation 840: Best = 2.900000e+03 FES = 84100 Diversity = 6.8219e-14 k_t
= [3, 3, 4, 4]
Generation 850: Best = 2.900000e+03 FES = 85100 Diversity = 6.8513e-14 k_t
= [1, 1, 1, 1]
Generation 860: Best = 2.900000e+03 FES = 86100 Diversity = 6.5510e-14 k_t
= [3, 3, 2, 4]
Generation 870: Best = 2.900000e+03 FES = 87100 Diversity = 5.9657e-14 k_t
= [1, 1, 1, 1]
Generation 880: Best = 2.900000e+03 FES = 88100 Diversity = 6.7985e-14 k_t
= [3, 4, 3, 3]
Generation 890: Best = 2.900000e+03 FES = 89100 Diversity = 5.9180e-14 k_t
= [1, 1, 1, 1]
Generation 900: Best = 2.900000e+03 FES = 90100 Diversity = 6.4478e-14 k_t
= [4, 4, 4, 3]
Generation 910: Best = 2.900000e+03 FES = 91100 Diversity = 5.9369e-14 k_t
= [1, 1, 1, 1]
Generation 920: Best = 2.900000e+03 FES = 92100 Diversity = 6.6089e-14 k_t
= [2, 3, 3, 4]
Generation 930: Best = 2.900000e+03 FES = 93100 Diversity = 6.4161e-14 k_t
= [1, 1, 1, 1]
Generation 940: Best = 2.900000e+03 FES = 94100 Diversity = 6.6623e-14 k_t
= [3, 4, 3, 4]
Generation 950: Best = 2.900000e+03 FES = 95100 Diversity = 6.6590e-14 k_t
= [1, 1, 1, 1]
Generation 960: Best = 2.900000e+03 FES = 96100 Diversity = 6.7158e-14 k_t
= [2, 2, 2, 2]
Generation 970: Best = 2.900000e+03 FES = 97100 Diversity = 6.9821e-14 k_t
= [1, 1, 1, 1]
Generation 980: Best = 2.900000e+03 FES = 98100 Diversity = 6.9787e-14 k_t
= [2, 4, 4, 3]
Generation 990: Best = 2.900000e+03 FES = 99100 Diversity = 6.9558e-14 k_t
= [1, 1, 1, 1]

```

Optimization completed!

Final Best Fitness = 2.900000e+03

Total Function Evaluations = 100000

=====

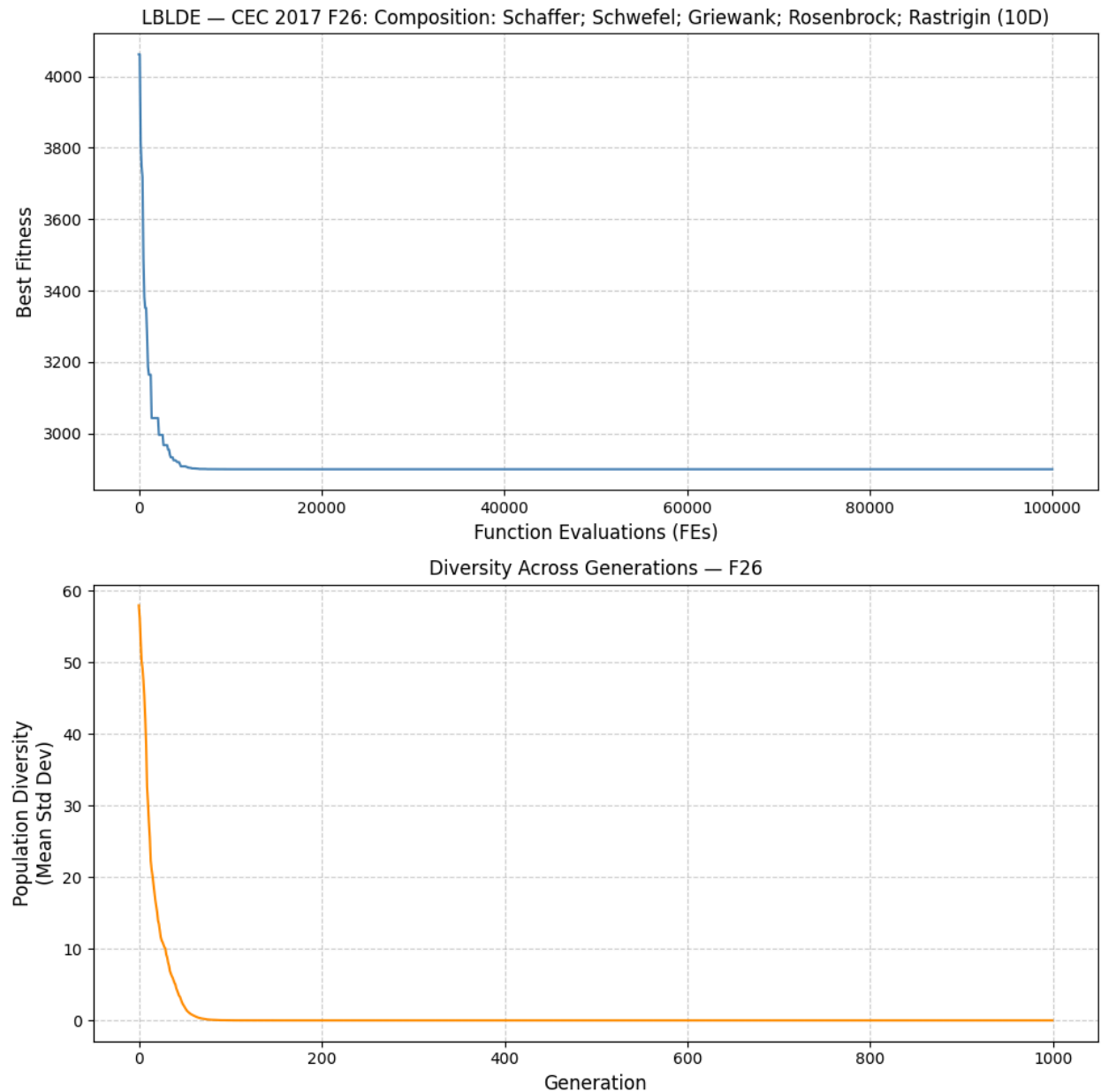
=

Results:

```
Best Fitness      : 2.900000e+03
Global Optimum    : 2.600000e+03
Error (|best-optimum|) : 3.000000e+02
Best Solution (5 dim) : [ 24.746414   -25.74629564   74.04615394  -62.35027147
35.80221832]
D^(0) (init. diversity): 5.7928e+01
```

=====

==



Graph saved as convergence_F26.png

```
In [43]: # =====
# F27 — Composition Function 7 (HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic)
# =====
run_and_plot('F27', make_wrapper(composition.f27), 2700.0, 'Composition: HGBat')
```

```

=====
=
LBLDE_s on CEC 2017 F27 – Single Run
=====
=
Function      : Composition: HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic;
Schaffer (10D)
Bounds       : [-100, 100]
Global optimum : 2700.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 3.284356e+03 | Diversity = 5.7928e+01
Generation 10: Best = 3.122769e+03 FES = 1100 Diversity = 4.0410e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.106440e+03 FES = 2100 Diversity = 1.6229e+01 k_t
= [4, 3, 2, 3]
Generation 30: Best = 3.100909e+03 FES = 3100 Diversity = 1.0555e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 3.096512e+03 FES = 4100 Diversity = 7.8556e+00 k_t
= [3, 4, 3, 2]
Generation 50: Best = 3.094903e+03 FES = 5100 Diversity = 7.7170e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 3.094599e+03 FES = 6100 Diversity = 7.0967e+00 k_t
= [2, 2, 2, 2]
Generation 70: Best = 3.092497e+03 FES = 7100 Diversity = 6.9220e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 3.092009e+03 FES = 8100 Diversity = 6.3248e+00 k_t
= [2, 4, 3, 2]
Generation 90: Best = 3.091332e+03 FES = 9100 Diversity = 5.6540e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 3.090607e+03 FES = 10100 Diversity = 4.3320e+00 k_t
= [4, 2, 3, 2]
Generation 110: Best = 3.090430e+03 FES = 11100 Diversity = 3.8276e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 3.089759e+03 FES = 12100 Diversity = 3.8043e+00 k_t
= [3, 2, 2, 3]
Generation 130: Best = 3.089496e+03 FES = 13100 Diversity = 4.0128e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 3.089332e+03 FES = 14100 Diversity = 2.9127e+00 k_t
= [2, 4, 4, 2]
Generation 150: Best = 3.089086e+03 FES = 15100 Diversity = 6.1429e-01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 3.089036e+03 FES = 16100 Diversity = 3.7141e-01 k_t
= [3, 4, 4, 3]
Generation 170: Best = 3.089022e+03 FES = 17100 Diversity = 2.1284e-01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.089009e+03 FES = 18100 Diversity = 1.1928e-01 k_t
= [4, 4, 2, 2]
Generation 190: Best = 3.089006e+03 FES = 19100 Diversity = 7.3225e-02 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.089006e+03 FES = 20100 Diversity = 4.0304e-02 k_t
= [4, 3, 3, 4]
Generation 210: Best = 3.089006e+03 FES = 21100 Diversity = 2.2946e-02 k_t
= [1, 1, 1, 1]
Generation 220: Best = 3.089005e+03 FES = 22100 Diversity = 1.2011e-02 k_t

```

```

= [2, 4, 3, 2]
Generation 230: Best = 3.089005e+03 FES = 23100 Diversity = 5.2943e-03 k_t
= [1, 1, 1, 1]
Generation 240: Best = 3.089005e+03 FES = 24100 Diversity = 2.6020e-03 k_t
= [4, 2, 3, 4]
Generation 250: Best = 3.089005e+03 FES = 25100 Diversity = 1.4271e-03 k_t
= [1, 1, 1, 1]
Generation 260: Best = 3.089005e+03 FES = 26100 Diversity = 7.2494e-04 k_t
= [4, 3, 4, 4]
Generation 270: Best = 3.089005e+03 FES = 27100 Diversity = 4.1281e-04 k_t
= [1, 1, 1, 1]
Generation 280: Best = 3.089005e+03 FES = 28100 Diversity = 1.9063e-04 k_t
= [3, 4, 3, 4]
Generation 290: Best = 3.089005e+03 FES = 29100 Diversity = 1.0933e-04 k_t
= [1, 1, 1, 1]
Generation 300: Best = 3.089005e+03 FES = 30100 Diversity = 6.8324e-05 k_t
= [3, 4, 4, 4]
Generation 310: Best = 3.089005e+03 FES = 31100 Diversity = 3.9911e-05 k_t
= [1, 1, 1, 1]
Generation 320: Best = 3.089005e+03 FES = 32100 Diversity = 2.0622e-05 k_t
= [3, 3, 3, 4]
Generation 330: Best = 3.089005e+03 FES = 33100 Diversity = 1.0632e-05 k_t
= [1, 1, 1, 1]
Generation 340: Best = 3.089005e+03 FES = 34100 Diversity = 6.0487e-06 k_t
= [4, 3, 4, 2]
Generation 350: Best = 3.089005e+03 FES = 35100 Diversity = 3.0217e-06 k_t
= [1, 1, 1, 1]
Generation 360: Best = 3.089005e+03 FES = 36100 Diversity = 1.5386e-06 k_t
= [4, 3, 2, 4]
Generation 370: Best = 3.089005e+03 FES = 37100 Diversity = 9.5158e-07 k_t
= [1, 1, 1, 1]
Generation 380: Best = 3.089005e+03 FES = 38100 Diversity = 6.9885e-07 k_t
= [3, 3, 3, 4]
Generation 390: Best = 3.089005e+03 FES = 39100 Diversity = 5.9505e-07 k_t
= [1, 1, 1, 1]
Generation 400: Best = 3.089005e+03 FES = 40100 Diversity = 4.3345e-07 k_t
= [4, 3, 4, 4]
Generation 410: Best = 3.089005e+03 FES = 41100 Diversity = 3.5864e-07 k_t
= [1, 1, 1, 1]
Generation 420: Best = 3.089005e+03 FES = 42100 Diversity = 3.5043e-07 k_t
= [3, 3, 2, 3]
Generation 430: Best = 3.089005e+03 FES = 43100 Diversity = 3.4734e-07 k_t
= [1, 1, 1, 1]
Generation 440: Best = 3.089005e+03 FES = 44100 Diversity = 3.6225e-07 k_t
= [3, 3, 3, 2]
Generation 450: Best = 3.089005e+03 FES = 45100 Diversity = 3.4001e-07 k_t
= [1, 1, 1, 1]
Generation 460: Best = 3.089005e+03 FES = 46100 Diversity = 3.5487e-07 k_t
= [3, 4, 2, 2]
Generation 470: Best = 3.089005e+03 FES = 47100 Diversity = 3.4974e-07 k_t
= [1, 1, 1, 1]
Generation 480: Best = 3.089005e+03 FES = 48100 Diversity = 3.4574e-07 k_t
= [4, 3, 2, 3]
Generation 490: Best = 3.089005e+03 FES = 49100 Diversity = 3.3333e-07 k_t

```



```

= [1, 1, 1, 1]
Generation 500: Best = 3.089005e+03 FES = 50100 Diversity = 3.3646e-07 k_t
= [2, 2, 2, 2]
Generation 510: Best = 3.089005e+03 FES = 51100 Diversity = 3.2921e-07 k_t
= [1, 1, 1, 1]
Generation 520: Best = 3.089005e+03 FES = 52100 Diversity = 3.0445e-07 k_t
= [4, 4, 2, 4]
Generation 530: Best = 3.089005e+03 FES = 53100 Diversity = 2.8121e-07 k_t
= [1, 1, 1, 1]
Generation 540: Best = 3.089005e+03 FES = 54100 Diversity = 2.9832e-07 k_t
= [4, 4, 4, 4]
Generation 550: Best = 3.089005e+03 FES = 55100 Diversity = 2.9204e-07 k_t
= [1, 1, 1, 1]
Generation 560: Best = 3.089005e+03 FES = 56100 Diversity = 3.2086e-07 k_t
= [2, 2, 3, 3]
Generation 570: Best = 3.089005e+03 FES = 57100 Diversity = 3.1367e-07 k_t
= [1, 1, 1, 1]
Generation 580: Best = 3.089005e+03 FES = 58100 Diversity = 3.1813e-07 k_t
= [3, 4, 3, 3]
Generation 590: Best = 3.089005e+03 FES = 59100 Diversity = 3.0220e-07 k_t
= [1, 1, 1, 1]
Generation 600: Best = 3.089005e+03 FES = 60100 Diversity = 3.1382e-07 k_t
= [3, 2, 4, 4]
Generation 610: Best = 3.089005e+03 FES = 61100 Diversity = 3.2152e-07 k_t
= [1, 1, 1, 1]
Generation 620: Best = 3.089005e+03 FES = 62100 Diversity = 3.1601e-07 k_t
= [4, 3, 4, 3]
Generation 630: Best = 3.089005e+03 FES = 63100 Diversity = 3.1396e-07 k_t
= [1, 1, 1, 1]
Generation 640: Best = 3.089005e+03 FES = 64100 Diversity = 3.0918e-07 k_t
= [3, 4, 4, 3]
Generation 650: Best = 3.089005e+03 FES = 65100 Diversity = 3.2609e-07 k_t
= [1, 1, 1, 1]
Generation 660: Best = 3.089005e+03 FES = 66100 Diversity = 3.1852e-07 k_t
= [2, 4, 2, 3]
Generation 670: Best = 3.089005e+03 FES = 67100 Diversity = 3.0784e-07 k_t
= [1, 1, 1, 1]
Generation 680: Best = 3.089005e+03 FES = 68100 Diversity = 3.3148e-07 k_t
= [2, 2, 3, 3]
Generation 690: Best = 3.089005e+03 FES = 69100 Diversity = 3.3546e-07 k_t
= [1, 1, 1, 1]
Generation 700: Best = 3.089005e+03 FES = 70100 Diversity = 3.2623e-07 k_t
= [4, 4, 3, 4]
Generation 710: Best = 3.089005e+03 FES = 71100 Diversity = 2.9647e-07 k_t
= [1, 1, 1, 1]
Generation 720: Best = 3.089005e+03 FES = 72100 Diversity = 3.1447e-07 k_t
= [4, 3, 4, 2]
Generation 730: Best = 3.089005e+03 FES = 73100 Diversity = 3.2545e-07 k_t
= [1, 1, 1, 1]
Generation 740: Best = 3.089005e+03 FES = 74100 Diversity = 2.9272e-07 k_t
= [2, 2, 3, 2]
Generation 750: Best = 3.089005e+03 FES = 75100 Diversity = 3.0893e-07 k_t
= [1, 1, 1, 1]
Generation 760: Best = 3.089005e+03 FES = 76100 Diversity = 2.9728e-07 k_t

```

```

= [4, 2, 2, 4]
Generation 770: Best = 3.089005e+03  FES = 77100  Diversity = 2.9046e-07  k_t
= [1, 1, 1, 1]
Generation 780: Best = 3.089005e+03  FES = 78100  Diversity = 3.0577e-07  k_t
= [2, 4, 3, 2]
Generation 790: Best = 3.089005e+03  FES = 79100  Diversity = 3.0939e-07  k_t
= [1, 1, 1, 1]
Generation 800: Best = 3.089005e+03  FES = 80100  Diversity = 2.9197e-07  k_t
= [4, 3, 4, 2]
Generation 810: Best = 3.089005e+03  FES = 81100  Diversity = 2.9304e-07  k_t
= [1, 1, 1, 1]
Generation 820: Best = 3.089005e+03  FES = 82100  Diversity = 2.9468e-07  k_t
= [4, 3, 3, 2]
Generation 830: Best = 3.089005e+03  FES = 83100  Diversity = 2.9101e-07  k_t
= [1, 1, 1, 1]
Generation 840: Best = 3.089005e+03  FES = 84100  Diversity = 3.0925e-07  k_t
= [4, 2, 2, 4]
Generation 850: Best = 3.089005e+03  FES = 85100  Diversity = 3.0036e-07  k_t
= [1, 1, 1, 1]
Generation 860: Best = 3.089005e+03  FES = 86100  Diversity = 2.9567e-07  k_t
= [3, 4, 4, 4]
Generation 870: Best = 3.089005e+03  FES = 87100  Diversity = 3.3292e-07  k_t
= [1, 1, 1, 1]
Generation 880: Best = 3.089005e+03  FES = 88100  Diversity = 3.4744e-07  k_t
= [2, 4, 2, 2]
Generation 890: Best = 3.089005e+03  FES = 89100  Diversity = 3.3473e-07  k_t
= [1, 1, 1, 1]
Generation 900: Best = 3.089005e+03  FES = 90100  Diversity = 3.4300e-07  k_t
= [2, 3, 2, 4]
Generation 910: Best = 3.089005e+03  FES = 91100  Diversity = 3.5711e-07  k_t
= [1, 1, 1, 1]
Generation 920: Best = 3.089005e+03  FES = 92100  Diversity = 3.3945e-07  k_t
= [4, 4, 4, 2]
Generation 930: Best = 3.089005e+03  FES = 93100  Diversity = 3.4395e-07  k_t
= [1, 1, 1, 1]
Generation 940: Best = 3.089005e+03  FES = 94100  Diversity = 3.2904e-07  k_t
= [2, 3, 2, 3]
Generation 950: Best = 3.089005e+03  FES = 95100  Diversity = 3.2232e-07  k_t
= [1, 1, 1, 1]
Generation 960: Best = 3.089005e+03  FES = 96100  Diversity = 3.1860e-07  k_t
= [4, 2, 3, 3]
Generation 970: Best = 3.089005e+03  FES = 97100  Diversity = 3.1792e-07  k_t
= [1, 1, 1, 1]
Generation 980: Best = 3.089005e+03  FES = 98100  Diversity = 2.9051e-07  k_t
= [2, 2, 2, 3]
Generation 990: Best = 3.089005e+03  FES = 99100  Diversity = 2.8671e-07  k_t
= [1, 1, 1, 1]

```

Optimization completed!

Final Best Fitness = 3.089005e+03

Total Function Evaluations = 100000

```

=====
=

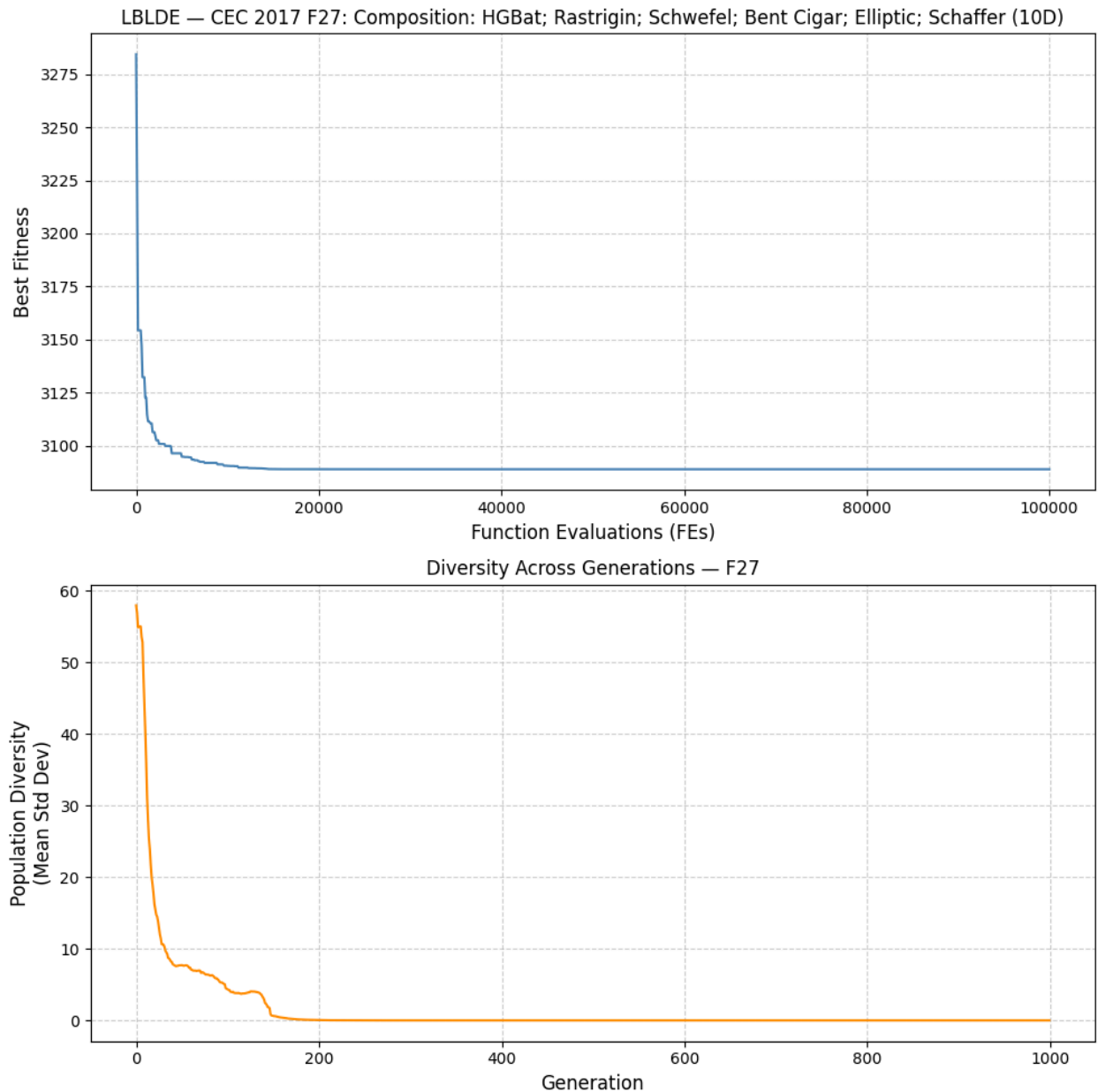
```

Results:

```
Best Fitness      : 3.089005e+03
Global Optimum    : 2.700000e+03
Error (|best-optimum|) : 3.890054e+02
Best Solution (5 dim) : [-84.4919629 -97.15006245 -89.64802745 -82.33315241
100.          ]
D^(0) (init. diversity): 5.7928e+01
```

=====

==



Graph saved as convergence_F27.png

```
In [44]: # =====
# F28 — Composition Function 8 (Ackley; Griewank; Discus; Rosenbrock; HappyCat)
# =====
run_and_plot('F28', make_wrapper(composition.f28), 2800.0, 'Composition: Ackle
```

```

=====
=
LBLDE_s on CEC 2017 F28 – Single Run
=====
=
Function      : Composition: Ackley; Griewank; Discus; Rosenbrock; HappyCat; S
chaffer (10D)
Bounds       : [-100, 100]
Global optimum : 2800.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 4.188047e+03 | Diversity = 5.7928e+01
Generation 10: Best = 3.300481e+03 FES = 1100 Diversity = 4.5652e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.242772e+03 FES = 2100 Diversity = 3.2120e+01 k_t
= [4, 3, 4, 2]
Generation 30: Best = 3.197963e+03 FES = 3100 Diversity = 1.9112e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 3.178863e+03 FES = 4100 Diversity = 1.1576e+01 k_t
= [3, 3, 4, 4]
Generation 50: Best = 3.172927e+03 FES = 5100 Diversity = 7.6343e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 3.169028e+03 FES = 6100 Diversity = 4.5915e+00 k_t
= [4, 2, 4, 4]
Generation 70: Best = 3.167262e+03 FES = 7100 Diversity = 2.6030e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 3.165626e+03 FES = 8100 Diversity = 1.7055e+00 k_t
= [4, 3, 2, 4]
Generation 90: Best = 3.165267e+03 FES = 9100 Diversity = 1.1815e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 3.164681e+03 FES = 10100 Diversity = 8.4709e-01 k_t
= [4, 4, 2, 4]
Generation 110: Best = 3.163958e+03 FES = 11100 Diversity = 7.9478e-01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 3.163222e+03 FES = 12100 Diversity = 7.3521e-01 k_t
= [4, 3, 4, 4]
Generation 130: Best = 3.161606e+03 FES = 13100 Diversity = 8.7978e-01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 3.159833e+03 FES = 14100 Diversity = 1.0680e+00 k_t
= [4, 2, 3, 4]
Generation 150: Best = 3.156364e+03 FES = 15100 Diversity = 9.5518e-01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 3.149110e+03 FES = 16100 Diversity = 1.2988e+00 k_t
= [2, 3, 2, 4]
Generation 170: Best = 3.128086e+03 FES = 17100 Diversity = 1.9141e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.110198e+03 FES = 18100 Diversity = 1.2805e+00 k_t
= [3, 4, 2, 4]
Generation 190: Best = 3.103270e+03 FES = 19100 Diversity = 6.3035e-01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.101490e+03 FES = 20100 Diversity = 2.8108e-01 k_t
= [3, 2, 2, 4]
Generation 210: Best = 3.100536e+03 FES = 21100 Diversity = 1.0356e-01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 3.100205e+03 FES = 22100 Diversity = 3.8864e-02 k_t

```

```

= [2, 2, 4, 4]
Generation 230: Best = 3.100059e+03 FES = 23100 Diversity = 1.5449e-02 k_t
= [1, 1, 1, 1]
Generation 240: Best = 3.100032e+03 FES = 24100 Diversity = 5.4877e-03 k_t
= [2, 3, 4, 4]
Generation 250: Best = 3.100007e+03 FES = 25100 Diversity = 2.4175e-03 k_t
= [1, 1, 1, 1]
Generation 260: Best = 3.100004e+03 FES = 26100 Diversity = 8.6984e-04 k_t
= [2, 2, 3, 4]
Generation 270: Best = 3.100002e+03 FES = 27100 Diversity = 3.6823e-04 k_t
= [1, 1, 1, 1]
Generation 280: Best = 3.100001e+03 FES = 28100 Diversity = 1.3526e-04 k_t
= [4, 4, 3, 2]
Generation 290: Best = 3.100000e+03 FES = 29100 Diversity = 4.9731e-05 k_t
= [1, 1, 1, 1]
Generation 300: Best = 3.100000e+03 FES = 30100 Diversity = 1.8290e-05 k_t
= [2, 4, 3, 4]
Generation 310: Best = 3.100000e+03 FES = 31100 Diversity = 7.0466e-06 k_t
= [1, 1, 1, 1]
Generation 320: Best = 3.100000e+03 FES = 32100 Diversity = 2.4409e-06 k_t
= [2, 4, 4, 3]
Generation 330: Best = 3.100000e+03 FES = 33100 Diversity = 9.7064e-07 k_t
= [1, 1, 1, 1]
Generation 340: Best = 3.100000e+03 FES = 34100 Diversity = 3.6570e-07 k_t
= [2, 2, 4, 3]
Generation 350: Best = 3.100000e+03 FES = 35100 Diversity = 1.3823e-07 k_t
= [1, 1, 1, 1]
Generation 360: Best = 3.100000e+03 FES = 36100 Diversity = 5.7027e-08 k_t
= [3, 4, 4, 4]
Generation 370: Best = 3.100000e+03 FES = 37100 Diversity = 2.2124e-08 k_t
= [1, 1, 1, 1]
Generation 380: Best = 3.100000e+03 FES = 38100 Diversity = 8.5845e-09 k_t
= [2, 2, 2, 2]
Generation 390: Best = 3.100000e+03 FES = 39100 Diversity = 3.3905e-09 k_t
= [1, 1, 1, 1]
Generation 400: Best = 3.100000e+03 FES = 40100 Diversity = 1.3988e-09 k_t
= [2, 4, 2, 3]
Generation 410: Best = 3.100000e+03 FES = 41100 Diversity = 5.5821e-10 k_t
= [1, 1, 1, 1]
Generation 420: Best = 3.100000e+03 FES = 42100 Diversity = 1.9951e-10 k_t
= [4, 3, 3, 4]
Generation 430: Best = 3.100000e+03 FES = 43100 Diversity = 6.7244e-11 k_t
= [1, 1, 1, 1]
Generation 440: Best = 3.100000e+03 FES = 44100 Diversity = 2.5724e-11 k_t
= [2, 3, 3, 2]
Generation 450: Best = 3.100000e+03 FES = 45100 Diversity = 1.0257e-11 k_t
= [1, 1, 1, 1]
Generation 460: Best = 3.100000e+03 FES = 46100 Diversity = 3.4462e-12 k_t
= [3, 4, 2, 2]
Generation 470: Best = 3.100000e+03 FES = 47100 Diversity = 1.3155e-12 k_t
= [1, 1, 1, 1]
Generation 480: Best = 3.100000e+03 FES = 48100 Diversity = 4.9869e-13 k_t
= [3, 3, 4, 3]
Generation 490: Best = 3.100000e+03 FES = 49100 Diversity = 2.1553e-13 k_t

```

```

= [1, 1, 1, 1]
Generation 500: Best = 3.100000e+03 FES = 50100 Diversity = 1.1082e-13 k_t
= [3, 3, 3, 4]
Generation 510: Best = 3.100000e+03 FES = 51100 Diversity = 8.7138e-14 k_t
= [1, 1, 1, 1]
Generation 520: Best = 3.100000e+03 FES = 52100 Diversity = 6.6515e-14 k_t
= [3, 3, 2, 4]
Generation 530: Best = 3.100000e+03 FES = 53100 Diversity = 6.5700e-14 k_t
= [1, 1, 1, 1]
Generation 540: Best = 3.100000e+03 FES = 54100 Diversity = 6.3912e-14 k_t
= [4, 3, 3, 3]
Generation 550: Best = 3.100000e+03 FES = 55100 Diversity = 5.6248e-14 k_t
= [1, 1, 1, 1]
Generation 560: Best = 3.100000e+03 FES = 56100 Diversity = 5.2277e-14 k_t
= [3, 4, 4, 2]
Generation 570: Best = 3.100000e+03 FES = 57100 Diversity = 6.0877e-14 k_t
= [1, 1, 1, 1]
Generation 580: Best = 3.100000e+03 FES = 58100 Diversity = 5.8719e-14 k_t
= [2, 3, 3, 3]
Generation 590: Best = 3.100000e+03 FES = 59100 Diversity = 5.7316e-14 k_t
= [1, 1, 1, 1]
Generation 600: Best = 3.100000e+03 FES = 60100 Diversity = 5.7702e-14 k_t
= [2, 3, 2, 4]
Generation 610: Best = 3.100000e+03 FES = 61100 Diversity = 5.4677e-14 k_t
= [1, 1, 1, 1]
Generation 620: Best = 3.100000e+03 FES = 62100 Diversity = 5.7307e-14 k_t
= [2, 4, 2, 3]
Generation 630: Best = 3.100000e+03 FES = 63100 Diversity = 5.0294e-14 k_t
= [1, 1, 1, 1]
Generation 640: Best = 3.100000e+03 FES = 64100 Diversity = 5.5067e-14 k_t
= [2, 3, 4, 3]
Generation 650: Best = 3.100000e+03 FES = 65100 Diversity = 5.1211e-14 k_t
= [1, 1, 1, 1]
Generation 660: Best = 3.100000e+03 FES = 66100 Diversity = 5.0262e-14 k_t
= [3, 3, 4, 4]
Generation 670: Best = 3.100000e+03 FES = 67100 Diversity = 5.3706e-14 k_t
= [1, 1, 1, 1]
Generation 680: Best = 3.100000e+03 FES = 68100 Diversity = 5.1972e-14 k_t
= [4, 2, 2, 3]
Generation 690: Best = 3.100000e+03 FES = 69100 Diversity = 4.8348e-14 k_t
= [1, 1, 1, 1]
Generation 700: Best = 3.100000e+03 FES = 70100 Diversity = 5.5974e-14 k_t
= [4, 2, 4, 2]
Generation 710: Best = 3.100000e+03 FES = 71100 Diversity = 5.4065e-14 k_t
= [1, 1, 1, 1]
Generation 720: Best = 3.100000e+03 FES = 72100 Diversity = 5.3852e-14 k_t
= [2, 2, 3, 4]
Generation 730: Best = 3.100000e+03 FES = 73100 Diversity = 4.8707e-14 k_t
= [1, 1, 1, 1]
Generation 740: Best = 3.100000e+03 FES = 74100 Diversity = 5.3302e-14 k_t
= [2, 3, 2, 3]
Generation 750: Best = 3.100000e+03 FES = 75100 Diversity = 4.8115e-14 k_t
= [1, 1, 1, 1]
Generation 760: Best = 3.100000e+03 FES = 76100 Diversity = 4.9766e-14 k_t

```

```

= [2, 2, 2, 2]
Generation 770: Best = 3.100000e+03 FES = 77100 Diversity = 5.1001e-14 k_t
= [1, 1, 1, 1]
Generation 780: Best = 3.100000e+03 FES = 78100 Diversity = 5.5056e-14 k_t
= [3, 3, 4, 2]
Generation 790: Best = 3.100000e+03 FES = 79100 Diversity = 4.8401e-14 k_t
= [1, 1, 1, 1]
Generation 800: Best = 3.100000e+03 FES = 80100 Diversity = 4.8821e-14 k_t
= [4, 3, 2, 4]
Generation 810: Best = 3.100000e+03 FES = 81100 Diversity = 4.6279e-14 k_t
= [1, 1, 1, 1]
Generation 820: Best = 3.100000e+03 FES = 82100 Diversity = 5.0379e-14 k_t
= [2, 2, 2, 2]
Generation 830: Best = 3.100000e+03 FES = 83100 Diversity = 4.7501e-14 k_t
= [1, 1, 1, 1]
Generation 840: Best = 3.100000e+03 FES = 84100 Diversity = 4.3978e-14 k_t
= [2, 2, 2, 2]
Generation 850: Best = 3.100000e+03 FES = 85100 Diversity = 4.7877e-14 k_t
= [1, 1, 1, 1]
Generation 860: Best = 3.100000e+03 FES = 86100 Diversity = 5.3133e-14 k_t
= [2, 2, 3, 4]
Generation 870: Best = 3.100000e+03 FES = 87100 Diversity = 4.6194e-14 k_t
= [1, 1, 1, 1]
Generation 880: Best = 3.100000e+03 FES = 88100 Diversity = 5.1289e-14 k_t
= [2, 4, 3, 2]
Generation 890: Best = 3.100000e+03 FES = 89100 Diversity = 5.2592e-14 k_t
= [1, 1, 1, 1]
Generation 900: Best = 3.100000e+03 FES = 90100 Diversity = 5.1317e-14 k_t
= [3, 4, 4, 4]
Generation 910: Best = 3.100000e+03 FES = 91100 Diversity = 5.3014e-14 k_t
= [1, 1, 1, 1]
Generation 920: Best = 3.100000e+03 FES = 92100 Diversity = 4.8919e-14 k_t
= [2, 3, 3, 4]
Generation 930: Best = 3.100000e+03 FES = 93100 Diversity = 4.6475e-14 k_t
= [1, 1, 1, 1]
Generation 940: Best = 3.100000e+03 FES = 94100 Diversity = 5.3063e-14 k_t
= [3, 4, 3, 4]
Generation 950: Best = 3.100000e+03 FES = 95100 Diversity = 5.0320e-14 k_t
= [1, 1, 1, 1]
Generation 960: Best = 3.100000e+03 FES = 96100 Diversity = 5.3769e-14 k_t
= [2, 3, 2, 3]
Generation 970: Best = 3.100000e+03 FES = 97100 Diversity = 5.0351e-14 k_t
= [1, 1, 1, 1]
Generation 980: Best = 3.100000e+03 FES = 98100 Diversity = 4.9335e-14 k_t
= [2, 4, 4, 3]
Generation 990: Best = 3.100000e+03 FES = 99100 Diversity = 5.0204e-14 k_t
= [1, 1, 1, 1]

```

Optimization completed!

Final Best Fitness = 3.100000e+03

Total Function Evaluations = 100000

=====

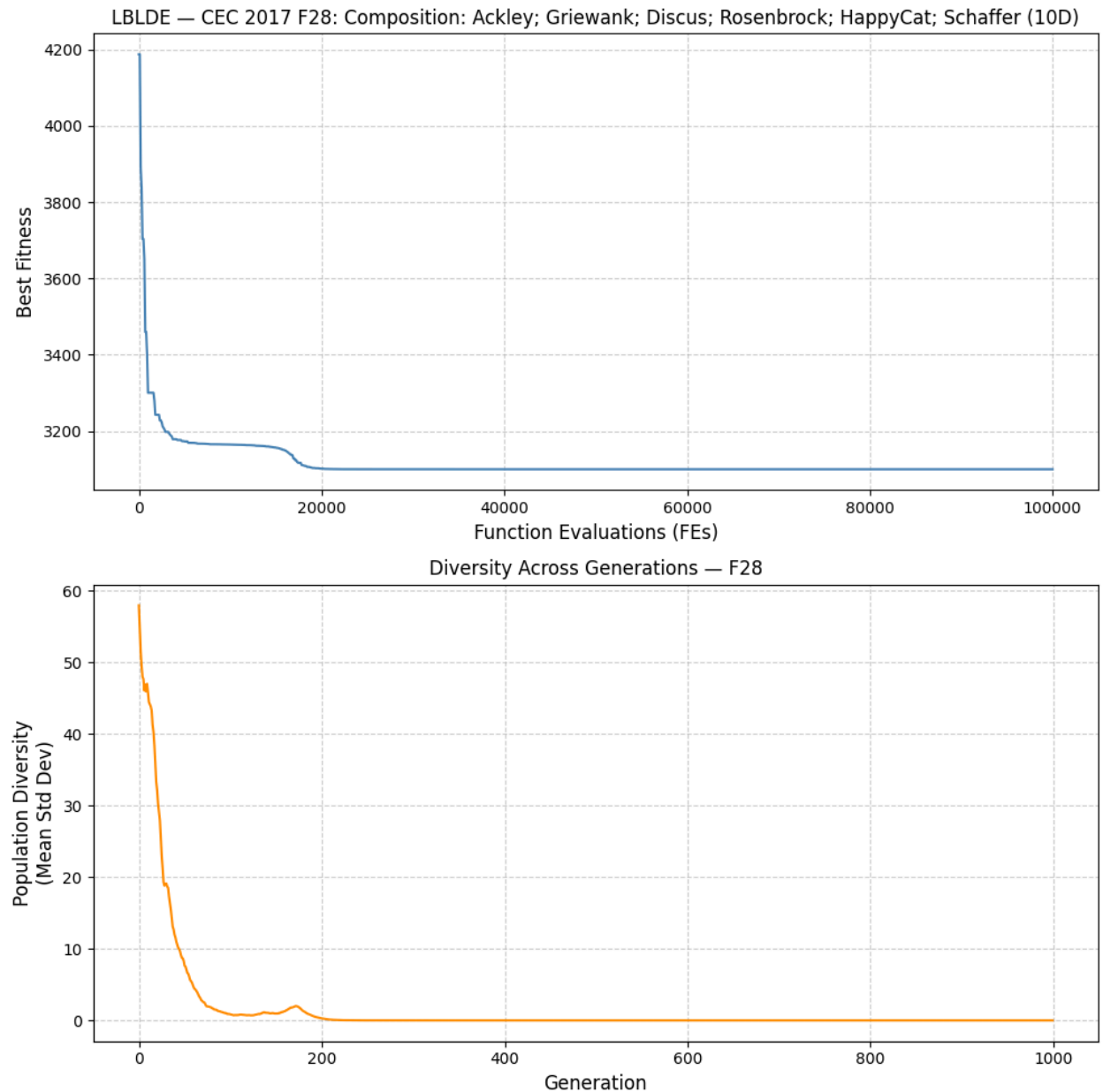
=

Results:

```
Best Fitness      : 3.100000e+03
Global Optimum    : 2.800000e+03
Error (|best-optimum|) : 3.000000e+02
Best Solution (5 dim) : [-26.37193452 -34.00844064 -49.92961782  47.6166024
-66.54538272]
D^(0) (init. diversity): 5.7928e+01
```

=====

==



Graph saved as convergence_F28.png

```
In [45]: # =====
# F29 — Composition Function 9 (F15; F16; F17)
# =====
run_and_plot('F29', make_wrapper(composition.f29), 2900.0, 'Composition: F15;
```



```

=====
=
LBLDE_s on CEC 2017 F29 – Single Run
=====
=
Function      : Composition: F15; F16; F17 (10D)
Bounds       : [-100, 100]
Global optimum : 2900.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 3.805371e+03 | Diversity = 5.7928e+01
Generation 10: Best = 3.450223e+03 FES = 1100 Diversity = 5.1419e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.266647e+03 FES = 2100 Diversity = 4.5581e+01 k_t
= [4, 3, 2, 3]
Generation 30: Best = 3.260750e+03 FES = 3100 Diversity = 4.3483e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 3.260750e+03 FES = 4100 Diversity = 4.2832e+01 k_t
= [4, 3, 4, 3]
Generation 50: Best = 3.226848e+03 FES = 5100 Diversity = 4.1744e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 3.226848e+03 FES = 6100 Diversity = 4.0739e+01 k_t
= [3, 4, 4, 3]
Generation 70: Best = 3.204778e+03 FES = 7100 Diversity = 3.9637e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 3.202800e+03 FES = 8100 Diversity = 3.6753e+01 k_t
= [4, 4, 3, 3]
Generation 90: Best = 3.179792e+03 FES = 9100 Diversity = 3.6260e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 3.179792e+03 FES = 10100 Diversity = 3.6084e+01 k_t
= [2, 3, 4, 4]
Generation 110: Best = 3.179792e+03 FES = 11100 Diversity = 3.6173e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 3.179792e+03 FES = 12100 Diversity = 3.6108e+01 k_t
= [2, 2, 4, 4]
Generation 130: Best = 3.179792e+03 FES = 13100 Diversity = 3.4882e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 3.179792e+03 FES = 14100 Diversity = 3.4534e+01 k_t
= [4, 4, 4, 2]
Generation 150: Best = 3.167376e+03 FES = 15100 Diversity = 3.3843e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 3.167376e+03 FES = 16100 Diversity = 3.3658e+01 k_t
= [2, 3, 3, 3]
Generation 170: Best = 3.167376e+03 FES = 17100 Diversity = 3.3631e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.167376e+03 FES = 18100 Diversity = 3.2785e+01 k_t
= [4, 2, 4, 4]
Generation 190: Best = 3.167268e+03 FES = 19100 Diversity = 3.2414e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.161744e+03 FES = 20100 Diversity = 3.2001e+01 k_t
= [3, 2, 2, 3]
Generation 210: Best = 3.152027e+03 FES = 21100 Diversity = 3.1714e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 3.149777e+03 FES = 22100 Diversity = 3.1149e+01 k_t
= [4, 3, 3, 3]

```

Generation 230:	Best = 3.149777e+03	FES = 23100	Diversity = 3.0913e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 3.149777e+03	FES = 24100	Diversity = 3.0038e+01	k_t = [4, 3, 2, 2]
Generation 250:	Best = 3.149777e+03	FES = 25100	Diversity = 2.9506e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 3.149777e+03	FES = 26100	Diversity = 2.9263e+01	k_t = [4, 3, 3, 3]
Generation 270:	Best = 3.149777e+03	FES = 27100	Diversity = 2.9529e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 3.149777e+03	FES = 28100	Diversity = 2.9557e+01	k_t = [2, 2, 2, 3]
Generation 290:	Best = 3.149777e+03	FES = 29100	Diversity = 3.0044e+01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 3.149777e+03	FES = 30100	Diversity = 2.9978e+01	k_t = [4, 3, 3, 4]
Generation 310:	Best = 3.149513e+03	FES = 31100	Diversity = 2.9908e+01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 3.149513e+03	FES = 32100	Diversity = 2.9664e+01	k_t = [3, 4, 4, 3]
Generation 330:	Best = 3.149513e+03	FES = 33100	Diversity = 2.9561e+01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 3.149513e+03	FES = 34100	Diversity = 2.9262e+01	k_t = [4, 4, 3, 4]
Generation 350:	Best = 3.149084e+03	FES = 35100	Diversity = 2.7720e+01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 3.148218e+03	FES = 36100	Diversity = 2.7489e+01	k_t = [3, 4, 3, 4]
Generation 370:	Best = 3.148218e+03	FES = 37100	Diversity = 2.7358e+01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 3.146367e+03	FES = 38100	Diversity = 2.7313e+01	k_t = [3, 2, 4, 3]
Generation 390:	Best = 3.146211e+03	FES = 39100	Diversity = 2.7098e+01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 3.146211e+03	FES = 40100	Diversity = 2.7082e+01	k_t = [4, 2, 2, 2]
Generation 410:	Best = 3.145595e+03	FES = 41100	Diversity = 2.7029e+01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 3.145595e+03	FES = 42100	Diversity = 2.6200e+01	k_t = [4, 3, 3, 2]
Generation 430:	Best = 3.145595e+03	FES = 43100	Diversity = 2.4646e+01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 3.145595e+03	FES = 44100	Diversity = 2.4271e+01	k_t = [2, 4, 3, 3]
Generation 450:	Best = 3.145595e+03	FES = 45100	Diversity = 2.3549e+01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 3.145595e+03	FES = 46100	Diversity = 2.2928e+01	k_t = [2, 3, 4, 4]
Generation 470:	Best = 3.145595e+03	FES = 47100	Diversity = 2.2028e+01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 3.145556e+03	FES = 48100	Diversity = 2.0976e+01	k_t = [3, 4, 3, 4]
Generation 490:	Best = 3.145556e+03	FES = 49100	Diversity = 2.0734e+01	k_t = [1, 1, 1, 1]

Generation 500:	Best = 3.145556e+03	FES = 50100	Diversity = 2.0336e+01	k_t = [3, 4, 4, 4]
Generation 510:	Best = 3.144029e+03	FES = 51100	Diversity = 1.9954e+01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 3.144029e+03	FES = 52100	Diversity = 2.0361e+01	k_t = [3, 4, 4, 2]
Generation 530:	Best = 3.142775e+03	FES = 53100	Diversity = 1.8843e+01	k_t = [1, 1, 1, 1]
Generation 540:	Best = 3.142342e+03	FES = 54100	Diversity = 1.8698e+01	k_t = [4, 4, 4, 3]
Generation 550:	Best = 3.142182e+03	FES = 55100	Diversity = 1.8568e+01	k_t = [1, 1, 1, 1]
Generation 560:	Best = 3.141575e+03	FES = 56100	Diversity = 1.7883e+01	k_t = [4, 3, 4, 4]
Generation 570:	Best = 3.140504e+03	FES = 57100	Diversity = 1.7076e+01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 3.139929e+03	FES = 58100	Diversity = 1.7122e+01	k_t = [3, 4, 3, 4]
Generation 590:	Best = 3.137378e+03	FES = 59100	Diversity = 1.6256e+01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 3.136521e+03	FES = 60100	Diversity = 1.4486e+01	k_t = [4, 3, 3, 4]
Generation 610:	Best = 3.135876e+03	FES = 61100	Diversity = 1.0971e+01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 3.135209e+03	FES = 62100	Diversity = 5.5697e+00	k_t = [4, 2, 4, 3]
Generation 630:	Best = 3.134563e+03	FES = 63100	Diversity = 2.5576e+00	k_t = [1, 1, 1, 1]
Generation 640:	Best = 3.134303e+03	FES = 64100	Diversity = 9.0492e-01	k_t = [3, 3, 2, 3]
Generation 650:	Best = 3.133737e+03	FES = 65100	Diversity = 1.0371e+00	k_t = [1, 1, 1, 1]
Generation 660:	Best = 3.133737e+03	FES = 66100	Diversity = 1.0605e+00	k_t = [3, 3, 4, 3]
Generation 670:	Best = 3.132781e+03	FES = 67100	Diversity = 1.1247e+00	k_t = [1, 1, 1, 1]
Generation 680:	Best = 3.132359e+03	FES = 68100	Diversity = 1.2138e+00	k_t = [3, 4, 3, 4]
Generation 690:	Best = 3.132295e+03	FES = 69100	Diversity = 1.2123e+00	k_t = [1, 1, 1, 1]
Generation 700:	Best = 3.132295e+03	FES = 70100	Diversity = 1.3175e+00	k_t = [2, 3, 3, 2]
Generation 710:	Best = 3.132295e+03	FES = 71100	Diversity = 1.3928e+00	k_t = [1, 1, 1, 1]
Generation 720:	Best = 3.132295e+03	FES = 72100	Diversity = 1.4381e+00	k_t = [3, 2, 4, 2]
Generation 730:	Best = 3.132087e+03	FES = 73100	Diversity = 1.5120e+00	k_t = [1, 1, 1, 1]
Generation 740:	Best = 3.132087e+03	FES = 74100	Diversity = 1.4950e+00	k_t = [3, 3, 3, 2]
Generation 750:	Best = 3.132087e+03	FES = 75100	Diversity = 1.5266e+00	k_t = [1, 1, 1, 1]
Generation 760:	Best = 3.132050e+03	FES = 76100	Diversity = 1.4815e+00	k_t = [2, 4, 4, 3]

Generation 770:	Best = 3.131617e+03	FES = 77100	Diversity = 1.5088e+00	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 3.131617e+03	FES = 78100	Diversity = 1.4923e+00	k_t
= [2, 3, 3, 4]				
Generation 790:	Best = 3.130509e+03	FES = 79100	Diversity = 1.5733e+00	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 3.130509e+03	FES = 80100	Diversity = 1.5429e+00	k_t
= [4, 4, 2, 2]				
Generation 810:	Best = 3.130509e+03	FES = 81100	Diversity = 1.5406e+00	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 3.130509e+03	FES = 82100	Diversity = 1.5330e+00	k_t
= [4, 4, 2, 2]				
Generation 830:	Best = 3.130509e+03	FES = 83100	Diversity = 1.5470e+00	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 3.130509e+03	FES = 84100	Diversity = 1.5833e+00	k_t
= [4, 3, 3, 4]				
Generation 850:	Best = 3.130509e+03	FES = 85100	Diversity = 1.6029e+00	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 3.130509e+03	FES = 86100	Diversity = 1.5835e+00	k_t
= [2, 4, 2, 3]				
Generation 870:	Best = 3.130509e+03	FES = 87100	Diversity = 1.5677e+00	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 3.130507e+03	FES = 88100	Diversity = 1.5607e+00	k_t
= [3, 3, 4, 3]				
Generation 890:	Best = 3.130500e+03	FES = 89100	Diversity = 1.5530e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 3.130500e+03	FES = 90100	Diversity = 1.5140e+00	k_t
= [2, 3, 3, 3]				
Generation 910:	Best = 3.130500e+03	FES = 91100	Diversity = 1.4538e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 3.130500e+03	FES = 92100	Diversity = 1.4543e+00	k_t
= [4, 2, 4, 2]				
Generation 930:	Best = 3.130463e+03	FES = 93100	Diversity = 1.4228e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 3.130453e+03	FES = 94100	Diversity = 1.4010e+00	k_t
= [4, 2, 3, 4]				
Generation 950:	Best = 3.130449e+03	FES = 95100	Diversity = 1.3856e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 3.130420e+03	FES = 96100	Diversity = 1.3704e+00	k_t
= [2, 4, 2, 3]				
Generation 970:	Best = 3.130420e+03	FES = 97100	Diversity = 1.3778e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 3.130417e+03	FES = 98100	Diversity = 1.3766e+00	k_t
= [4, 2, 3, 4]				
Generation 990:	Best = 3.130410e+03	FES = 99100	Diversity = 1.3259e+00	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 3.130410e+03

Total Function Evaluations = 100000

=====

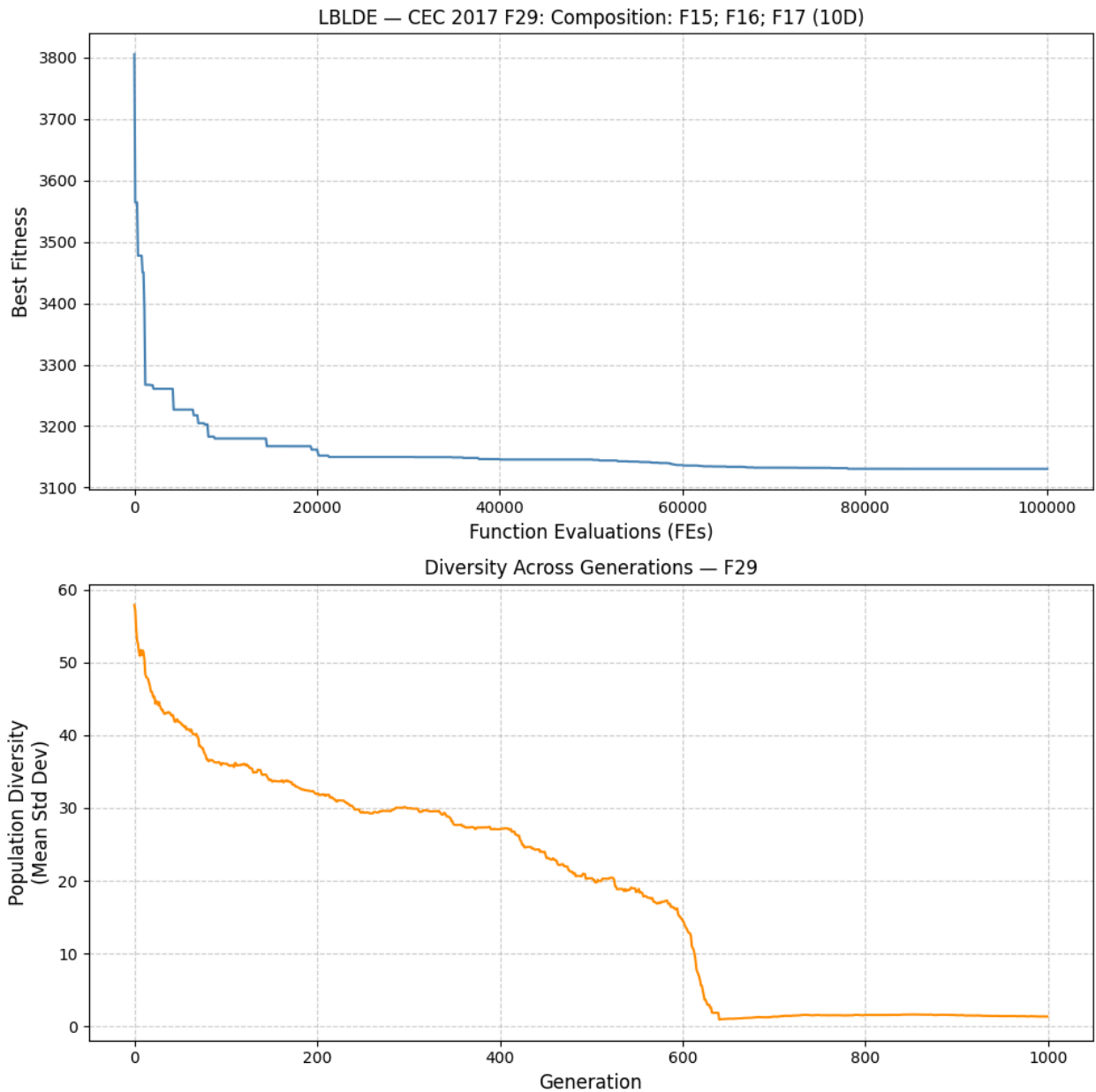
=

Results:

```

Best Fitness      : 3.130410e+03
Global Optimum    : 2.900000e+03
Error (|best-optimum|) : 2.304099e+02
Best Solution (5 dim) : [-76.27955177 -89.31982483 -70.27714802  95.63728074
-16.46353033]
D^(0) (init. diversity): 5.7928e+01
=====
=

```



Graph saved as convergence_F29.png

```

In [46]: # =====
# F30 — Composition Function 10 (F15; F18; F19)
# =====
run_and_plot('F30', make_wrapper(composition.f30), 3000.0, 'Composition: F15;

```

```

=====
=
LBLDE_s on CEC 2017 F30 – Single Run
=====
=
Function      : Composition: F15; F18; F19 (10D)
Bounds       : [-100, 100]
Global optimum : 3000.0
Initial D^(0) = 5.7928e+01
Generation 0: Best Fitness = 1.165390e+08 | Diversity = 5.7928e+01
Generation 10: Best = 2.122833e+07 FES = 1100 Diversity = 5.2644e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.964703e+06 FES = 2100 Diversity = 4.5793e+01 k_t
= [4, 3, 2, 3]
Generation 30: Best = 2.964703e+06 FES = 3100 Diversity = 3.8544e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.560880e+06 FES = 4100 Diversity = 3.3766e+01 k_t
= [4, 2, 3, 2]
Generation 50: Best = 1.446030e+06 FES = 5100 Diversity = 2.9258e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 4.589767e+05 FES = 6100 Diversity = 2.4587e+01 k_t
= [3, 4, 3, 2]
Generation 70: Best = 4.589767e+05 FES = 7100 Diversity = 2.0368e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.681061e+05 FES = 8100 Diversity = 1.6231e+01 k_t
= [2, 3, 4, 2]
Generation 90: Best = 2.681061e+05 FES = 9100 Diversity = 1.5995e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.576481e+05 FES = 10100 Diversity = 1.5739e+01 k_t
= [3, 3, 2, 3]
Generation 110: Best = 2.477160e+05 FES = 11100 Diversity = 1.7602e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.270226e+05 FES = 12100 Diversity = 1.8633e+01 k_t
= [2, 2, 4, 4]
Generation 130: Best = 1.710135e+05 FES = 13100 Diversity = 2.0093e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.710135e+05 FES = 14100 Diversity = 2.3427e+01 k_t
= [3, 3, 4, 2]
Generation 150: Best = 1.685753e+05 FES = 15100 Diversity = 2.5424e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.583119e+05 FES = 16100 Diversity = 2.6294e+01 k_t
= [4, 4, 4, 4]
Generation 170: Best = 1.583119e+05 FES = 17100 Diversity = 1.7944e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 5.538240e+04 FES = 18100 Diversity = 9.3309e+00 k_t
= [4, 3, 3, 2]
Generation 190: Best = 2.243236e+04 FES = 19100 Diversity = 1.2356e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.609541e+04 FES = 20100 Diversity = 1.3288e+01 k_t
= [4, 3, 2, 2]
Generation 210: Best = 6.205356e+03 FES = 21100 Diversity = 1.5249e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 5.322386e+03 FES = 22100 Diversity = 1.4005e+01 k_t
= [4, 3, 4, 4]

```

Generation 230:	Best = 4.684974e+03	FES = 23100	Diversity = 1.3098e+01	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 4.054112e+03	FES = 24100	Diversity = 1.2624e+01	k_t
= [2, 3, 2, 4]				
Generation 250:	Best = 4.007778e+03	FES = 25100	Diversity = 1.1592e+01	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 3.839039e+03	FES = 26100	Diversity = 1.0332e+01	k_t
= [3, 2, 4, 4]				
Generation 270:	Best = 3.669030e+03	FES = 27100	Diversity = 1.0651e+01	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 3.668286e+03	FES = 28100	Diversity = 9.7338e+00	k_t
= [4, 3, 3, 3]				
Generation 290:	Best = 3.553560e+03	FES = 29100	Diversity = 8.0266e+00	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 3.537808e+03	FES = 30100	Diversity = 7.3501e+00	k_t
= [4, 2, 2, 3]				
Generation 310:	Best = 3.514169e+03	FES = 31100	Diversity = 6.2785e+00	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 3.510125e+03	FES = 32100	Diversity = 5.8453e+00	k_t
= [3, 4, 4, 3]				
Generation 330:	Best = 3.494032e+03	FES = 33100	Diversity = 5.7120e+00	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 3.488456e+03	FES = 34100	Diversity = 5.0185e+00	k_t
= [2, 4, 3, 2]				
Generation 350:	Best = 3.485135e+03	FES = 35100	Diversity = 4.8194e+00	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 3.473952e+03	FES = 36100	Diversity = 4.7818e+00	k_t
= [3, 4, 2, 3]				
Generation 370:	Best = 3.473533e+03	FES = 37100	Diversity = 4.8178e+00	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 3.467061e+03	FES = 38100	Diversity = 4.5950e+00	k_t
= [4, 3, 3, 2]				
Generation 390:	Best = 3.462712e+03	FES = 39100	Diversity = 3.7864e+00	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 3.462712e+03	FES = 40100	Diversity = 4.3639e+00	k_t
= [2, 2, 2, 3]				
Generation 410:	Best = 3.456467e+03	FES = 41100	Diversity = 4.7990e+00	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 3.449745e+03	FES = 42100	Diversity = 4.7601e+00	k_t
= [3, 4, 3, 2]				
Generation 430:	Best = 3.443755e+03	FES = 43100	Diversity = 4.1106e+00	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 3.443755e+03	FES = 44100	Diversity = 3.9228e+00	k_t
= [2, 3, 3, 4]				
Generation 450:	Best = 3.443755e+03	FES = 45100	Diversity = 3.5839e+00	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 3.443202e+03	FES = 46100	Diversity = 3.4232e+00	k_t
= [3, 3, 3, 2]				
Generation 470:	Best = 3.439315e+03	FES = 47100	Diversity = 3.4232e+00	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 3.439315e+03	FES = 48100	Diversity = 3.3988e+00	k_t
= [4, 3, 3, 3]				
Generation 490:	Best = 3.437131e+03	FES = 49100	Diversity = 3.2178e+00	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 3.435911e+03	FES = 50100	Diversity = 3.1061e+00	k_t = [2, 3, 2, 4]
Generation 510:	Best = 3.435835e+03	FES = 51100	Diversity = 2.8624e+00	k_t = [1, 1, 1, 1]
Generation 520:	Best = 3.429657e+03	FES = 52100	Diversity = 2.5473e+00	k_t = [4, 4, 4, 2]
Generation 530:	Best = 3.424910e+03	FES = 53100	Diversity = 2.1447e+00	k_t = [1, 1, 1, 1]
Generation 540:	Best = 3.421925e+03	FES = 54100	Diversity = 1.8446e+00	k_t = [3, 2, 2, 2]
Generation 550:	Best = 3.421925e+03	FES = 55100	Diversity = 1.5367e+00	k_t = [1, 1, 1, 1]
Generation 560:	Best = 3.418138e+03	FES = 56100	Diversity = 1.0434e+00	k_t = [3, 2, 3, 4]
Generation 570:	Best = 3.418138e+03	FES = 57100	Diversity = 9.7396e-01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 3.418138e+03	FES = 58100	Diversity = 9.5321e-01	k_t = [4, 3, 4, 4]
Generation 590:	Best = 3.415744e+03	FES = 59100	Diversity = 8.3639e-01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 3.415744e+03	FES = 60100	Diversity = 7.5730e-01	k_t = [3, 2, 3, 4]
Generation 610:	Best = 3.413344e+03	FES = 61100	Diversity = 7.0884e-01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 3.412895e+03	FES = 62100	Diversity = 6.1451e-01	k_t = [4, 2, 4, 4]
Generation 630:	Best = 3.412851e+03	FES = 63100	Diversity = 5.2824e-01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 3.412851e+03	FES = 64100	Diversity = 4.9425e-01	k_t = [4, 2, 2, 3]
Generation 650:	Best = 3.411732e+03	FES = 65100	Diversity = 4.9615e-01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 3.411382e+03	FES = 66100	Diversity = 4.8253e-01	k_t = [3, 3, 2, 3]
Generation 670:	Best = 3.411277e+03	FES = 67100	Diversity = 4.4218e-01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 3.410227e+03	FES = 68100	Diversity = 4.3447e-01	k_t = [2, 4, 3, 3]
Generation 690:	Best = 3.410227e+03	FES = 69100	Diversity = 4.2043e-01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 3.410227e+03	FES = 70100	Diversity = 4.1788e-01	k_t = [4, 4, 4, 4]
Generation 710:	Best = 3.410227e+03	FES = 71100	Diversity = 4.1135e-01	k_t = [1, 1, 1, 1]
Generation 720:	Best = 3.410227e+03	FES = 72100	Diversity = 4.0538e-01	k_t = [4, 4, 2, 3]
Generation 730:	Best = 3.409710e+03	FES = 73100	Diversity = 3.7803e-01	k_t = [1, 1, 1, 1]
Generation 740:	Best = 3.409332e+03	FES = 74100	Diversity = 3.7979e-01	k_t = [4, 3, 2, 3]
Generation 750:	Best = 3.408663e+03	FES = 75100	Diversity = 3.9528e-01	k_t = [1, 1, 1, 1]
Generation 760:	Best = 3.408612e+03	FES = 76100	Diversity = 4.0290e-01	k_t = [3, 4, 4, 4]

Generation 770:	Best = 3.408612e+03	FES = 77100	Diversity = 4.0741e-01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 3.408277e+03	FES = 78100	Diversity = 4.0944e-01	k_t
= [3, 2, 2, 4]				
Generation 790:	Best = 3.408277e+03	FES = 79100	Diversity = 3.7438e-01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 3.408277e+03	FES = 80100	Diversity = 3.7518e-01	k_t
= [4, 4, 4, 2]				
Generation 810:	Best = 3.408277e+03	FES = 81100	Diversity = 3.7741e-01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 3.408277e+03	FES = 82100	Diversity = 3.8207e-01	k_t
= [2, 4, 2, 4]				
Generation 830:	Best = 3.407903e+03	FES = 83100	Diversity = 3.6099e-01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 3.407722e+03	FES = 84100	Diversity = 3.5047e-01	k_t
= [4, 3, 2, 2]				
Generation 850:	Best = 3.407720e+03	FES = 85100	Diversity = 3.2933e-01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 3.407695e+03	FES = 86100	Diversity = 3.1609e-01	k_t
= [2, 2, 3, 3]				
Generation 870:	Best = 3.407695e+03	FES = 87100	Diversity = 2.9210e-01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 3.407535e+03	FES = 88100	Diversity = 2.6928e-01	k_t
= [3, 3, 4, 2]				
Generation 890:	Best = 3.407535e+03	FES = 89100	Diversity = 2.5531e-01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 3.407535e+03	FES = 90100	Diversity = 2.5065e-01	k_t
= [2, 4, 2, 2]				
Generation 910:	Best = 3.407535e+03	FES = 91100	Diversity = 2.5213e-01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 3.407521e+03	FES = 92100	Diversity = 2.5153e-01	k_t
= [4, 3, 4, 3]				
Generation 930:	Best = 3.407519e+03	FES = 93100	Diversity = 2.4804e-01	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 3.407279e+03	FES = 94100	Diversity = 2.4404e-01	k_t
= [2, 4, 4, 2]				
Generation 950:	Best = 3.407279e+03	FES = 95100	Diversity = 2.3249e-01	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 3.407279e+03	FES = 96100	Diversity = 2.2491e-01	k_t
= [2, 2, 4, 4]				
Generation 970:	Best = 3.407279e+03	FES = 97100	Diversity = 2.1619e-01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 3.407279e+03	FES = 98100	Diversity = 2.0857e-01	k_t
= [4, 2, 4, 3]				
Generation 990:	Best = 3.407279e+03	FES = 99100	Diversity = 2.1166e-01	k_t
= [1, 1, 1, 1]				

Optimization completed!

Final Best Fitness = 3.407279e+03

Total Function Evaluations = 100000

=====

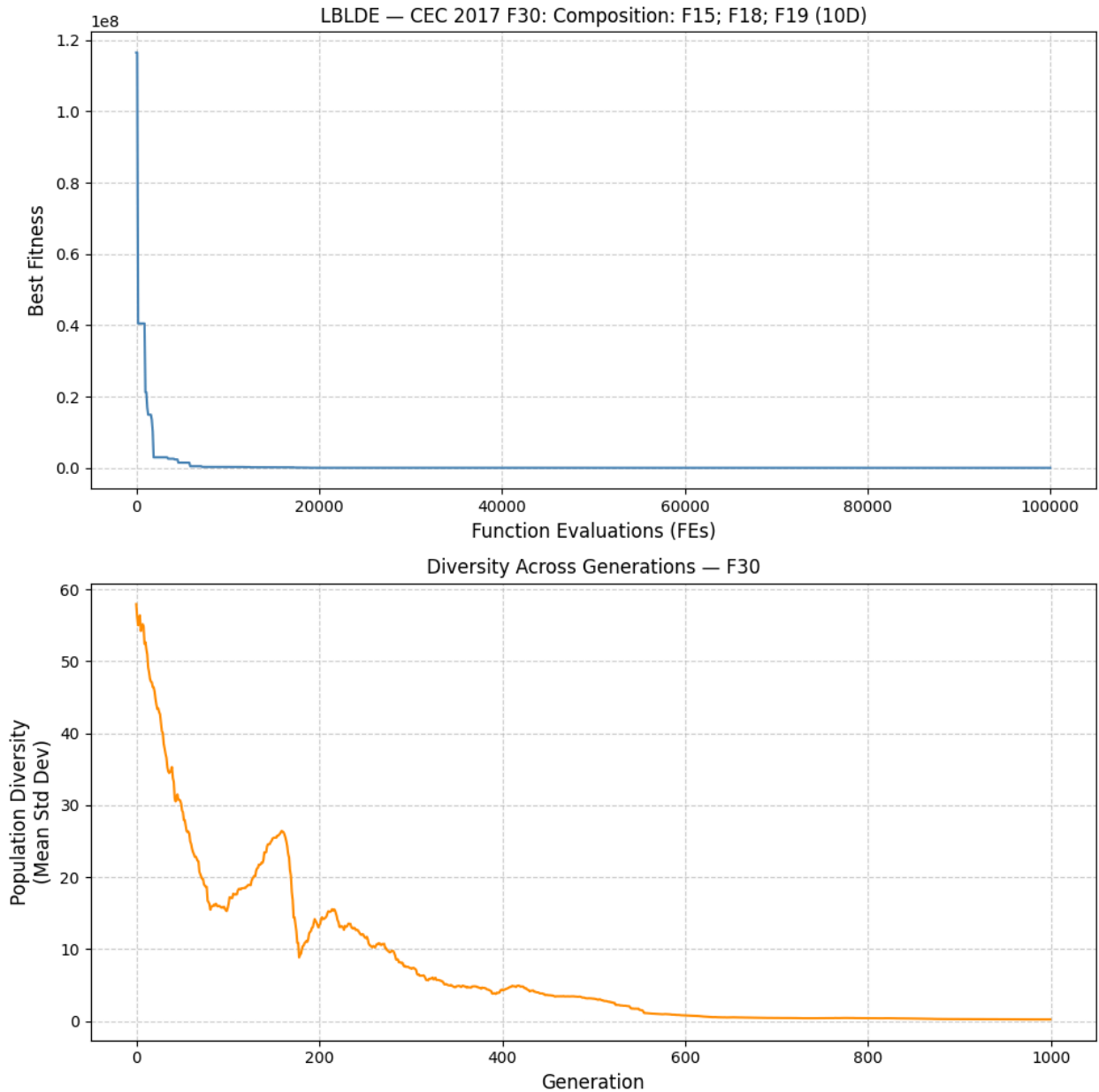
=

Results:

```

Best Fitness          : 3.407279e+03
Global Optimum        : 3.000000e+03
Error (|best-optimum|) : 4.072791e+02
Best Solution (5 dim)  : [-46.3194115   72.36478874  99.98732989  99.94834968
-94.14321017]
D^(0) (init. diversity): 5.7928e+01

```



Graph saved as convergence_F30.png

Section 7. Compiled Results

Print out a summary of all results in a tabular format

```
In [47]: print('=' * 80)
print('SUMMARY – LBLDE_s on CEC 2017 (10D), Single Run, Seed=42')
print('=' * 80)
print(f'{"Function":<10} {"Best Fitness":>18} {"Global Optimum":>18} {"Error |
print('-' * 70)
for fn in [f'F{i}' for i in range(1, 31)]:
    if fn not in all_results:
        continue
    r = all_results[fn]
    if r['best_fit'] is None:
        print(f'{fn:<10} {"N/A (deprecated)":>18} {r["f_global"]:>18.2f} {"N/A
    else:
        print(f'{fn:<10} {r["best_fit"]:>18.6e} {r["f_global"]:>18.2f} {r["err
print('=' * 80)
```

```
=====
=
SUMMARY – LBLDE_s on CEC 2017 (10D), Single Run, Seed=42
=====
=
```

Function	Best Fitness	Global Optimum	Error best-opt
F1	1.000000e+02	100.00	0.000000e+00
F2	2.000000e+02	200.00	0.000000e+00
F3	3.000000e+02	300.00	0.000000e+00
F4	4.005515e+02	400.00	5.514667e-01
F5	5.009950e+02	500.00	9.949591e-01
F6	6.000000e+02	600.00	0.000000e+00
F7	7.127291e+02	700.00	1.272909e+01
F8	8.000000e+02	800.00	0.000000e+00
F9	9.000000e+02	900.00	0.000000e+00
F10	1.018472e+03	1000.00	1.847221e+01
F11	1.100000e+03	1100.00	0.000000e+00
F12	1.200000e+03	1200.00	3.825973e-05
F13	1.300000e+03	1300.00	8.890311e-11
F14	1.400000e+03	1400.00	0.000000e+00
F15	1.500002e+03	1500.00	1.624597e-03
F16	1.600430e+03	1600.00	4.301387e-01
F17	1.700000e+03	1700.00	2.545672e-05
F18	1.800003e+03	1800.00	2.647263e-03
F19	1.900010e+03	1900.00	9.715910e-03
F20	2.000000e+03	2000.00	2.545513e-05
F21	2.304700e+03	2100.00	2.046999e+02
F22	2.300000e+03	2200.00	1.000000e+02
F23	2.604150e+03	2300.00	3.041496e+02
F24	2.734337e+03	2400.00	3.343373e+02
F25	2.899585e+03	2500.00	3.995850e+02
F26	2.900000e+03	2600.00	3.000000e+02
F27	3.089005e+03	2700.00	3.890054e+02
F28	3.100000e+03	2800.00	3.000000e+02
F29	3.130410e+03	2900.00	2.304099e+02
F30	3.407279e+03	3000.00	4.072791e+02

```
=====
=
```

```
In [48]: !zip plots.zip convergence_F{1..30}.png
```

adding: convergence_F1.png (deflated 21%)
adding: convergence_F2.png (deflated 21%)
adding: convergence_F3.png (deflated 21%)
adding: convergence_F4.png (deflated 20%)
adding: convergence_F5.png (deflated 19%)
adding: convergence_F6.png (deflated 21%)
adding: convergence_F7.png (deflated 20%)
adding: convergence_F8.png (deflated 18%)
adding: convergence_F9.png (deflated 21%)
adding: convergence_F10.png (deflated 18%)
adding: convergence_F11.png (deflated 20%)
adding: convergence_F12.png (deflated 19%)
adding: convergence_F13.png (deflated 19%)
adding: convergence_F14.png (deflated 19%)
adding: convergence_F15.png (deflated 19%)
adding: convergence_F16.png (deflated 18%)
adding: convergence_F17.png (deflated 17%)
adding: convergence_F18.png (deflated 19%)
adding: convergence_F19.png (deflated 19%)
adding: convergence_F20.png (deflated 18%)
adding: convergence_F21.png (deflated 19%)
adding: convergence_F22.png (deflated 21%)
adding: convergence_F23.png (deflated 19%)
adding: convergence_F24.png (deflated 19%)
adding: convergence_F25.png (deflated 21%)
adding: convergence_F26.png (deflated 20%)
adding: convergence_F27.png (deflated 20%)
adding: convergence_F28.png (deflated 19%)
adding: convergence_F29.png (deflated 17%)
adding: convergence_F30.png (deflated 18%)